

Strict growth and the Catalan limit for adjacency-bounded 132-avoiding permutations

A proof of the remaining strict-growth conjecture
and a two-term large-bound asymptotic

Research report prepared for Vladimir Reshetnikov

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Abstract

Let $a_n^{(m)}$ count permutations $\pi \in S_n$ avoiding the pattern 132 and satisfying $|\pi_{i+1} - \pi_i| \leq m$. Nadler conjectured that their exponential growth constants increase strictly from 1 to 4. Mayama and Akita established a finite-state enumeration, the existence of the growth constants, non-strict monotonicity, and convergence to 4. We prove the missing strict inequality for every adjacency bound. The argument embeds each of the two irreducible cyclic state components at bound m into a proper principal submatrix of its counterpart at bound $m + 1$ by increasing both finite endpoint thresholds by one. Perron–Frobenius comparison then gives strict growth in *each* component, independently of which component dominates.

A separate counting argument gives

$$4 - \alpha_m = \frac{2 \log m + 4 \log \log m + O(1)}{m}.$$

It compares the full class with a scalar Catalan majorant and an injectively encoded family of variable-length skew blocks. A truncated-series root lemma also bounds the possible constant term. Complete proofs, exact generating functions for $m \leq 4$, integer sequences, and independently checked rational Perron certificates accompany the report.

Addition of 28 September 2026 (batch 38). Part II, from a later manuscript, proves $\alpha_m < \beta_m$ for the root β_m^{-1} of $\sum_{k=1}^m C_{k-1}x^k = 1$ and $\beta_m - \alpha_m = O((\log m)^2/m^2)$; hence the constant term converges, $m(4 - \alpha_m) - 2 \log m - 4 \log \log m \rightarrow 2 \log \pi$, and the growth deficit has an expansion to every inverse-logarithmic order.

Addition of 29 September 2026 (batch 43). Part III, from a further manuscript, takes the bound near its trivial value, $m = n - d$. The deficit $n - M_n$ of the largest adjacent jump of a uniform 132-avoider converges in total variation, at the sharp rate $n^{-1/2}$, to an explicit algebraic law with $\mathbb{P}(D > k) \sim 3/\sqrt{\pi k}$; uniformly, $a_n^{(n-d)}/C_n = 3/\sqrt{\pi d} + O(d^{-3/2} + n^{-1/2})$. This answers the joint limit in n and m near the boundary only; the bulk regime stays open.

Addition of 29 September 2026 (batch 52). Part IV, merged from two manuscripts, finds the limit profile of $\sqrt{n} \mathbb{P}(n - M_n > un)$ and the leading constant of every moment of order $p > 1/2$; for example $\mathbb{E}(n - M_n) \sim 3\sqrt{n/\pi}$.

Addition of 30 September 2026 (batch 64). Part V shows that $\mathbb{E}\sqrt{n - M_n} - (3/(2\sqrt{\pi})) \log n$ converges to an explicit constant and finds the first correction to every moment of order $p < 1/2$.

Addition of 30 September 2026 (batch 68). Part VI proves $\mathbb{P}(M_n \leq \lfloor \theta n \rfloor) = \Theta(n^{-\lfloor 1/\theta \rfloor/2})$ for every fixed $0 < \theta < 1$; at $m = n/2$ the order is n^{-1} and it changes on the scale $\sqrt{n}$.

Addition of 1 October 2026 (batch 71). Part VII proves $n \mathbb{P}(M_n \leq \lfloor n/2 + t\sqrt{n} \rfloor) \rightarrow 6 + 12t_+/\sqrt{\pi}$ and $\mathbb{P}(M_n \leq \lfloor \theta n \rfloor) \sim I_3(\theta)/(2\pi n)$ for $1/3 < \theta \leq 1/2$.

Addition of 1 October 2026 (batch 72). Part VIII finds every fixed phase amplitude, $\mathbb{P}(M_n \leq \lfloor \theta n \rfloor) \sim \kappa_p I_p(\theta) n^{-(p-1)/2}$ for $1/p < \theta < 1/(p-1)$, and the crossover limit $\mathfrak{R}_r + \mathfrak{Q}_r t_+^{r-1}$ at every $1/r$; Part VII's theorems are its first cases.

Addition of 1 October 2026 (batch 72). Part IX, merged from six manuscripts, takes $m \rightarrow \infty$ with $n/m \rightarrow \infty$: uniformly for $n \geq c_0 m \log m$, $\log \mathbb{P}(M_n \leq m) = -(n/m)t_m + O(n(\log m)^2/m^2 + \log n)$, with relative equivalents (a saddle sum, a theta modulation, Gamma sums) where $n(\log m)^2/m^2 \rightarrow 0$.

Scope and attribution. The finite-state system and fixed- m spectral description are established work, not claimed as new here. The contributions developed in this report are the all- m strict comparison and the refined asymptotic bounds. A targeted literature check found no later resolution of the strict inequality in the sources inspected. This is a research proof with reproducible checks, not a claim of external peer review or an exhaustive novelty certification.

Added 1 October 2026, batch 72. A. C. Li publicly announced on 8 September 2026, eleven days before this Part's date, a proof that the exponential growth constants are strictly increasing in m , the strict-inequality part of Nadler's Conjecture 2 that Part I proves [7] (abstract inspected, full manuscript not inspected). Part I does not cite it, its proof does not use it, and the sources inspected by its literature check did not include it. The strict inequality is therefore not claimed as first proved here. This records an earlier external announcement; it is not a retraction (section 101.5).

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1 The problem and the results

A permutation $\pi \in S_n$ avoids 132 when there are no indices $i < j < k$ for which $\pi_i < \pi_k < \pi_j$. Fix an integer $m \geq 1$ and put

$$\mathcal{A}_n^{(m)} = \{\pi \in \text{Av}_n(132) : |\pi_{i+1} - \pi_i| \leq m \text{ for } 1 \leq i < n\}, \quad a_n^{(m)} = |\mathcal{A}_n^{(m)}|.$$

We set $a_0^{(m)} = 1$ and use the ordinary generating function

$$A_m(x) = \sum_{n \geq 0} a_n^{(m)} x^n.$$

The growth constant is

$$\alpha_m = \lim_{n \rightarrow \infty} (a_n^{(m)})^{1/n}. \quad (1)$$

The order of limits matters: $n \rightarrow \infty$ is taken for each fixed m , and only afterwards do we let $m \rightarrow \infty$. The trivial identity $a_n^{(m)} = C_n$ for $m \geq n - 1$ does not by itself settle this iterated limit. Here $C_n = \binom{2n}{n}/(n+1)$ is the n th Catalan number. [Added 29 September 2026, batch 43: Part III studies the joint regime next to that identity, $m = n - d$ with d fixed or $d = o(n)$ (theorems 22.1 and 22.2); it does not bear on the iterated limit taken here.]

1.1 Which part was still open?

Nadler's Conjecture 2 in Section 5.2 of [1] asks for the existence of (1), strict increase $1 = \alpha_1 < \alpha_2 < \dots < 4$, and limit 4. Mayama and Akita subsequently proved rationality for all m , existence of the limit, non-strict increase, and $4 - \alpha_m = O(\log m/m)$ [2, Theorems 2.4, 4.12, and 4.14]. Their monotonicity theorem states $\alpha_m \leq \alpha_{m+1}$, not strict inequality.

Merely finding additional permutations at bound $m+1$ is insufficient: sequences 2^n and $2^n + 1$ differ strictly at every index but have identical exponential growth. The needed argument must show that the extra freedom changes the exponential rate.

Theorem 1.1 (Strict increase). *For every integer $m \geq 1$,*

$$\alpha_{m+1} > \alpha_m.$$

In the two-component spectral representation described below, both component rates increase strictly: for every $m \geq 2$,

$$\lambda_U(m+1) > \lambda_U(m), \quad \lambda_V(m+1) > \lambda_V(m).$$

Consequently Nadler's growth-constant conjecture holds in full.

Theorem 1.2 (Two-term approach to the Catalan limit). *As $m \rightarrow \infty$ through integers,*

$$4 - \alpha_m = \frac{2 \log m + 4 \log \log m + O(1)}{m}. \quad (2)$$

In particular, $4 - \alpha_m \sim 2 \log m/m$. All logarithms in this report are natural logarithms.

Theorem 1.3 (A window for the next constant). *Define $E_m = m(4 - \alpha_m) - 2 \log m - 4 \log \log m$ for $m \geq 2$. Then*

$$2 \log \pi - 4 \log 2 \leq \liminf_{m \rightarrow \infty} E_m \leq \limsup_{m \rightarrow \infty} E_m \leq 2 \log \pi. \quad (3)$$

No convergence of E_m is asserted.

[Added 28 September 2026, batch 38: the last sentence describes the report at its pinned state 6c9175a54. Part II proves that E_m converges to the upper endpoint, $E_m \rightarrow 2 \log \pi$ (theorem 10.2 and corollary 15.3).]

The strict comparison uses a finite-state embedding. The asymptotic results use scalar comparison series and do not require deciding which finite-state component has the larger rate. The distinction is important: the all- m comparison of $\lambda_U(m)$ with $\lambda_V(m)$ is a separate question.

2 The exact endpoint recursion

This section develops the endpoint representation of Mayama and Akita [2, Section 2], including the counting details needed later. It is included to make the subsequent proofs independently readable.

2.1 Decomposition at the maximum

If $\pi = L n R$ avoids 132, every entry of L exceeds every entry of R : otherwise an entry of L , then n , then an entry of R forms 132. Conversely, if this separation holds and both standardized sides avoid 132, their concatenation with n avoids 132.

Suppose R is nonempty and the maximum is at position k . Then L has length $k - 1$, its values are $n - k + 1, \dots, n - 1$, and R has values $1, \dots, n - k$. The jump from n to the first entry of R is at least k . Therefore

$$R \neq \emptyset \implies 1 \leq k \leq m. \quad (4)$$

The alternative $R = \emptyset$ means appending the maximum to an avoider of length $n - 1$.

2.2 Cumulative endpoint thresholds

Let

$$B_m = \{0, 1, \dots, m - 1, \infty\}.$$

For $p, q \in B_m$, let $T_{p,q}^{(m)}(n)$ count the permutations in $\mathcal{A}_n^{(m)}$ satisfying

$$n - \pi_1 \leq p, \quad n - \pi_n \leq q.$$

A threshold ∞ imposes no restriction. Define

$$F_{p,q}^{(m)}(x) = \sum_{n \geq 1} T_{p,q}^{(m)}(n) x^n.$$

A negative threshold gives the zero series, and $\infty - k = \infty$. We suppress the superscript (m) where it is fixed.

For $k \geq 2$ define the nonnegative integer

$$c_{k,p} = |\{\sigma \in \text{Av}_{k-1}(132) : k - \sigma_1 \leq p\}|, \quad c_{1,p} = 1 \quad (5)$$

for nonnegative or infinite p ; for negative p set $c_{k,p} = 0$. These coefficients do not otherwise depend on m . Two properties will be crucial:

$$c_{k,p+1} \geq c_{k,p}, \quad c_{k,\infty} = C_{k-1}. \quad (6)$$

For finite $p \geq 0$ and $k \geq 2$, an explicit formula is

$$c_{k,p} = \sum_{j=1}^{\min(p,k-1)} \frac{j}{k-1} \binom{2k-j-3}{k-2}. \quad (7)$$

A generating-function proof of this formula appears in section 5.3; no numerical fitting is involved.

Proposition 2.1 (Endpoint recurrence). *For all $p, q \in B_m$,*

$$F_{p,q} = x + \sum_{k=1}^m c_{k,p} x^k F_{m-k,q-k} + x F_{p-1,m-1}. \quad (8)$$

Moreover $A_m(x) = 1 + F_{\infty,\infty}(x)$.

Proof. The singleton permutation contributes x to every nonnegative threshold state. For length at least two, first suppose $R \neq \emptyset$ in the maximum decomposition. By (4), $k \leq m$. The left block has $k - 1$ values in an interval of that length, so its internal jumps, and its jump to n , are automatically at most m . Its first-entry restriction gives $c_{k,p}$ choices; for $k = 1$ the first entry is n and the left block is empty.

The right block has length $n - k$. Writing its first and last entries as R_1 and R_{n-k} , the two remaining restrictions are exactly

$$(n - k) - R_1 \leq m - k, \quad (n - k) - R_{n-k} \leq q - k.$$

The first inequality is the jump bound from n to R_1 ; the second is the last-entry threshold for the whole permutation. This gives $c_{k,p} x^k F_{m-k, q-k}$.

If $R = \emptyset$, remove the final maximum. The first deficiency increases by one when the maximum is appended, giving threshold $p - 1$ for the smaller permutation. Its last deficiency must be at most $m - 1$ to permit the last jump. The final entry of the whole permutation is n , so any nonnegative q is satisfied. This is $x F_{p-1, m-1}$. The cases are disjoint and reversible. $\square$

Let $W_m(x)$ have rows and columns indexed by B_m^2 , with

$$(W_m)_{(p,q),(r,s)} = \sum_{k=1}^m c_{k,p} x^k \mathbf{1}_{(r,s)=(m-k, q-k)} + x \mathbf{1}_{(r,s)=(p-1, m-1)}. \quad (9)$$

Invalid negative thresholds are omitted. Rows are the left-hand states of (8); columns are the right-hand states. Thus

$$F = x\mathbf{1} + W_m(x)F, \quad F = x(I - W_m(x))^{-1}\mathbf{1}. \quad (10)$$

Every entry of W_m has zero constant term, so the inverse is well-defined as a formal power series. This also proves rationality and gives an exact algorithm for computing A_m .

3 Cyclic components and growth constants

The following fixed- m structure is established in [2, Proposition 2.7 and Theorem 4.12]. We supply elementary component and transfer-graph arguments to make the spectral input explicit.

3.1 The two relevant components

For $m \geq 2$ put

$$U_m = \{(p, \infty) : 0 \leq p \leq m - 1\}, \quad (11)$$

$$V_m = \{(p, q) : 0 \leq q < p \leq m - 1\} \cup \{(p, m - 1) : 0 \leq p \leq m - 2\}. \quad (12)$$

Their sizes are m and $(m - 1)(m + 2)/2$. Write W_{U_m} and W_{V_m} for the corresponding principal matrices. A directed edge means a nonzero term in a row of (9), directed from its row state to its column state.

Lemma 3.1 (Component structure). *At every real $x > 0$, both $W_{U_m}(x)$ and $W_{V_m}(x)$ are irreducible. The only other cyclic component of the full state graph is the singleton $I_m = (\infty, m - 1)$, whose internal matrix is (x) . Each of these three components is reachable from (∞, ∞) .*

Proof. Within U_m , every state has a $k = 1$ edge to $(m - 1, \infty)$. From that state, the $k = 1, \dots, m$ edges reach every state of U_m , since $c_{k, m-1} > 0$. Hence U_m is strongly connected.

For V_m , call the states with $q < p$ lower states and the states $(p, m - 1)$ top states. Every lower state has an append edge to a top state. Every top state has a $k = 1$ edge to

$$e = (m - 1, m - 2).$$

It remains to show that e reaches every state. For $m = 2$ this is the immediate two-state cycle. Suppose $m \geq 3$. Appending at e reaches $(m - 2, m - 1)$; its split edges reach all lower states $(p, p - 1)$, $1 \leq p \leq m - 1$. Their append edges reach every top state. Inductively, suppose all lower states of gap at most $d - 1$ are reachable, where $2 \leq d \leq m - 1$. The state $(m - 1, m - d)$ is then reachable. Its split edge with $k = m - p$ reaches $(p, p - d)$ for every $d \leq p \leq m - 1$. Thus every lower gap is reached.

For finite q , every valid split edge ends at a lower state, because $(m - k) - (q - k) = m - q > 0$. Every valid append edge with finite p ends at a top state. Therefore a finite-finite state outside V_m cannot be on a cycle. A state (∞, q) with finite q has append target I_m and split targets in V_m ; among these states only I_m has a self-loop. No edge returns to (∞, ∞) . Edges with second coordinate ∞ stay in U_m or leave it by an append edge, and cannot return from a finite second coordinate. This exhausts all possibilities.

Finally, the $k = 1$ split from (∞, ∞) reaches U_m , its append edge reaches I_m , and a $k = 1$ split from I_m reaches $e \in V_m$. $\square$

3.2 The strict matrix comparison used in the proof

We use the Perron–Frobenius theorem for finite nonnegative matrices: an irreducible matrix has positive left and right eigenvectors at its spectral radius; a primitive matrix has powers asymptotic to a positive rank-one matrix times the corresponding power of that radius.

Lemma 3.2 (Proper principal comparison). *Let M be an irreducible nonnegative matrix and let S be a nonempty proper subset of its indices. If $B = M[S, S]$, then $\text{spr}(B) < \text{spr}(M)$. If $0 \leq N \leq B$ entrywise, then $\text{spr}(N) < \text{spr}(M)$ as well.*

Proof. Take a nonnegative nonzero Perron vector v of B and extend it by zeros to a vector w on all indices of M . Then $Mw \geq \text{spr}(B)w$. There is strict inequality in at least one coordinate: the support of w is nonempty and proper, and irreducibility supplies an edge from its complement to that support. If u is a positive left Perron vector of M , multiplication by $u^\top$ gives

$$\text{spr}(M)u^\top w > \text{spr}(B)u^\top w.$$

Since $u^\top w > 0$, the first assertion follows. Entrywise monotonicity of the spectral radius gives $\text{spr}(N) \leq \text{spr}(B)$; for example, this monotonicity follows by comparing powers and then taking matrix-norm roots. $\square$

For $C = U, V$, there is a unique $r_C(m) > 0$ satisfying

$$\text{spr}(W_{C_m}(r_C(m))) = 1, \quad \lambda_C(m) = r_C(m)^{-1}. \quad (13)$$

Indeed the spectral radius is continuous, tends to zero as $x \downarrow 0$, and tends to infinity as $x \rightarrow \infty$ because the component has a positive cycle. If $y > x > 0$, every monomial has degree at least one, so $W_{C_m}(y) \geq (y/x)W_{C_m}(x)$. Its positive spectral radius is therefore strictly increasing.

Proposition 3.3 (Fixed-bound growth). *For $m \geq 2$ the limit (1) exists and*

$$\alpha_m = \max\{\lambda_U(m), \lambda_V(m)\}. \quad (14)$$

Also $\lambda_U(m) > 1$.

Proof. The state $(m-1, \infty)$ has an internal self-loop of weight x . At $x = 1$ its singleton principal matrix is (1) , and it is a proper principal matrix of $W_{U_m}(1)$. By lemma 3.2, the latter has spectral radius greater than one. Thus $r_U(m) < 1$ and $\lambda_U(m) > 1$.

To justify the actual limit rather than only a limsup, replace each edge of weight cx^k by a directed path of k unit-length edges with total multiplicity c . Use fresh intermediate vertices for every edge. Let H be the resulting finite nonnegative adjacency matrix, and let b be one on original endpoint states and zero on intermediate states. With o the state (∞, ∞) , (10) gives

$$a_n^{(m)} = e_o^T H^{n-1} b \quad (n \geq 1).$$

For an expanded irreducible component with Perron eigenvalue λ , eliminating the intermediate coordinates of a positive eigenvector gives $z = W_{C_m}(1/\lambda)z$. Its eigenvalue is therefore $\lambda_C(m)$. The expanded singleton component has eigenvalue one. The component classification and $\lambda_U(m) > 1$ show that $\text{spr}(H)$ is the right-hand side of (14).

The expanded U_m component is primitive, since it is strongly connected and has a length-one loop. For $m \geq 3$, the expanded V_m component has closed walks of lengths two and five. The first is

$$e \xrightarrow{1} (m-2, m-1) \xrightarrow{1} e.$$

For the second, split with $k = 2$ on the middle state and then append and split with $k = 1$:

$$e \xrightarrow{1} (m-2, m-1) \xrightarrow{2} (m-2, m-3) \xrightarrow{1} (m-3, m-1) \xrightarrow{1} e.$$

The labels are edge lengths. All coefficients are positive, including when $m = 3$. Their greatest common divisor is one, so this component is primitive. When $m = 2$, V_m is a two-cycle with rate one and is not dominant.

A dominant primitive component is reachable from o , and b is positive on its original states. Fixing an entrance path and then applying the primitive Perron theorem supplies $a_n^{(m)} \geq c \text{spr}(H)^n$ for all sufficiently large n , with $c > 0$. A finite-matrix power bound gives $a_n^{(m)} \leq Cn^D \text{spr}(H)^n$ for some fixed C, D (or, equivalently, an upper bound by $(\text{spr}(H) + \varepsilon)^n$ for each $\varepsilon > 0$). Taking n th roots proves (14) and the existence of the limit. $\square$

4 A shift of endpoint thresholds proves strict increase

The key comparison preserves the split length k , not merely the underlying unweighted graph. Define

$$\phi(p) = \begin{cases} p+1, & p < \infty, \\ \infty, & p = \infty, \end{cases} \quad \Phi(p, q) = (\phi(p), \phi(q)).$$

The next lemma is the essential new observation.

Lemma 4.1 (Coefficientwise shift embedding). *For every $m \geq 2$ and $C = U, V$, the map Φ embeds C_m into a proper subset of C_{m+1} . Under this identification,*

$$W_{C_m}(x) \leq W_{C_{m+1}}(x)[\Phi(C_m), \Phi(C_m)] \quad (15)$$

coefficientwise in every matrix entry.

Proof. A state (p, ∞) of U_m maps to $(p+1, \infty)$ of U_{m+1} ; the state $(0, \infty)$ of the latter is missing. A lower state (p, q) of V_m maps to a lower state $(p+1, q+1)$ of V_{m+1} . A top state $(p, m-1)$ maps to $(p+1, m)$, a top state at the new bound. The new top state $(0, m)$ is missing, so that inclusion is also proper.

Consider a split term at row (p, q) of the old matrix, with column $(m - k, q - k)$ and coefficient $c_{k,p}x^k$. At the new row $\Phi(p, q)$, the split with the same k has column

$$(m + 1 - k, \phi(q) - k) = \Phi(m - k, q - k),$$

whenever the old term is valid. Its coefficient is $c_{k,\phi(p)}x^k \geq c_{k,p}x^k$. For $p = \infty$ the coefficients are equal. For a valid append term, the old column $(p - 1, m - 1)$ maps to

$$(\phi(p) - 1, m),$$

which is exactly the new append column, and the weight remains x . Each term is preserved with at least its old multiplicity. Adding terms with the same row and column proves the coefficientwise assertion. $\square$

Proof of theorem 1.1. Fix $m \geq 2$, $C \in \{U, V\}$, and a positive real x . By lemmas 3.1, 3.2 and 4.1,

$$\begin{aligned} \text{spr}(W_{C_m}(x)) &\leq \text{spr}(W_{C_{m+1}}(x)[\Phi(C_m), \Phi(C_m)]) \\ &< \text{spr}(W_{C_{m+1}}(x)). \end{aligned}$$

At $x = r_C(m)$ the first quantity is one. Strict increase of the new spectral radius as a function of x gives $r_C(m + 1) < r_C(m)$, hence $\lambda_C(m + 1) > \lambda_C(m)$.

Choose a component C realizing the maximum at bound m . Then

$$\alpha_{m+1} \geq \lambda_C(m + 1) > \lambda_C(m) = \alpha_m.$$

This works even if the dominant component changes between the two bounds.

For $m = 1$, any adjacent difference is ± 1 , and reversing direction would repeat a value. The only permutations of length $n \geq 2$ are the increasing and decreasing ones, both avoiding 132. Thus $\alpha_1 = 1$. At $m = 2$, in the order $(0, \infty), (1, \infty)$,

$$W_{U_2}(x) = \begin{pmatrix} 0 & x \\ x^2 & x \end{pmatrix}, \quad \det(I - W_{U_2}(x)) = 1 - x - x^3.$$

Its positive root is below one, so $\alpha_2 \geq \lambda_U(2) > 1$. This covers the remaining step. $\square$

Remark 4.2. Strictness comes from a *proper principal extension of an irreducible component*. It does not require finding a strictly larger entry inside the old image. This is useful for the U component, where many coefficients are preserved exactly. Nor is a numerical lower bound on the gap between consecutive rates needed.

5 Scalar upper bounds and injective skew-block lower bounds

We next obtain two scalar comparison series whose root asymptotics have the same first two terms. Write $P \leq Q$ for coefficientwise inequality of formal power series with nonnegative coefficients.

5.1 A Catalan majorant

Define

$$R_m(x) = x + \sum_{k=1}^m C_{k-1}x^k. \tag{16}$$

The separate initial x accounts for appending a maximum, whereas the $k = 1$ term in the sum accounts for a maximum in the first position with a nonempty right block. Thus the coefficient of x in R_m is two.

Proposition 5.1 (Scalar upper bound). *Let $q_m > 0$ be the unique positive root of $R_m(q_m) = 1$. Then*

$$A_m(x) \leq 1 + \frac{x}{1 - R_m(x)}, \quad \alpha_m \leq q_m^{-1} < 4. \quad (17)$$

Proof. The maximum decomposition, with all endpoint restrictions discarded, gives for $n \geq 2$

$$a_n^{(m)} \leq a_{n-1}^{(m)} + \sum_{k=1}^{\min(m, n-1)} C_{k-1} a_{n-k}^{(m)}.$$

The upper endpoint $n - 1$ is essential: the right block in a split must be nonempty. Starting with $a_1^{(m)} = 1$, induction yields the generating-function majorant. Its radius is q_m : the denominator is nonzero for $|x| < q_m$ by positivity, and it vanishes at q_m without numerator cancellation. Cauchy–Hadamard therefore gives $\alpha_m \leq q_m^{-1}$.

Finally,

$$R_m(1/4) < \frac{1}{4} + \sum_{k \geq 1} C_{k-1} 4^{-k} = \frac{1}{4} + \frac{1}{2} = \frac{3}{4} < 1.$$

Hence $q_m > 1/4$ and the strict upper bound by four follows. $\square$

5.2 Variable-length blocks with recoverable boundaries

Fix $d \geq 1$ and $m > d$. An admissible block of length k , where $1 \leq k \leq m - d$, is a permutation $\beta \in \text{Av}_k(132)$ such that

$$\beta_k = k, \quad k - \beta_1 \leq d.$$

There are $c_{k,d}$ such blocks: for $k \geq 2$, deleting the last maximum leaves the permutations counted in (5); the singleton contributes $c_{1,d} = 1$.

For a sequence of blocks, assign consecutive value intervals in decreasing order. That is, every entry of an earlier block exceeds every entry of a later block. This operation is their *skew sum*. Let

$$K_{m,d}(x) = \sum_{k=1}^{m-d} c_{k,d} x^k. \quad (18)$$

Proposition 5.2 (Injective block lower bound). *Let $s_{m,d} > 0$ be the unique positive root of $K_{m,d}(s_{m,d}) = 1$. Then*

$$\frac{1}{1 - K_{m,d}(x)} \leq A_m(x), \quad s_{m,d}^{-1} \leq \alpha_m \leq q_m^{-1}. \quad (19)$$

Proof. A skew sum of 132-avoiding blocks avoids 132. If three indices are not all in one block, their first and last indices lie in different blocks, so the first value exceeds the last; this is incompatible with 132. Within a block, the largest possible jump is $k - 1 < m$.

For a boundary, let the earlier block have length k and let the next block have first-entry deficiency $j \leq d$. The earlier block ends at its maximum. Since the two value intervals are consecutive and decreasing, the boundary jump is exactly $k + j \leq m$. This also covers a singleton next block, where $j = 0$.

It remains to prove injectivity; this is not automatic for variable-length concatenations. A skew cut in a permutation is a proper cut for which every value on the left exceeds every value on the right. A block ending at its maximum has no internal skew cut, because its maximum lies to the right of every proper cut. Every actual boundary between the constructed blocks is a skew cut. Any other proposed cut is internal to a block and is not a skew cut. Consequently the complete set of skew cuts recovers exactly the block boundaries, and standardization recovers each block. Distinct block sequences cannot yield the same permutation.

The empty sequence and all nonempty sequences of admissible blocks are therefore counted by $1 + K + K^2 + \dots = (1 - K)^{-1}$ and inject into the desired class. Its radius is $s_{m,d}$ by the same positivity argument as above, which proves the growth lower bound. $\square$

For $d = 1$ this becomes particularly simple:

$$K_{m,1}(x) = x + \sum_{k=2}^{m-1} C_{k-2} x^k. \quad (20)$$

This one-parameter subfamily already proves theorem 1.2. Allowing arbitrary fixed d sharpens the bounded remainder in theorem 1.3.

5.3 Exact block weights and their Catalan tail

Let $h(k, j)$ count blocks of length $k \geq 2$ ending at their maximum and having first entry $k - j$, where $1 \leq j \leq k - 1$. Put

$$C(x) = \sum_{n \geq 0} C_n x^n = \frac{1 - \sqrt{1 - 4x}}{2x}, \quad C(x) = 1 + xC(x)^2.$$

Lemma 5.3 (First-entry generating function). *For each fixed $j \geq 1$,*

$$\sum_{k \geq j+1} h(k, j) x^k = x^{j+1} C(x)^j, \quad h(k, j) = \frac{j}{k-1} \binom{2k-j-3}{k-2}. \quad (21)$$

Consequently

$$K_{\infty,d}(x) := \sum_{k \geq 1} c_{k,d} x^k = x \sum_{j=0}^d (xC(x))^j. \quad (22)$$

Proof. Delete the final maximum from such a block. Its first entry is $a = k - j$. The $j - 1$ larger entries $a + 1, \dots, k - 1$ must appear in increasing order: a decreasing pair together with the initial a would form 132. The smaller entries lie in the j intervening slots, giving the form

$$a, \tau_0, a + 1, \tau_1, a + 2, \dots, k - 1, \tau_{j-1}.$$

Every value in an earlier small slot must exceed every value in a later small slot. Otherwise those two values and a separating large entry form 132. Each slot must itself avoid 132. Conversely these conditions are sufficient: triples meeting different small slots cannot form 132, the large entries are increasing, and any triple entirely in one small slot is excluded by its own avoidance condition. Once the slot lengths are specified, their value intervals are forced in decreasing order. There are therefore j independent Catalan slots and $j + 1$ fixed entries when the deleted final maximum is restored. This gives the first identity.

For completeness, put $Y = C(x) - 1$, so $Y = x(1 + Y)^2$. For $N \geq 1$, Lagrange inversion gives

$$[x^N] C(x)^j = \frac{j}{N} [u^{N-1}] (1 + u)^{2N+j-1} = \frac{j}{2N+j} \binom{2N+j}{N}.$$

The same final expression is one for $N = 0$. Substituting $N = k - j - 1$ and cancelling factorials gives the displayed formula for $h(k, j)$. Summing over $1 \leq j \leq d$, with the singleton added, proves (22) and (7). $\square$

Since $C(1/4) = 2$, (22) yields

$$B_d := \sum_{k \geq 1} c_{k,d} 4^{-k} = \frac{1}{2} (1 - 2^{-(d+1)}). \quad (23)$$

To obtain the tail constant without singularity-analysis machinery, compare each first-entry count directly with a Catalan number:

$$\frac{h(k, j)}{C_{k-2}} = j \prod_{r=0}^{j-2} \frac{k-2-r}{2k-4-r} \longrightarrow \frac{j}{2^{j-1}} \quad (k \rightarrow \infty). \quad (24)$$

The empty product for $j = 1$ is one. Stirling's formula gives $C_n \sim 4^n / (\sqrt{\pi} n^{3/2})$. For fixed d it follows that

$$c_{k,d} 4^{-k} \sim \frac{a_d}{4\sqrt{\pi}} k^{-3/2}, \quad a_d := \sum_{j=1}^d \frac{j}{2^{j+1}} = 1 - \frac{d+2}{2^{d+1}}. \quad (25)$$

Both a_d and B_d are strictly positive for $d \geq 1$, and $B_d < 1$.

6 A truncated-series root lemma

The following elementary estimate is the analytic step responsible for the $\log \log m$ term. It applies to both sides of (19).

Lemma 6.1 (A tilted $k^{-3/2}$ tail). *Let $b_k \geq 0$ satisfy*

$$b_k \sim c k^{-3/2}, \quad c > 0, \quad B := \sum_{k \geq 1} b_k < 1.$$

Fix an integer $h \geq 0$. For all sufficiently large m , let $r_m > 0$ be the unique positive solution of

$$\sum_{k=1}^{m-h} b_k (4r_m)^k = 1.$$

Then $r_m > 1/4$ and

$$m \log(4r_m) = \frac{1}{2} \log m + \log \log m + \log \frac{1-B}{2c} + o(1). \quad (26)$$

Proof. At $r = 1/4$ the truncated sum is at most $B < 1$. For large enough m the polynomial is nonconstant with a positive coefficient, strictly increases on the positive real axis, and tends to infinity. Thus the stated root exists uniquely and exceeds $1/4$.

For a fixed real s , write

$$t = t_m(s) = \frac{1}{2} \log m + \log \log m + s.$$

We first prove the limiting profile

$$\sum_{k=1}^{m-h} b_k e^{tk/m} \longrightarrow B + 2ce^s. \quad (27)$$

The untilted truncated sum tends to B . It remains to evaluate the increment from exponential tilting.

For $k \leq m/2$, use $e^u - 1 \leq ue^u$ for $u \geq 0$ to obtain

$$\begin{aligned} 0 \leq \sum_{k \leq m/2} b_k (e^{tk/m} - 1) &\leq \frac{t}{m} e^{t/2} \sum_{k \leq m/2} kb_k \\ &= O\left(\frac{te^{t/2}}{\sqrt{m}}\right) = O\left(m^{-1/4}(\log m)^{3/2}\right) = o(1). \end{aligned}$$

Here $b_k = O(k^{-3/2})$ implies $\sum_{k \leq M} kb_k = O(\sqrt{M})$.

For $k > m/2$, the untilted tail is $O(m^{-1/2}) = o(1)$. The assumed asymptotic for b_k holds uniformly in this range as $m \rightarrow \infty$. Put $j = m - k$. The tilted upper tail is therefore asymptotic to

$$cm^{-3/2}e^t \sum_{h \leq j < m/2} (1 - j/m)^{-3/2} e^{-tj/m}. \quad (28)$$

The sum in (28) is asymptotic to m/t . To check this precisely, for $j \leq m/\sqrt{t}$ the factor $(1 - j/m)^{-3/2}$ is $1 + O(t^{-1/2})$ uniformly. The geometric sum without that factor is

$$\sum_{j \geq h} e^{-tj/m} = \frac{e^{-th/m}}{1 - e^{-t/m}} \sim \frac{m}{t}.$$

The omitted part $j > m/\sqrt{t}$, with $j < m/2$, is bounded by a constant times $e^{-\sqrt{t}}/(1 - e^{-t/m}) = o(m/t)$, since $(1 - j/m)^{-3/2} \leq 2^{3/2}$. The tail beyond $m/2$ in the geometric sum is negligible by the same estimate. A fixed lower cutoff h has no effect on the asymptotic.

Thus (28) is asymptotic to

$$\frac{ce^t}{\sqrt{m}t} = \frac{ce^s \log m}{\frac{1}{2} \log m + \log \log m + s} \longrightarrow 2ce^s,$$

which proves (27).

Let $s_0 = \log((1 - B)/(2c))$. For each fixed $\varepsilon > 0$, the limiting value in (27) is below one at $s = s_0 - \varepsilon$ and above one at $s = s_0 + \varepsilon$. Monotonicity of the polynomial in t traps $m \log(4r_m)$ between $t_m(s_0 - \varepsilon)$ and $t_m(s_0 + \varepsilon)$ for all sufficiently large m . Since ε is arbitrary, (26) follows. $\square$

Remark 6.2 (Why the second logarithm appears). The effective large- k contribution comes from a boundary layer of width about m/t . Each coefficient near the cutoff is of order $m^{-3/2}$, and the exponential factor is e^t . Its total mass is consequently $e^t/(\sqrt{m}t)$. Requiring a nonzero limiting mass gives $t = \frac{1}{2} \log m + \log t + O(1)$, hence the additional $\log \log m$. The proof above supplies the constants and controls the discarded ranges.

7 The refined asymptotic and its constant window

Put

$$L_m = \frac{1}{2} \log m + \log \log m, \quad t_m^* = m \log \frac{4}{\alpha_m}.$$

We apply lemma 6.1 with two different nonnegative coefficient sequences.

For the upper-growth majorant R_m , write

$$R_m(x) = \sum_{k=1}^m b_k (4x)^k, \quad b_1 = \frac{1}{2}, \quad b_k = C_{k-1} 4^{-k} \quad (k \geq 2).$$

Then

$$B = \frac{3}{4}, \quad c = \frac{1}{4\sqrt{\pi}}, \quad h = 0.$$

Therefore

$$m \log(4q_m) = L_m + \log \frac{\sqrt{\pi}}{2} + o(1). \quad (29)$$

For the block lower bound, take $b_k = c_{k,d}4^{-k}$ and $h = d$, holding d fixed as $m \rightarrow \infty$. Equations (23) and (25) give

$$m \log(4s_{m,d}) = L_m + \gamma_d + o(1), \quad \gamma_d = \log \left(\frac{\sqrt{\pi}(1 + 2^{-(d+1)})}{1 - (d+2)2^{-(d+1)}} \right). \quad (30)$$

The $o(1)$ in this formula is for each fixed d ; no uniformity in a growing d is used.

Proof of theorem 1.2. From (19),

$$m \log(4q_m) \leq t_m^* \leq m \log(4s_{m,d}). \quad (31)$$

Already $d = 1$, for which $\gamma_1 = \log(5\sqrt{\pi})$, shows $t_m^* = L_m + O(1)$. Hence $\alpha_m = 4e^{-t_m^*/m}$ and

$$m(4 - \alpha_m) = 4m(1 - e^{-t_m^*/m}) = 4t_m^* + O\left(\frac{(t_m^*)^2}{m}\right) = 4t_m^* + o(1). \quad (32)$$

Since $4L_m = 2 \log m + 4 \log \log m$, the theorem follows. This also independently proves $\alpha_m \rightarrow 4$. $\square$

Proof of theorem 1.3. The lower side of (31), together with (29) and (32), gives

$$\liminf_{m \rightarrow \infty} E_m \geq 4 \log(\sqrt{\pi}/2) = 2 \log \pi - 4 \log 2.$$

For every fixed $d \geq 1$, the upper side gives

$$\limsup_{m \rightarrow \infty} E_m \leq 4\gamma_d.$$

Now let $d \rightarrow \infty$ after taking the limsup in m . Because $\gamma_d \rightarrow \log \sqrt{\pi}$, the upper bound becomes $2 \log \pi$. The middle inequality between liminf and limsup is automatic. $\square$

Remark 7.1 (The constants are not finite- m error bars). The endpoints in (3) are asymptotic bounds, not a claim that every E_m lies in that interval. Numerically the scalar comparisons approach their limits slowly. Even at $m = 10^6$, the best of the four computed fixed- d lower bounds gives a remainder upper bound above $2 \log \pi$. This is compatible with the theorem: a fixed d has limit $4\gamma_d$, and the root expansion also has a slowly vanishing error.

[Added 28 September 2026, batch 38: with the sharper upper bound and the growing cutoff of Part II, the computed corridor for E_m at $m = 10^6$ is about $[3.1495, 3.1503]$, still far above the proved limit $2 \log \pi$; see remark 17.1.]

8 Exact computations and reproducible certificates

The proofs above hold for all m and do not rest on a finite search. The computations serve three different purposes: checking the combinatorial implementation against direct enumeration, providing exact certificates for displayed finite cases, and illustrating the scalar bounds at large m . These evidence levels are kept separate.

Table 1: Initial counts computed from the exact endpoint recurrence. The empty permutation is included at $n = 0$.

n	$m = 1$	$m = 2$	$m = 3$	$m = 4$	$m = 5$	$m = 6$
0	1	1	1	1	1	1
1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	2	5	5	5	5	5
4	2	8	14	14	14	14
5	2	12	28	42	42	42
6	2	18	55	95	132	132
7	2	26	108	216	322	429
8	2	37	214	494	799	1101
9	2	53	412	1135	2005	2892
10	2	76	787	2613	5054	7714
11	2	109	1497	5892	12758	20714
12	2	157	2841	13218	32140	55697

8.1 Small sequences and rational generating functions

For example, exact symbolic elimination gives

$$A_1(x) = \frac{1+x^2}{1-x}, \quad A_2(x) = \frac{1-2x+2x^2-x^4-x^6}{(1-x)^2(1-x-x^3)}. \quad (33)$$

Thus, with our convention $a_0^{(2)} = 1$,

$$a_n^{(2)} = 3a_{n-1}^{(2)} - 3a_{n-2}^{(2)} + 2a_{n-3}^{(2)} - 2a_{n-4}^{(2)} + a_{n-5}^{(2)} \quad (n \geq 7).$$

The lower threshold on n comes from the numerator degree, not just the denominator degree. The archive includes exact numerator and denominator coefficient arrays for $m = 1, 2, 3, 4$, and coefficients through $n = 200$ for $m = 1, \dots, 12$.

These rational functions are obtained by exact polynomial-matrix inversion, not by guessing recurrences from the initial terms. More importantly, *the verifier does not need to trust the symbolic solver*. For every $m \leq 4$, the supplied certificate gives a common denominator $D(x)$ and numerator polynomials $N_{p,q}(x)$ for all endpoint states, and verifies exactly that

$$(I - W_m(x))N(x) = xD(x)\mathbf{1}, \quad D(0) = 1. \quad (34)$$

Uniqueness of the formal solution then proves all coefficients, not merely the first 200 checked by an additional test. Integer cross multiplication also verifies the reported reduced expression for $1 + F_{\infty, \infty}$.

8.2 Certified spectral intervals

For each $m = 2, \dots, 20$ and each component, a numerical eigensolver proposes an approximate positive Perron vector and an approximate radius. These proposals are converted to positive integer vectors and rational endpoints. They are accepted only when integer arithmetic verifies

$$W_{C_m}(\ell)v < v, \quad W_{C_m}(u)w > w \quad \text{coordinatewise}, \quad v, w > 0. \quad (35)$$

Positive left Perron vectors show that these imply $\text{spr}(W_{C_m}(\ell)) < 1 < \text{spr}(W_{C_m}(u))$. Consequently

$$\ell < r_C(m) < u, \quad \frac{1}{u} < \lambda_C(m) < \frac{1}{\ell}.$$

The verifier uses no floating-point operations for these assertions. If $\ell = p/D$, an edge of degree k contributes the integer factor $p^k D^{m-k}$ after clearing the common denominator D^m .

For example, the $m = 2$, U certificate gives

$$\frac{6823278035}{10^{10}} < r_U(2) < \frac{6823278042}{10^{10}}.$$

All 38 component certificates pass. Their disjoint intervals additionally certify the component comparison for every $m = 2, \dots, 20$: U dominates for $m = 2, 3, 4$ and V for $m = 5, \dots, 20$. This finite conclusion is not extrapolated into an all- m theorem.

Table 2: Component growth rates, rounded to six decimals. The component ordering is certified by rational intervals; displayed decimals are rounded numerical values.

m	$\lambda_U(m)$	$\lambda_V(m)$	α_m	Dominant
2	1.465571	1.000000	1.465571	U
3	1.826516	1.691403	1.826516	U
4	2.100456	2.091477	2.100456	U
5	2.312119	2.351509	2.351509	V
6	2.479693	2.535762	2.535762	V
8	2.727583	2.785681	2.785681	V
10	2.902175	2.952689	2.952689	V
12	3.032016	3.074770	3.074770	V
15	3.174821	3.208325	3.208325	V
20	3.333400	3.356796	3.356796	V

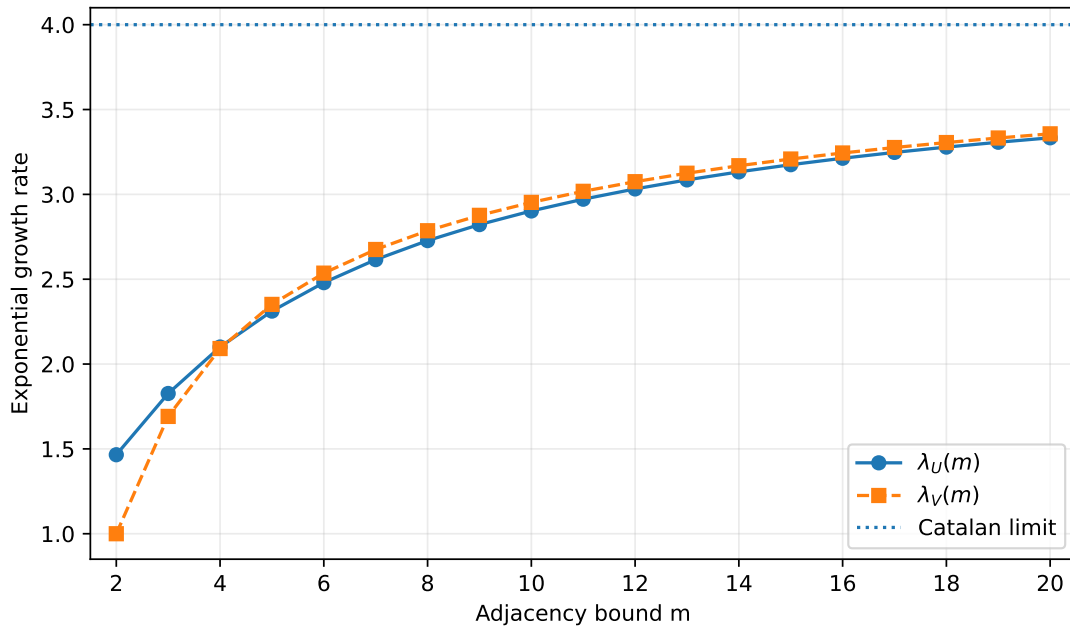


Figure 1: The two component rates for $2 \leq m \leq 20$. Both increase, and the component attaining the maximum changes between $m = 4$ and $m = 5$. The proof of strict growth does not depend on this crossover pattern.

8.3 Large- m scalar comparisons

The scalar bounds can be evaluated without matrices of order m^2 . The implementation stores normalized Catalan coefficients $C_k 4^{-k}$, avoiding enormous intermediate Catalan integers in this

numerical calculation. It solves for $t = m \log(4x)$, not for a root whose distance from $1/4$ is tiny.

Table 3: Approximate evaluation of the rigorous scalar root bounds. The lower column is $\max_{d \in \{1,2,4,8\}} s_{m,d}^{-1}$; the upper is q_m^{-1} . These decimal evaluations are floating-point results, not outward-rounded interval certificates.

m	Lower-bound value	Upper-bound value
10	2.7001330054	3.2847596398
20	3.2358742549	3.5390478302
50	3.6448786614	3.7589497022
100	3.8065234928	3.8581087164
1000	3.9751398963	3.9790979985
10000	3.9969422752	3.9972999790
100000	3.9996392834	3.9996731933
1000000	3.9999586173	3.9999618975

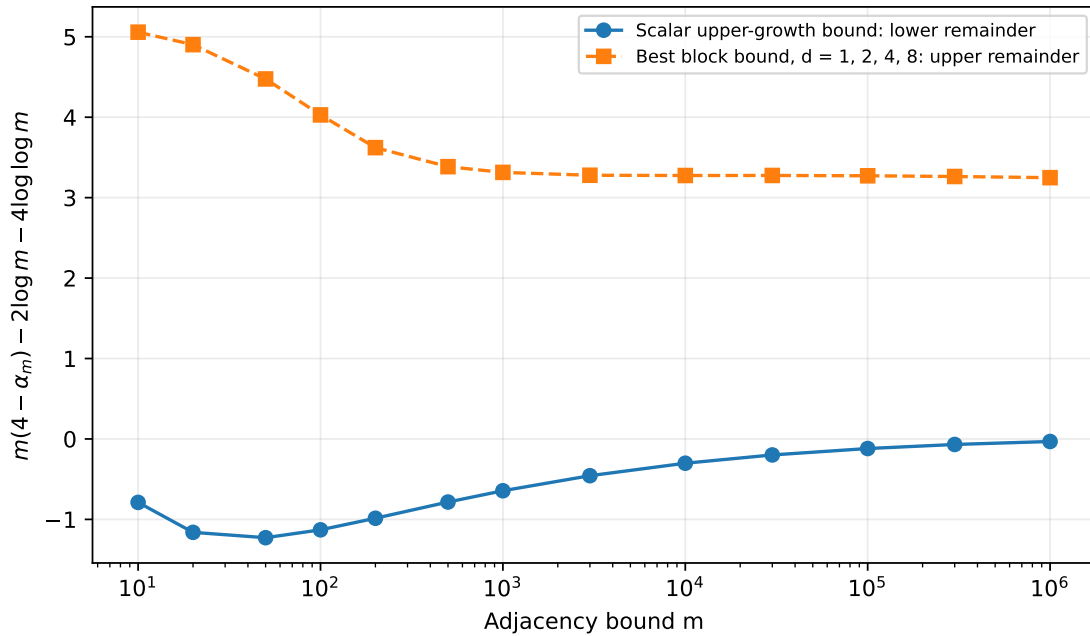


Figure 2: Numerical evaluations of the scalar bounds on the centered remainder E_m . The plotted upper envelope uses only $d = 1, 2, 4, 8$. Neither curve is the true remainder, and the asymptotic constants in (3) are not finite- m interval bounds.

8.4 What was actually checked

The bundled `verification_log.txt` records a successful complete run of the standard-library verifier. Its checks include direct triple-definition testing of all permutations through length seven against the Catalan generator; the first-entry formula for every Catalan avoider through length ten; 2,840 cumulative endpoint counts for $m = 1, \dots, 8$; and 53,534 weighted-edge comparisons under the shift for $m = 2, \dots, 30$. Both component graphs are also tested for strong connectivity in those finite cases.

The verifier checks coefficientwise scalar majorants and block lower bounds through $n = 100$ for $m = 2, \dots, 12$ and the admissible tested values of d . It validates the 38 rational Perron intervals using 3,458 strict integer row inequalities, then compares their endpoints as exact fractions. Finally, it checks all four full polynomial-matrix certificates and compares the resulting

generating-function coefficients with the recurrence through $n = 200$.

For certificate checking, the verifier reconstructs states and weighted edges separately, using Catalan convolutions in place of the binomial formula used by the numerical matrix builder. It also checks agreement between these two implementations. This reduces the risk that a common implementation error merely reproduces itself. Nevertheless, finite computational checks are supplements to, not replacements for, the universal arguments in sections 4, 6 and 7.

9 What is resolved, and what remains

The strict inequality $\alpha_{m+1} > \alpha_m$ is proved for every positive integer m . Together with the existing fixed-bound results, it completes the growth-constant conjecture selected here. The scalar comparison proof also determines the leading constant and next logarithmic term in the approach to four.

Several nearby questions are not resolved by these arguments. The first is whether $\lambda_V(m) > \lambda_U(m)$ for *every* $m \geq 5$; the reverse inequality holds for $m = 2, 3, 4$. This is the separate component-comparison problem formulated in [2, Problem P3]. Strict increase in both sequences is compatible with crossings, so it cannot settle their ordering. The finite certificates here establish the pattern only through $m = 20$. [Added 28 September 2026, batch 38: this question remains open. Part II shows only that a dominant V can lead by at most $(\lambda_V(m) - \lambda_U(m))_+ = O((\log m)^2/m^2)$ (corollary 15.4), which is why the ordering is invisible to every inverse-logarithmic term; see section 19.]

A second question is whether E_m converges, and, if so, to which constant. The scalar majorant discards endpoint compatibility, while the lower construction permits only a special recoverable family of blocks. Their limiting constants leave a window of width $4 \log 2$. Closing that gap requires controlling more of the endpoint information, not merely taking more terms of the same scalar expansions. A growing- d analysis could improve finite bounds, but our argument deliberately takes d fixed and then sends d to infinity only after a limsup. [Added 28 September 2026, batch 38: the preceding paragraph describes the report at its pinned state 6c9175a54. The question is answered in Part II: $E_m \rightarrow 2 \log \pi$ (corollary 15.3). As anticipated here, the upper side keeps last-endpoint information, through a Catalan-weighted positive vector (lemma 12.2), and the lower side lets the cutoff grow, $d = \lceil \log_2 m \rceil$, using a tail bound uniform in d (lemma 13.2 and theorem 15.2).]

Finally, the minimal recurrence order and uniform denominator factorization of $A_m(x)$ remain outside the scope of this report. The exact $m \leq 4$ artifacts are useful examples, not a claimed solution of that general algebraic problem. The new strict-growth argument is valuable precisely because it avoids needing such a formula.

Part II. The exact constant: an endpoint renewal comparison

Sections 1–9 and the appendices form the original report (Part I) of 19 September 2026, unchanged except for dated pointers to this addition. Part II was added on 28 September 2026 in batch 38 of ProvelT’s incoming-report intake, from one manuscript written against the pinned state of Part I. It answers the question Part I left open in section 9: E_m converges, and its limit is $2\log \pi$. Section 10 records the provenance and fixes the notation of the addition.

10 Provenance, conventions and the answer

10.1 One manuscript, one addition

Part II is batch-38 manuscript 04, *Closing the Catalan growth-constant gap: Endpoint renewal comparison for adjacency-bounded 132-avoiding permutations*, an AI-assisted research manuscript prepared for Vladimir Reshetnikov and dated September 28, 2026 (23 pages as delivered). It arrived as the archive `Closing_Catalan_Growth_Gap.zip` (inner directory `Adjacency_Growth_Constant/`) in ProvelT commit `fbba58593` and was placed in `938b2f74e`. It was written against ProvelT commit `6c9175a54`, at which Part I’s `article.tex` had the Git blob `5ec75ee4`; that blob is unchanged at the placement commit, so every statement the manuscript makes about Part I refers to the text printed above. Its code and data are shipped beside Part I’s files with the prefix `02-centered-remainder-` (Part I’s unprefix files count as 01); its manuscript, README and PDF are not shipped and survive in the arrival commit. Its two figures were added to `figures/` in the write phase (section 20.5).

The manuscript contributes, in the order printed here: a Catalan-weighted spectral upper bound $\alpha_m < \beta_m$ (section 12); a uniform first-deficiency tail bound and a lower bound for the first component rate (section 13); a uniform endpoint expansion of the truncated Catalan root (section 14); a corridor of width $O((\log m)^2/m^2)$ from a logarithmically growing cutoff, and with it the exact constant (section 15); an all-orders inverse-logarithmic expansion (section 16); exact checks and numerical illustrations (section 17); an $m = 2$ example with a non-implication (section 18); nine research directions (section 19); and its proof boundaries and provenance (section 20). Every result, proof, example, remark, limitation and question of the manuscript is printed. The manuscript re-derives the finite endpoint structure, the first-entry lemma and the injective block construction of Part I; those are printed once, in Part I, and credited where Part II uses them.

Status (manuscript 04). The manuscript presents mathematical proofs of a repository-open asymptotic question and stronger approximation results. It is not externally refereed and contains no new Lean certification. Novelty is assessed against the inspected repository snapshot and the primary papers cited here, not against an exhaustive literature census. Nothing in Part II is formalized in any proof assistant.

10.2 Notation of Part II

Part I’s symbols keep their meanings: $a_n^{(m)}$, A_m , α_m , C_j , C , B_m , $c_{k,p}$, W_m , U_m , V_m , $r_C(m)$, $\lambda_C(m)$, $h(k, j)$, $K_{m,d}$, $s_{m,d}$, $\leq$, and the centered remainder E_m of theorem 1.3. The manuscript uses the same symbols with the same meanings. Its new symbols, and the ones renamed here to avoid a clash with Part I, are listed in table 4.

No normalization changed: every renamed symbol denotes exactly the manuscript’s quantity. Logarithms are natural.

Table 4: Symbols of Part II. “Renamed” gives the manuscript’s own symbol. The last column prints the tempting false reading.

Symbol	Meaning in Part II	Not to be confused with
$K_m(x)$ (one index)	$\sum_{k=1}^m C_{k-1}x^k$	Part I’s block polynomial $K_{m,d}$ (two indices). In particular K_{m-d} is the Catalan polynomial truncated at $m-d$, and $K_{m,d} \leq K_{m-d}$. Part I’s majorant is $R_m = x + K_m$.
$r_m, \beta_m = r_m^{-1}$	positive root of $K_m(r_m) = 1$	the component roots $r_U(m), r_V(m)$, and the local r_m inside lemma 6.1. Since $R_m = x + K_m$, $q_m < r_m$ and $\beta_m < q_m^{-1}$.
t_m	$m \log(4r_m) = m \log(4/\beta_m)$	Part I’s $t_m^* = m \log(4/\alpha_m)$; $t_m < t_m^*$ for $m \geq 2$.
b_k	$C_{k-1}4^{-k}$ for all $k \geq 1$, so $b_1 = \frac{1}{4}$	the coefficients of R_m in section 7, where $b_1 = \frac{1}{2}$.
Λ (renamed from L)	$\frac{1}{2} \log m$	Part I’s $L_m = \frac{1}{2} \log m + \log \log m$.
z	$\log \Lambda + \log(2\sqrt{\pi}) = \log \log m + \frac{1}{2} \log \pi$	—
$\mathcal{P}_j(z)$ (renamed from P_j)	correction polynomials, proposition 16.1	the numerator polynomials P_3, P_4 of section A.
$\mathcal{B}(w), \mathcal{B}_N(t)$ (renamed from B_N)	the formal series $\sum_j (3/2)_j w^j$ and its truncation $\sum_{j < N} (3/2)_j t^{-j}$	B_m (thresholds), B_d of (23), B of lemma 6.1.
$H_q(x)$	$\sum_{j=0}^q C_j x^j$	the expanded adjacency matrix H in the proof of proposition 3.3.
ε_d	$(d+2)2^{-d}$	Part I’s a_d ; $1 - a_d = \varepsilon_d/2$ (remark 13.3).
$d = d_m$	$\lceil \log_2 m \rceil$ in section 15	Part I, where d is fixed while $m \rightarrow \infty$.
Δ_m (renamed from T_m)	mass of the last d tilted terms	Part I’s endpoint counts $T_{p,q}^{(m)}(n)$.
C_*, c_* (renamed from C, c)	absolute constants	Catalan numbers C_j ; the component letter $C \in \{U, V\}$.
W_{U_m}, W_{V_m} (manuscript: W_U, W_V)	component matrices of Part I	—

10.3 The question and the answer

Part I proved $4 - \alpha_m = (2 \log m + 4 \log \log m + O(1))/m$ (theorem 1.2) and, for the centered remainder $E_m = m(4 - \alpha_m) - 2 \log m - 4 \log \log m$, the window (3) of width $4 \log 2$. Its concluding discussion asks whether E_m converges and, if so, to which constant. That is the question resolved here. It was an open question in the inspected report (Part I), not a claim that every open problem in the original enumeration papers has been settled.

The older upper comparison was based on $R_m(x) = x + \sum_{k=1}^m C_{k-1}x^k$. The extra x accounts for appending a maximum when endpoint restrictions are discarded. One cannot simply delete that term from the counting recurrence. Instead, we retain the last-endpoint information and assign a different positive weight to each endpoint threshold. A Catalan convolution then incorporates the append transition without paying for it as an independent unconstrained scalar choice. Part I's diagnosis in section 9, that closing the window needs more endpoint information than the scalar series carry, is exactly what this weighting supplies on the upper side; on the lower side a cutoff d growing with m is used, made legitimate by a bound uniform in d .

Write

$$C_j = \frac{1}{j+1} \binom{2j}{j}, \quad K_m(x) = \sum_{k=1}^m C_{k-1}x^k, \quad K_m(r_m) = 1, \quad \beta_m = r_m^{-1}. \quad (36)$$

The polynomial K_m is strictly increasing on the positive real axis. Thus r_m is well defined. Since $K_m(1/4) < \sum_{k \geq 1} C_{k-1}4^{-k} = 1/2$, we have $r_m > 1/4$. For $m \geq 2$, $K_m(1) > 1$, so $r_m < 1$.

Theorem 10.1 (Finite renewal upper bound). *For every $m \geq 2$,*

$$\alpha_m < \beta_m < 4.$$

For $m = 1$, $\alpha_1 = \beta_1 = 1$.

Theorem 10.2 (Quantitative scalar universality). *As $m \rightarrow \infty$,*

$$0 \leq \beta_m - \alpha_m \leq \beta_m - \lambda_U(m) = O\left(\frac{(\log m)^2}{m^2}\right), \quad (37)$$

where $\lambda_U(m)$ is the first cyclic-component growth rate of (13). In particular,

$$\lim_{m \rightarrow \infty} E_m = 2 \log \pi. \quad (38)$$

For the all-orders statement, put

$$\Lambda = \frac{1}{2} \log m, \quad z = \log \Lambda + \log(2\sqrt{\pi}) = \log \log m + \frac{1}{2} \log \pi. \quad (39)$$

Theorem 10.3 (All inverse-logarithmic orders). *There are explicitly computable polynomials $\mathcal{P}_j \in \mathbb{Q}[z]$, of degree j , such that for every fixed integer $J \geq 0$,*

$$m(4 - \alpha_m) = 4\Lambda + 4z + 4 \sum_{j=1}^J \frac{\mathcal{P}_j(z)}{\Lambda^j} + O_J\left(\frac{(1+|z|)^{J+1}}{\Lambda^{J+1}}\right). \quad (40)$$

The first three are

$$\begin{aligned} \mathcal{P}_1(z) &= z - \frac{3}{2}, \\ \mathcal{P}_2(z) &= -\frac{1}{2}z^2 + \frac{5}{2}z - \frac{33}{8}, \\ \mathcal{P}_3(z) &= \frac{1}{3}z^3 - 3z^2 + \frac{43}{4}z - 15. \end{aligned} \quad (41)$$

The same expansion holds with α_m replaced by $\lambda_U(m)$ or β_m .

Since $4\Lambda + 4z = 2 \log m + 4 \log \log m + 2 \log \pi$, the case $J = 0$ is again $E_m \rightarrow 2 \log \pi$: the upper endpoint of Part I's window (3). Part II refutes nothing in Part I; it sharpens the window to a point.

10.4 What is inherited and what is added

Table 5: Attribution and scope. “Here” means proved in Part II (manuscript 04).

Ingredient or conclusion	Provenance
Endpoint recurrence and two cyclic components	Mayama–Akita; rederived in Part I and again in manuscript 04
Strict increase of α_m and the window (3)	Part I; not claimed anew
Injective skew blocks with fixed endpoint cutoff	Part I; recalled
Catalan-weighted spectral upper bound $\alpha_m < \beta_m$	Here
Uniform tail control and $O((\log m)^2/m^2)$ approximation	Here
Exact constant $2\log \pi$ and the transferred all-orders expansion	Here
Ordering of U and V for every $m \geq 5$	Not resolved here

The inspected repository path, commit and source blob are recorded in section 20.4. This matters because the repository is an active research collection: an unpinned statement about what it contains would not be a reproducible novelty comparison.

11 The finite interface

This section collects the input to the new comparison. Its finite-state structure is established work [2] and is proved in Part I; manuscript 04 rederives it, to make its argument independent of undocumented implementation details, and its proofs coincide with Part I’s except where a second route is marked below.

The endpoint recurrence is proposition 2.1. In the language of edges, $W_m(x)$ of (9) has a split edge

$$(p, q) \longrightarrow (m - k, q - k) \quad \text{with weight } c_{k,p}x^k, \quad (42)$$

and an append edge

$$(p, q) \longrightarrow (p - 1, m - 1) \quad \text{with weight } x, \quad (43)$$

invalid negative thresholds being omitted. By (6),

$$0 \leq c_{k,p} \leq C_{k-1}, \quad c_{k,p} \leq c_{k,p+1}, \quad c_{k,\infty} = C_{k-1}. \quad (44)$$

Proposition 11.1 (Spectral interface). *For $m \geq 2$, both $W_{U_m}(x)$ and $W_{V_m}(x)$ are irreducible for $x > 0$. Their spectral radii are continuous and strictly increasing from zero to infinity. With $r_C(m)$ and $\lambda_C(m)$ as in (13),*

$$\alpha_m = \max\{\lambda_U(m), \lambda_V(m)\}, \quad \lambda_U(m) > 1. \quad (45)$$

Proof. This is lemma 3.1, the paragraph after lemma 3.2, and proposition 3.3. Manuscript 04’s Appendix A gives the same graph and transfer-matrix proof: the same strong-connectivity induction on the gap $p - q$, the same classification of cyclic components, the same expansion of every edge into unit-length edges, and the same closed walks of lengths two and five for primitivity. Its argument for $\lambda_U(m) > 1$ differs and is printed next. $\square$

Second proof of $\lambda_U(m) > 1$ (manuscript 04). Let $e_U = (m - 1, \infty)$. At $x = 1$, the column of W_{U_m} indexed by e_U consists of ones, from the $k = 1$ edges. Consequently $W_{U_m}(1)\delta_{e_U} = \mathbf{1} \geq \delta_{e_U}$, strictly in another coordinate when $m \geq 2$. A positive left Perron vector gives $\text{spr}(W_{U_m}(1)) > 1$, and hence $r_U(m) < 1$. $\square$

Remark 11.2 (Constants depend on m). All constants in the proof of [proposition 3.3](#) (the c , C and D of its last paragraph) may depend on m ; no joint-limit uniformity is claimed, in Part I or here. [Added 29 September 2026, batch 43: this remains true of Parts I and II. Part III proves a joint estimate near the boundary $m = n - d$ by a separate route that does not use this interface ([theorem 22.2](#)).] The interface is all that Part II uses: not explicit determinants, a conjectural ordering of components, or a simple-pole assertion about A_m .

We will repeatedly use two elementary consequences of Perron–Frobenius positivity.

- Lemma 11.3** (Positivity tools). (i) *If M is irreducible and nonnegative, and $v > 0$ satisfies $Mv \leq v$ with a strict inequality in at least one coordinate, then $\text{spr}(M) < 1$.*
(ii) *For any nonnegative matrix, the spectral radius is at least its minimum row sum, and it is monotone under entrywise increase and under principal extension.*

Proof. For (i), multiply by a positive left Perron vector u : $\text{spr}(M)u^\top v = u^\top Mv < u^\top v$. For (ii), the row-sum bound follows by iterating $M\mathbf{1} \geq a\mathbf{1}$; the monotonicity follows by comparing powers. The mechanism of (i) is the one used in [lemma 3.2](#). $\square$

12 The endpoint renewal comparison

The essential new finite observation is a positive vector adapted to the last threshold. Before applying it, it is useful to isolate its algebraic mechanism.

12.1 A renewal system with a reset

Lemma 12.1 (Renewal–reset identity). *Let $w_1, \dots, w_{m-1} \geq 0$ and $a \geq 0$. On indices $0, \dots, m-1$, let L be the strictly lower triangular matrix with $L_{q,q-k} = w_k$ for $1 \leq k \leq q$, and put*

$$N = L + a\mathbf{1}e_{m-1}^\top.$$

Define $u_0 = 1$, $u_n = \sum_{k=1}^n w_k u_{n-k}$ for $1 \leq n < m$, and $H_q = \sum_{j=0}^q u_j$. Then

$$(I - L)H = \mathbf{1}, \tag{46}$$

$$NH = H + (aH_{m-1} - 1)\mathbf{1}, \tag{47}$$

$$\det(I - N) = 1 - aH_{m-1}. \tag{48}$$

Proof. For each q ,

$$\sum_{k=1}^q w_k H_{q-k} = \sum_{n=1}^q \sum_{k=1}^n w_k u_{n-k} = \sum_{n=1}^q u_n = H_q - 1.$$

This proves (46); the reset term gives (47). Since L is strictly triangular, $\det(I - L) = 1$. The determinant identity for a rank-one update, or direct elimination of $(I - L)$, gives

$$\det(I - N) = \det(I - L)(1 - ae_{m-1}^\top(I - L)^{-1}\mathbf{1}) = 1 - aH_{m-1}.$$

All identities also hold over a commutative coefficient ring when the nonnegativity assertions are omitted. $\square$

For the Catalan application, take $w_k = C_{k-1}x^k$ and $a = x$. The Catalan recurrence gives $u_j = C_j x^j$. Thus

$$H_q(x) = \sum_{j=0}^q C_j x^j, \quad xH_{m-1}(x) = K_m(x). \tag{49}$$

Equivalently, the formal identity $C(x) = 1 + xC(x)^2$ gives $(1 - xC(x))^{-1} = C(x)$. The finite version we need is

$$\sum_{k=1}^q C_{k-1} x^k H_{q-k}(x) = H_q(x) - 1. \quad (50)$$

There is no convergence issue in this polynomial identity.

12.2 A positive vector for the true two-endpoint component

Lemma 12.2 (Last-threshold supersolution). *For a state $(p, q) \in V_m$, set $v_{p,q}(x) = H_q(x)$. For every $x > 0$,*

$$(W_{V_m}(x)v(x))_{p,q} \leq H_q(x) - 1 + K_m(x). \quad (51)$$

At $x = r_m$, at least one coordinate inequality is strict.

Proof. A valid split from (p, q) has $k \leq q$, and its target has last threshold $q - k$. We may bound $c_{k,p}$ by C_{k-1} and include every $1 \leq k \leq q$. If an append edge exists, its target has last threshold $m - 1$; including such an edge even when it is absent can only increase the result. Therefore

$$\begin{aligned} (W_{V_m}v)_{p,q} &\leq \sum_{k=1}^q C_{k-1} x^k H_{q-k}(x) + xH_{m-1}(x) \\ &= H_q(x) - 1 + K_m(x), \end{aligned}$$

by (50). At the state $(0, m - 1)$, which belongs to V_m for $m \geq 2$, there is no append edge. The positive term $xH_{m-1}(x)$ was consequently added strictly. This proves strictness, in fact for every $x > 0$. $\square$

Proof of theorem 10.1. At $x = r_m$, lemma 12.2 gives $W_{V_m}v \leq v$, with a strict coordinate and $v > 0$. Irreducibility and lemma 11.3(i) imply $\text{spr}(W_{V_m}(r_m)) < 1$.

In U_m , append edges leave the component. Its row at first threshold p has sum $\sum_{k=1}^m c_{k,p} r_m^k \leq K_m(r_m) = 1$. The row $p = 0$ has sum $r_m < 1$, since only the $k = 1$ coefficient survives. Thus $\text{spr}(W_{U_m}(r_m)) < 1$ by the same positivity argument, now with vector $\mathbf{1}$.

Strict increase of each spectral radius gives $r_U(m), r_V(m) > r_m$. Taking reciprocals and using (45) proves $\alpha_m < \beta_m$. The already established $r_m > 1/4$ gives $\beta_m < 4$. For $m = 1$, the only permutations of length at least two with jumps of absolute value one are the increasing and decreasing permutations; a change of direction would repeat a value. Hence $\alpha_1 = 1 = \beta_1$. $\square$

Remark 12.3 (Comparison with Part I's majorant). Since $R_m = x + K_m$, one has $R_m(r_m) = 1 + r_m > 1$, so $q_m < r_m$ and $\beta_m < q_m^{-1}$ for every $m \geq 1$: theorem 10.1 is strictly sharper than proposition 5.1. At $m = 100$, for instance, the upper bound drops from 3.8581... (table 3) to 3.8201... (table 7). Part I's bound is coefficientwise, theorem 10.1 only spectral; see the next remark.

Remark 12.4 (What this comparison does not say). The scalar polynomial is not asserted to be a denominator of A_m , and $\det(I - W_{V_m})$ is not asserted to equal $1 - K_m$. The determinant identity (48) belongs to the *majorizing reset system*, not to the full endpoint system. Nor does the proof assert the coefficientwise inequality $A_m(x) \leq (1 - K_m(x))^{-1}$; it is false (section 18). Our comparison is a spectral one, witnessed by a positive vector. These distinctions prevent a valid growth bound from being misread as a solution of state minimization or exact enumeration.

13 A uniform lower approximation from recoverable blocks

The block construction itself is that of proposition 5.2 in Part I. The additional ingredients of manuscript 04 are a component version of the lower bound, an estimate uniform in the cutoff, and its use with a cutoff growing with m .

13.1 Block and component lower bounds

Fix integers $1 \leq d < m$, and let $K_{m,d}$ and $s_{m,d}$ be as in (18) and proposition 5.2:

$$K_{m,d}(x) = \sum_{k=1}^{m-d} c_{k,d} x^k, \quad K_{m,d}(s_{m,d}) = 1. \quad (52)$$

The coefficient of x is one, so a unique positive root exists.

Proposition 13.1 (Block and component lower bounds). *For every $1 \leq d < m$,*

$$\frac{1}{1 - K_{m,d}(x)} \leq A_m(x), \quad s_{m,d}^{-1} \leq \lambda_U(m) \leq \alpha_m < \beta_m. \quad (53)$$

Proof. The coefficientwise inequality is proposition 5.2: skew sums of blocks ending at their maximum, with the boundary jump $k + j \leq m$, and injectivity because the skew cuts recover the block boundaries. Manuscript 04 reproves it by the same argument.

For the sharper component statement, restrict $W_{U_m}(x)$ to first thresholds $p \in \{d, \dots, m-1\}$. For every $1 \leq k \leq m-d$, the split target $m-k$ is in that subset and its coefficient satisfies $c_{k,p} \geq c_{k,d}$. Every row sum of this principal submatrix is at least $K_{m,d}(x)$. At $x = s_{m,d}$ its spectral radius is at least one. Principal comparison (lemma 11.3(ii)) implies $\text{spr}(W_{U_m}(s_{m,d})) \geq 1$, so $r_U(m) \leq s_{m,d}$ and $\lambda_U(m) \geq s_{m,d}^{-1}$. The remaining inequalities are (45) and theorem 10.1. $\square$

13.2 First-entry counts and a uniform tail

The first-entry formula is lemma 5.3: $h(k, j) = \frac{j}{k-1} \binom{2k-j-3}{k-2}$ for $1 \leq j < k$, with generating function (21). Manuscript 04 reproves it with the same slot decomposition and the same Lagrange inversion. Consequently $c_{k,d} = \sum_{j=1}^{\min(d, k-1)} h(k, j)$ for $k \geq 2$, as in (7).

Lemma 13.2 (Uniform deficiency-tail estimate). *For all $d \geq 1$ and $k \geq 1$,*

$$0 \leq C_{k-1} - c_{k,d} \leq \varepsilon_d C_{k-1}, \quad \varepsilon_d = (d+2)2^{-d}. \quad (54)$$

Hence, whenever $1 \leq d < m$ and $\varepsilon_d < 1$,

$$(1 - \varepsilon_d)K_{m-d}(x) \leq K_{m,d}(x) \leq K_{m-d}(x) \quad (x \geq 0). \quad (55)$$

The corresponding derivative inequalities also hold for $x \geq 0$.

Proof. There is nothing to prove if $k = 1$ or $k \leq d+1$. For the remaining cases, formula (21) gives

$$\frac{h(k, j)}{C_{k-2}} = j \prod_{r=0}^{j-2} \frac{k-2-r}{2k-4-r} \leq \frac{j}{2^{j-1}}. \quad (56)$$

The product is empty for $j = 1$. Each displayed factor is at most $1/2$. For a missing deficiency $j \geq d+1 \geq 2$, we have $k \geq 3$ and

$$\frac{C_{k-2}}{C_{k-1}} = \frac{k}{4k-6} \leq \frac{1}{2}.$$

Thus $h(k, j)/C_{k-1} \leq j2^{-j}$. Summing missing deficiencies and extending the tail to infinity gives

$$\frac{C_{k-1} - c_{k,d}}{C_{k-1}} \leq \sum_{j=d+1}^{\infty} \frac{j}{2^j} = \frac{d+2}{2^d}.$$

For the final equality, differentiate the finite-tail geometric sum or subtract its successive terms. Multiplying coefficientwise by x^k and summing proves (55). Differentiation preserves a coefficientwise inequality between these nonnegative polynomials. $\square$

The identity in (56) is Part I's (24); the inequality is new.

Remark 13.3 (Merge note: the uniform bound against Part I's limit). By (25) and $C_{k-1}4^{-k} \sim (4\sqrt{\pi})^{-1}k^{-3/2}$, the missing fraction $1 - c_{k,d}/C_{k-1}$ tends to $1 - a_d = (d+2)2^{-(d+1)} = \varepsilon_d/2$ as $k \rightarrow \infty$. The uniform bound ε_d of lemma 13.2 therefore loses at most a factor two against the limit, the factor coming from $C_{k-2}/C_{k-1} \leq \frac{1}{2}$ where the limit is $\frac{1}{4}$. This observation is the merge's, not the manuscript's.

The role of uniformity is decisive. A fixed- d asymptotic for $c_{k,d}$, such as (25), would not justify substituting $d = d(m)$. Inequality (54) holds simultaneously for all k, d , so the growing cutoff used below requires no interchange of unproved uniform limits.

14 The truncated Catalan root at all orders

The remaining analysis begins with the scalar root alone. Normalize its coefficients by

$$b_k = C_{k-1}4^{-k}, \quad \sum_{k \geq 1} b_k = \frac{1}{2}, \quad b_k = \frac{1}{4\sqrt{\pi}}k^{-3/2}(1 + O(k^{-1})). \quad (57)$$

The last estimate follows from Stirling's formula. In particular, $b_k \leq C_*k^{-3/2}$ with one absolute constant C_* . If

$$t_m = m \log(4r_m), \quad (58)$$

then

$$\sum_{k=1}^m b_k e^{t_m k/m} = 1, \quad \beta_m = 4e^{-t_m/m}. \quad (59)$$

The tail is summable but has total mass only $1/2$. A positive tilt must supply the missing half. It does so in a thin layer near $k = m$.

14.1 A uniform endpoint expansion

For $a \in \mathbb{R}$ and $j \geq 0$, write $(a)_0 = 1$ and $(a)_j = a(a+1) \cdots (a+j-1)$.

Lemma 14.1 (Tilted Catalan endpoint expansion). *Fix $M > 0$ and an integer $N \geq 1$. Uniformly for*

$$t = \frac{1}{2} \log m + \log \log m + s, \quad |s| \leq M, \quad (60)$$

as $m \rightarrow \infty$,

$$\sum_{k=1}^m b_k e^{tk/m} = \frac{1}{2} + \frac{e^t}{4\sqrt{\pi m} t} \left(\sum_{j=0}^{N-1} \frac{(3/2)_j}{t^j} + O_{M,N}(t^{-N}) \right). \quad (61)$$

Proof. Split the sum at $m/2$. For the lower half, the unweighted partial sum is $1/2 + O(m^{-1/2})$. Since $e^u - 1 \leq ue^u$ for $u \geq 0$, its tilt increment is at most

$$\begin{aligned} \sum_{k \leq m/2} b_k (e^{tk/m} - 1) &\leq \frac{C_* t}{m} e^{t/2} \sum_{k \leq m/2} k^{-1/2} \\ &= O\left(\frac{te^{t/2}}{\sqrt{m}}\right) = O_M\left(m^{-1/4}(\log m)^{3/2}\right). \end{aligned}$$

This is smaller than every fixed negative power of t . Thus the lower half contributes $1/2$ plus an error that can be absorbed into the remainder in (61).

On the upper half, Stirling's estimate is uniform:

$$b_k = \frac{1}{4\sqrt{\pi}}k^{-3/2}(1 + O(m^{-1})).$$

A Riemann-sum comparison, using that $y^{-3/2}e^{ty}$ is increasing on $[1/2, 1]$ for all sufficiently large t , gives

$$\begin{aligned} \sum_{m/2 < k \leq m} b_k e^{tk/m} &= \frac{1}{4\sqrt{\pi m}} \int_{1/2}^1 y^{-3/2} e^{ty} dy + O\left(\frac{e^t}{m^{3/2}}\right) \\ &= \frac{e^t}{4\sqrt{\pi m} t} \int_0^{t/2} (1 - v/t)^{-3/2} e^{-v} dv + O\left(\frac{e^t}{m^{3/2}}\right). \end{aligned}$$

A possible half-integer lower endpoint changes only the same error term. For $0 \leq v \leq t/2$, Taylor's theorem with a uniform remainder gives

$$(1 - v/t)^{-3/2} = \sum_{j=0}^{N-1} \frac{(3/2)_j}{j!} (v/t)^j + O_N((v/t)^N).$$

Integrating each monomial against e^{-v} and extending its upper limit to infinity produces $j!$; the extension error is exponentially small in t . The integral is therefore $\sum_{j=0}^{N-1} (3/2)_j t^{-j} + O_N(t^{-N})$.

In the window (60), the prefactor $e^t/(\sqrt{m} t)$ is bounded above and below by positive constants depending on M . The relative Riemann error is $O(t/m)$, and all lower-half errors are smaller than any fixed t^{-N} . Combining the estimates proves the uniform expansion. $\square$

Corollary 14.2 (Scalar root location). *As $m \rightarrow \infty$,*

$$t_m = \frac{1}{2} \log m + \log \log m + \frac{1}{2} \log \pi + o(1). \quad (62)$$

In particular, $t_m = O(\log m)$, $e^{t_m} = O(\sqrt{m} \log m)$, and $r_m \rightarrow 1/4$.

Proof. For a fixed real s , the $N = 1$ case of lemma 14.1 gives

$$\sum_{k=1}^m b_k \exp\left(\frac{k}{m} \left[\frac{1}{2} \log m + \log \log m + s\right]\right) \rightarrow \frac{1}{2} + \frac{e^s}{2\sqrt{\pi}}.$$

The limit equals one precisely at $s_0 = \frac{1}{2} \log \pi$. For every $\delta > 0$, its values at $s_0 - \delta$ and $s_0 + \delta$ lie strictly on opposite sides of one. The finite sum is strictly increasing in t , so its root lies between those trial values for all sufficiently large m . Let $\delta \downarrow 0$. The remaining assertions follow directly from (62) and (58). $\square$

Second proof (merge note, from Part I). Apply lemma 6.1 to $b_k = C_{k-1} 4^{-k}$, $h = 0$: then $B = \frac{1}{2}$, $c = 1/(4\sqrt{\pi})$, and $\log \frac{1-B}{2c} = \log \sqrt{\pi}$, which is (62). Part I applied the same lemma to the coefficients of R_m ($B = \frac{3}{4}$), which gave the lower endpoint of its window; the extra x of R_m alone accounts for the loss of $4 \log 2$ there, as $4 \log \frac{1-3/4}{1-1/2} = -4 \log 2$. $\square$

Remark 14.3 (An immediate resolution using the earlier window). Already theorem 10.1 and corollary 14.2 imply $\liminf E_m \geq 2 \log \pi$. Combined with the upper bound on $\limsup E_m$ in (3), this closes Part I's question. The next section proves a substantially stronger quantitative result and provides an independent lower argument rather than relying on that asymptotic window.

15 A logarithmically growing cutoff gives a sharp corridor

Take, for all sufficiently large m ,

$$d = d_m = \lceil \log_2 m \rceil, \quad \varepsilon_d = (d+2)2^{-d} = O(\log m/m). \quad (63)$$

In particular, $d < m$, $d = O(\log m)$, and $\varepsilon_d < 1$.

Lemma 15.1 (Value deficit and slope). *At the scalar root r_m , with d as in (63),*

$$0 \leq 1 - K_{m,d}(r_m) = O\left(\frac{(\log m)^2}{m}\right), \quad (64)$$

$$K'_{m,d}(r_m) \geq c_* m \quad \text{for all sufficiently large } m, \quad (65)$$

for an absolute constant $c_* > 0$.

Proof. Set $t = t_m$. The lost last d terms of $K_m(r_m)$ satisfy

$$\begin{aligned} \Delta_m &:= K_m(r_m) - K_{m-d}(r_m) = \sum_{k=m-d+1}^m b_k e^{tk/m} \\ &\leq C_* d(m-d)^{-3/2} e^t = O\left(\frac{d \log m}{m}\right) = O\left(\frac{(\log m)^2}{m}\right), \end{aligned} \quad (66)$$

using corollary 14.2. Since $K_m(r_m) = 1$, lemma 13.2 gives

$$K_{m,d}(r_m) \geq (1 - \varepsilon_d)(1 - \Delta_m), \quad 1 - K_{m,d}(r_m) \leq \varepsilon_d + \Delta_m.$$

This proves (64).

The lower-half estimate in the proof of lemma 14.1, now evaluated at t_m , shows

$$\sum_{k \leq m/2} b_k e^{t_m k/m} \rightarrow \frac{1}{2}.$$

The whole sum is one, and $\Delta_m \rightarrow 0$. Consequently

$$\sum_{m/2 < k \leq m-d} b_k e^{t_m k/m} \rightarrow \frac{1}{2}. \quad (67)$$

Every term in this range has $k > m/2$, so

$$K'_{m-d}(r_m) = \frac{1}{r_m} \sum_{k=1}^{m-d} k b_k e^{t_m k/m} \geq \frac{m}{2r_m} \sum_{m/2 < k \leq m-d} b_k e^{t_m k/m}.$$

As $r_m \rightarrow 1/4$, the right side is bounded below by a positive constant times m . The derivative version of (55), together with $\varepsilon_d \rightarrow 0$, proves (65). $\square$

Theorem 15.2 (Quantitative scalar corridor). *With $d = \lceil \log_2 m \rceil$,*

$$0 \leq s_{m,d} - r_m = O\left(\frac{(\log m)^2}{m^2}\right), \quad 0 \leq \beta_m - s_{m,d}^{-1} = O\left(\frac{(\log m)^2}{m^2}\right). \quad (68)$$

In particular, (37) holds.

Proof. Since $K_{m,d} \leq K_m$, we have $s_{m,d} \geq r_m$. Its derivative is nondecreasing on the positive axis, since all coefficients are nonnegative. Writing $s = s_{m,d}$, integration or the mean-value inequality gives

$$1 - K_{m,d}(r_m) = K_{m,d}(s) - K_{m,d}(r_m) \geq K'_{m,d}(r_m)(s - r_m).$$

Divide the estimate (64) by (65) to obtain the first part of (68). Because $s, r_m \geq 1/4$,

$$0 \leq r_m^{-1} - s^{-1} = \frac{s - r_m}{r_m s} \leq 16(s - r_m),$$

which proves the second part. Finally, use the entire chain (53):

$$0 \leq \beta_m - \alpha_m \leq \beta_m - \lambda_U(m) \leq \beta_m - s_{m,d}^{-1}. \quad \square$$

Corollary 15.3 (The exact constant). *The centered remainder of theorem 1.3 satisfies $E_m \rightarrow 2 \log \pi$. Equivalently,*

$$4 - \alpha_m = \frac{2 \log m + 4 \log \log m + 2 \log \pi + o(1)}{m}. \quad (69)$$

Proof. By theorem 15.2 and $\beta_m = 4e^{-t_m/m}$,

$$\begin{aligned} m(4 - \alpha_m) &= 4m(1 - e^{-t_m/m}) + O((\log m)^2/m) \\ &= 4t_m + O((\log m)^2/m). \end{aligned} \quad (70)$$

Here Taylor's theorem applies since $t_m = O(\log m)$. Substitute (62). This proves (69) and theorem 10.2 in full. $\square$

Corollary 15.4 (A one-sided constraint on component competition). *Regardless of which component dominates,*

$$0 \leq \alpha_m - \lambda_U(m) = (\lambda_V(m) - \lambda_U(m))_+ = O\left(\frac{(\log m)^2}{m^2}\right). \quad (71)$$

This is deliberately a one-sided conclusion. We have not proved a corresponding lower approximation for $\lambda_V(m)$ when it is below $\lambda_U(m)$, and do not claim an absolute bound on $|\lambda_V(m) - \lambda_U(m)|$ in that case. The result does show that whenever V wins, its advantage is invisible to every inverse-logarithmic term in the scaled deficit.

16 Inverting the scalar expansion to every logarithmic order

The quantitative corridor is what makes a scalar asymptotic expansion a statement about the actual permutation class. We now derive the coefficients, including their error interpretation.

16.1 Finite asymptotic equations, not a convergent series

Introduce the *formal* power series

$$\mathcal{B}(w) = \sum_{j \geq 0} (3/2)_j w^j. \quad (72)$$

Its coefficients grow factorially. The series has radius of convergence zero and is never evaluated as an ordinary convergent series at $w > 0$. For each fixed $N \geq 1$, instead use the finite polynomial in $1/t$

$$\mathcal{B}_N(t) = \sum_{j=0}^{N-1} (3/2)_j t^{-j}.$$

Substituting $t = t_m$ into lemma 14.1 gives

$$t_m - \log t_m + \log \mathcal{B}_N(t_m) = \frac{1}{2} \log m + \log(2\sqrt{\pi}) + O_N(t_m^{-N}). \quad (73)$$

Indeed the missing mass is $1/2$, so $e^{t_m} \mathcal{B}_N(t_m)/t_m = 2\sqrt{\pi m} (1 + O_N(t_m^{-N}))$. The polynomial $\mathcal{B}_N(t)$ is positive for $t > 0$, and its logarithm is unambiguous.

Formal logarithms are used below only as a finite coefficient-extraction device. To compute a coefficient of degree j , only coefficients through degree j of (72) are needed. Thus the divergent notation organizes an unlimited sequence of finite, justified asymptotic statements; it is not an analytic summation prescription.

16.2 A triangular recursion for all coefficients

Proposition 16.1 (Coefficient recursion). *Set $\mathcal{P}_0$ aside and define $\mathcal{P}_j$ recursively as follows. At step $j \geq 1$, put*

$$T_{j-1}(u, z) = 1 + zu + \sum_{h=1}^{j-1} \mathcal{P}_h(z) u^{h+1},$$

and define

$$\mathcal{P}_j(z) = [u^j] \left\{ \log T_{j-1}(u, z) - \log \mathcal{B}\left(\frac{u}{T_{j-1}(u, z)}\right) \right\}. \quad (74)$$

These are well-defined rational polynomials of degree j , with leading coefficient $(-1)^{j+1}/j$. For every fixed $J \geq 0$,

$$t_m = \Lambda + z + \sum_{j=1}^J \frac{\mathcal{P}_j(z)}{\Lambda^j} + O_J\left(\frac{(1+|z|)^{J+1}}{\Lambda^{J+1}}\right), \quad (75)$$

with Λ, z as in (39).

Proof. Write $u = 1/\Lambda$ and seek $t = \Lambda T$, with

$$T = 1 + zu + \sum_{j=1}^J \mathcal{P}_j(z) u^{j+1}.$$

Since $z = \log \Lambda + \log(2\sqrt{\pi})$, substitution into the formal version of (73) leaves the residual

$$\sum_{j=1}^J \mathcal{P}_j(z) u^j - \log T + \log \mathcal{B}(u/T). \quad (76)$$

The coefficient of u^j in the last two terms depends only on $\mathcal{P}_1, \dots, \mathcal{P}_{j-1}$: $\mathcal{P}_j$ first enters T at order u^{j+1} . Cancelling the coefficient of u^j therefore gives exactly (74), proving existence and uniqueness of the recursion.

Induction shows $\deg \mathcal{P}_j \leq j$. The only term of degree j in z at order u^j comes from $\log(1 + zu)$ and equals $(-1)^{j+1} z^j / j$. Terms involving an earlier $\mathcal{P}_h u^{h+1}$ spend at least one more power of u than their possible degree in z ; terms from $\log \mathcal{B}(u/T)$ have the same strict deficit. Thus the degree is exactly j and its leading coefficient has the stated value.

To justify the error, fix J and take $N \geq J + 2$ in the finite equation (73). For the trial value t_* obtained by truncating (75), we have $T = 1 + O_J((1 + |z|)/\Lambda)$ and $z = O(\log \Lambda)$. Finite Taylor expansions of the logarithms, with remainders uniform for $|T - 1| \leq 1/2$, turn the formal cancellations into the numerical residual estimate

$$t_* - \log t_* + \log \mathcal{B}_N(t_*) - \Lambda - \log(2\sqrt{\pi}) = O_J((1 + |z|)^{J+1} \Lambda^{-J-1}).$$

Both t_* and t_m lie in a window with bounded shift of the form (60), by corollary 14.2 and $\sum_{j \leq J} \mathcal{P}_j(z) \Lambda^{-j} = o(1)$. The derivative of $t - \log t + \log \mathcal{B}_N(t)$ is

$$1 - \frac{1}{t} + O_N(t^{-2}),$$

so it is at least $1/2$ throughout the interval between them for large m . Apply the mean-value theorem to the residuals of t_* and t_m . The $O_N(t_m^{-N})$ term is smaller than the claimed bound. This proves (75). $\square$

The first four polynomials are

$$\begin{aligned} \mathcal{P}_1(z) &= z - \frac{3}{2}, \\ \mathcal{P}_2(z) &= -\frac{1}{2}z^2 + \frac{5}{2}z - \frac{33}{8}, \\ \mathcal{P}_3(z) &= \frac{1}{3}z^3 - 3z^2 + \frac{43}{4}z - 15, \\ \mathcal{P}_4(z) &= -\frac{1}{4}z^4 + \frac{10}{3}z^3 - \frac{153}{8}z^2 + \frac{223}{4}z - \frac{4461}{64}. \end{aligned} \tag{77}$$

For an independent algebraic check, the formal logarithm begins

$$\log \mathcal{B}(w) = \frac{3}{2}w + \frac{21}{8}w^2 + \frac{69}{8}w^3 + \frac{2529}{64}w^4 + O(w^5). \tag{78}$$

The bundled symbolic verifier constructs the polynomials from the recursion and confirms that (76) vanishes through degree four using exact rational polynomial arithmetic.

16.3 Transfer to the true growth constant

Proof of theorem 10.3. Equation (70) gives $m(4 - \alpha_m) = 4t_m + O((\log m)^2/m)$. For each fixed J , this error is smaller than every fixed power of $1/\log m$, and hence is absorbed by the remainder in (75). Multiply that expansion by four. The same proof applies to $\lambda_U(m)$ by (37). For β_m , the direct Taylor expansion of $4e^{-t_m/m}$ supplies the transfer. $\square$

With $z = \log \log m + \frac{1}{2} \log \pi$, the first three corrections can also be written in powers of $1/\log m$ (the manuscript abbreviates $\log m$ as H , which Part II avoids):

$$\begin{aligned} m(4 - \alpha_m) &= 2 \log m + 4 \log \log m + 2 \log \pi + \frac{8z - 12}{\log m} \\ &\quad + \frac{-8z^2 + 40z - 66}{(\log m)^2} \\ &\quad + \frac{\frac{32}{3}z^3 - 96z^2 + 344z - 480}{(\log m)^3} + O\left(\frac{(1 + |\log \log m|)^4}{(\log m)^4}\right). \end{aligned} \tag{79}$$

The assertion is asymptotic for each fixed truncation order. It does not promise that adding another term improves a numerical approximation at a particular small or moderate value of m .

16.4 Two different error scales

There are two conceptually distinct approximations. The scalar surrogate β_m differs from α_m by at most order $(\log m)^2/m^2$. A fixed truncation of the logarithmic expansion generally has a much larger error, of order a power of $1/\log m$ after multiplying the growth deficit by m . Thus a scalar root can be numerically extremely accurate while its first few asymptotic terms are still inaccurate.

This also explains the component issue. The proof does not need to decide whether $\alpha_m = \lambda_U(m)$ or $\lambda_V(m)$. When V is dominant, corollary 15.4 confines its extra growth to a scale smaller than all the terms in (79). A calculation retaining only those terms has discarded the very information needed to settle that component ordering.

17 Exact checks and numerical evidence

The universal results above are proved analytically. The companion computations have three separate roles: exact checks of finite algebra and combinatorics, exact root enclosures in finite examples, and numerical illustrations. They are deliberately not merged into a single ambiguous notion of verification. File names below are the shipped ones; the scripts themselves still read and write their delivered names, so they are rerun on a copy (section 20.2).

17.1 Exact finite checks

The exact verifier, the standard-library program code/02-centered-remainder-verify.py, uses integers and fractions.Fraction; it needs no floating-point library. Its successful run is recorded in data/02-centered-remainder-verification_report.json. The row check verifies coefficient-

Table 6: Executed exact checks (manuscript 04). Finite ranges do not replace the all- m proofs.

Check	Executed range or count
Direct triple-definition avoidance versus recursive Catalan generation	All permutations for $0 \leq n \leq 7$
First-entry formula against explicit maximum-ending blocks	55 counts, $1 \leq k \leq 10$
Actual V -row polynomial domination by the Catalan witness	4,930 rows, $2 \leq m \leq 30$
Uniform deficiency-tail inequality	9,000 pairs, $k \leq 300$, $d \leq 30$
Restricted U -row coefficient comparisons	67,425 comparisons, $m \leq 30$
Exact scalar-root corridors	Five (m, d) pairs

twice that $H_q - (W_{V_m} v)_{p,q} - (1 - K_m)$ has nonnegative coefficients. This is stronger than testing the inequality at a list of decimal values of x . The tail check clears the denominator 2^d and verifies an integer inequality. The direct avoidance test provides a separate combinatorial check of the recursive generator before its blocks are used to test the first-entry formula. The first two rows repeat, with independent code and the same ranges, checks of Part I's verifier (section 8); the other four are new.

The symbolic program code/02-centered-remainder-verify_expansion.py uses SymPy but no floating-point coefficients. It verifies (77), (78), and the vanishing residual through degree four. Its output is data/02-centered-remainder-expansion_verification.json. Symbolic cancellation proves these finite coefficient identities, not the analytic error estimate; that estimate is the content of lemma 14.1 and proposition 16.1.

17.2 Certified finite corridors

For a positive polynomial P with zero constant term, exact bisection encloses the root of $P(x) = 1$ by fractions ℓ, u satisfying $P(\ell) \leq 1 \leq P(u)$. Apply this separately to K_m and $K_{m,d}$. If the enclosures are $[\ell_r, u_r]$ and $[\ell_s, u_s]$, then

$$\frac{1}{u_s} \leq \alpha_m \leq \frac{1}{\ell_r}. \quad (80)$$

The rational certificate file data/02-centered-remainder-rational_certificates.json retains the full fractions and can be regenerated by the exact verifier. The following decimal endpoints are rounded *outwards* from those fractions, not nearest-rounded approximations to unknown

Table 7: Rigorous finite intervals from exact scalar bisection.

m	d	Lower endpoint for α_m	Upper endpoint for α_m
5	1	2.0000000000	2.4580325859
10	2	2.7001330054	2.9828910906
20	3	3.2335773659	3.3677623965
50	5	3.6481708456	3.6845734707
100	7	3.8074760932	3.8201232724

values of α_m . These intervals are not advertised as optimal small- m estimates. Their purpose is to provide simple, independently checkable examples of the new all- m scalar corridor without computing a large characteristic polynomial. For $m \leq 20$, Part I's rational Perron certificates (section 8) are much sharper.

17.3 Floating-point component and large-bound calculations

The separate program code/02-centered-remainder-compute.py evaluates sparse component matrices and scalar roots. Its component values reproduce the known small-bound pattern (compare table 2) and illustrate the new upper bound. For larger m , no matrix eigensolver is

Table 8: Numerical growth rates; displayed digits are not interval certificates. The source file data/02-centered-remainder-component_numerics.csv also has a row for $m = 30$.

m	$\lambda_U(m)$	$\lambda_V(m)$	α_m	β_m
2	1.465571232	1.000000000	1.465571232	1.618033989
3	1.826515772	1.691402554	1.826515772	2.000000000
4	2.100456128	2.091476748	2.100456128	2.263408655
5	2.312119493	2.351509000	2.351509000	2.458032586
10	2.902174743	2.952689154	2.952689154	2.982891091
20	3.333400105	3.356795518	3.356795518	3.367762396
50	3.675866778	3.681914231	3.681914231	3.684573471

needed to evaluate the scalar corridor. Normalized coefficients b_k are generated by

$$b_1 = \frac{1}{4}, \quad \frac{b_{k+1}}{b_k} = \frac{2k-1}{2(k+1)}.$$

Solving for $t = m \log(4x)$ avoids subtracting a tiny root displacement from $1/4$. The block coefficients are also normalized, using the consecutive first-deficiency ratio obtained from (21).

Table 9: Floating-point evaluations with $d = \lceil \log_2 m \rceil$. The mathematical endpoints are rigorous bounds; these nearest-rounded numerical evaluations are not themselves certified intervals. From data/02-centered-remainder-scalar_numerics.csv (63 values of m).

m	d	$s_{m,d}^{-1}$	β_m	Corridor width
10^2	7	3.807476093297	3.820123272395	$1.265 \cdot 10^{-2}$
10^3	10	3.975171751563	3.975423985146	$2.522 \cdot 10^{-4}$
10^4	14	3.996950887245	3.996955062448	$4.175 \cdot 10^{-6}$
10^5	17	3.999640264705	3.999640325443	$6.074 \cdot 10^{-8}$
10^6	20	3.999958715503	3.999958716333	$8.301 \cdot 10^{-10}$

At $m = 10^6$, the two numerical bounds for E_m are approximately 3.149478 and 3.150308, although $2 \log \pi \approx 2.289460$. This is compatible with the theorem: the leading correction to the

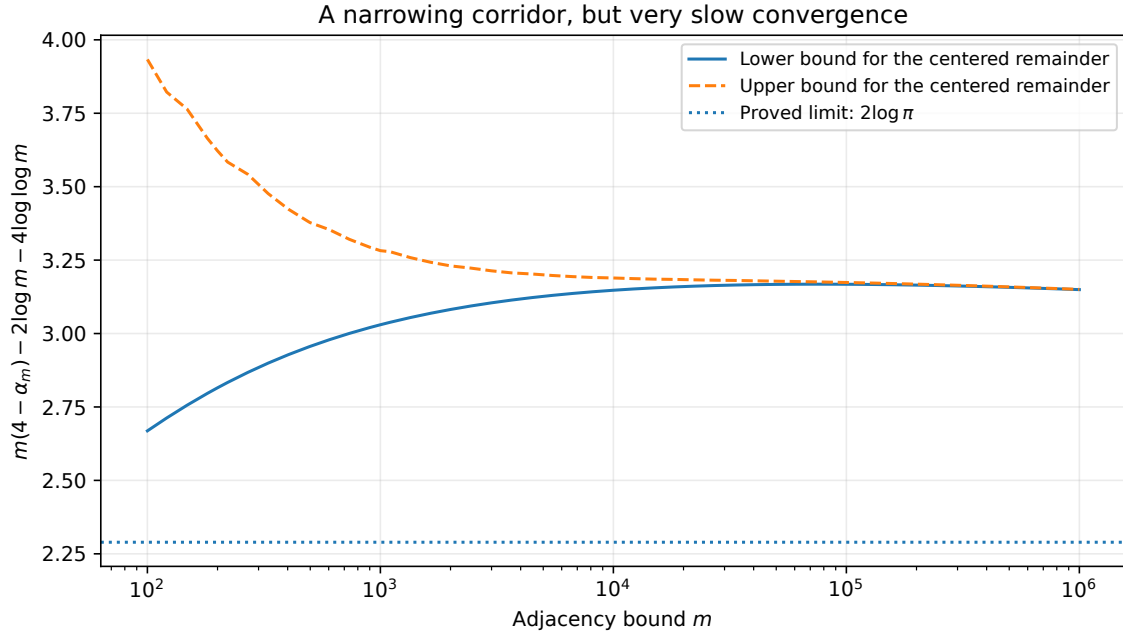


Figure 3: Numerical evaluations of the lower and upper bounds for E_m , not direct evaluations of the unknown large- m component maximum. The dotted line is the proved limiting constant. The asymptotic bounds need not be monotone, and the visible range has not reached the limit. (Manuscript 04’s figure; compare figure 2, whose bounds are Part I’s.)

constant is $(8z - 12)/\log m$, which decays very slowly. A close upper and lower corridor is not evidence that its center is already near its limit.

Remark 17.1 (Merge note: the numerics are not evidence for the value of the limit). The corridor in figure 3 sits about 0.86 above $2 \log \pi$ at $m = 10^6$, and its lower endpoint is not even monotone in the computed range: it rises to about 3.168 near $m = 73,920$ and then decreases slowly. At $m = 10^6$ the first correction term $(8z - 12)/\log m$ of (79) is about 0.983, and the three displayed corrections together give $2 \log \pi + 0.983 - 0.104 - 0.005 \approx 3.164$, within 0.015 of the corridor. That agreement shows that the computations are consistent with the inverse-logarithmic corrections. It is *not* evidence for the value $2 \log \pi$ of the limit: no computation in the range $m \leq 10^6$ distinguishes that constant from nearby ones, and a slowly drifting corridor centred near 3.15 would look the same whatever the limit were. The value $2 \log \pi$ rests on the proofs of sections 12 to 15 alone. The decimals quoted in this remark were computed for the merge from data/02-centered-remainder-scalar_numerics.csv.

The computations do not estimate an exact leading constant for the scalar defect, certify component ordering beyond the tested values, or validate an infinite asymptotic expansion by numerical fitting.

18 A complete small example and a useful non-implication

At $m = 2$, order the U states by $p = 0, 1$, and the V states as $(1, 0), (0, 1)$. Then

$$W_{U_2}(x) = \begin{pmatrix} 0 & x \\ x^2 & x \end{pmatrix}, \quad W_{V_2}(x) = \begin{pmatrix} 0 & x \\ x & 0 \end{pmatrix}.$$

Thus $\lambda_U(2)$ is the positive root of $\lambda^3 - \lambda^2 - 1 = 0$, while $\lambda_V(2) = 1$ (the matrix W_{U_2} also appears in the proof of theorem 1.1). The scalar polynomial is $K_2(x) = x + x^2$, and $\beta_2 = (1 + \sqrt{5})/2$.

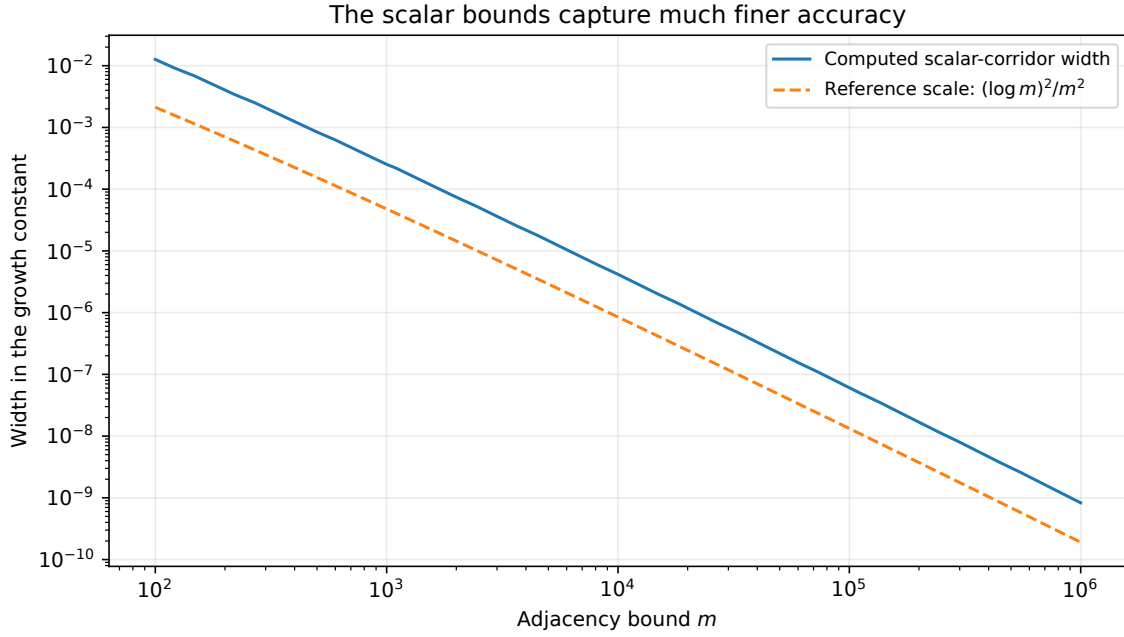


Figure 4: The numerical width $\beta_m - s_{m,d}^{-1}$ and the comparison scale $(\log m)^2/m^2$. The dashed curve has coefficient one solely for scale comparison; no claim is made that it is a finite- m upper bound or the optimal asymptotic equivalent.

The last-threshold vector is $v = (1, 1 + x)^\top$. At the root $K_2(r_2) = 1$,

$$W_{V_2}(r_2)v = \begin{pmatrix} r_2 + r_2^2 \\ r_2 \end{pmatrix} = \begin{pmatrix} 1 \\ r_2 \end{pmatrix} <_{\text{some}} \begin{pmatrix} 1 \\ 1 + r_2 \end{pmatrix} = v,$$

where $<_{\text{some}}$ means coordinatewise inequality with at least one strict coordinate. The separate reset comparator of lemma 12.1 is

$$N(x) = \begin{pmatrix} 0 & x \\ x & x \end{pmatrix}, \quad \det(I - N(x)) = 1 - x - x^2.$$

This differs from $\det(I - W_{V_2}(x)) = 1 - x^2$, illustrating exactly why a spectral comparison is not an equality of state systems.

There is also an immediate counterexample to the tempting coefficientwise interpretation. For length three and $m = 2$, every 132-avoider meets the adjacency bound, so $[x^3]A_2(x) = C_3 = 5$ (table 1). But

$$[x^3] \frac{1}{1 - K_2(x)} = [x^3] \frac{1}{1 - x - x^2} = 3.$$

The upper bound on exponential growth is valid even though this proposed coefficientwise majorant is false. This example is a useful diagnostic when transporting the renewal argument to another combinatorial class.

19 Further research questions and proposed approaches

The exact constant is no longer the first obstruction. The quantitative corridor changes which questions are mathematically distinguishable and which tools are likely to be useful. The second question below is Part I's first open question (section 9), now constrained by corollary 15.4; the eighth continues Part I's remark on minimal recurrence orders and denominators.

19.1 The first term beyond scalar universality

Determine the true order and leading behavior of $\beta_m - \alpha_m$. Our upper bound is $O((\log m)^2/m^2)$, obtained by discarding a logarithmic number of terminal block lengths and controlling a deficiency tail uniformly. Those losses need not be optimal. A sharper approach would keep the compatibility between a block length and the next first deficiency rather than impose a single worst-case cutoff. A small matrix of deficiency classes, with a positive-vector comparison analogous to lemma 12.2, is a concrete candidate. No asymptotic equivalent for this defect is asserted here.

19.2 The eventual ordering of the two components

Resolve whether $\lambda_V(m) > \lambda_U(m)$ for every $m \geq 5$, the ordering suggested by finite computations and related to Problem P3 in [2]. By corollary 15.4, a proof cannot be obtained merely by comparing their shared inverse-logarithmic expansion with the scalar root. It must detect finer endpoint losses or construct a comparison preserving more information than the present supersolution. One possibility is to compare first-return operators after isolating the large-threshold states, where long blocks are compatible with nearly all first entries.

19.3 The geometry of near-critical Perron vectors

Describe the positive eigenvectors of W_{U_m} and W_{V_m} at their critical roots as $m \rightarrow \infty$. The scalar tilted mass is concentrated in a layer of width about m/t_m near the maximal block length, while the chosen cutoff is only of order $\log m$. A useful question is whether endpoint coordinates admit a scaling limit connecting these two regimes. Such a limit could explain the loss of the scalar upper bound and supply an analytic route to the sign of the component gap. The present test vector is a supersolution, not a proved approximation to the true eigenvector.

19.4 Poles, amplitudes, and a uniform spectral gap

For each fixed m , determine when the dominant pole of A_m is simple and obtain a usable expression for the leading amplitude. An equality of component rates could interact with paths between components and alter pole order. Our argument identifies only their maximal exponential rate. A uniform analysis of subdominant singularities would be needed to turn fixed- m spectral information into estimates with useful error constants as m grows. This is distinct from the scalar root's own simple zero, which follows immediately from $K'_m(r_m) > 0$.

19.5 Joint growth of length and adjacency bound

Study $a_n^{(m(n))}/C_n$ when $m = m(n)$ grows with n . The fixed- m limit used here is not automatically uniform, so substituting $m(n)$ into α_m^n is not justified. Candidate regimes should compare the exponential penalty $n \log(4/\alpha_m)$ with the finite-state mixing scale and the Catalan polynomial factor. A satisfactory result would establish an explicit range of uniformity before proposing a transition law. The identity $a_n^{(m)} = C_n$ when $m \geq n - 1$ provides an exact boundary condition for such a theory. [Added 29 September 2026, batch 43: the boundary end of this direction is answered in Part III. For fixed d , $a_n^{(n-d)}/C_n$ converges to an explicit rational limit, and uniformly for $n \geq 8$, $2 \leq d \leq n - 1$, $a_n^{(n-d)}/C_n = 3/\sqrt{\pi d} + O(d^{-3/2} + n^{-1/2})$, an equivalent whenever $d \rightarrow \infty$ and $d = o(n)$ (theorem 22.2 and equation (136)). The direction stays open for $n - m \asymp n$ (remark 29.3) and for any interpolation with the fixed- m regime (section 32.10).] [Added 29 September 2026, batch 52: Part IV answers the linear regime $m = \lfloor \theta n \rfloor$ for $1/2 < \theta < 1$ to first order, $a_n^{(\lfloor \theta n \rfloor)}/C_n \sim \frac{3(2\theta-1)}{\sqrt{\pi\theta(1-\theta)}} n^{-1/2}$ (theorem 38.1); for $0 < \theta \leq 1/2$ it proves only $o(n^{-1/2})$ and polynomial lower bounds (theorem 45.1). The interpolation with the fixed- m regime stays open.] [Added 30 September 2026, batch 68: Part VI proves the matching order $a_n^{(\lfloor \theta n \rfloor)}/C_n =$

$\Theta(n^{-\lfloor 1/\theta \rfloor/2})$ for every fixed $0 < \theta < 1$ (theorem 68.1); no equivalent below half size is proved, and the interpolation with the fixed- m regime stays open.] [Added 1 October 2026, batch 71: Part VII proves the equivalent $I_3(\theta)/(2\pi n)$ for $1/3 < \theta \leq 1/2$ (theorem 80.2); below $1/3$ no equivalent is proved, and the interpolation stays open.] [Added 1 October 2026, batch 72: Part VIII proves the equivalent for every fixed $0 < \theta \leq 1/3$ as well, $\kappa_p I_p(\theta) n^{-(p-1)/2}$ in the phase $1/p < \theta \leq 1/(p-1)$ (theorem 91.1), so every fixed $0 < \theta < 1$ now has one; the interpolation with the fixed- m regime stays open.] [Added 1 October 2026, batch 72: Part IX treats the side $m \rightarrow \infty$, $n/m \rightarrow \infty$, each theorem with an explicit range of uniformity: a logarithmic bridge $\log(a_n^{(m)}/C_n) = n \log(\alpha_m/4) + O(n(\log m)^2/m^2 + \log n)$ for all $n \geq c_0 m \log m$ (theorem 105.1), so that substituting $m(n)$ into α_m^n is justified there at logarithmic precision; relative equivalents whenever also $n(\log m)^2/m^2 \rightarrow 0$ (theorems 106.1 and 108.2); and Gamma-sum equivalents below and at $m \log m$ (theorems 109.1 and 109.3). A relative equivalent for $n(\log m)^2/m^2 \not\rightarrow 0$ and the regime of bounded $\lceil (n-1)/m \rceil$ stay open (section 113).]

19.6 Renewal comparisons for other constrained tree classes

The reset identity lemma 12.1 is not specifically Catalan. It applies whenever a monotone threshold is lowered by weighted steps and restored by a reset. Identify other pattern-avoidance or tree-enumeration problems where the associated renewal sequence is explicitly controlled. If a truncated scalar coefficient law has a summable power tail, an analogous endpoint tilt should be investigated. The necessary research task is to prove a matching combinatorial lower construction; a scalar majorant alone does not make its asymptotic constant universal.

19.7 Weighted statistics and multivariate perturbations

Refine the endpoint recurrence with variables recording descents, records, or block statistics, in the direction of Problem P4 of [2]. The central question is whether a positive last-threshold witness persists under a parameter perturbation. A successful extension might permit derivative bounds for a weighted growth rate and limit laws at fixed m . Positivity and irreducibility must be checked in the perturbed system; formal differentiation of an unproved multivariate analogue of K_m would be insufficient.

19.8 State minimization versus a spectral majorant

Investigate whether the renewal witness can guide an exact reduction of the endpoint system without confusing domination with equality. Problems P1 and P2 in [2] concern issues such as state complexity and rational denominators. Our m -state reset comparator has determinant $1 - K_m$, but it forgets first-endpoint compatibility. One concrete goal is a characterization of the lost information as a correction operator of small rank or controlled support. An exact operator identity would be a substantive advance beyond the present scalar comparison.

19.9 Certified computation and formalization

Build an adaptive exact or interval algorithm choosing d to meet a requested error tolerance, using K_m and $K_{m,d}$ directly. This would avoid large component determinants and would certify large- m growth intervals with clear cost accounting. For formalization, the finite renewal polynomial identity, the spectral supersolution argument, the skew-cut injection, and the uniform deficiency-tail inequality are natural first targets. The analytic tail splitting and all-orders inversion can then be formalized as separate lemmas with explicit uniform quantifiers. None of this formalization has been started.

20 Proof boundaries, reproducibility and provenance

20.1 What the manuscript has established

The proof route can be read as a dependency chain:

endpoint system $\implies \alpha_m < \beta_m \implies$ sharp scalar corridor $\implies$ exact constant and all orders.

The lower arrow uses both recoverable skew blocks and the uniform first-entry tail; the analytic arrow uses the finite endpoint expansion with its stated remainder. No repository assertion about strict monotonicity is needed for the new theorems: [theorem 1.1](#) is not used. The finite-state growth interface is proved in Part I and was rederived in the manuscript, so it is also not being accepted solely on the authority of a software claim.

Table 10: Trust and scope ledger of Part II.

Statement or artifact	Validation status
All- m upper bound and asymptotic theorems	Written mathematical proofs here
Finite row, block, and tail tests	Executed exact integer/rational checks
First four inverse-log polynomials	Executed exact symbolic checks
Large- m numerical figures and component table	Floating-point illustrations only
New Lean or Rocq theorem declarations	None supplied or claimed
External peer review or exhaustive novelty verification	Not performed
Component ordering, optimal scalar defect, joint limits	Open directions (section 19) [joint limits near $m = n$: Part III, added 29 September 2026; for $1/2 < m/n < 1$: Part IV, added the same day]

20.2 Reproducing Part II's computations

Part II's files are

code/02-centered-remainder-model.py	Exact Catalan, endpoint, and block objects
code/02-centered-remainder-verify.py	Standard-library exact verification
code/02-centered-remainder-verify_expansion.py	Exact symbolic coefficient check
code/02-centered-remainder-compute.py	Floating-point numerical illustrations
code/02-centered-remainder-plot_results.py	The two figures from the CSV data
data/02-centered-remainder-*	Check reports, rational certificates, CSVs, environment, requirements, records of the delivered PDF
figures/02-centered-remainder-*.pdf	The two figures of this part

The scripts are shipped byte-identical to the delivery and still use the delivered names. They write

data/verification_report.json	data/rational_certificates.json
data/expansion_verification.json	data/scalar_numerics.csv
data/component_numerics.csv	figures/remainder_corridor.{pdf,png}
	figures/corridor_width.{pdf,png}

and the verifier and the numerical program execute from `model import`. Run in place, that import would load Part I's different `code/model.py` and fail with `ImportError`, and the outputs would land beside Part I's unprefix files. The report's README gives a recipe that restores the delivered layout in a scratch copy. There, the exact checks need Python 3.10 or newer and no third-party packages (`python code/verify.py`; do not use `-0`, the verifier uses assertions); the symbolic check needs SymPy; the numerical programs need NumPy, SciPy and, for the plots, Matplotlib (versions in `data/02-centered-remainder-environment.json`). A POSIX `OPENBLAS_NUM_THREADS=1` prefix on the numerical command is optional; it limits numerical-library threading rather than changing any mathematical assertion. The retained figures suffice to rebuild the article without rerunning any computation.

20.3 Points checked in the written argument

The sharp upper bound is not obtained by dropping a legitimate append term: that term is incorporated by the last-threshold vector. The first-entry tail estimate is uniform in both parameters, which permits a growing cutoff. The conversion from a polynomial value deficit to a root displacement uses a proved derivative lower bound, not a heuristic linearization. The order of inequalities is preserved when reciprocal roots become growth bounds and when growth bounds become deficit bounds. Finally, the factorially divergent formal series is used only to define finite coefficient calculations, while each analytic statement has its own fixed-order remainder.

These checks identify the parts of the proof most worth independent scrutiny. They do not replace that scrutiny, but they make its targets explicit and keep finite experiments distinct from universal arguments.

20.4 Repository provenance and novelty boundary

The manuscript inspected Part I at the repository-relative path

```
SetTheory/Cardinals/docs/reports/
enumerative-combinatorics/
adjacency-bounded-132-avoiders/article.tex
```

at the ProveIt commit

```
6c9175a54bcaad3bfc2d416257d2b7d3dff2c1e0
```

where its Git blob identifier was

```
5ec75ee48df94b6d12804a26ba109c08564df5c7
```

Its section “What is resolved, and what remains” (section 9) explicitly asks whether E_m converges and describes the unresolved constant window; the opening theorem statements specify the window in (3). The finite endpoint system and lower block construction used here are attributed to their prior sources, with independent proofs supplied in the manuscript.

The manuscript’s primary-paper check used the arXiv version-one records and HTML sources for [1, 2], accessed on September 28, 2026. The inspected materials did not contain the last-threshold supersolution, the sharp scalar corridor, or the exact constant proved here. This is a bounded provenance statement. It does not certify the absence of an independent unpublished argument, a later repository change, or an uninspected paper.

20.5 Where the merge had to choose

- *Placement.* Part II follows Part I’s last main section and precedes Part I’s appendices, which contain no numbered equations, tables or figures; no number of Part I changed.
- *Duplicates.* The endpoint recurrence, the component classification and spectral interface (the manuscript’s Section 2 and Appendix A), the first-entry lemma and the injective skew-block construction are printed once, in Part I, and credited in sections 11 and 13. The manuscript’s different argument for $\lambda_U(m) > 1$ is kept as a second proof. The second proof of corollary 14.2 and remarks 12.3, 13.3 and 17.1 are the merge’s.
- *Notation.* $L \rightarrow \Lambda$, $P_j \rightarrow \mathcal{P}_j$, $B_N \rightarrow \mathcal{B}_N$, $T_m \rightarrow \Delta_m$, $C, c \rightarrow C_*, c_*$ (absolute constants), $W_U, W_V \rightarrow W_{U_m}, W_{V_m}$, and $H = \log m$ written out (table 4). No normalization changed.
- *References to Part I.* The manuscript cites Part I as a repository report; here those citations are internal cross-references, and the manuscript’s bibliography entries for [1, 2] are Part I’s.
- *Reproduction.* The manuscript’s archive listing and commands are replaced by the shipped names and the copy recipe (section 20.2); its `article.tex`, `article.pdf`, `README.md` and the PNG versions of its figures are not shipped.

- *Figures.* In the write phase the manuscript's two figure PDFs were copied byte-identically into figures/, under their delivered names with the prefix 02-centered-remainder- (figures 3 and 4).
- *Stale sentences in Part I.* The abstract, theorem 1.3, the remark after theorem 1.3's proof, section 9 and section B carry dated pointers to Part II; their original wording is kept.

Part III. The largest jump near the boundary

Part III was added on 29 September 2026 in batch 43 of ProveIt's incoming-report intake, from one manuscript written against the state of Parts I and II printed above. Parts I and II take $n \rightarrow \infty$ with the adjacency bound m fixed. Part III takes the other side of the joint problem: the bound sits a distance d below its trivial value, $m = n - d$, with d fixed or growing slower than n . It answers Part II's research direction "Joint growth of length and adjacency bound" (section 19) in that boundary regime only; the bulk regime $n - m \asymp n$ and the fixed- m spectral questions stay open (section 21.3).

21 Provenance, conventions and the answer

21.1 One manuscript, one addition

Part III is batch-43 manuscript 02, *The Largest Jump Is Almost Maximal: An algebraic boundary law for random 132-avoiding permutations*, an AI-assisted research manuscript prepared for Vladimir Reshetnikov and dated September 29, 2026 (26 pages, US Letter, as delivered). It arrived as the archive `ProveIt_Largest_Jump_Research.zip` (inner directory `ProveIt_Largest_Jump/`, main file `article.tex`) in ProveIt commit `06bcc37a8` and was placed in `faef2ed2a`. It was written against ProveIt commit `8d936ee23`, at which this report's `README.md` and `article.tex` had the Git blobs `7f56b878` and `f24ea59a`. Both blobs were unchanged at the placement commit, so every statement the manuscript makes about this report refers to Parts I and II as printed above. Its code, data and audit notes are shipped beside the earlier parts' files with the prefix `03-largest-jump-` (Part I's unprefix files count as 01, Part II's `02-centered-remainder-` files as 02); its manuscript, README and PDF are not shipped and survive in the arrival commit. Its two figures were added to `figures/` in the write phase (section 36.2).

The manuscript contributes, in the order printed here: its main theorems and their scope (section 22); first- and last-entry identities for 132-avoiders (section 23); a countable probability space of finite absorbing skeletons (section 24); a deterministic reconstruction lemma (section 25); a total-variation bound by counting domination (section 26); the exact algebraic probability generating function of the limit law (section 27); singularity analysis and a sharp rate (section 28); the joint limits for the adjacency bound (section 29); a moment transition (section 30); exact probabilities and finite checks (section 31); ten research questions (section 32); a proof audit (section 33); and, as the last three sections of this part, its coefficient recursions, an algebraic certificate and its provenance (sections 34 to 36), which were appendices in the manuscript. Every result, proof, example, remark, limitation and question of the manuscript is printed.

Status (manuscript 02). The theorems of Part III have written proofs and reproducible exact finite checks. They are not Lean- or Rocq-verified and have not undergone external peer review. The contribution is an extension of the report's subject, not a verified claim of publication priority. No asymptotic theorem from an unrefereed repository report is assumed: Part III uses none of the growth-constant theorems of Parts I and II. Nothing in Part III is formalized in any proof assistant.

21.2 Notation of Part III

The symbols $a_n^{(m)}$, C_n , C and $\text{Av}_n(132)$ keep the meanings of Parts I and II, and m is again the adjacency bound and n the length. Part III is otherwise self-contained, and several of its letters are used in Parts I and II for other objects. Table 11 lists them, with the tempting false reading in the last column; five symbols are renamed from the manuscript.

No normalization changed: every renamed symbol denotes exactly the manuscript's quantity. Probability, expectation, total variation and law are written $\mathbb{P}$, $\mathbb{E}$, d_{TV} , $\mathcal{L}$, as in the manuscript.

Table 11: Symbols of Part III. “Renamed” gives the manuscript’s own symbol. The last column prints the tempting false reading.

Symbol	Meaning in Part III	Not to be confused with
$M(\pi), M_n$	largest adjacent absolute difference; that of a uniform $\Pi_n \in \text{Av}_n(132)$	the matrix M of lemma 11.3.
$D(\pi), D_n, D$	the deficit $n - M(\pi)$, its value $n - M_n$, and the limit law of D_n	Part I’s common denominator $D(x)$ of (34), its denominators $D_U(x), D_V(x)$ (section A), the integer D of its Perron certificates, and the exponent D in the proof of proposition 3.3.
d	a deficit threshold, $m = n - d$; also the first coordinate of the endpoint state (d, ℓ)	Part I’s fixed cutoff d and Part II’s $d_m = \lceil \log_2 m \rceil$; neither is tied to n .
$F(\pi) = \pi_1$	first entry	Part I’s series $F_{p,q}$. Part I’s first deficiency is $n - F(\pi)$.
$N_{N,f}$	avoiders of size N with first entry f	the same count as Part I’s $h(k, j)$, with $k = N + 1, j = N + 1 - f$ (remark 23.2).
$P(x, z)$	bivariate first-entry series	Part I’s numerators P_3, P_4 ; Part II’s $\mathcal{P}_j$.
R_β, A, T_τ	skeleton operations; β, τ are avoiders	$R_m = x + K_m, A_m(x)$, the counts $T_{p,q}^{(m)}(n)$, and Part II’s number $\beta_m = r_m^{-1}$.
t, T	cost of a skeleton; the random cost	Part II’s $t_m = m \log(4r_m)$; $T_{p,q}^{(m)}(n)$.
$b_t, B(x)$	number of skeletons of cost t ; $B = (C - 1)^2$	Part II’s $b_k = C_{k-1}4^{-k}$; Part I’s threshold set B_m and B_d .
c_j (one index)	$C_j/4^j$	Part I’s $c_{k,p}$ (two indices).
h (no argument)	the cutoff $\lfloor n/4 \rfloor$, or a cutoff below $n/2$	Part I’s first-entry count $h(k, j)$.
L	random last-entry state, in $\mathbb{N}_{\geq 0} \cup \{\infty\}$	the left block L of $\pi = L n R$ (also used here), Part I’s L_m .
ω_n (renamed from ε_n)	$n(n + 1)(2n + 1)2^{-n/2}$	Part II’s $\varepsilon_d = (d + 2)2^{-d}$, a different bound for a related fraction (remark 23.2).
$z; s, r$	generating-function variable; $\sqrt{1 - z}$, $\sqrt{(1 + s)/2}$	Part II’s $z = \log \log m + \frac{1}{2} \log \pi$; Part I’s $s_{m,d}$; Part II’s root r_m and r_U, r_V .
$q = Q(z)$	$(1 - \sqrt{1 - z})/2$	Part I’s root q_m and threshold index q .
G, H, J	generating function of D ; its parts with $L < \infty$ and $L = \infty$	Part II’s $H_q(x)$, the matrix H in the proof of proposition 3.3, Part II’s order index J .
$\mathcal{K}(z)$	clipping series	$K_{m,d}$ (Part I), K_m (Part II).
$\Upsilon(z)$ (renamed from $\mathcal{A}$)	clipping correction	Part I’s class $\mathcal{A}_n^{(m)}$, the manuscript’s own glyph.
$\kappa_{1-\varepsilon}$ (renamed from $q_{1-\varepsilon}$)	quantile of D	$q = Q(z)$.
C_{TV} (renamed from C, C_0), $\underline{c}_p$ (renamed from c_p)	constants in two proofs	the Catalan numbers, $C_0 = 1$; c_j ; $c_{k,p}$; Part II’s C_*, c_* .
p	a moment order	Part I’s threshold $p \in B_m$.

The equations of Part III are numbered consecutively with those of Parts I and II, not by section as in the manuscript.

21.3 Which question is answered, and how far

Part II's research direction "Joint growth of length and adjacency bound" (section 19) asks for $a_n^{(m(n))}/C_n$ when $m = m(n)$ grows with n , warns that the fixed- m limit is not automatically uniform, asks that "a satisfactory result" establish an explicit range of uniformity before a transition law, and names the identity $a_n^{(m)} = C_n$ for $m \geq n - 1$ as a boundary condition. The report's README lists "no joint limit in n and m " among the things not claimed. Part III settles the regime next to that boundary condition:

- for each fixed d , $a_n^{(n-d)}/C_n \rightarrow \mathbb{P}(D \geq d)$, an explicit rational number (equation (136));
- uniformly for $n \geq 8$ and $2 \leq d \leq n - 1$, $a_n^{(n-d)}/C_n = 3/\sqrt{\pi d} + O(d^{-3/2} + n^{-1/2})$ with an absolute constant (theorem 22.2), which is an explicit range of uniformity of the kind Part II asks for;
- hence $a_n^{(n-d)}/C_n \sim 3/\sqrt{\pi d}$ whenever $d \rightarrow \infty$ and $d = o(n)$.

It does not settle the question. When $n - m$ is a fixed fraction of n , the additive error is as large as the main term and no equivalent is claimed (remark 29.3); for fixed m the growth-constant regime of Parts I and II is untouched; and nothing connects the two regimes yet (section 32.10). Part II's direction is therefore re-scoped, not closed: its boundary end is answered here, and its bulk and interpolation questions remain open. Part III refutes nothing in Parts I and II and uses none of their theorems. [Added 29 September 2026, batch 52: Part IV settles the bulk regime for $1/2 < m/n < 1$ to first order and determines the exact supercritical moment constants (section 37.3); the regime $m/n \leq 1/2$ and the interpolation with fixed m remain open.]

22 The question, its context, and the main results

22.1 A different order of limits

As in Part I, a permutation $\pi \in S_n$ avoids 132 if no three indices $i < j < k$ satisfy $\pi_i < \pi_k < \pi_j$. Write $\text{Av}_n(132)$ for this class, and let

$$C_n = \frac{1}{n+1} \binom{2n}{n}, \quad C(x) = \sum_{n \geq 0} C_n x^n = \frac{1 - \sqrt{1 - 4x}}{2x}.$$

For $n \geq 2$, set

$$M(\pi) = \max_{1 \leq i < n} |\pi_{i+1} - \pi_i|, \quad D(\pi) = n - M(\pi). \quad (81)$$

If Π_n is uniform on $\text{Av}_n(132)$, write $M_n = M(\Pi_n)$ and $D_n = D(\Pi_n)$. Thus $1 \leq D_n \leq n - 1$.

Let $a_n^{(m)}$ count the avoiders with every adjacent difference at most m . The identity connecting enumeration and probability is

$$\frac{a_n^{(m)}}{C_n} = \mathbb{P}(M_n \leq m) = \mathbb{P}(D_n \geq n - m). \quad (82)$$

The fixed- m problem studies $n \rightarrow \infty$ with m unchanged. Part III studies the boundary $m = n - d$, and then allows d to grow with n . Neither problem is a substitution into the other.

Nadler initiated the adjacency-constrained enumeration and analyzed $m = 2$ [1]. Mayama and Akita supplied a uniform endpoint-state system, rational generating functions for every fixed m , and a spectral characterization of growth [2]. The ProveIt report *Strict growth and the Catalan limit for adjacency-bounded 132-avoiding permutations*, including its September 28 continuation, that is, Parts I and II above, addresses strict growth and increasingly precise fixed-bound growth

asymptotics. Its README explicitly excludes a joint limit in n and m from its claims (at the pin recorded in section 36).

Our connection to that report is the same counting class and the same maximum decomposition, not a dependence on its claimed growth-constant estimates. The proofs below derive their own finite combinatorics. The infinite-tree viewpoint is standard background in conditioned tree probability [3]; here it is made into an explicit finite counting argument. Other statistics on random 132-avoiders have been studied through tree and excursion methods [4]; those results are not substituted for the extreme-jump analysis.

22.2 The law and the rate

For real $0 \leq z \leq 1$, put

$$s = \sqrt{1-z}, \quad r = \sqrt{\frac{1+s}{2}}. \quad (83)$$

Both square roots are the nonnegative ones. Around $z = 0$, they denote the analytic branches with $s(0) = r(0) = 1$.

Theorem 22.1 (Algebraic boundary law). *There is a positive integer-valued random variable D with probability generating function*

$$G(z) = \mathbb{E}z^D = \frac{(1-s)(4+8s+3s^2-s^2r)}{4(1+2s)(1+s)^2}. \quad (84)$$

Moreover,

$$d_{\text{TV}}(\mathcal{L}(D_n), \mathcal{L}(D)) = \Theta(n^{-1/2}). \quad (85)$$

With $h = \lfloor n/4 \rfloor$, an explicit bound for $n \geq 8$ is

$$d_{\text{TV}}(\mathcal{L}(D_n), \mathcal{L}(D)) \leq \min \left\{ 1, \frac{2(h+1)(2h+1)}{h+2} \frac{C_h}{4^h} + n(n+1)(2n+1)2^{-n/2} \right\}. \quad (86)$$

In particular, $D_n = O_{\mathbb{P}}(1)$ and $M_n/n \rightarrow 1$ in probability.

The convention for total variation is $d_{\text{TV}}(\mu, \nu) = \sup_B |\mu(B) - \nu(B)|$, equivalently half the ℓ^1 distance on a countable space. The explicit bound is designed for a transparent proof rather than a small finite- n constant. The matching lower bound follows from the limiting heavy tail and the finite support of D_n , as shown in corollary 28.2.

Theorem 22.2 (Tail and joint enumeration). *As $k \rightarrow \infty$,*

$$\mathbb{P}(D = k) = \frac{3}{2\sqrt{\pi}} k^{-3/2} + O(k^{-5/2}), \quad (87)$$

$$\mathbb{P}(D > k) = \frac{3}{\sqrt{\pi}} k^{-1/2} + O(k^{-3/2}). \quad (88)$$

Uniformly for integers $n \geq 8$ and $2 \leq d \leq n-1$,

$$\frac{a_n^{(n-d)}}{C_n} = \frac{3}{\sqrt{\pi d}} + O(d^{-3/2} + n^{-1/2}), \quad (89)$$

with an absolute implied constant. Consequently, if $d \rightarrow \infty$ and $d = o(n)$, then

$$\frac{a_n^{(n-d)}}{C_n} \sim \frac{3}{\sqrt{\pi d}}, \quad a_n^{(n-d)} \sim \frac{3}{\pi} \frac{4^n}{n^{3/2} \sqrt{d}}. \quad (90)$$

Theorem 22.3 (Moment transition). *For each fixed real $p > 0$,*

$$0 < p < \frac{1}{2} : \mathbb{E}D_n^p = \mathbb{E}D^p + O(n^{p-1/2}), \quad (91)$$

$$p = \frac{1}{2} : \mathbb{E}\sqrt{D_n} = \frac{3}{2\sqrt{\pi}} \log n + O(1), \quad (92)$$

$$p > \frac{1}{2} : \mathbb{E}D_n^p = \Theta(n^{p-1/2}). \quad (93)$$

The constants in the first and third lines may depend on p . In particular,

$$\mathbb{E}M_n = n - \Theta(\sqrt{n}), \quad \text{Var}(D_n) = \Theta(n^{3/2}). \quad (94)$$

22.3 What is and is not resolved

The results determine the entire fixed-deficit boundary law, give a uniform quantitative approximation to it, and establish a growing-deficit regime. They do not determine the asymptotic equivalent of $\mathbb{E}D_n$, nor the fraction $a_n^{(\lfloor \theta n \rfloor)} / C_n$ to relative precision for fixed $0 < \theta < 1$. The additive error in (89) has the same order as its displayed term when d is proportional to n ; it must not be advertised as an equivalent there. The component-ordering question in the fixed- m finite-state system is also untouched.

23 Catalan decompositions and endpoint identities

23.1 Decomposition at the maximum

If $\pi = L n R$ avoids 132, every value of L is greater than every value of R . Conversely, two avoiding standardized blocks with this separation give an avoiding permutation. If $|R| = b$, the left block is obtained by adding b to each entry of an avoider of size $n - b - 1$. This proves

$$C_n = \sum_{b=0}^{n-1} C_b C_{n-b-1}, \quad C(x) = 1 + xC(x)^2.$$

All the reconstruction operations used below are instances of this bijection; no approximate counting is involved. (The decomposition is the one of Part I, section 2, where the separation property is also proved; manuscript 02 re-derives it independently.)

23.2 First entries and first deficiencies

For a nonempty avoider π , let $F(\pi) = \pi_1$. Define

$$P(x, z) = \sum_{n \geq 1} \sum_{\pi \in \text{Av}_n(132)} x^n z^{F(\pi)}.$$

When the left block at the maximum is empty, the first value is the maximum. When it is nonempty, its first value is shifted by the size of the right block. Hence

$$P(x, z) = xzC(xz) + xC(xz)P(x, z),$$

and therefore

$$P(x, z) = \frac{xzC(xz)}{1 - xC(xz)}. \quad (95)$$

Replacing x by xu and z by u^{-1} gives the useful deficiency form

$$\sum_{n \geq 1} \sum_{\pi \in \text{Av}_n(132)} x^n u^{n-F(\pi)} = \frac{x C(x)}{1 - u x C(x)}. \quad (96)$$

In particular, for every $j \geq 0$,

$$\sum_{n \geq j+1} \#\{\pi \in \text{Av}_n(132) : n - F(\pi) = j\} x^n = (xC(x))^{j+1}. \quad (97)$$

Lemma 23.1 (First-entry count). *For $1 \leq f \leq N$,*

$$N_{N,f} := \#\{\pi \in \text{Av}_N(132) : F(\pi) = f\} = \frac{N-f+1}{N} \binom{N+f-2}{N-1}. \quad (98)$$

Moreover, for $1 \leq t \leq N$,

$$\frac{\#\{\pi \in \text{Av}_N(132) : F(\pi) \leq t\}}{C_N} \leq t(N+1)(2N+1)2^{t-N-2}. \quad (99)$$

Proof. The coefficient formula

$$[x^a]C(x)^b = \frac{b}{2a+b} \binom{2a+b}{a}$$

follows by applying Lagrange inversion to $C = 1 + xC^2$. Apply it to (97), with $j = N - f$, to obtain (98); the two binomial forms are equal by cancellation of factorials. For the bound, discard the prefactor, use $\binom{u}{v} \leq 2^u$, and sum over $f \leq t$:

$$\sum_{f \leq t} N_{N,f} \leq t2^{N+t-2}.$$

Since the largest binomial coefficient in the expansion of $(1+1)^{2N}$ is central, $\binom{2N}{N} \geq 4^N/(2N+1)$. Dividing proves the claim. $\square$

Remark 23.2 (Merge note: the same count in Parts I and II). Deleting the final maximum of a block of length k that ends at its maximum and has first entry $k - j$ leaves an avoider of size $N = k - 1$ with first entry $f = N + 1 - j$. Hence $N_{N,f} = h(N + 1, N + 1 - f)$, and (98) is Part I's formula (21) for $h(k, j)$, reindexed. Part I proves it by a slot decomposition at the first entry (lemma 5.3), manuscript 02 by the maximum decomposition and the bivariate series (95); both finish with Lagrange inversion. The manuscript's route is kept here, because (95) and (97) are used again in section 27.

The fraction bounded in (99) is also bounded by Part II's uniform tail estimate. With $k = N + 1$ and $d = N - t \geq 1$, the difference $C_N - c_{N+1,N-t}$ counts exactly the avoiders of size N with first entry at most t , so (54) gives

$$\frac{\#\{\pi \in \text{Av}_N(132) : F(\pi) \leq t\}}{C_N} \leq (N - t + 2)2^{-(N-t)} \quad (1 \leq t \leq N - 1).$$

This is sharper than (99) by a factor of order N^3 . In the setting of lemma 26.1, $N - t = n - 2t \geq n/2$, and $x \mapsto (x + 2)2^{-x}$ decreases for $x \geq 0$, so the exceptional fraction is at most $(n/2 + 2)2^{-n/2}$, which is below 1 for every $n \geq 8$; the term $n(n+1)(2n+1)2^{2h-n}$ in the proof of corollary 28.2 can likewise be replaced by $(n - 2h + 2)2^{2h-n}$. Neither replacement changes an order of magnitude or any statement of Part III, which prints the manuscript's bounds. This remark is the merge's, not the manuscript's; the bound was checked by exhaustive enumeration for $N \leq 8$.

23.3 Last entries

A particularly simple decomposition occurs at the last value ℓ . Every entry greater than ℓ must precede every entry smaller than ℓ : otherwise the smaller entry, the greater entry, and the final ℓ form 132. Thus, uniquely,

$$\pi = (\tau + \ell) \sigma \ell, \quad |\tau| = n - \ell, \quad |\sigma| = \ell - 1, \quad (100)$$

where τ and σ are arbitrary avoiders of those sizes. The converse is verified directly, or by the maximum decomposition. Consequently,

$$\#\{\pi \in \text{Av}_n(132) : \pi_n = \ell\} = C_{n-\ell} C_{\ell-1}. \quad (101)$$

An additional consequence will control the clipping operation: if the last value is not the maximum, then the first value is greater than the last value. Indeed, τ in (100) is then nonempty.

24 A finite absorbing decomposition

24.1 Three operations with one unexpanded block

We use a finite decorated word, called a *skeleton*, whose outer operations end at one terminal operation. The unexpanded block will later be filled with a large avoiding permutation. There are three operation types:

R_β : a large right block.

Here β is any avoider of size $b \geq 0$. Applied to a core σ of size N , it returns $(\beta + N)(N + b + 1)\sigma$. Its cost is $b + 1$.

A : append a maximum.

Applied to a core of size N , it returns $\sigma(N + 1)$. Its cost is 1.

T_τ : a terminal finite right block.

Here τ is any nonempty avoider of size $b \geq 1$. Applied to the hole σ of size N , it returns $(\sigma + b)(N + b + 1)\tau$. Its cost is $b + 1$.

A skeleton is a word in R_β and A , followed by exactly one T_τ . Operations act from the terminal outward. Its cost t is the sum of their costs, so inserting a hole of size N gives a permutation of size $N + t$. All resulting permutations avoid 132.

For example, the word $AR_{(1)}T_{(2,1)}$ has cost $1 + 2 + 3 = 6$. The terminal will initialize the pair $(2, 1)$, $R_{(1)}$ changes it to $(4, 1)$, and A changes it to $(1, \infty)$. Thus this skeleton produces a maximal possible jump, once a sufficiently large hole with a sufficiently large first value is inserted.

24.2 A probability measure on skeletons

Give each individually decorated operation of cost c weight 4^{-c} . The total weights are

$$\sum_{\beta} 4^{-(|\beta|+1)} = \frac{1}{2}, \quad w(A) = \frac{1}{4}, \quad \sum_{\tau \neq \emptyset} 4^{-(|\tau|+1)} = \frac{1}{4}. \quad (102)$$

Thus one can repeatedly choose an operation using these weights, stopping when a terminal is chosen. The probability of not stopping after j choices is $(3/4)^j$. Every skeleton is finite almost surely, and a specified decorated skeleton of cost t has probability 4^{-t} .

This is a probability measure on a countable set, not a heuristic “typical giant block” assertion. The finite comparison in the next two sections will connect it to the uniform permutation measure.

24.3 Exact cost distribution

Let b_t be the number of skeletons of cost t . Their ordinary generating function is

$$B(x) = \sum_{t \geq 0} b_t x^t = \frac{x(C(x) - 1)}{1 - xC(x) - x} = (C(x) - 1)^2. \quad (103)$$

For the last equality, note that $1 - xC(x) - x = C(x)^{-2}$ and $xC(x)^2 = C(x) - 1$. For $t \geq 1$,

$$b_t = C_{t+1} - 2C_t = \frac{2(t-1)}{t+2}C_t, \quad b_0 = b_1 = 0. \quad (104)$$

If T denotes the random cost, then

$$\mathbb{E}z^T = B(z/4) = \left(\frac{1 - \sqrt{1-z}}{1 + \sqrt{1-z}} \right)^2. \quad (105)$$

Lemma 24.1 (Exact cost tail). *For every integer $h \geq 0$,*

$$\mathbb{P}(T > h) = \frac{2(h+1)(2h+1)}{h+2} \frac{C_h}{4^h} \sim \frac{4}{\sqrt{\pi h}}. \quad (106)$$

Proof. Put $c_j = C_j/4^j$. The Catalan ratio is $c_{j+1}/c_j = (2j+1)/(2j+4)$, and induction gives $\sum_{j=0}^h c_j = 2 - (2h+1)c_h$. Using $b_t 4^{-t} = 4c_{t+1} - 2c_t$ for $t \geq 1$, with the constant-term adjustment at $t = 0$, yields

$$\sum_{t=0}^h b_t 4^{-t} = 1 - \frac{2(h+1)(2h+1)}{h+2} c_h.$$

Stirling's formula gives the equivalent. □

24.4 The endpoint state and its transformations

The state is a pair (d, ℓ) , where d is a positive integer and $\ell \in \mathbb{N}_{>0} \cup \{\infty\}$. The symbol ∞ records a last entry close to the high end of a large block; it is not an actual entry of a finite permutation. Starting at the terminal, define

$$T_\tau : (d, \ell) = (F(\tau), \tau_{|\tau|}), \quad (107)$$

$$R_\beta : (d, \ell) \mapsto (d + |\beta| + 1, \ell), \quad (108)$$

$$A : (d, \ell) \mapsto (\min\{d+1, \ell\}, \infty), \quad (109)$$

where $\min(d+1, \infty) = d+1$. Let (D, L) be the resulting pair on the probability space of skeletons. It is finite in its first coordinate almost surely and obeys

$$1 \leq D \leq T - 1. \quad (110)$$

This defines the candidate law probabilistically before any generating function is simplified; coefficient nonnegativity will therefore not need to be guessed from the radical expression.

25 The deterministic reconstruction lemma

The state rules need proof: an endpoint heuristic would not justify a law for the largest jump. The following statement is the central finite lemma.

Lemma 25.1 (A large hole determines the extreme jump). *Fix a skeleton s of cost t and state (d, ℓ) . Insert an avoider σ of size $N > t$ whose first entry satisfies $F(\sigma) > t$. Let π be the resulting permutation of size $n = N + t$. Then*

$$D(\pi) = d. \quad (111)$$

If $\ell < \infty$, then $\pi_n = \ell$. If $\ell = \infty$, then $\pi_n \geq N + 1$. At every intermediate reconstruction stage, the first entry is greater than t and the largest jump is at least $N + 1$.

Proof. Proceed from the terminal outward. Suppose the terminal decoration has size $b \geq 1$, first entry f , and last entry ℓ . The reconstructed terminal block is

$$(\sigma + b)(N + b + 1)\tau.$$

The jump from its maximum to f has size $N + b + 1 - f \geq N + 1$. The jumps internal to σ are at most $N - 1$; the jump into the maximum from the left is at most N ; and the jumps internal to τ are at most $b - 1 < N + 1$. Thus that downward jump is largest, its deficit is exactly f , and the last value is ℓ . The first value is $F(\sigma) + b > t$.

Maintain the following stronger invariants. If c is the cost used so far, the current length is $N + c$, its deficit d_c satisfies $d_c \leq c - 1$, its first entry exceeds t , and its last entry is either the recorded finite value, at most t , or at least $N + 1$ when the state records ∞ . The largest jump is then at least $N + 1$.

For R_β , let $b = |\beta|$ and let the old length be a . The old internal maximum jump has new deficit $d_c + b + 1$. The finite left block contributes only jumps of size at most b , including its boundary with the new maximum. These cannot exceed the old maximum jump. The jump from the new maximum to the old first value is strictly less than $a + b + 1 - t$, whereas the old maximum is

$$a - d_c = (a + b + 1) - (d_c + b + 1) \geq (a + b + 1) - t + 1,$$

because the new accumulated cost is at most t . So the old internal maximum remains largest, giving (108). The last entry is unchanged. The new first entry is either the new maximum or a value exceeding a , hence exceeds t since $a \geq N > t$.

For A , appending the new maximum changes the deficit of the old largest jump from d_c to $d_c + 1$. The new boundary jump has deficit equal to the old last value. When that value is finite, the new deficit is precisely $\min(d_c + 1, \ell)$. When it is recorded as ∞ , the actual value is at least $N + 1 > t$, while $d_c + 1 \leq t - 1$; the new jump cannot win. The last value becomes the new maximum, at least $N + 1$, and the first value is unchanged. These are (109) and all the invariants. $\square$

Lemma 25.2 (Disjoint reconstruction families). *Fix a total size n and a cutoff $h < n/2$. Reconstructions from distinct skeletons of cost $t \leq h$, with holes of size $n - t$, are disjoint. For each fixed skeleton, reconstruction is injective in the hole.*

Proof. At every maximum decomposition, the block containing the hole is larger than its finite sibling: its size is at least $n - t > t$, while the sibling has size at most t . Hence it is identifiable from the final permutation. A large right block specifies an R operation. A large left block with empty right block specifies A . A large left block with nonempty finite right block specifies the terminal T , where parsing stops. Each finite decoration is recovered from the actual sibling values by standardization. The remaining large left block at that terminal recovers the hole. Thus the whole decorated word and the hole are uniquely determined. $\square$

Remark 25.3. The first-entry hypothesis cannot simply be deleted. A very small first entry in the hole can create an additional large downward jump at an outer R operation. The next section bounds precisely this exceptional set. Likewise, using only the deficit and omitting the last-entry state would lose the clipping at an append operation.

26 Counting domination and total variation

26.1 Why each short skeleton has enough finite mass

The normalized Catalan numbers $c_j = C_j/4^j$ decrease strictly, because $c_{j+1}/c_j = (2j+1)/(2j+4) < 1$. Consequently, whenever $0 \leq t < n$,

$$\frac{C_{n-t}}{C_n} = 4^{-t} \frac{c_{n-t}}{c_n} \geq 4^{-t}. \quad (112)$$

The left side is the probability mass of the reconstruction family of one fixed skeleton of cost t under the uniform size- n measure, before the first-entry restriction is imposed. The right side is exactly that skeleton's probability under the limiting construction. This one-sided comparison is stronger for our purpose than an unquantified local convergence statement.

Lemma 26.1 (Exceptional first entries). *For $n \geq 8$, $t \leq \lfloor n/4 \rfloor$, and $N = n - t$, the fraction of holes in $\text{Av}_N(132)$ with first entry at most t is at most*

$$\omega_n = n(n+1)(2n+1)2^{-n/2}. \quad (113)$$

Proof. Apply (99). Since $t - N = 2t - n \leq -n/2$ and $t \leq n$, $N \leq n$, its right side is at most the displayed bound. The deliberately enlarged prefactor avoids unnecessary finite-size casework. If it exceeds 1, the assertion is still valid. $\square$

26.2 A common subprobability measure

For probability measures on a countable space, a common subprobability measure of mass $1 - \delta$ implies total-variation distance at most δ . This follows immediately from $d_{\text{TV}}(\mu, \nu) = 1 - \sum_x \min(\mu\{x\}, \nu\{x\})$.

Proof of the rate in theorem 22.1. Let $h = \lfloor n/4 \rfloor$ and suppose first that $\omega_n \leq 1$. For each skeleton s of cost $t \leq h$, the good reconstruction family has probability at least

$$\frac{C_{n-t}}{C_n}(1 - \omega_n) \geq 4^{-t}(1 - \omega_n).$$

The families are disjoint by lemma 25.2. Every good member has deficit $d(s)$ by lemma 25.1. The limiting deficit law also puts the skeleton mass 4^{-t} at that same value. After summing over skeletons, even when multiple skeletons yield the same deficit, the two deficit laws have a common subprobability measure of mass at least

$$(1 - \omega_n) \sum_{\text{cost}(s) \leq h} 4^{-\text{cost}(s)} = (1 - \omega_n) \mathbb{P}(T \leq h).$$

It follows that

$$d_{\text{TV}}(\mathcal{L}(D_n), \mathcal{L}(D)) \leq \mathbb{P}(T > h) + \omega_n. \quad (114)$$

If $\omega_n > 1$, the same displayed upper bound is trivial. Insert the exact tail from lemma 24.1 and combine with the universal bound $d_{\text{TV}} \leq 1$. Stirling's formula gives

$$\mathbb{P}(T > \lfloor n/4 \rfloor) = \frac{8}{\sqrt{\pi n}} + O(n^{-3/2}),$$

while ω_n is exponentially small up to its polynomial factor. This proves the upper bound in (85) and proves (86); the lower bound is given in corollary 28.2. Since D is a proper finite-valued random variable, the laws of D_n are tight, and $M_n/n = 1 - D_n/n \rightarrow 1$ in probability. $\square$

Remark 26.2 (The order is sharp; the leading constant remains open). The cutoff $n/4$ makes every inequality transparent. After deriving the limiting tail, corollary 28.2 will prove the matching order and improve the upper constant by taking a cutoff approaching $n/2$. These arguments do not identify an exact asymptotic constant for the total-variation error.

27 Evaluation of the algebraic probability generating function

27.1 Renewal notation

For $0 \leq z \leq 1$, write

$$q = Q(z) = \frac{z}{4}C(z/4) = \frac{1 - \sqrt{1-z}}{2}. \quad (115)$$

Then

$$z = 4q(1-q), \quad \frac{1}{1-q} = C(z/4), \quad s = 1 - 2q. \quad (116)$$

An R_β operation adds $|\beta| + 1$ to the deficit and has weight $4^{-(|\beta|+1)}$. Its deficit-increment generating function is therefore q . A sequence of such operations has generating function

$$\frac{1}{1-q} = \sum_{b \geq 0} \frac{C_b}{4^b} z^b. \quad (117)$$

Here b records the *sum of increments*, not the number of operations and not the size of a single decoration.

Define two subprobability generating functions

$$H(z) = \mathbb{E}[z^D; L < \infty], \quad J(z) = \mathbb{E}[z^D; L = \infty], \quad G = H + J. \quad (118)$$

The last-entry state is finite exactly when there is no A operation: it starts finite at the terminal, R preserves it, and A changes it permanently to ∞ .

27.2 The finite-last part

The terminal first-entry series is $\frac{1}{4}P(\frac{1}{4}, z)$. By (95),

$$P(\frac{1}{4}, z) = \frac{z}{1 + 2\sqrt{1-z}} = \frac{z}{3-4q}.$$

Adding an arbitrary outer sequence of R operations gives

$$H(z) = \frac{P(\frac{1}{4}, z)}{4(1-q)} = \frac{q}{3-4q}. \quad (119)$$

We also need the last-entry marginal, rather than its joint relation with D . From (101),

$$\sum_{n \geq 1} \sum_{\pi \in \text{Av}_n(132)} x^n z^{\pi_n} = C(x) x z C(xz).$$

At $x = 1/4$ the terminal last-entry series is $q/2$. An arbitrary sequence of R operations has total weight $1/(1-Q(1)) = 2$. Therefore

$$\mathbb{E}[z^L; L < \infty] = q. \quad (120)$$

In particular, $H(1) = J(1) = 1/2$ and $\mathbb{P}(L = \ell) = C_{\ell-1}/4^\ell$ for finite $\ell \geq 1$.

27.3 Why the append operation is not independent

Let

$$\mathcal{K}(z) = \mathbb{E}[z^{\min(D+1, L)}; L < \infty]. \quad (121)$$

The outermost-operation decomposition gives

$$J(z) = qJ(z) + \frac{z}{4}J(z) + \frac{1}{4}\mathcal{K}(z).$$

The first term is an outer R , the second an outer A applied to an infinite-last state, and the third an outer A applied to a finite-last state. Hence

$$J(z) = \frac{\mathcal{K}(z)}{4(1-q-z/4)} = \frac{\mathcal{K}(z)}{4(1-q)^2}. \quad (122)$$

The dependence between D and L is concentrated in $\mathcal{K}$. Replacing L by an independent variable with its marginal law would be incorrect. The next calculation retains the full dependence.

27.4 An exact clipping calculation

If a terminal permutation τ does not end in its maximum, then $F(\tau) > \tau_{|\tau|}$ by (100). Every outer R only increases the first state coordinate, so clipping returns the terminal last value. The only correction to the marginal (120) arises from terminal permutations ending in their maximum.

Write such a non-singleton terminal as $\sigma(k+1)$, where $|\sigma| = k \geq 1$, $F(\sigma) = f$, and $j = k - f$. If the outer R sequence has total increment b , the state before clipping is $(b+f, k+1)$. Clipping is strict exactly when $b < j$. The singleton terminal produces no correction. Its non-singleton correction therefore has generating function

$$\mathcal{K}(z) - q = \frac{z}{16} \sum_{j \geq 1} q^{j+1} \sum_{0 \leq b < j} \frac{C_b}{4^b} (z^{b-j} - 1). \quad (123)$$

To see the factor $1/16$, the terminal size is $k+1$, hence its weight is $4^{-(k+2)}$. Formula (97), evaluated at $x = z/4$, sums its possible σ to q^{j+1} . The factor z records the final maximum; (117) supplies $C_b/4^b$.

For $0 < z < 1$, the geometric sums converge absolutely; in particular $q/z = 1/(4(1-q)) < 1$. Interchanging the sums in (123) gives

$$\begin{aligned} \mathcal{K}(z) - q &= \frac{z}{16} \sum_{b \geq 0} \frac{C_b}{4^b} q^{b+2} \left(\frac{1}{z-q} - \frac{1}{1-q} \right) \\ &= \frac{z(1-z)q^2 C(q/4)}{16(z-q)(1-q)} \\ &= (1-z) \frac{q^2 C(q/4)}{4(3-4q)}. \end{aligned} \quad (124)$$

Both sides extend analytically to $z = 0$. As $qC(q/4) = 2(1 - \sqrt{1-q})$, define

$$\Upsilon(z) = \frac{q(1 - \sqrt{1-q})}{2(3-4q)}. \quad (125)$$

We have proved the particularly useful representation

$$\boxed{\mathcal{K}(z) = q + (1-z)\Upsilon(z)}. \quad (126)$$

The series Υ also has the direct tail interpretation

$$[z^k]\Upsilon(z) = \mathbb{P}(L > k, D \leq k-1, L < \infty). \quad (127)$$

Indeed, for finite L , the tail of $\min(D+1, L)$ at k differs from the tail of L precisely on this event; use the standard probability-generating-function tail identity.

27.5 Completion of the formula

Combining (119), (122), and (126),

$$G(z) = \frac{q}{3-4q} + \frac{q + (1-z)q(1-\sqrt{1-q})/(2(3-4q))}{4(1-q)^2}. \quad (128)$$

Substitute $q = (1-s)/2$ and $\sqrt{1-q} = r$. The result is (84), proving the generating-function assertion of theorem 22.1. This derivation also verifies $G(0) = 0$ and $G(1) = 1$. The probabilistic construction has already proved that its coefficients are nonnegative and sum to one.

Remark 27.1 (Algebraicity, but not a finite constant-coefficient recurrence). The expression lies in the field obtained from $\mathbb{Q}(z)$ by adjoining s and then r , so its algebraic degree is at most four. It is not rational: the nonzero square-root term at $z = 1$, computed next, excludes rationality. Thus the limiting probability sequence does not satisfy an eventual constant-coefficient linear recurrence. This is compatible with rationality of each fixed- m generating function $A_m(x)$: it concerns a different sequence and a different limit.

28 Singularity analysis and the heavy tail

28.1 The dominant singularity is genuinely unique

The square-root branches in (84) extend to $\mathbb{C} \setminus [1, \infty)$. On this domain the principal value of $s = \sqrt{1-z}$ has positive real part. Consequently $1+s$ and $1+2s$ do not vanish, and $(1+s)/2$ never meets the negative real axis or zero. Thus r is analytic there as well. In particular, G is analytic in a slit neighborhood of the closed unit disk except at $z = 1$. There are no competing unit-circle singularities or periodicity factors hidden in the coefficient asymptotics.

Expansion of the explicit formula at $s = 0$ gives

$$\begin{aligned} G(z) = 1 - 3s + \left(\frac{23}{4} - \frac{\sqrt{2}}{8}\right)s^2 + \left(-\frac{43}{4} + \frac{9\sqrt{2}}{16}\right)s^3 \\ + \left(\frac{81}{4} - \frac{99\sqrt{2}}{64}\right)s^4 + \left(-\frac{155}{4} + \frac{461\sqrt{2}}{128}\right)s^5 + O(s^6). \end{aligned} \quad (129)$$

This calculation uses only $r = 2^{-1/2}(1+s)^{1/2}$ and ordinary Taylor expansion. An optional symbolic checker in the package (code/03-largest-jump-verify_symbolic.py) verifies the displayed coefficients. The term $-3\sqrt{1-z}$ is the source of both the infinite mean of D and the critical exponent $1/2$.

28.2 Coefficient extraction, including the second term

For a fixed nonintegral γ , the exact generalized binomial formula and the asymptotic expansion of the gamma quotient give

$$[z^k](1-z)^\gamma = \frac{k^{-\gamma-1}}{\Gamma(-\gamma)} \left(1 + \frac{\gamma(\gamma+1)}{2k} + O(k^{-2})\right). \quad (130)$$

Analytic continuation in the slit neighborhood just established permits termwise singularity transfer; this is the standard univariate transfer principle of [5]. Integer powers of $1-z$ are locally analytic terms and do not contribute the displayed algebraic orders. For completeness, the nonanalytic terms needed for the first two orders are $(1-z)^{1/2}$ and $(1-z)^{3/2}$, whose leading coefficient constants are respectively $-1/(2\sqrt{\pi})$ and $3/(4\sqrt{\pi})$.

Proposition 28.1 (Two-term probability and survival expansions). *As $k \rightarrow \infty$,*

$$\mathbb{P}(D = k) = \frac{3}{2\sqrt{\pi}} k^{-3/2} + \frac{-480 + 27\sqrt{2}}{64\sqrt{\pi}} k^{-5/2} + O(k^{-7/2}), \quad (131)$$

$$\mathbb{P}(D > k) = \frac{3}{\sqrt{\pi}} k^{-1/2} + \frac{-184 + 9\sqrt{2}}{32\sqrt{\pi}} k^{-3/2} + O(k^{-5/2}). \quad (132)$$

Proof. For the point probabilities, apply (130) to the s and s^3 terms of (129). The coefficient of $k^{-5/2}$ is

$$\frac{9}{16\sqrt{\pi}} + \frac{3}{4\sqrt{\pi}} \left(-\frac{43}{4} + \frac{9\sqrt{2}}{16} \right) = \frac{-480 + 27\sqrt{2}}{64\sqrt{\pi}}.$$

For the survival probabilities use the exact identity

$$\sum_{k \geq 0} \mathbb{P}(D > k) z^k = \frac{1 - G(z)}{1 - z}. \quad (133)$$

Writing $a_2 = 23/4 - \sqrt{2}/8$ and $a_3 = -43/4 + 9\sqrt{2}/16$, its expansion begins $3(1 - z)^{-1/2} - a_2 - a_3(1 - z)^{1/2} + O(1 - z)$. The second coefficient is

$$-\frac{3}{8\sqrt{\pi}} + \frac{a_3}{2\sqrt{\pi}} = \frac{-184 + 9\sqrt{2}}{32\sqrt{\pi}}.$$

Continuing the expansion two half-orders further gives the stated remainders. The analyticity argument above supplies the hypotheses of singularity transfer throughout. $\square$

Corollary 28.2 (Sharp total-variation order). *The error in the boundary law has order exactly $n^{-1/2}$. More precisely,*

$$\frac{3}{\sqrt{\pi}} \leq \liminf_{n \rightarrow \infty} \sqrt{n} d_{\text{TV}}(\mathcal{L}(D_n), \mathcal{L}(D)) \leq \limsup_{n \rightarrow \infty} \sqrt{n} d_{\text{TV}}(\mathcal{L}(D_n), \mathcal{L}(D)) \leq \frac{4\sqrt{2}}{\sqrt{\pi}}. \quad (134)$$

Proof. Because $D_n \leq n - 1$, the event $\{n, n + 1, \dots\}$ gives

$$d_{\text{TV}}(\mathcal{L}(D_n), \mathcal{L}(D)) \geq \mathbb{P}(D \geq n) = \frac{3}{\sqrt{\pi n}} + O(n^{-3/2}).$$

For the improved upper bound, repeat the common-submeasure proof with any integer $2 \leq h < n/2$. The first-entry estimate gives

$$d_{\text{TV}}(\mathcal{L}(D_n), \mathcal{L}(D)) \leq \mathbb{P}(T > h) + n(n + 1)(2n + 1)2^{2h-n}.$$

For all sufficiently large n , choose $h = \lfloor n/2 - 2 \log_2 n \rfloor$. The second term is $O(n^{-1})$, while the exact cost tail is $(4\sqrt{2}/\sqrt{\pi})n^{-1/2}(1 + o(1))$. This proves the three inequalities and completes the sharp-rate claim in theorem 22.1. $\square$

Corollary 28.3 (Quantiles and limiting moments). *The variable D has finite moment of order $p > 0$ exactly when $p < 1/2$. If $\kappa_{1-\epsilon} = \min\{k : \mathbb{P}(D \leq k) \geq 1 - \epsilon\}$, then*

$$\kappa_{1-\epsilon} \sim \frac{9}{\pi\epsilon^2} \quad (\epsilon \downarrow 0). \quad (135)$$

Proof. The moment criterion follows by summing $k^p \mathbb{P}(D = k) \sim (3/(2\sqrt{\pi}))k^{p-3/2}$. For the quantile statement, bracket the survival function between $(1 \pm \eta)3/\sqrt{\pi k}$ for all sufficiently large k , invert these inequalities, and then let $\eta \downarrow 0$. Rounding the resulting bounds to integers has no effect on the equivalent. $\square$

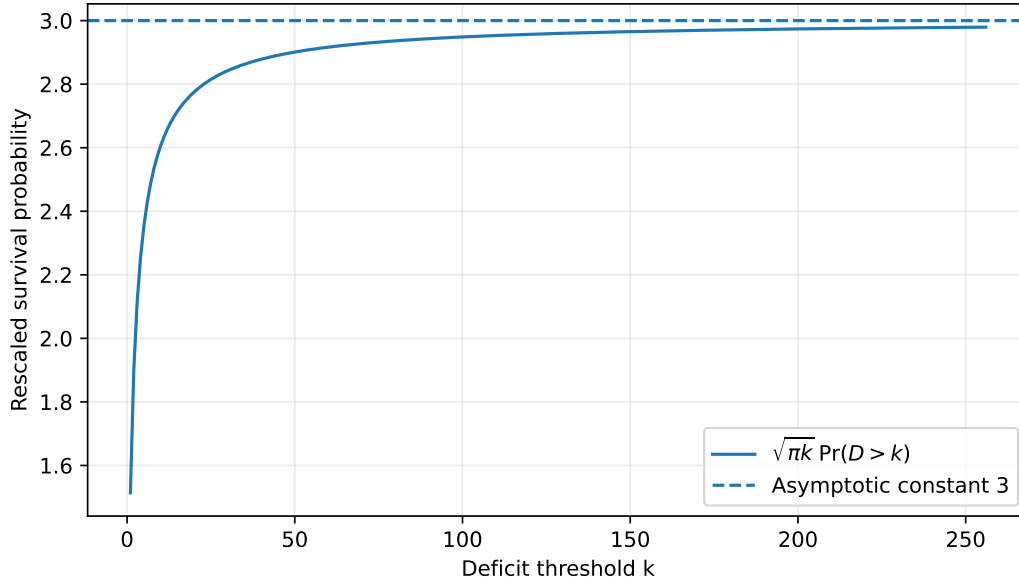


Figure 5: The scaled survival function $\sqrt{\pi k} \mathbb{P}(D > k)$, computed from exact rational coefficients of G . The horizontal reference is 3. The graph illustrates (132); it is not used as evidence in its proof.

29 Joint limits for the adjacency bound

29.1 A fixed distance from the maximal possible jump

For every fixed integer $d \geq 1$, (82) and total-variation convergence give

$$\lim_{n \rightarrow \infty} \frac{a_n^{(n-d)}}{C_n} = \mathbb{P}(D \geq d) = 1 - \sum_{k=1}^{d-1} [z^k]G(z). \quad (136)$$

Thus the limiting profile across the boundary consists of explicit rational numbers. At $d = 1$ the fraction is identically one. At $d = 2$, its limit is $41/48$: asymptotically $7/48$ of all avoiders contain a jump of the largest possible size, namely $n - 1$.

This is not a smooth limiting distribution for M_n/n . That normalized statistic tends to the constant 1. The nontrivial distribution is visible only after retaining the integer distance $n - M_n$.

29.2 A uniform estimate and its domain of relative validity

Proof of theorem 22.2. The point and survival estimates follow from proposition 28.1. For $2 \leq d \leq n - 1$, total variation gives

$$\left| \frac{a_n^{(n-d)}}{C_n} - \mathbb{P}(D \geq d) \right| \leq C_{TV} n^{-1/2}$$

for an absolute constant C_{TV} . Also

$$\mathbb{P}(D \geq d) = \mathbb{P}(D > d - 1) = \frac{3}{\sqrt{\pi d}} + O(d^{-3/2}).$$

Combining these inequalities proves (89). Dividing the error by the main term gives

$$\frac{a_n^{(n-d)}}{C_n} = \frac{3}{\sqrt{\pi d}} \left(1 + O\left(d^{-1} + \sqrt{d/n}\right) \right). \quad (137)$$

When $d \rightarrow \infty$ and $d = o(n)$ the relative error tends to zero. Finally $C_n \sim 4^n / (\sqrt{\pi} n^{3/2})$ gives the second equivalent in (90). $\square$

Example 29.1 (Polynomial and logarithmic boundary windows). For any fixed $0 < \alpha < 1$ and $d_n = \lfloor n^\alpha \rfloor$,

$$\frac{a_n^{(n-d_n)}}{C_n} \sim \frac{3}{\sqrt{\pi}} n^{-\alpha/2}.$$

Taking $d_n = \lfloor \log n \rfloor$ gives the equivalent $3/\sqrt{\pi \log n}$. More generally, the same formula with d_n in place of these choices holds along every unbounded sublinear integer sequence. Neither monotonicity nor regular variation of d_n is required.

Corollary 29.2 (All diverging boundary deficits are negligible). *If $d_n \rightarrow \infty$ and $1 \leq d_n \leq n-1$, then*

$$a_n^{(n-d_n)} / C_n \rightarrow 0.$$

For every fixed $0 < \theta < 1$,

$$a_n^{(\lfloor \theta n \rfloor)} / C_n = O(n^{-1/2}).$$

Proof. The first assertion follows either from tightness of D_n or directly from the absolute estimate. In the second, $d_n = n - \lfloor \theta n \rfloor$ is of order n , and both terms of (89) are $O(n^{-1/2})$. $\square$

Remark 29.3 (The unresolved bulk regime). When d is a fixed positive fraction of n , the total-variation error is of the same order as the proposed leading term in (89). The equation is still a valid absolute estimate, but it does *not* prove a relative equivalent there. In particular, Part III does not assert $a_n^{(\lfloor \theta n \rfloor)} / C_n \sim 3/\sqrt{\pi(1-\theta)n}$. An additional finite-size scaling analysis would be needed, and it could yield a different function of θ . [Added 29 September 2026, batch 52: it does. Part IV proves $a_n^{(\lfloor \theta n \rfloor)} / C_n \sim \frac{3(2\theta-1)}{\sqrt{\pi\theta(1-\theta)}} n^{-1/2}$ for $1/2 < \theta < 1$ and $o(n^{-1/2})$ for $\theta \leq 1/2$ (theorem 38.1); the formula above is therefore false for every $0 < \theta < 1$.]

30 The moment transition and its consequences

30.1 Truncation converts total variation into moment bounds

For two probability measures μ, ν and a measurable function with $0 \leq f \leq B$, the layer-cake representation gives

$$\left| \int f d\mu - \int f d\nu \right| \leq B d_{\text{TV}}(\mu, \nu). \quad (138)$$

Apply this to $f(x) = \min(x, n)^p$. Since $D_n < n$,

$$|\mathbb{E} D_n^p - \mathbb{E} \min(D, n)^p| = O(n^{p-1/2}). \quad (139)$$

The finite support of D_n matters: total-variation convergence without a rate or a truncation bound would not control these unbounded moments.

For any positive integer-valued X , summation by parts gives

$$\mathbb{E} \min(X, n)^p = \sum_{k=0}^{n-1} ((k+1)^p - k^p) \mathbb{P}(X > k). \quad (140)$$

Using (132) and $(k+1)^p - k^p = pk^{p-1} + O(k^{p-2})$ for $k \geq 1$, we obtain all three moment regimes.

30.2 Subcritical and critical moments

Proof of theorem 22.3, for $p \leq 1/2$. If $0 < p < 1/2$, the tail of the sum (140) is $O(n^{p-1/2})$, and therefore

$$\mathbb{E} \min(D, n)^p = \mathbb{E} D^p + O(n^{p-1/2}).$$

Equation (139) proves (91).

If $p = 1/2$, the summand in (140) is

$$(\sqrt{k+1} - \sqrt{k})\mathbb{P}(D > k) = \frac{3}{2\sqrt{\pi}} \frac{1}{k} + O(k^{-2}).$$

Consequently

$$\mathbb{E} \min(D, n)^{1/2} = \frac{3}{2\sqrt{\pi}} \log n + O(1).$$

The error in (139) is now $O(1)$, proving the critical statement. $\square$

30.3 Supercritical moments: the order, not the constant

Completion of the proof of theorem 22.3. Let $p > 1/2$. The sum in (140) is $O(n^{p-1/2})$, so (139) gives the upper bound. For the lower bound choose a constant C_{TV} such that $d_{TV}(\mathcal{L}(D_n), \mathcal{L}(D)) \leq C_{TV} n^{-1/2}$ for all sufficiently large n . Fix a sufficiently small $\epsilon > 0$ such that

$$\frac{3}{2\sqrt{\pi\epsilon}} > C_{TV} + 1.$$

Set $d = \lfloor \epsilon n \rfloor$. The limiting tail estimate implies, for all sufficiently large n ,

$$\mathbb{P}(D \geq d) \geq \frac{3}{2\sqrt{\pi\epsilon}} n^{-1/2}.$$

Hence $\mathbb{P}(D_n \geq d) \geq n^{-1/2}$, and

$$\mathbb{E} D_n^p \geq d^p \mathbb{P}(D_n \geq d) \geq \underline{c}_p n^{p-1/2}$$

for some $\underline{c}_p > 0$. This proves the claimed two-sided order. $\square$

Corollary 30.1 (Mean and variance of the largest-jump deficit). *As $n \rightarrow \infty$,*

$$\mathbb{E} D_n = \Theta(\sqrt{n}), \quad \mathbb{E} M_n = n - \Theta(\sqrt{n}), \quad \text{Var}(D_n) = \Theta(n^{3/2}). \quad (141)$$

The same variance identity holds for M_n .

Proof. Use theorem 22.3 with $p = 1, 2$. The squared mean is $O(n)$, whereas the second moment is bounded above and below by positive constants times $n^{3/2}$. Subtracting the squared mean preserves this order. Finally $M_n = n - D_n$. $\square$

Remark 30.2 (Why a bounded typical deficit and an unbounded mean coexist). The limit has median 9, but an infinite mean. Rare permutations with much larger deficits carry an increasing fraction of the finite-size mean. There is therefore no contradiction between $D_n = O_{\mathbb{P}}(1)$ and $\mathbb{E} D_n = \Theta(\sqrt{n})$. Nor can the constant in the mean be read off from the tail constant $3/\sqrt{\pi}$: the range D_n of order n can contribute to the mean at the same order, and that bulk range has not been identified asymptotically in Part III. [Added 29 September 2026, batch 52: Part IV identifies it (theorem 38.1); the mean constant is $3/\sqrt{\pi}$, and half of it comes from deficits above $n/5$ (theorems 38.3 and 38.4). The value coincides with the tail constant, but it is not read off from it: extrapolating the boundary tail $3/\sqrt{\pi x}$ over $0 < x < 1$ would give $6/\sqrt{\pi}$. This comparison is the merge's.]

Table 12: Initial point probabilities and cumulative probabilities of the limiting deficit. The full first 256 point probabilities are included as exact numerator–denominator pairs in the package (data/03-largest-jump-limit_probabilities.csv).

k	$\mathbb{P}(D = k)$	$\mathbb{P}(D \leq k)$
1	$7/48 = 0.1458333333$	0.1458333333
2	$223/2304 = 0.0967881944$	0.2426215278
3	$7657/110592 = 0.0692364728$	0.3118580006
4	$139913/2654208 = 0.0527136532$	0.3645716538
5	$10681477/254803968 = 0.0419203715$	0.4064920253
6	0.0344074271	0.4408994524
7	0.0289238487	0.4698233011
8	0.0247727295	0.4945960306
9	0.0215382164	0.5161342470
10	0.0189582188	0.5350924658
11	0.0168600735	0.5519525393
12	0.0151257982	0.5670783376

31 Exact probabilities, finite checks, and reproducibility

31.1 The first probabilities and the median

The probability coefficients are rational: the branches of the two square roots at $z = 0$ have rational Taylor coefficients, as do all the rational operations in (84). Table 12 shows the first twelve values. The decimals are rounded displays of exact fractions, not floating-point coefficients used in the proof.

The exact inequalities $\mathbb{P}(D \leq 8) < 1/2 < \mathbb{P}(D \leq 9)$, checked with rational arithmetic, show that 9 is the unique integer median. Total-variation convergence also implies that, for every sufficiently large n , the finite deficit D_n has the same integer median. This is an eventual statement; the article does not give a minimal such n .

31.2 An independent exact check of the first mass

Proposition 31.1 (The largest possible jump). *For $n \geq 3$,*

$$\#\{\pi \in \text{Av}_n(132) : M(\pi) = n - 1\} = C_{n-2} + \sum_{j=0}^{n-2} C_j. \quad (142)$$

Consequently $\mathbb{P}(D_n = 1) \rightarrow 7/48$.

Proof. A jump of size $n - 1$ is exactly an adjacency of the values 1 and n . If it is the ascent 1, n , this pair must be at the end: any later entry would form 132 with the pair. The preceding entries, standardized, are an arbitrary avoider of length $n - 2$, giving C_{n-2} choices.

If it is the descent $n, 1$, decompose at the maximum. The right block starts with 1 and must be increasing, since a later descent together with that first 1 would form 132. For each right-block length $r \in \{1, \dots, n - 1\}$, the left block is an arbitrary avoider of length $n - r - 1$. This gives the sum in (142). The two cases are disjoint.

Finally $C_{n-2}/C_n \rightarrow 1/16$. Also $\sum_{j=0}^{n-2} C_j \sim (4/3)C_{n-2}$, as follows either by reversing this sum and using the Catalan ratio, or by Stolz–Cesàro applied to $\sum_{j=0}^N C_j$ and C_N . Thus the second term divided by C_n tends to $1/12$. Their sum is $7/48$. $\square$

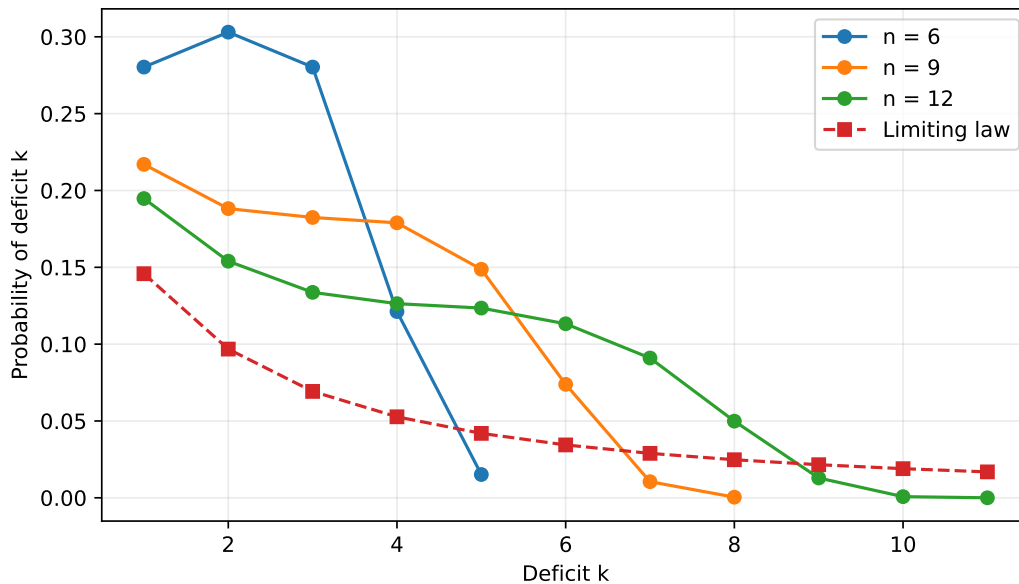


Figure 6: Exact finite point probabilities of D_n for selected small sizes, compared with the limiting coefficients. Finite support strongly affects the far end of each finite distribution. The curves are illustrations generated from exhaustive Catalan enumeration, not Monte Carlo samples.

31.3 What the verifier actually checks

The standard-library verifier performs the following separate tests. The wording is deliberately finite: passing them supports implementation correctness, not the infinite or asymptotic claims by itself.

1. It computes the first 40 point probabilities exactly, verifies positivity, the first five printed fractions, and the median. It compares 31 clipping-correction coefficients computed from the radical formula with a separate finite-endpoint tail summation.
2. For sizes 1 through 8, it compares the Catalan recursive generator with all ordinary permutations filtered by a direct three-index 132 test. This checks 2,055 avoiders and verifies every first-entry count and the extreme-jump formula in that range.
3. For every one of the 804 decorated skeletons of costs 2 through 7, it tests every hole of size $t + 1$ with first entry greater than t . There are 269,402 reconstruction tests. Each checks the exact deficit, the last-entry state, and absence of collisions within the fixed-cost reconstruction family. The counts of skeletons and their probability masses are checked against the exact Catalan formulas as well.

All tests passed in the recorded run. The cross-cost disjointness needed in the proof is established by lemma 25.2; the finite collision check is not described as a test of every pair of costs.

31.4 Rebuilding and rerunning

Part III's files are

code/03-largest-jump-model.py	Exact Catalan, endpoint, skeleton and rational-series code
code/03-largest-jump-verify.py	Standard-library exact finite verifier
code/03-largest-jump-verify_symbolic.py	Optional SymPy identity and expansion checks
code/03-largest-jump-make_data.py	Regenerates the three CSV files
code/03-largest-jump-plot_results.py	Optional Matplotlib figures
code/03-largest-jump-build.sh	The manuscript's three-pass build

data/03-largest-jump-*	Recorded runs, 256 exact limiting probabilities, finite histograms n = 2..12, extreme-jump checks, optional requirements
figures/03-largest-jump-*.pdf	The two figures of this part
03-largest-jump-{PROOF_STATUS,SOURCES,VALIDATION}.md	Audit notes

The scripts are shipped byte-identical to the delivery and still use the delivered names: they write `data/verification.json`, `data/symbolic_verification.json`, the three unprefix CSV files and `figures/tail_scaling.*`, `figures/finite_probabilities.*`, and the verifier and the data generator execute from `model import`. Run in place, that import would load Part I's different `code/model.py` and fail with `ImportError`, and `code/03-largest-jump-build.sh` would look for an `article.tex` in `code/`. The report's README gives a recipe that restores the delivered layout in a scratch copy. There, the manuscript's instructions apply unchanged. The exact verification needs Python 3.10 or later and no third-party package:

```
python code/verify.py
python code/make_data.py
```

Do not use Python's `-O` option, which disables the verifier's assertions. The optional symbolic checks need SymPy, and optional plot regeneration needs Matplotlib:

```
python code/verify_symbolic.py
python code/plot_results.py
```

The retained figures make Python unnecessary for rebuilding the article. The manuscript's build script placed auxiliary files in a temporary directory and copied back only its `article.pdf`; this report is built as described in its README. The included data files record exact fractions; rounded display columns and plots are not interval-arithmetic certificates.

32 Further research questions

The results above settle the bounded and unbounded sublinear deficit windows. They also reveal several questions whose answers would extend the theory rather than merely recompute its constants.

32.1 Bulk finite-size scaling

For $0 < u < 1$, determine whether the limit

$$\Psi(u) = \lim_{n \rightarrow \infty} \sqrt{n} \mathbb{P}(D_n > \lfloor un \rfloor) \quad (143)$$

exists, and if so compute it, including its endpoint behavior. This is a proposed problem, not a theorem or an asserted formula. Part III proves an upper bound on the expression, and a positive lower bound for sufficiently small fixed u . It does not establish a limit. A likely starting point is to retain the finite Catalan ratio C_{n-t}/C_n instead of replacing it by 4^{-t} , and to analyze the regime where both sides at a maximum decomposition are macroscopic. [Added 29 September 2026, batch 52: answered in Part IV. The limit exists for every $0 < u \leq 1$, uniformly on $[\delta, 1]$ for each $\delta > 0$, and $\Psi(u) = \frac{3}{\sqrt{\pi}} \frac{1-2u}{\sqrt{u(1-u)}}$ for $u < 1/2$, $\Psi(u) = 0$ for $u \geq 1/2$ (theorem 38.1). At the endpoints, $\Psi(u)\sqrt{u} \rightarrow 3/\sqrt{\pi}$ as $u \rightarrow 0$, matching theorem 22.2, and $\Psi(1/2 - \varepsilon) \sim 12\varepsilon/\sqrt{\pi}$ (section 43.1). Both manuscripts behind Part IV analyze exactly the regime suggested here, where both sides of a maximum decomposition are macroscopic.]

32.2 Exact constants for the mean and higher moments

Determine whether $n^{1/2-p}\mathbb{E}D_n^p$ converges for each $p > 1/2$, especially whether $\mathbb{E}D_n/\sqrt{n}$ has a limit. A bulk profile with sufficient uniform integrability would make these constants integrals of that profile. At the critical point, ask whether

$$\mathbb{E}\sqrt{D_n} - \frac{3}{2\sqrt{\pi}} \log n$$

converges. The $O(1)$ result proved here does not imply convergence of the centered quantity. [Added 29 September 2026, batch 52: answered in Part IV for $p > 1/2$: $n^{1/2-p}\mathbb{E}D_n^p \rightarrow m_p = \frac{3}{\sqrt{\pi}} \int_0^1 (z^2/(1+z^2))^{p-1} dz$, so $\mathbb{E}D_n/\sqrt{n} \rightarrow 3/\sqrt{\pi}$, and the constants are indeed integrals of the bulk profile (theorem 38.3). The critical question stays open (section 47.6).] [Added 30 September 2026, batch 64: the critical question is answered in Part V: $\mathbb{E}\sqrt{D_n} - \frac{3}{2\sqrt{\pi}} \log n \rightarrow c_\star \approx -0.942121414$, an explicit convergent constant (theorem 53.1).]

32.3 The sharp total-variation constant and local corrections

The order $n^{-1/2}$ is now proved sharp, but the interval in (134) leaves the leading constant open. Does $\sqrt{n} d_{TV}(\mathcal{L}(D_n), \mathcal{L}(D))$ converge? Is the missing limiting mass beyond the finite support the whole leading discrepancy, or does an additional contribution arise inside the support? Separately, for each fixed k , determine the first correction to $\mathbb{P}(D_n = k)$. A pointwise expansion can be much sharper than the global total-variation bound and would give effective finite-size estimates for the first several adjacency thresholds.

32.4 Ascending and descending extreme jumps jointly

Define the largest ascent and largest descent by

$$M_n^+ = \max_i (\Pi_n(i+1) - \Pi_n(i))_+, \quad M_n^- = \max_i (\Pi_n(i) - \Pi_n(i+1))_+.$$

Determine their joint limiting behavior, allowing the possibility that one coordinate requires an extended state space or a different scale. The current clipping step mixes an old extreme jump with a newly created ascent, so its explicit dependence on the terminal endpoint should provide useful information. Independence must not be assumed.

32.5 Near-maximal jumps as a point process

Instead of retaining only the minimum deficit, retain the multiset

$$\{n - |\Pi_n(i+1) - \Pi_n(i)| : 1 \leq i < n\}$$

near the finite end of the positive integers. Determine its joint count law on every fixed finite set, including ties for the maximum jump. A decorated skeleton records where a finite number of new extreme candidates are created. It may therefore support a marked extension of the present generating-function calculation. A proof must also control unrecorded jumps inside the large hole.

32.6 Weighted Catalan models

Replace the uniform measure by weights tracking descents, leaves of the associated binary tree, or another statistic with an explicit bivariate Catalan generating function. Determine when an absorbing finite skeleton still exists, whether the moment threshold remains $1/2$, and which critical parameter values change the tail exponent. Both the operation probabilities and the endpoint clipping law would have to be recomputed; a formal substitution into (84) is not justified.

32.7 Other avoidance classes

Develop an analogous boundary analysis for other pattern classes, with particular attention to classes whose recursive specifications have several large-block types. Before treating a new pattern as a distinct case, account for the reversal and complement symmetries that preserve absolute adjacent differences. The difficult extension is not a mere renaming of a symmetry-equivalent class, but a new decomposition in which the competition between large adjacent jumps has a different structure.

32.8 An exact sampler with controlled work

The skeleton probability space is an exact description of the limit, but its cost has infinite expectation by (106). Sampling full finite decorations literally can therefore be expensive on rare runs. Design an exact sampler for D that manipulates only the endpoint information and clipping events it needs. Determine its expected bit complexity, and compare a state-based sampler with inversion of the rational coefficient sequence. Almost-sure termination alone is not a finite expected-cost theorem.

32.9 A formalization boundary

A feasible first formal target is the deterministic reconstruction lemma: define 132-avoidance, the maximum decomposition, decorated skeletons, and the endpoint state, then prove reconstruction and injectivity by induction. Next formalize the first-entry formula and the common-submeasure inequality. These steps isolate the combinatorial content from the later analytic transfer theorem. A complete formalization would still need countable probability, formal power series or analytic identities, and asymptotic coefficient transfer; the present package does not claim to supply them.

32.10 Reconnect to the fixed-bound spectral questions

The boundary law does not decide which of the two irreducible endpoint components controls the fixed- m growth rate, nor the optimal defect between that rate and the Catalan scalar comparison rate discussed in Parts I and II (sections 9 and 19). Study whether a two-parameter endpoint system can interpolate between the fixed- m spectral regime and the boundary regime here. Any proposed interpolation must specify its uniform error before taking a moving-bound limit; fixed-parameter rationality alone is insufficient. [Added 1 October 2026, batch 72: from the growing-cap side, Part IX connects the fixed- m rate with moving bounds: uniformly for $n \geq c_0 m \log m$, $\log \mathbb{P}(M_n \leq m) = n \log(\alpha_m/4) + O(n(\log m)^2/m^2 + \log n)$ (theorem 105.1), with relative equivalents where $n(\log m)^2/m^2 \rightarrow 0$ (section 101.2). It does not decide the component question, and no interpolation down to the boundary regime is proved.]

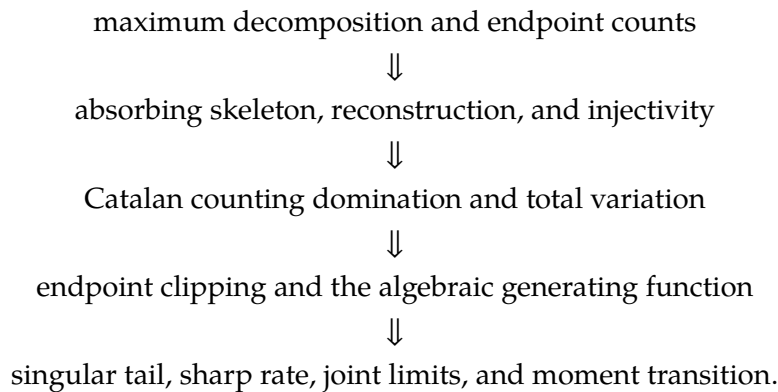
33 Proof audit, scope, and conclusion

Table 13: Claim-by-claim status of Part III (manuscript 02).

Claim	Status and limitation
Algebraic law for D_n	Written proof from finite counting and exact state equations; not formally verified.
Sharp order $n^{-1/2}$ in total variation	Upper bound by common mass; lower bound by the limiting mass beyond finite support. Exact constant open.
Uniform estimate for $m = n - d$	Absolute estimate for $2 \leq d \leq n - 1$; relative equivalent only when $d \rightarrow \infty$ and $d = o(n)$.
Moment transition	Exact critical logarithmic coefficient and subcritical convergence; only two-sided orders above the critical exponent.
Finite verification	Exact rational and integer regression tests over specified finite ranges; not a substitute for the proofs.
Novelty and priority	Targeted comparison with the cited papers and repository report; no exhaustive literature or priority certification.
Bulk profile, exact mean constant	Proposed research questions, not claimed results. [Answered in Part IV, added 29 September 2026.]

33.1 The dependency chain

The argument proceeds through explicit finite objects:



The only analytic coefficient-transfer input is standard singularity analysis, with the needed continuation and local expansion checked explicitly. No growth-constant estimate from the existing ProveIt report (Parts I and II) is needed, and no general conditioned-tree convergence theorem is used as a black box.

33.2 What is resolved

The adjacency-bounded enumeration has a nontrivial joint-limit boundary at distance $O(1)$ below n . Part III identifies the entire limiting profile there, gives a sharp-order convergence rate, and extends it uniformly to every growing sublinear deficit window. The algebraic law has a heavy enough tail to force a moment transition, so typical and average deficits live on different scales. These conclusions answer a specific missing regime in the repository's program while leaving its fixed-bound spectral problems and the bulk finite-size profile separate.

The reusable mechanism is the common-mass comparison: a finite stopping skeleton of cost t has limiting weight 4^{-t} , while the corresponding finite Catalan family has at least that weight. Once exceptional first entries are removed, the extreme statistic agrees exactly. This finite argument simultaneously supplies convergence, its quantitative rate, and a controlled bridge from limiting probabilities to moving-bound counts.

34 Exact coefficient recursions

This section and the next two were the manuscript's Appendices A–C. They are numbered here as sections of Part III, so that Part I's appendices keep their letters. The package computes all coefficients with rational arithmetic. To degree N , begin with

$$q(z) = \sum_{k=1}^N \frac{C_{k-1}}{4^k} z^k + O(z^{N+1}).$$

For a series $a(z) = 1 + \sum_{k \geq 1} a_k z^k$, the square root $b(z) = 1 + \sum_{k \geq 1} b_k z^k$ is obtained from

$$b_k = \frac{1}{2} \left(a_k - \sum_{j=1}^{k-1} b_j b_{k-j} \right). \quad (144)$$

For a series a with nonzero constant term, its inverse is obtained from

$$b_0 = a_0^{-1}, \quad b_k = -a_0^{-1} \sum_{j=1}^k a_j b_{k-j}. \quad (145)$$

Use these recursions to compute $r = \sqrt{1-q}$, $H = q/(3-4q)$, $\Upsilon = q(1-r)/(2(3-4q))$, and $J = (q + (1-z)\Upsilon)/(4(1-q)^2)$. Their sum is G . Only the first N coefficients of each input are needed. Ordinary convolution gives $O(N^2)$ rational arithmetic operations; this is not a bit-complexity bound, because numerators and denominators grow.

The independent clipping computation uses instead the endpoint counts (98) and their tails. Writing $P(z) = P(1/4, z) = z/(1 + 2\sqrt{1-z})$, it extracts coefficients of P without the formula for Υ , subtracts finite first-entry counts, and sums over the possible finite R -increments. Agreement is therefore not merely two evaluations of the same line of code.

35 An algebraic certificate without branch ambiguity

One may describe G without nested radical notation by introducing q and r and imposing

$$4q^2 - 4q + z = 0, \quad q(0) = 0, \quad (146)$$

$$r^2 - (1-q) = 0, \quad r(0) = 1. \quad (147)$$

In addition,

$$8(3-4q)(1-q)^2 G = q(15-24q+8q^2-z-(1-z)r). \quad (148)$$

Equations (146)–(148), with the stated initial values, select the same analytic series as (84). Eliminating r first gives

$$\begin{aligned} & [q(15-24q+8q^2-z) - 8(3-4q)(1-q)^2 G]^2 \\ & - q^2(1-z)^2(1-q) = 0. \end{aligned} \quad (149)$$

The resultant of (149) and $4q^2 - 4q + z$ eliminates q . Squaring can add branches, so the conditions at zero must be retained; an arbitrary root of the eliminated polynomial is not automatically the probability generating function. Part III claims degree at most four, not an independently proved irreducibility statement for a minimal polynomial.

36 Repository pin, literature scope and merge

36.1 The manuscript's pin and literature scope

The inspected repository head was

8d936ee2357f9decf78c2ecc7d9baf3100a6787d.

The relevant report is in the directory

```
SetTheory/Cardinals/docs/reports/
enumerative-combinatorics/
adjacency-bounded-132-avoiders/.
```

The inspected README blob was 7f56b878e420c9fb0a0df299f54f9c154c68780e; its LaTeX article blob was f24ea59acf0f9809462f7fb0f055e78d38d4740a. The README identifies two source parts, from September 19 and September 28, 2026, and explicitly says that no joint limit in n and m is claimed there. That exclusion motivates the present target; it is not presented as a named classical conjecture.

The literature comparison used the arXiv records and available HTML of Nadler and Mayama-Akita, the records of the two Janson papers, and targeted searches for extreme adjacent differences in 132-avoiders. The searches did not establish exhaustive coverage. In particular, absence of a matching result from the returned material is not evidence sufficient to certify priority. The full metadata ledger is retained in SOURCES.md (shipped as 03-largest-jump-SOURCES.md). No external PDF or source article is redistributed in this package.

Both blobs above are the ones this report still had when the manuscript was placed (section 21.1); in particular the README sentence quoted above is the one Part III re-scopes, and it now carries a dated pointer here, as does Part II's research direction "Joint growth of length and adjacency bound" (section 19).

36.2 Where the merge had to choose

- *Placement.* Part III follows Part II and precedes Part I's appendices, which contain no numbered equations, tables or figures; no number of Part I or Part II changed. The manuscript's three appendices are printed as sections 34 to 36. Its title page and abstract are replaced by the opening note of Part III, the status box of section 21.1 (which prints its research-status box) and a paragraph of the report's abstract; every claim of its abstract is stated in section 22.
- *Scope.* The manuscript describes its target as the joint limit the report's README excluded. Here it is recorded as answering that question, and Part II's corresponding research direction, only in the boundary regime $m = n - d$ with $d = o(n)$ (section 21.3); the README and Part II's text keep their wording, with dated pointers.
- *Duplicates.* The maximum decomposition is Part I's (section 2) and is credited there; the first-entry count is Part I's (21), reindexed, and the manuscript's different derivation is kept (remark 23.2). No other result of Part III is proved in Parts I or II; elementary Catalan identities such as $\sum_{k \geq 1} C_{k-1} 4^{-k} = \frac{1}{2}$ are used in both. Remark 23.2's comparison with Part II's tail bound is the merge's, as are section 21 and this list.
- *Notation.* $\varepsilon_n \rightarrow \omega_n$, $\mathcal{A} \rightarrow \Upsilon$, $q_{1-\varepsilon} \rightarrow \kappa_{1-\varepsilon}$, $C, C_0 \rightarrow C_{TV}$, $c_p \rightarrow \underline{c}_p$ (table 11). The deficit D_n , its law D , the threshold d and the cutoff h keep the manuscript's letters, because each collides only with objects of a different kind (polynomials, constants, two-argument counts) that are listed in the table. No normalization changed.
- *References to this report.* The manuscript cites Parts I and II as a repository report (its reference "Repo"); here those citations are internal cross-references. Its entries for Nadler and Mayama-Akita are Part I's; its Janson and Flajolet–Odlyzko entries are added to the bibliography.

- *Reproduction.* The manuscript's package listing and commands are replaced by the shipped names and the copy recipe (section 31.4); its `article.tex`, `article.pdf`, `README.md` and the PNG versions of its figures are not shipped.
- *Figures.* In the write phase the manuscript's two figure PDFs were copied byte-identically into `figures/`, under their delivered names with the prefix `03-largest-jump-` (figures 5 and 6).
- *Stale sentences elsewhere.* The abstract, Part I's remark on the order of limits (section 1), Part II's remark 11.2, its research direction on joint growth, its scope ledger (table 10) and Part I's section B carry dated pointers to Part III; their original wording is kept.

Part IV. The macroscopic jump law

Part IV was added on 29 September 2026 in batch 52 of ProveIt's incoming-report intake, merged from two manuscripts that were written independently of each other against the state of Parts I–III printed above. Part III determined the law of the largest-jump deficit $D_n = n - M_n$ at bounded distance from the trivial bound $m = n - 1$, and asked in its Sections 32.1 and 32.2 for the macroscopic profile of D_n and the exact constants of its supercritical moments. Both manuscripts answer both questions, with different proofs. Part IV prints the shared results once, keeps both proofs, and adds what only one of them proved (section 37.1).

37 Provenance, conventions and the answer

37.1 Two manuscripts, one addition

Part IV is built from two AI-assisted research manuscripts prepared for Vladimir Reshetnikov, both dated September 29, 2026:

- *Manuscript 10* of batch 52, *The Macroscopic Jump Law: Exact rare-event profiles and moment constants for random 132-avoiding permutations* (23 pages, US Letter, as delivered). It arrived as the archive `ProveIt_Macroscopic_Jump.zip` (inner directory `ProveIt_Macroscopic_Jump/`, main file `article.tex`) in ProveIt commit 458ccfb80. It was written against ProveIt commit 1ee53d57d, whose root tree c772efcde0 it records. For the proofs and questions of Part III in contiguous form it also read the standalone source of Part III's manuscript, `article(20260929-161704).tex`, from Vladimir's file library; that file is not in the repository, and its own pin is Part III's 8d936ee23 (section 36).
- *Manuscript 11* of batch 52, *Macroscopic Deficits of the Largest Jump: A rare-event law and exact moment constants for 132-avoiding permutations* (24 pages, US Letter, as delivered). It arrived as the archive `ProveIt_Macroscopic_Jump_Research.zip` (inner directory also `ProveIt_Macroscopic_Jump/`, main file `article.tex`) in ProveIt commit 11e1e9001. It was written against the same commit 1ee53d57d and quotes the blob aeadaf364b of this report's `article.tex`.

At 1ee53d57d this report's `article.tex` and `README.md` had the blobs aeadaf364b and 815242ed6, and both were unchanged at the placement commit a701d9098. Every statement either manuscript makes about this report therefore refers to Parts I–III as printed above. Neither manuscript saw the other. Their code, data and provenance notes are shipped beside the earlier parts' files with the prefixes `04-macroscopic-jump-` (manuscript 10) and `05-macroscopic-deficits-` (manuscript 11), continuing the report's local sequence after Part III's `03-largest-jump-`. Their manuscripts, READMEs and PDFs are not shipped and survive in the arrival commits; their checksum ledgers were verified at placement and retired. Their four figures were regenerated in the write phase (section 50.3).

Manuscript 10 is the base: it arrived first and covers more regimes. The two agree on every shared statement and neither contradicts Parts I–III. They contribute, in the order printed here:

- *both*: the macroscopic profile Ψ and the moving-bound equivalent for $1/2 < m/n < 1$ (theorem 38.1), the deficit rare-event measure (theorem 38.2, last assertion), every supercritical moment constant with the mean and variance constants (theorem 38.3), a uniform finite-size tail envelope (proposition 40.3, one explicit and one non-explicit form), the first-entry and last-entry estimates of section 39, the matching with Part III's boundary tail and the linear vanishing at $u = 1/2$ (section 43.1), the critical pole of the moment constants, and seven of the research questions of section 47;
- *manuscript 10 only*: the exact deficit recursion at the maximum (proposition 39.1), the stopping-word envelope, the marked rare-event measure recording the last entry and the renewal proof of the profile (section 41), the conditional three-channel law (corollary 43.1), the

power-biased and size-biased limits (theorem 38.4), polynomial lower bounds below the half-size cutoff (theorem 45.1), a uniform Catalan sampler with Monte Carlo illustrations (section 46.6), and its small exact examples (section 49);

- *manuscript 11 only*: the canonical threshold parser and its rare-scale cost tail (section 40.4), the second proof of the profile through fixed outer words (section 42), the universal outer-word and terminal-split law with its three endpoint modes (theorem 43.2, corollary 43.3, and proposition 43.4), the moment constants at $p = 3/2$ and $p = 4$ with a recurrence (section 44.2) and their large-order asymptotic (proposition 44.1), an exact shortcut for the endpoint recurrence, and exact counts up to $n = 320$ (section 46.3).

Every result, proof, example, remark, limitation and question of both manuscripts is printed.

Status (manuscripts 10 and 11). The theorems of Part IV have written proofs, exact finite regression checks, and (manuscript 10) explicitly labelled Monte Carlo illustrations. They are not Lean- or Rocq-verified and have not been independently refereed. Publication priority has not been established: the advances are stated relative to the inspected repository report, whose explicit scope they extend. No asymptotic theorem from an unreferenced repository report is assumed: Part IV rederives the finite facts it uses and uses none of the asymptotic theorems of Parts I–III, and no fixed-bound spectral asymptotic is substituted into a varying-bound problem. Nothing in Part IV is formalized in any proof assistant.

37.2 Notation of Part IV

The symbols $a_n^{(m)}$, C_n , C , $Av_n(132)$, $M(\pi)$, $D(\pi)$, M_n , D_n , $F(\pi)$, $c_j = C_j/4^j$, the operations R_β , A , T_r , the cost t and the cutoff h keep their meanings from Part III; D without index is Part III's limit law. Manuscript 10 also sets $M = 0$ and $D = 1$ for the one-element permutation, which makes its recursions exact. The two manuscripts name the same objects differently, and several of their letters are used in Parts I–III for other objects. Part IV fixes one name for each notion (table 14). In particular the threshold fraction is always u (manuscript 11 writes x), the split fraction always y (manuscript 11 writes u), and the profile always Ψ , the name Part III's question (143) gave it (manuscript 11 writes Φ).

No normalization changed: every renamed symbol denotes exactly its manuscript's quantity. Probability, expectation, total variation, law and cost are written $\mathbb{P}$, $\mathbb{E}$, d_{TV} , $\mathcal{L}$, cost as in Part III, and variance Var . The letters a , b (block sizes), N (a core size), σ , τ , β (blocks), η , δ , ϵ , ε , and r and q inside single proofs are local and declared where used. Equations are numbered consecutively with those of Parts I–III.

37.3 Which questions are answered, and how far

Part III's Section 32.1 asks whether $\Psi(u) = \lim_n \sqrt{n} \mathbb{P}(D_n > \lfloor un \rfloor)$ exists for $0 < u < 1$, and for its value and endpoint behaviour. Its Section 32.2 asks whether $n^{1/2-p} \mathbb{E} D_n^p$ converges for each $p > 1/2$, especially $\mathbb{E} D_n / \sqrt{n}$, and whether $\mathbb{E} \sqrt{D_n} - \frac{3}{2\sqrt{\pi}} \log n$ converges. Part IV settles:

- the first question completely: the limit exists for every $0 < u \leq 1$, uniformly away from $u = 0$, and equals $\frac{3}{\sqrt{\pi}} \frac{1-2u}{\sqrt{u(1-u)}}$ for $u < 1/2$ and 0 for $u \geq 1/2$; near $u = 0$ it matches Part III's boundary tail, and it vanishes linearly at $u = 1/2$ (theorem 38.1, section 43.1);
- the supercritical half of the second: $n^{1/2-p} \mathbb{E} D_n^p \rightarrow m_p$, an explicit integral, for every fixed $p > 1/2$; in particular $\mathbb{E} D_n \sim 3\sqrt{n/\pi}$ and $\text{Var}(D_n) \sim \frac{3(4-\pi)}{4\sqrt{\pi}} n^{3/2}$ (theorem 38.3);
- Part III's remark 29.3 and Part II's research direction “Joint growth of length and adjacency bound” (section 19) for $m = \lfloor \theta n \rfloor$ with $1/2 < \theta < 1$: $a_n^{(\lfloor \theta n \rfloor)} / C_n \sim \frac{3(2\theta-1)}{\sqrt{\pi\theta(1-\theta)}} n^{-1/2}$. The equivalent $3/\sqrt{\pi(1-\theta)n}$, which Part III declined to assert, is indeed false for every such θ :

Table 14: Symbols of Part IV. “From” gives the manuscripts’ symbols (10, 11); the last column prints the tempting false reading.

Symbol	Meaning in Part IV	From	Not to be confused with
$L(\pi), L_n$	last value π_n ; that of Π_n	10: L ; 11: L	Part III’s state L and its left block L in $\pi = L n R$ (not used here), Part I’s L_m .
$J(\pi), J_n$	first deficiency $ \pi - F(\pi)$; that of Π_n	both	Part III’s series J , Part II’s order J .
$u; y; x$	threshold fraction d/n ; split fraction a/n ; deficit variable	10: u, y, x ; 11: x, u, x	Part II’s $u = 1/\Lambda$.
$\Psi(u)$	the profile (150)	10: Ψ ; 11: Φ	Part I’s shift map Φ .
$\mu; \mu_D$	marked measure on $\mathcal{X}$; its deficit marginal	10: μ ; 11: μ ($= \mu_D$)	the generic measures of (138).
$\mathcal{X}, \Xi_n$	$(0, 1] \times [0, 1]$; the pair $(D_n/n, L_n/n)$	10: E, Y_n	Part I’s E_m .
$\rho, v, v(y)$	deficit density; source functional; $\min(y, 1 - y)$	10	—
$S, S^* f$	endpoint map; $f \circ S$	10: $S, P f$	Part III’s $P(x, z); \mathbb{P}$.
$r_a, \bar{r}_H$	$C_a 4^{-a-1}$; $\sum_{a \leq H} r_a$	10: r_a, R_H	Part II’s r_m ; Part III’s r ; the operations R_β ; Part I’s R_m .
m_p	moment constant, $p > 1/2$	both: κ_p	Part III’s quantile $\kappa_{1-\epsilon}$ (index also in $(0, 1)$).
$\mathcal{I}(p); I(u)$	$\int_0^1 (z^2/(1+z^2))^{p-1} dz$; $\int_u^{1-u} \mathfrak{f}$	11: $I(p)$; 10: $I(u)$, 11: I_x	each other.
$\mathfrak{f}(y)$	$[y(1-y)]^{-3/2}$	11: $k(u)$	an index k .
$H^\star, \mathbb{P}_n^\star$	size-biased distribution function and measure	10: $H, \mathbb{P}_n^\star$	Part III’s $H(z)$.
$\mathcal{W}, \mathcal{W}_R; w_t, w_t^R$	outer-word series; their counts	11: $W, W_R; w_t, r_t$	Part I’s matrix W_m and W_{U_m} , W_{V_m} .
W_n, W	canonical outer word; its limit	11	the series $\mathcal{W}$.
$T_{n,m}, T_W$	canonical cost; $\text{cost}(W)$	11: $T_{n,m}, T$	Part III’s skeleton cost T (which is also manuscript 10’s T).
$\text{len}(W)$	number of operations of W	11: K	the constants K .
$\chi(w)$	endpoint acceptance weight	11: $\alpha(w)$	Part I’s α_m .
$\ell_n(w)$	append threshold	11: $r_n(w)$	r_a .
$\mathcal{E}_{n,m}, \mathcal{E}_n(w)$	the events $M_n \leq m$, and with canonical word w	11: $E_{n,m}, E_n(w)$	Part I’s E_m .
$Q_{N,m}$	raw terminal-pair count	11	Part III’s Q .
$U_n, U^{(u)}$	terminal split fraction; its limit	11: U_n, U_x ; 10: Y	—
ξ, V, ζ	channel, deficit, last-entry mark in corollary 43.1	10: B, V, W	$B(x); W_n$.
$\mathcal{N}_s(u)$	avoiders of size s with first deficiency $\leq u$	11: $H_s(u)$	Part III’s $N_{N,f}$ ($F = f$ exactly).
$\Omega_r(\theta)$	conjectured constant of section 47.1	10: $A_r(\theta)$	Part I’s $A_m(x)$.
$\underline{c}_{\theta,r}, \underline{c}'_{\theta,r}$	constants of theorem 45.1	10: $c_{\theta,r}, c'_{\theta,r}$	Part I’s $c_{k,p}$; Part III’s $\underline{c}_p$.
$K, K_f, K_\eta, \dots$	constants, absolute unless indexed	both	Part I’s $K_{m,d}$, Part II’s K_m , Part III’s $\mathcal{K}$.

the true constant is smaller by the factor $(2\theta - 1)/\sqrt{\theta} < 1$ (the merge's comparison).

It does not settle: the critical centered half-moment, which stays open (section 47.6) [answered in Part V, added 30 September 2026: theorems 53.1 and 53.2]; an equivalent for $0 < \theta \leq 1/2$, where only $o(n^{-1/2})$ and polynomial lower bounds are proved (theorems 38.1 and 45.1); the window at $m = n/2$; a local limit; the exact total-variation constant of Part III's corollary 28.2; and any interpolation with the fixed- m regime of Parts I and II. Part IV refutes nothing in Parts I–III. [Added 30 September 2026, batch 68: Part VI proves the order below half size, $\Theta(n^{-\lfloor 1/\theta \rfloor/2})$ for every fixed $0 < \theta \leq 1/2$, and at $m = n/2$ the order $\Theta(n^{-1})$ with order-crossover scale $\sqrt{n}$ (theorem 68.1 and corollary 68.4); equivalents below half size and a limiting transition function at $m = n/2$ stay open.] [Added 1 October 2026, batch 71: Part VII proves the equivalent for $1/3 < \theta \leq 1/2$ and the limiting transition function at $m = n/2$ on the scale $\sqrt{n}$ (theorems 80.1 and 80.2); equivalents for $\theta \leq 1/3$ stay open.] [Added 1 October 2026, batch 72: Part VIII proves them, and the limiting transition function at every $m = n/q$ (theorems 91.1 and 91.2).]

38 The questions and the main results

38.1 From a boundary distribution to macroscopic rare events

Recall from Part III that $M(\pi) = \max_i |\pi_{i+1} - \pi_i|$ and $D(\pi) = n - M(\pi)$ for $\pi \in \text{Av}_n(132)$, $n \geq 2$, that Π_n is uniform on $\text{Av}_n(132)$ with $M_n = M(\Pi_n)$, $D_n = D(\Pi_n)$, and that (82) reads $a_n^{(m)}/C_n = \mathbb{P}(M_n \leq m) = \mathbb{P}(D_n \geq n - m)$. For a nonempty permutation write $F(\pi) = \pi_1$, $L(\pi) = \pi_{|\pi|}$ and $J(\pi) = |\pi| - F(\pi)$, and write F_n, L_n, J_n for their values at Π_n . An inequality $D_n \geq un$ means $D_n \geq \lceil un \rceil$. All moment orders are fixed before n tends to infinity.

The finite-state, fixed- m problem was developed by Nadler [1], who treated $m = 2$, and Mayama–Akita [2], who supplied endpoint states, rational generating functions for each fixed bound and growth-constant results; their maximum-position restriction is especially useful here, and their finite-state recursion is the basis of manuscript 11's exact numerical checks, not a new result of it. Part III proves a proper integer-valued boundary limit for D_n , a sharp-order total-variation bound, a limiting tail proportional to $k^{-1/2}$, a growing sublinear-deficit equivalent and a moment transition at $p = 1/2$. It explicitly does *not* infer a relative asymptotic at D_n of order n from its additive boundary estimate (remark 29.3), and its Sections 32.1 and 32.2 ask for the bulk profile and the exact supercritical moment constants. Those are the starting points of Part IV.

That distinction is essential. A boundary tail of the form $c/\sqrt{k}$ does not by itself identify $\mathbb{E}D_n/\sqrt{n}$: events with D_n proportional to n contribute at precisely the same order. Part IV analyzes those events directly, rather than extrapolating the boundary approximation. On the ordinary probability scale $D_n/n \rightarrow 0$; on the scale of probabilities multiplied by $\sqrt{n}$, macroscopic deficits remain visible, and it is this second scale that governs all moments of order greater than $1/2$. Part IV builds on the subject and elementary decompositions of Parts I–III but assumes none of their probabilistic or spectral theorems; this makes the dependency on the existing research particularly narrow and auditable.

38.2 The macroscopic profile and moving-bound enumeration

Define, for $0 < u \leq 1$,

$$\Psi(u) = \begin{cases} \frac{3}{\sqrt{\pi}} \frac{1-2u}{\sqrt{u(1-u)}}, & 0 < u < \frac{1}{2}, \\ 0, & \frac{1}{2} \leq u \leq 1. \end{cases} \quad (150)$$

Ψ is continuous on $(0, 1]$. The zero at $1/2$ and beyond is part of the definition, not a claim that the finite probability vanishes there.

Theorem 38.1 (Complete first-order macroscopic profile; both manuscripts). *For every $0 < u \leq 1$,*

$$\sqrt{n} \mathbb{P}(D_n > un) \longrightarrow \Psi(u). \quad (151)$$

More generally, for every integer sequence d_n with $d_n/n \rightarrow u \in (0, 1)$,

$$\sqrt{n} \mathbb{P}(D_n \geq d_n) \longrightarrow \Psi(u). \quad (152)$$

For each fixed $\delta > 0$ the convergence in (151) is uniform in $u \in [\delta, 1]$, and so is that of $u \mapsto \sqrt{n} \mathbb{P}(D_n \geq \lceil nu \rceil)$. Consequently, for fixed $1/2 < \theta < 1$,

$$\frac{a_n^{(\lfloor \theta n \rfloor)}}{C_n} \sim \frac{3(2\theta - 1)}{\sqrt{\pi\theta(1 - \theta)}} n^{-1/2}, \quad (153)$$

$$a_n^{(\lfloor \theta n \rfloor)} \sim \frac{3(2\theta - 1)}{\pi\sqrt{\theta(1 - \theta)}} \frac{4^n}{n^2}. \quad (154)$$

For fixed $0 < \theta \leq 1/2$ the ratio is $o(n^{-1/2})$ and the count is $o(4^n/n^2)$; no positive asymptotic equivalent in that range is asserted.

Manuscript 10 proves (151) with uniformity on $[\delta, 1]$ and both consequences; manuscript 11 proves the sequence form (152), compact uniformity and (153). The first proof (manuscript 10) solves a renewal equation for a marked measure (section 41); the second (manuscript 11) sums fixed outer words of a canonical threshold decomposition (section 42). The zero part of Ψ is a first-order *asymptotic* cutoff, not a finite combinatorial prohibition. The increasing permutation, for example, has $M_n = 1$ and $D_n = n - 1$. Such permutations lie beyond the cutoff, but have negligible mass on the $n^{-1/2}$ scale.

38.3 The marked measure behind the coefficient three

Put

$$X = (0, 1] \times [0, 1], \quad \Xi_n = (D_n/n, L_n/n), \quad v(y) = \min(y, 1 - y).$$

The following locally finite measure is infinite near the missing boundary $x = 0$ of the deficit coordinate; it is not a probability measure.

Theorem 38.2 (Marked rare-event measure; manuscript 10, last assertion both). *For every continuous function f on $[0, 1]^2$ that vanishes on some strip $\{0 \leq x \leq \delta\}$ of the deficit coordinate, $\delta > 0$,*

$$\sqrt{n} \mathbb{E} f(\Xi_n) \longrightarrow \mu(f), \quad (155)$$

$$\mu(f) = \frac{1}{4\sqrt{\pi}} \int_0^1 \frac{f(v(y), 0) + f(v(y), 1 - y) + f(v(y), 1)}{[y(1 - y)]^{3/2}} dy. \quad (156)$$

Equivalently, $\sqrt{n} \mathcal{L}(\Xi_n)$ converges vaguely on X to μ . The deficit marginal of μ has density

$$\rho(x) = \frac{3}{2\sqrt{\pi}} [x(1 - x)]^{-3/2} \mathbf{1}_{(0, 1/2)}(x). \quad (157)$$

That is, on the locally compact space $(0, 1]$ the measures $\mu_{D,n} = \sqrt{n} \mathcal{L}(D_n/n)$ converge vaguely to

$$\mu_D(dx) = \rho(x) dx = \frac{3}{2\sqrt{\pi}} \frac{\mathbf{1}_{(0, 1/2)}(x)}{[x(1 - x)]^{3/2}} dx, \quad (158)$$

and $\mu_D([u, 1]) = \Psi(u)$ for $u > 0$. The measure μ_D has infinite mass near zero and is not a probability measure.

The three summands describe a last entry close to the low end, a last entry in the interior, and a last entry close to the high end. Their equal weights are a theorem of the renewal calculation, not an independence assumption about endpoints. The last assertion is manuscript 11's Theorem 1.2, proved there from the profile (section 42.4); manuscript 10 obtains it as the deficit marginal (section 41.6).

38.4 Exact moment constants and the mean-carrying law

Theorem 38.3 (Every supercritical moment; both manuscripts). *For every fixed real $p > 1/2$,*

$$\mathbb{E}D_n^p \sim m_p n^{p-1/2}, \quad (159)$$

where

$$m_p = \frac{3}{2\sqrt{\pi}} \int_0^{1/2} x^{p-3/2}(1-x)^{-3/2} dx \quad (160)$$

$$= \frac{3}{\sqrt{\pi}} \int_0^1 \left(\frac{z^2}{1+z^2} \right)^{p-1} dz. \quad (161)$$

In particular,

$$\mathbb{E}M_n = n - \frac{3}{\sqrt{\pi}}\sqrt{n} + o(\sqrt{n}), \quad (162)$$

$$\text{Var}(D_n) = \text{Var}(M_n) \sim \frac{3(4-\pi)}{4\sqrt{\pi}} n^{3/2} = \frac{3}{\sqrt{\pi}} \left(1 - \frac{\pi}{4}\right) n^{3/2}. \quad (163)$$

The two printed forms of the variance constant are manuscript 10's and manuscript 11's; they are equal. Both manuscripts state both integral forms of m_p ; manuscript 10 adds the size-biased law below, which at $p = 1$ carries the same information as the integral.

Define deficit-size bias by

$$\mathbb{P}_n^\star(B) = \frac{\mathbb{E}[D_n \mathbf{1}_B]}{\mathbb{E}D_n}. \quad (164)$$

This is a probability measure on the same finite permutation space.

Theorem 38.4 (A uniform law under deficit-size bias; manuscript 10). *Under $\mathbb{P}_n^\star$,*

$$\sqrt{\frac{D_n}{n - D_n}} \Rightarrow \text{Uniform}[0, 1]. \quad (165)$$

Equivalently, the limiting distribution function of D_n/n under this bias is

$$H^\star(x) = \begin{cases} 0, & x \leq 0, \\ \sqrt{x/(1-x)}, & 0 < x < 1/2, \\ 1, & x \geq 1/2. \end{cases} \quad (166)$$

Thus half the leading mean contribution comes from deficits above $n/5$:

$$\frac{\mathbb{E}[D_n; D_n > n/5]}{\mathbb{E}D_n} \longrightarrow \frac{1}{2}. \quad (167)$$

Because $m_1 = 3/\sqrt{\pi}$, manuscript 11's moment formula at $p = 1$ and manuscript 10's size-biased law carry the same information: the size-biased density $x\rho(x)/m_1$ is the derivative of $H^\star$, whose total mass one is the statement $\mathcal{I}(1) = 1$ of section 44.2 (the merge's comparison).

38.5 What makes the constant three

Both manuscripts explain the coefficient 3 in Ψ , from different decompositions.

Manuscript 11: outer words. The combinatorial proof isolates one split into two macroscopic blocks, preceded by a finite outer word of two possible operations. One operation adds a finite block on the left of a maximum; the other appends a new maximum. The total critical weight of

all outer words is 4. However, words containing an append operation are accepted only when the terminal right block has a last entry near its high endpoint. That event has asymptotic probability $1/2$. Words without an append operation have total weight 2; words with one or more appends also have total weight 2. The accepted weight is therefore

$$2 + \frac{1}{2} \cdot 2 = 3. \quad (168)$$

Neither ignoring append operations nor treating every append as harmless gives the correct constant. The endpoint condition is not a secondary error term.

Manuscript 10: a renewal with three endpoint channels. Integrating the Catalan root-split density alone gives only the source term ν of (202) (remark 41.2): small outer decorations recur, and appending a maximum can erase a rare deficit. The resulting renewal equation has the solution (156), whose three equally weighted summands are the three endpoint channels; section 49 separates the folding factor two from the three channels. (On tests of the deficit alone the source is a third of the answer; section 43.4 identifies the three channels with manuscript 11's three endpoint modes. Both observations are the merge's.)

38.6 Scope of the advance

The main profile and moment results answer the two explicit repository questions just identified. The marked endpoint limit, the size-biased uniform law, the lower bounds of section 45, and the conditional outer-word and split laws of section 43 are further results. All needed finite facts are rederived. In particular, no asymptotic claim from an unrefereed repository report is assumed in the proofs; the finite-state recurrence is used only for computations and is attributed to its prior source. Manuscript 10's proof is self-contained at the level of finite Catalan decompositions, elementary generating functions, Stirling's formula, and convergence of measures: an exact root identity eliminates the left block's internal jump statistic, and a uniform counting bound controls the otherwise dangerous mesoscopic remainder. Manuscript 11's proof is by exact Catalan counting, endpoint estimates, and a tail bound at the rare-event scale, a different asymptotic decomposition for a bound proportional to n ; in neither is a fixed-bound spectral asymptotic substituted into a varying-bound problem.

Neither manuscript claims the exact total-variation constant, a constant term in the critical half-moment, a local limit for individual macroscopic lattice masses, or the optimal scale below the half-size cutoff. [Added 30 September 2026, batch 64: Part V proves the constant term (theorem 53.1).] The positive bulk equivalent concerns adjacency bounds $m/n \in (1/2, 1)$; the smaller-bound bulk regime and the critical window at $m = n/2$ remain open. [Added 30 September 2026, batch 68: Part VI proves the optimal polynomial order below the half-size cutoff and the order and order-crossover scale at $m = n/2$ (theorem 68.1 and corollary 68.4), but no equivalent there.] [Added 1 October 2026, batch 71: Part VII proves the equivalent for $1/3 < \theta \leq 1/2$ and the limit in the window at $m = n/2$ (theorems 80.1 and 80.2).] [Added 1 October 2026, batch 72: Part VIII proves the equivalent for every fixed $\theta \leq 1/3$ and the limit in the window at every $m = n/q$ (theorems 91.1 and 91.2).] Nor does this work resolve the fixed- m component-ordering problem of Mayama–Akita.

39 Exact identities at the maximum and at the endpoints

39.1 The Catalan maximum decomposition

Both manuscripts start from the maximum decomposition of Part I (section 2), which Part III also rederives (section 23): every $\pi \in \text{Av}_n(132)$ is uniquely

$$\pi = (\sigma + b) n \tau, \quad \sigma \in \text{Av}_a(132), \quad \tau \in \text{Av}_b(132), \quad a + b + 1 = n,$$

and every such pair produces an avoider. Hence, for the left and right block sizes of Π_n (manuscript 10),

$$\mathbb{P}(\text{left size } a, \text{ right size } b) = \frac{C_a C_b}{C_n}, \quad (169)$$

and, conditional on their sizes, the standardized blocks are independent and uniform. Summing gives $C = 1 + xC^2$. If π is m -bounded and $b > 0$, then $a + 1 \leq m$ (manuscript 11's positional restriction), because the entry after n is at most b : this is Part I's (4). Both manuscripts rederive the decomposition and the restriction by the argument of Part I; it is not repeated.

39.2 The exact deficit recursion

Proposition 39.1 (The exact deficit recursion; manuscript 10). *Let $\pi = (\sigma + b)n\tau$ as above. If $b \geq 1$, then*

$$M(\pi) = \max\{n - F(\tau), M(\tau)\}, \quad (170)$$

$$D(\pi) = \min\{F(\tau), a + 1 + D(\tau)\}, \quad (171)$$

$$L(\pi) = L(\tau). \quad (172)$$

If $b = 0$ and $n \geq 2$, then

$$D(\pi) = \min\{D(\sigma) + 1, L(\sigma)\}, \quad L(\pi) = n. \quad (173)$$

For $a > 0$, the first-entry deficiency satisfies

$$J(\pi) = n - F(\pi) = a + 1 - F(\sigma); \quad (174)$$

for $a = 0$, it is zero.

Proof. Suppose $b \geq 1$. The downward boundary jump from n to $F(\tau)$ is $n - F(\tau) \geq n - b = a + 1$. Every jump inside the left block is at most $a - 1$, and the jump from the left block into n , when present, is at most a . Therefore none of these can be largest. Only the downward boundary and jumps internal to the right block remain, proving (170). Subtracting the maximum from n gives (171); the last value is unchanged. If $b = 0$, appending n introduces the single new jump $n - L(\sigma)$, and subtraction gives (173). Finally, $F(\pi) = b + F(\sigma)$ when $a > 0$, proving (174). $\square$

This is stronger than an asymptotic simplification. It explains why an arbitrary left decoration contributes only its Catalan count in the later analysis. The last endpoint cannot likewise be discarded: the minimum in (173) is exactly where a putative large deficit can be clipped away. (The recursion is the one-step form of the state transformations (108)–(109) of Part III, which hold there under a first-entry hypothesis on a large hole; (171) and (173) hold for every avoider.)

39.3 First entries: a uniformly exponential tail

The first-entry count

$$\#\{\pi \in \text{Av}_N(132) : F(\pi) = f\} = \frac{N - f + 1}{N} \binom{N + f - 2}{N - 1}, \quad 1 \leq f \leq N, \quad (175)$$

is Part III's (98), which is Part I's (21) reindexed (remark 23.2). Manuscripts 10 and 11 rederive it exactly as Part III does: the maximum decomposition gives the bivariate first-value series (95), the substitution $(x, z) \mapsto (xu, u^{-1})$ gives the deficiency form (96), deficiency j has generating function $(xC(x))^{j+1}$, and Lagrange inversion applied to $C = 1 + xC^2$ gives $[x^k]C(x)^q = \frac{q}{2k+q} \binom{2k+q}{k}$. The new content is a pair of bounds uniform in the size, which remain useful when the event being studied has probability only of order $n^{-1/2}$.

Lemma 39.2 (A uniform first-deficiency bound; both manuscripts). *For $n \geq 1$ and $0 \leq j \leq n - 1$,*

$$\mathbb{P}(J_n = j) \leq (j + 1)2^{-j-1} \leq (j + 1)2^{-j}, \quad \mathbb{E}J_n \leq 8, \quad (176)$$

and for every integer $h \geq 0$,

$$\mathbb{P}(J_n \geq h) \leq (h + 2)2^{-h}. \quad (177)$$

In particular $\mathbb{P}(J_n \geq h)$ is bounded by an absolute constant times $(h + 1)2^{-h}$, uniformly in n .

Manuscript 10 states the weaker bound $(j + 1)2^{-j}$, $\mathbb{E}J_n \leq 8$ and the last sentence; manuscript 11 states $(j + 1)2^{-j-1}$ and (177). The proofs are the same computation.

Proof. For $n \geq 2$, divide the count (175) with $f = n - j$ by C_n :

$$\mathbb{P}(J_n = j) = (j + 1) \frac{n + 1}{2(2n - 1)} \prod_{r=0}^{j-1} \frac{n - 1 - r}{2n - 2 - r}.$$

Indeed the ratio $\binom{2n-2}{n-1} / \binom{2n}{n}$ is $n/(2(2n - 1))$, and lowering the first upper index a further j times multiplies it by the product. The empty product is one, each product factor is at most $1/2$, and $(n + 1)/(2(2n - 1)) \leq 1/2$. Thus $\mathbb{P}(J_n = j) \leq (j + 1)2^{-j-1}$ (manuscript 10 bounds the prefactor $(j + 1)(n + 1)/(2(2n - 1))$ by $j + 1$ instead). Summing the infinite geometric-arithmetic tail gives $\sum_{j \geq h} (j + 1)2^{-j-1} = (h + 2)2^{-h}$. For $n = 1$, the deficiency is zero and every inequality holds directly. Finally $\sum_{j \geq 0} j(j + 1)2^{-j} = 8$. $\square$

39.4 Last entries: two endpoint modes

The last-entry count

$$\#\{\pi \in \text{Av}_N(132) : L(\pi) = \ell\} = C_{N-\ell} C_{\ell-1}, \quad 1 \leq \ell \leq N, \quad (178)$$

is Part III's (101), from the unique decomposition $\pi = (\tau + \ell) \sigma \ell$ with $|\tau| = N - \ell$, $|\sigma| = \ell - 1$ ((100)); both manuscripts rederive it that way, noting that triples crossing the separated blocks cannot form 132.

Lemma 39.3 (The last entry has two endpoint limits; both manuscripts). *As $n \rightarrow \infty$,*

$$L_n/n \implies \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1. \quad (179)$$

More precisely, for every fixed $\eta \in (0, 1/2)$, uniformly for integer $r \in [\eta N, (1 - \eta)N]$,

$$\mathbb{P}(L(\Pi_N) \geq r) = \frac{1}{2} + O_\eta(N^{-1/2}), \quad (180)$$

and the probability that $L(\Pi_N)/N$ belongs to $[\eta, 1 - \eta]$ is $O_\eta(N^{-1/2})$.

Manuscript 10 proves (179), manuscript 11 the quantitative form; the two proofs differ.

Proof (manuscript 10). For fixed $j \geq 0$, Stirling's formula and (178) give

$$\mathbb{P}(L_n = j + 1) \longrightarrow C_j 4^{-j-1}, \quad \mathbb{P}(L_n = n - j) \longrightarrow C_j 4^{-j-1}.$$

Each family of limiting masses sums to $\frac{1}{4}C(1/4) = 1/2$. For a fixed endpoint cutoff the two families are disjoint for large n . Letting the cutoff tend to infinity leaves no asymptotic mass in the middle. This proves (179). $\square$

Second proof, with the uniform estimate (manuscript 11). The distribution (178) is symmetric under $\ell \mapsto N+1-\ell$. For $\eta N \leq \ell \leq (1-\eta)N$, the Catalan bounds (181) below bound each probability by $K_\eta N^{-3/2}$. Summing at most N terms gives the middle-mass bound. Symmetry puts one half of the remaining mass at each end, proving (180) uniformly in the stated range. $\square$

In particular, the first deficiency has an exponentially bounded tail, whereas the last entry converges after division by its size to an equal mixture of the endpoints zero and one. No independence between the two endpoints, or between D_n and L_n , has been asserted or will be needed. Later, the fact that $D_n/n \rightarrow 0$ in probability will make its macroscopic value deterministic inside a large unconditioned block, which is a different argument.

39.5 Uniform Catalan estimates

Manuscript 11 records the following bounds. Stirling's formula and adjustment of finitely many small indices give absolute constants $c_-, c_+ > 0$ such that

$$c_-(j+1)^{-3/2} \leq c_j \leq c_+(j+1)^{-3/2}, \quad c_j = \frac{1 + O(j^{-1})}{\sqrt{\pi} j^{3/2}} \quad (j \rightarrow \infty). \quad (181)$$

(Manuscript 10 proves $c_h \leq (h+1)^{-3/2}$ directly, in the proof of lemma 40.2.) All constants denoted by K are absolute unless a subscript specifies otherwise; they can be enlarged from line to line. For $N \geq n/2$, (181) implies

$$\frac{C_N}{C_n} \leq K 4^{N-n}. \quad (182)$$

These inequalities are used as bounds, and the asymptotic formula only when all involved block sizes tend to infinity uniformly. The numbers c_j decrease strictly ((112)), so $C_{n-t}/C_n \geq 4^{-t}$ for $0 \leq t < n$.

40 Tail envelopes at the rare-event scale

The proof of a rare-event limit needs an upper bound that survives multiplication by $\sqrt{n}$. Ordinary weak convergence of D_n/n is not enough. Both manuscripts obtain such a bound from finite counting, without using Part III's algebraic boundary law, by two different decompositions.

40.1 Stopping words and a deterministic deficit bound

Manuscript 10 uses Part III's three reconstruction operations R_β , A and T_τ (section 24), with costs $|\beta| + 1$, 1 and $|\tau| + 1$. It calls Part III's skeletons *stopping words*: a word in R_β, A followed by one T_τ , applied from the terminal operation outward. It has total cost t and one unexpanded hole; filling that hole with a size- N avoider produces size $N + t$.

Lemma 40.1 (Disjoint families and a deterministic deficit bound). *Fix n and $h < n/2$. Reconstructions from different stopping words of cost $t \leq h$ are disjoint, and the reconstruction map is injective in the size- $(n - t)$ hole. Every such reconstruction satisfies*

$$D(\pi) \leq t - 1. \quad (183)$$

Proof. Disjointness and injectivity are Part III's lemma 25.2; manuscript 10 rederives them by the same argument (the block containing the hole is larger than its finite sibling at every maximum decomposition, so the word and the hole are read off from the final permutation).

For (183): at the terminal, the jump from the maximum to the first value of τ has size at least $N + 1$. Every outer operation preserves all existing internal differences, so the final maximum jump is still at least $N + 1$. Thus $D \leq (N+t) - (N+1) = t - 1$. Notably, no first-entry condition on the hole is needed for this one-sided assertion, unlike the exact identity of Part III's lemma 25.1. $\square$

40.2 An elementary bound for the exact cost tail

The stopping words of cost t are counted by Part III's b_t , with generating function $B(x) = (C(x) - 1)^2$ ((103)): the numerator of $x(C - 1)/(1 - xC - x)$ counts terminals and the denominator an arbitrary prefix. In particular $B(1/4) = 1$, so the weights $b_t 4^{-t}$ form the probability law of Part III's cost T ; and $b_t = C_{t+1} - 2C_t$ for $t \geq 1$, $b_0 = 0$ ((104)). Manuscript 10 rederives all of this.

Lemma 40.2 (Exact tail and an elementary bound). *For every integer $h \geq 0$,*

$$\mathbb{P}(T > h) = \frac{2(h+1)(2h+1)}{h+2} \frac{C_h}{4^h} \leq \frac{4}{\sqrt{h+1}}. \quad (184)$$

Proof. The equality is Part III's lemma 24.1, proved there, and in manuscript 10, from $c_{j+1}/c_j = (2j+1)/(2j+4)$, $\sum_{j=0}^h c_j = 2 - (2h+1)c_h$ and $b_t 4^{-t} = 4c_{t+1} - 2c_t$ for $t \geq 1$. For the bound (manuscript 10), $c_h \leq (h+1)^{-3/2}$ by induction, because

$$(2h+1)^2(h+2) \leq 4(h+1)^3.$$

The prefactor in (184) is at most $4(h+1)$, proving the bound. $\square$

40.3 The uniform envelope

Proposition 40.3 (Uniform finite-size deficit envelope; both manuscripts). *For $1 \leq d \leq n$, put $h = \min\{d, \lfloor (n-1)/2 \rfloor\}$. Then*

$$\mathbb{P}(D_n \geq d) \leq \mathbb{P}(T > h) \quad (185)$$

$$\leq \min \left\{ 1, \frac{4\sqrt{2}}{\sqrt{d}} \right\}. \quad (186)$$

Consequently $D_n/n \rightarrow 0$ in probability and, for every $0 < u \leq 1$,

$$\sqrt{n} \mathbb{P}(D_n > un) \leq 4\sqrt{2} u^{-1/2}. \quad (187)$$

Manuscript 10 proves the explicit form given here. Manuscript 11 proves $\mathbb{P}(D_n \geq d) \leq K/\sqrt{d}$ for $n \geq 2$, $1 \leq d \leq n-1$, with an unspecified absolute constant (corollary 40.8), by a different route.

Proof. By lemma 40.1, the disjoint short-word families have total probability $\sum_{t \leq h} b_t C_{n-t}/C_n$ and all satisfy $D_n < d$. Since c_j decreases, $C_{n-t}/C_n = 4^{-t} c_{n-t}/c_n \geq 4^{-t}$. Their mass is therefore at least $\mathbb{P}(T \leq h)$, proving the first inequality. Because $h+1 \geq d/2$, lemma 40.2 proves the second. Use $d = \lfloor un \rfloor + 1 > un$ to obtain the scaled bound, with the zero event handled separately when $u = 1$. $\square$

The same stopping construction occurs in Part III's boundary-law proof. Its use here is deliberately weaker and cleaner: the preserved terminal jump supplies a tail bound directly, without reconstructing the exact limiting deficit or controlling an exceptional set of hole endpoints.

40.4 A canonical threshold parser

Manuscript 11 decomposes a bounded permutation differently. Its outer words use only the operations R_β (with β an arbitrary avoider, including $R_\emptyset$ at cost one) and A , with Part III's costs; an *outer word* is a finite word in these decorated operations, including the empty word, written outermost operation first, so that reconstruction applies it from right to left. Its cost is the sum of operation costs, and every operation preserves avoidance.

Fix integers n, m with

$$n/2 \leq m \leq n-1, \quad d = n - m. \quad (188)$$

Let $\mathcal{E}_{n,m}$ be the event $M_n \leq m$. For an m -bounded permutation, start at its maximum decomposition. As long as a child block has more than m entries, follow that block. A large right child records an R_β operation. A large left child must have an empty right sibling, by the positional restriction, and records an A operation. Stop at the first node whose two child blocks both have size at most m .

Lemma 40.4 (Canonical parsing and reconstruction; manuscript 11). *Every permutation in $\mathcal{E}_{n,m}$ has a unique representation*

$$w((\sigma + b)N\tau), \quad a = |\sigma|, \quad b = |\tau|, \quad (189)$$

where w is an outer word of cost t , and

$$0 \leq t \leq d-1, \quad N = n - t > m, \quad a + b = N - 1, \quad 0 \leq a, b \leq m. \quad (190)$$

The word, all its decorations, and the terminal pair are recoverable from the permutation. For any fixed word of cost $t \leq d-1$, reconstruction from terminal pairs satisfying (190) is injective. Whenever the reconstructed permutation is m -bounded, its canonical word is exactly the prescribed word.

Proof. At every stage the current block has more than m entries. Under (188), two children of more than m entries are impossible by their total size; for bounded permutations the positional restriction also rules out a nonempty right child next to a left child of more than m entries. The stated rule therefore gives a unique next operation whenever parsing continues. The followed size strictly decreases. At size $m+1$ both children have at most m entries, so parsing has stopped by then. This proves $N \geq m+1$ and $t \leq d-1$.

Each maximum decomposition recovers its standardized children, so it also recovers every decoration and the terminal pair. Conversely, the core in (189) and every intermediate core have more than m entries. An R decoration has size at most $t-1 \leq d-2 < m$, while an A operation has no right sibling. Hence the same parsing rule follows the reconstructed core at each operation and stops exactly at the specified terminal pair. This proves injectivity and the final assertion. $\square$

Remark 40.5 (The terminal pair is not a boundary-law skeleton). The absorbing skeleton of Part III (manuscript 10's stopping word) stops at a finite right block next to a giant left block. The threshold parser stops when *both* blocks lie below the adjacency threshold. Under a macroscopic deficit, these terminal blocks are themselves macroscopic. The two parsers serve different limits and must not be identified.

Let w_t count all individually decorated outer words of cost t , and let w_t^R count those containing no A operation. The generating function of one R operation is $zC(z)$, including $R_\emptyset$ at cost one. Therefore

$$\mathcal{W}(z) := \sum_{t \geq 0} w_t z^t = \frac{1}{1 - zC(z) - z} = C(z)^2, \quad (191)$$

$$\mathcal{W}_R(z) := \sum_{t \geq 0} w_t^R z^t = \frac{1}{1 - zC(z)} = C(z). \quad (192)$$

Both identities follow from $zC^2 = C - 1$. In particular,

$$w_t = C_{t+1}, \quad w_t^R = C_t. \quad (193)$$

At $z = 1/4$ these series converge, with $\mathcal{W}(1/4) = 4$ and $\mathcal{W}_R(1/4) = 2$. This critical summability will allow the finite outer word to be summed without a spectral approximation. (Part III's cost series factors as $B(z) = \mathcal{W}(z) \cdot z(C(z) - 1)$: an outer word followed by a terminal T_τ , so the two parsers share their outer operations. This remark is the merge's.)

40.5 Tightness of the canonical cost

The main technical point is not merely that the outer word is usually short under some limiting construction. Its tail must be negligible *relative to a probability of order $n^{-1/2}$* . The following exact majorant supplies that missing uniformity. Let

$$Q_{N,m} = \sum_{\substack{a+b=N-1 \\ 0 \leq a, b \leq m}} C_a C_b \quad (194)$$

count terminal pairs before checking the boundary jumps.

Lemma 40.6 (Terminal split bound; manuscript 11). *For $m < N \leq 2m + 1$, with $r = N - m \geq 1$,*

$$Q_{N,m} \leq \frac{K C_N}{\sqrt{r}}. \quad (195)$$

Proof. For $r = 1$, the full Catalan recurrence gives $Q_{N,m} \leq C_N$. For $r \geq 2$, every admissible pair has $a, b \geq r - 1$. By symmetry we may sum the half with $a \leq b$ and multiply by two. On that half, $b + 1 \geq N/2$, and the Catalan bounds give

$$\frac{C_a C_b}{C_N} \leq \frac{K}{(a + 1)^{3/2}}.$$

The sum over $a \geq r - 1$ is at most $K/\sqrt{r}$, by comparison with the integral of $u^{-3/2}$. This proves (195). $\square$

On $\mathcal{E}_{n,m}$, write $T_{n,m}$ for the cost of the canonical outer word. It need not be defined outside that event.

Lemma 40.7 (Uniform tail of the canonical cost; manuscript 11). *Under (188), for every integer $h \geq 0$,*

$$\mathbb{P}(\mathcal{E}_{n,m}, T_{n,m} > h) \leq K \left\{ \frac{1}{\sqrt{d}\sqrt{h+1}} + \frac{1}{d} \right\}. \quad (196)$$

Moreover,

$$\mathbb{P}(\mathcal{E}_{n,m}) \leq \frac{K}{\sqrt{d}}. \quad (197)$$

Proof. Discard every boundary constraint after applying the injective parser. For each cost t , there are w_t words and at most $Q_{n-t,m}$ terminal pairs per word. Since $n - t \geq m + 1 \geq n/2$, lemma 40.6, (182) and (193) give

$$\begin{aligned} \mathbb{P}(\mathcal{E}_{n,m}, T_{n,m} > h) &\leq \frac{1}{C_n} \sum_{h < t \leq d-1} C_{t+1} Q_{n-t,m} \\ &\leq K \sum_{h < t \leq d-1} \frac{1}{(t+1)^{3/2} \sqrt{d-t}}. \end{aligned} \quad (198)$$

For $t \leq d/2$, the sum is at most $Kd^{-1/2} \sum_{t > h} (t+1)^{-3/2}$, hence at most $K/\sqrt{d(h+1)}$. For $t > d/2$, put $r = d - t$. This part is at most

$$Kd^{-3/2} \sum_{1 \leq r \leq d/2} r^{-1/2} \leq Kd^{-1}.$$

These bounds prove (196). Starting the same sum at $t = 0$ proves (197), after enlarging K for $d = 1$. $\square$

Corollary 40.8 (An all-deficit envelope; manuscript 11). *For every $n \geq 2$ and $1 \leq d \leq n - 1$,*

$$\mathbb{P}(D_n \geq d) \leq \frac{K}{\sqrt{d}}. \quad (199)$$

In particular, $D_n/n \rightarrow 0$ in probability.

Proof, and second proof of proposition 40.3 up to the constant. For $d \leq n/2$, apply (197) with $m = n - d$. For $d > n/2$, use monotonicity and the bound at $\lfloor n/2 \rfloor$. Since $d \leq n$, the resulting $K/\sqrt{n}$ is at most a constant multiple of $d^{-1/2}$. Enlarge K to cover the small values of n . $\square$

Remark 40.9 (The order of truncations). If $d_n/n \rightarrow u > 0$, (196) yields

$$\limsup_{n \rightarrow \infty} \sqrt{n} \mathbb{P}(\mathcal{E}_{n,n-d_n}, T_{n,n-d_n} > h) \leq K_u (h + 1)^{-1/2}.$$

Thus one first fixes a finite cost cutoff, takes $n \rightarrow \infty$, and only then lets the cutoff grow. Ordinary unscaled tightness would not justify this interchange. This estimate is the reason the finite-word calculation of section 42 determines the complete leading constant.

41 First proof of the profile: a renewal equation for the marked measure

This section and the next are manuscript 10's proof of theorem 38.2, and with it of theorem 38.1.

41.1 Test functions and the append projection

It is enough initially to consider a Lipschitz function f on $[0, 1]^2$ that is zero whenever the deficit coordinate satisfies $x \leq \delta$, for some $\delta > 0$. Set

$$Z_n(f) = \sqrt{n} \mathbb{E} f(D_n/n, L_n/n). \quad (200)$$

By (187), this sequence is bounded, with a bound depending only on δ and $\|f\|_\infty$. Define

$$S(x, \ell) = (\min(x, \ell), 1), \quad S^* f = f \circ S. \quad (201)$$

Then $S^2 = S$, hence $(S^*)^2 = S^*$. Also $S^* f$ is Lipschitz and vanishes on the same low-deficit strip.

The macroscopic source functional is

$$v(f) = \frac{1}{8\sqrt{\pi}} \int_0^1 \frac{f(v(y), 0) + f(v(y), 1 - y)}{[y(1 - y)]^{3/2}} dy. \quad (202)$$

Its apparent endpoint singularities cause no problem: the numerator is zero when $v(y) \leq \delta$.

Lemma 41.1 (Truncated renewal with a controlled remainder). *Put $r_a = C_a 4^{-a-1}$. For each admissible Lipschitz f there is a finite constant K_f such that, for every sufficiently large integer H ,*

$$\limsup_{n \rightarrow \infty} \left| Z_n(f) - \sum_{a=0}^H r_a Z_{n-a-1}(f) - \frac{1}{4} Z_{n-1}(S^* f) - v(f) \right| \leq K_f (H + 1)^{-1/2}. \quad (203)$$

41.2 Proof of the truncated renewal

Proof of lemma 41.1: small left blocks, the identity transition. Fix H and use the exact root decomposition. If $0 \leq a \leq H$ and $b = n - a - 1$, proposition 39.1 gives

$$D_n = \min\{b - J_b, a + 1 + D_b\}, \quad L_n = L_b$$

inside the conditional expectation. The distance between this normalized state and $(D_b/b, L_b/b)$ is bounded by an absolute constant times $(a + 1 + J_b)/n$. By lemma 39.2, its expected value is $O_H(n^{-1})$. After multiplying by $\sqrt{n}$, the Lipschitz error tends to zero. Moreover

$$\frac{C_a C_b}{C_n} \sqrt{\frac{n}{b}} \longrightarrow C_a 4^{-a-1} = r_a.$$

Boundedness of $Z_b(f)$ therefore gives the small-left contribution $\sum_{a=0}^H r_a Z_{n-a-1}(f) + o(1)$.

An empty right block: the append transition. For $b = 0$, the exact state differs from $S(D_{n-1}/(n-1), L_{n-1}/(n-1))$ by $O(n^{-1})$. The coefficient $(C_{n-1}/C_n)\sqrt{n/(n-1)}$ tends to $1/4$. Thus this contribution is $\frac{1}{4}Z_{n-1}(S^*f) + o(1)$. If $1 \leq b \leq H$, then $D_n \leq F_b \leq H$ exactly, so its contribution is zero for all sufficiently large n .

Two macroscopic blocks: the source term. Choose a fixed $0 < \eta < \delta/4$ and consider $\eta n \leq a, b \leq (1 - \eta)n$. Set $y = a/n$. The exact recursion and the endpoint lemmas give, uniformly in these sizes,

$$D_n/n - \min(a/n, b/n) \longrightarrow 0 \quad \text{in probability,}$$

and the last coordinate is asymptotically 0 or b/n , each with probability $1/2$. Indeed $J_b/b \rightarrow 0$ and $D_b/b \rightarrow 0$ uniformly for $b \geq \eta n$, while L_b/b has the two-point limit of lemma 39.3. Because the other coordinates have deterministic limiting values, no endpoint independence is required. The uniformity follows directly from convergence of the corresponding scalar sequences as their integer index tends to infinity, and uniform continuity of f on a compact square.

Stirling's formula gives, uniformly on the same interval,

$$\frac{C_a C_b}{C_n} = \frac{1 + o(1)}{4\sqrt{\pi} n^{3/2}} [y(1-y)]^{-3/2}. \quad (204)$$

The replacement of $b/n = 1 - y - 1/n$ by $1 - y$ has vanishing effect. A Riemann sum therefore gives the source integral over $[\eta, 1 - \eta]$, which is exactly (202), since its integrand vanishes outside $[\delta, 1 - \delta]$.

Mesoscopic left blocks: the indispensable error bound. It remains to treat $H < a < \eta n$; the case $b < \eta n$ contributes zero because $D_n \leq b < \delta n$. On the support of f , the root identity forces

$$a + 1 + D_b > \delta n, \quad \text{hence} \quad D_b > \delta n/2$$

for large n . By proposition 40.3, its conditional probability is $O_\delta(n^{-1/2})$. In this range b is a fixed positive fraction of n , so the Catalan ratio also gives

$$\frac{C_a C_b}{C_n} \leq K r_a$$

for an absolute constant K depending at most on η . After multiplication by $\sqrt{n}$, the entire mesoscopic contribution is bounded by

$$K'_{f,\delta} \sum_{a>H} r_a = O_{f,\delta}((H+1)^{-1/2}).$$

Here the summable tail follows either from Stirling or from $c_a \leq (a+1)^{-3/2}$ proved earlier. Combining all cases proves (203). $\square$

Remark 41.2 (Why a formal split heuristic is insufficient). Simply integrating the Catalan root-split density would give the source ν , not the answer μ . Small outer decorations recur, and appending a maximum can erase a rare deficit. Moreover, without the uniform tail envelope, the range $H < a = o(n)$ cannot be discarded after multiplication by $\sqrt{n}$. The proof handles both issues explicitly.

41.3 Solving the renewal equation without assuming convergence

The Catalan identity gives

$$\sum_{a \geq 0} r_a = \frac{1}{4}C(1/4) = \frac{1}{2}. \quad (205)$$

Formally, a limit should satisfy $\mu(f) = \frac{1}{2}\mu(f) + \frac{1}{4}\mu(S^*f) + \nu(f)$. We now prove the limit exists rather than inserting this equation as an assumption.

Proof of theorem 38.2 for Lipschitz tests. First put $g = S^*f$, so that $S^*g = g$, and let $\bar{Z}_g = 4\nu(g)$. All $Z_n(g)$ are bounded. Write $\bar{r}_H = \sum_{a=0}^H r_a$ and $e_g = \limsup_n |Z_n(g) - \bar{Z}_g|$. Applying lemma 41.1 to g and subtracting $\bar{Z}_g$ yields

$$e_g \leq (\bar{r}_H + \frac{1}{4})e_g + K_g(H+1)^{-1/2} + (\frac{1}{2} - \bar{r}_H)|\bar{Z}_g|.$$

Let $H \rightarrow \infty$. By (205), $e_g \leq \frac{3}{4}e_g$. Since e_g is finite, it is zero. Thus

$$Z_n(S^*f) \rightarrow 4\nu(S^*f). \quad (206)$$

Next set $\bar{Z}_f = 2\nu(f) + 2\nu(S^*f)$ and $e_f = \limsup_n |Z_n(f) - \bar{Z}_f|$. The already established convergence of the projected sequence gives

$$e_f \leq \bar{r}_H e_f + K_f(H+1)^{-1/2} + (\frac{1}{2} - \bar{r}_H)|\bar{Z}_f|.$$

Letting $H \rightarrow \infty$ now gives $e_f \leq \frac{1}{2}e_f$, hence $e_f = 0$. Therefore

$$\mu(f) = 2\nu(f) + 2\nu(S^*f). \quad (207)$$

Since f vanishes at deficit zero and $\nu(y) \leq 1 - y$,

$$S^*f(\nu(y), 0) = f(0, 1) = 0, \quad S^*f(\nu(y), 1 - y) = f(\nu(y), 1).$$

Substituting into (202) proves precisely (156). $\square$

41.4 Extension to continuous tests

An admissible continuous f can be approximated uniformly by Lipschitz functions supported away from $x = 0$, using piecewise-linear approximation on a rectangular mesh and a Lipschitz cutoff in the first coordinate. The scaled finite measures have uniformly bounded mass on every strip $x \geq \delta$ by (187), and the limiting integral has finite mass there. Consequently the approximation error in both integrals is bounded by a constant times the uniform approximation error. Letting that error tend to zero proves the full theorem.

41.5 A reusable two-state resolvent

The calculation is an elementary renewal principle. If the total identity weight is w_{id} , the weight of an idempotent endpoint operator S^* is w_S , and $w_{\text{id}} + w_S < 1$, the analogous equation has the solution

$$\mu(f) = \frac{\nu(f)}{1 - w_{\text{id}}} + \frac{w_S}{(1 - w_{\text{id}})(1 - w_{\text{id}} - w_S)} \nu(S^*f). \quad (208)$$

The same bounded-sequence limsup proof establishes convergence whenever an analogue of lemma 41.1 is available. Here $w_{\text{id}} = 1/2$ and $w_S = 1/4$, so both coefficients in (207) equal 2. This algebraic principle is not itself advertised as a new general theorem; the work is in proving the appropriate source and uniform remainder for this statistic.

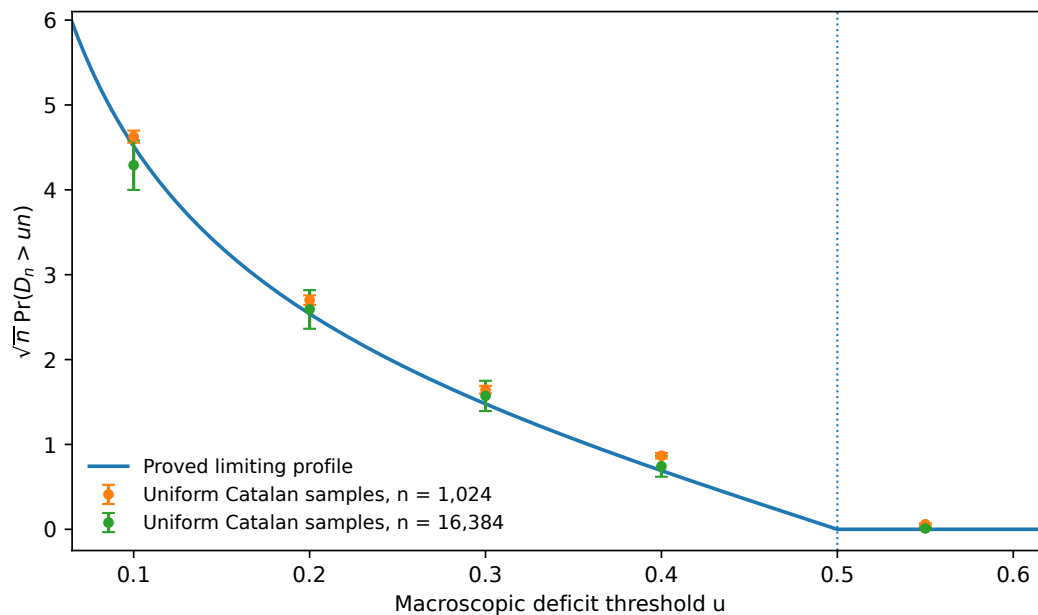


Figure 7: Manuscript 10: the proved macroscopic tail profile Ψ and uniform Catalan sampling illustrations at $n = 1,024$ and $16,384$, from the shipped simulation data of table 16. Error bars show two estimated Monte Carlo standard errors; they do not bound finite-size bias and are not rigorous confidence certificates. In particular, positive finite-size tails beyond $u = 1/2$ are compatible with the limiting zero there.

41.6 Evaluating the profile

For a test depending only on the deficit, all three summands in (156) agree. Folding y and $1 - y$ onto $(0, 1/2)$ gives the density (157). Thus

$$\mu\{x > u\} = \frac{3}{2\sqrt{\pi}} \int_u^{1/2} [x(1-x)]^{-3/2} dx \quad (0 < u < 1/2). \quad (209)$$

The derivative identity

$$\frac{d}{dx} \frac{1-2x}{\sqrt{x(1-x)}} = -\frac{1}{2[x(1-x)]^{3/2}} \quad (210)$$

evaluates the integral as (150). The limiting measure has no mass at any fixed positive deficit, including $1/2$. Continuous upper and lower approximations to the tail indicator therefore prove (151).

For uniformity on $[\delta, 1]$, both the finite tails and Ψ are nonincreasing, and Ψ is continuous. Choose a finite mesh on which its oscillation is small, use pointwise convergence at the mesh points, and sandwich between adjacent points. This proves the uniform assertion in theorem 38.1. Finally (82), continuity of Ψ , and $C_n \sim 4^n / (\sqrt{\pi} n^{3/2})$ yield the moving-bound formulas (153) and (154), including their floor conventions. (The sequence form (152) follows from (151) in the same way, by monotonicity and continuity of Ψ ; manuscript 11 proves it directly.)

42 Second proof of the profile: the canonical threshold decomposition

This section is manuscript 11's proof of theorem 38.1 and of the deficit part of theorem 38.2. It uses the parser of section 40.4 instead of the renewal equation.

42.1 The asymptotic contribution of one outer word

Fix a sequence $d_n/n \rightarrow u \in (0, 1/2)$ and put $m_n = n - d_n$. In this subsection w is one fixed, fully decorated outer word, of fixed cost t . Define its endpoint acceptance weight by

$$\chi(w) = \begin{cases} 1, & w \text{ contains no } A, \\ 1/2, & w \text{ contains at least one } A. \end{cases} \quad (211)$$

Let $\mathcal{E}_n(w)$ be the event that $M_n \leq m_n$ and the canonical word is w . On this event, let a be the left size at the terminal split and define $U_n = a/n$. Write

$$\mathfrak{f}(y) = [y(1-y)]^{-3/2}, \quad I(u) = \int_u^{1-u} \mathfrak{f}(y) dy. \quad (212)$$

Proposition 42.1 (Fixed-word contribution; manuscript 11). *For every continuous function g on $[0, 1]$,*

$$\sqrt{n} \mathbb{E}[\mathbf{1}_{\mathcal{E}_n(w)} g(U_n)] \rightarrow \frac{\chi(w)4^{-t}}{4\sqrt{\pi}} \int_u^{1-u} g(y) \mathfrak{f}(y) dy. \quad (213)$$

The random variable inside the expectation is set to zero off $\mathcal{E}_n(w)$.

Proof of proposition 42.1: the endpoint step. The terminal core has size $N = n - t$ and is $(\sigma + b)N\tau$, with $a + b = N - 1$ and $a, b \leq m_n$. Fix a small $\eta > 0$ such that $[u + \eta, 1 - u - \eta]$ is nonempty, and first consider splits with

$$u + \eta \leq a/n \leq 1 - u - \eta. \quad (214)$$

For large n , both a and b are bounded below by a positive multiple of n , and each lies below m_n by a positive multiple of n .

All internal jumps in the two terminal blocks are at most m_n simply because their sizes are at most m_n . The jump from the left block into N is at most a . The jump from N into the right block equals

$$N - F(\tau) = a + 1 + J(\tau). \quad (215)$$

By lemma 39.2, the fraction of terminal pairs for which $J(\sigma)$ or $J(\tau)$ exceeds a sufficiently small fixed multiple of n is exponentially small, uniformly over (214). On the complementary set, (215) is at most m_n .

The first deficiency of the terminal core is $J(\sigma) + 1$. An append operation increases a current first deficiency by one. An R_β operation either makes the first entry the new maximum, when β is empty, or makes its new first deficiency at most the operation cost, when β is nonempty. Therefore, throughout the fixed outer word, the current first deficiency is at most $J(\sigma) + t + 1$. Each new downward jump at an R operation is at most $J(\sigma) + 2t + 1$. Each finite decoration and its jump into the adjacent maximum have bounded size. These contributions are harmless on the same good set for large n .

If w contains no A , all constraints are now satisfied. Thus the accepted fraction of the $C_a C_b$ terminal pairs is $1 - o(1)$, uniformly on (214).

Suppose instead that w contains an A . Apply the word from the inside out and let s_w be the total cost of the R operations encountered before the first such A . Those operations preserve the terminal last entry $L(\tau)$. The first append creates a jump bounded by m_n exactly when

$$L(\tau) \geq \ell_n(w) := N + s_w + 1 - m_n = d_n - t + s_w + 1. \quad (216)$$

After this append, the last entry is a maximum at least N . Subsequent R operations preserve that value, and every later append creates a jump of size at most t . Thus no further macroscopic last-entry condition is introduced.

In the interior range, $\ell_n(w)/n \rightarrow u$, while both $\ell_n(w)$ and $b - \ell_n(w)$ are bounded below by positive multiples of n . By lemma 39.3, the fraction satisfying (216) is $1/2 + O_{u,\eta,w}(n^{-1/2})$. Removing the exponentially small bad first-deficiency set changes this by $o(1)$. We have not assumed that the first and last entries are independent: deleting any event of probability $o(1)$ changes the last-entry probability by at most $o(1)$.

We have proved that, uniformly over (214), the accepted fraction of terminal pairs is

$$\chi(w) + o(1). \quad (217)$$

By lemma 40.4 these are exactly the pairs counted by $\mathcal{E}_n(w)$, rather than merely a possibly overlapping construction. $\square$

Proof of proposition 42.1: counting and boundary strips. For fixed t and $a + b = n - t - 1$, Stirling's formula gives, uniformly when a/n stays in a compact subset of $(0, 1)$,

$$\frac{C_a C_b}{C_n} = \frac{4^{-(t+1)}}{\sqrt{\pi} n^{3/2}} \{f(a/n) + o(1)\}. \quad (218)$$

The exponent $t + 1$ is important: one unit is spent at the terminal maximum in addition to the t units of outer cost. Combining (217) with (218), summing the interior splits, and using the Riemann-sum mesh $1/n$, gives the asserted limit with integral restricted to $[u + \eta, 1 - u - \eta]$.

All raw terminal splits have

$$d_n - t - 1 \leq a \leq m_n.$$

Thus their scaled range tends to $[u, 1 - u]$. In the two excluded strips, the number of terms is at most $2\eta n + o(n)$; every normalized count is at most $K_{u,w} n^{-3/2}$ because both block sizes remain macroscopic. The contribution of the strips after multiplication by $\sqrt{n}$ is at most $K_{u,w} \|g\|_\infty \eta + o(1)$. Let $\eta \downarrow 0$. This proves (213), including the moving integer boundaries. $\square$

Remark 42.2 (Three distinct uniformities). The argument uses an exponentially uniform first-deficiency bound, a uniform half-mass estimate for the last entry away from both endpoints, and a Catalan asymptotic uniform on compact split intervals. The subsequent sum over words uses the separate rare-scale tail estimate of lemma 40.7. None of these four tasks can be replaced by a pointwise “typical block” statement.

42.2 Summing the endpoint-weighted words

Second proof of theorem 38.1, for $0 < u < 1/2$. Fix a cost cutoff h . There are finitely many decorated words of cost at most h . Apply proposition 42.1 with $g = 1$ and sum over them:

$$\sqrt{n} \mathbb{P}(\mathcal{E}_{n,m_n}, T_{n,m_n} \leq h) \longrightarrow \frac{I(u)}{4\sqrt{\pi}} \sum_{\text{cost}(w) \leq h} \chi(w) 4^{-\text{cost}(w)}.$$

By lemma 40.7, the limsup of the omitted scaled probability is at most $K_u(h + 1)^{-1/2}$. It tends to zero as $h \rightarrow \infty$. The total weighted mass is, by (191)–(192),

$$\begin{aligned} \sum_w \chi(w) 4^{-\text{cost}(w)} &= \mathcal{W}_R(1/4) + \frac{1}{2} \{\mathcal{W}(1/4) - \mathcal{W}_R(1/4)\} \\ &= 2 + \frac{1}{2}(4 - 2) = 3. \end{aligned} \quad (219)$$

Thus the limit is $3I(u)/(4\sqrt{\pi})$. To evaluate $I(u)$, use (210): the expression being differentiated changes sign under $y \mapsto 1 - y$. Consequently,

$$I(u) = \frac{4(1 - 2u)}{\sqrt{u(1 - u)}}, \quad (220)$$

and the resulting limit is exactly (150). $\square$

42.3 The edge at one half and compact uniformity

Completion of the second proof of theorem 38.1. Let $d_n/n \rightarrow u \geq 1/2$. For any fixed $y < 1/2$, eventually $d_n \geq \lceil yn \rceil$. Therefore

$$\limsup_{n \rightarrow \infty} \sqrt{n} \mathbb{P}(D_n \geq d_n) \leq \Psi(y).$$

Letting $y \uparrow 1/2$ gives zero because $\Psi(y) \rightarrow 0$. This proves the pointwise assertion at and above one half.

For compact uniformity, fix $\delta > 0$ and consider the nonincreasing functions $f_n(u) = \sqrt{n} \mathbb{P}(D_n \geq \lceil nu \rceil)$ on $[\delta, 1]$. They converge pointwise to the continuous nonincreasing function Ψ . Given $\varepsilon > 0$, choose a finite partition of $[\delta, 1]$ on which the oscillation of Ψ in each interval is at most ε . Pointwise convergence holds at every partition point, simultaneously for sufficiently large n . Monotonicity sandwiches f_n between its endpoint values on each interval, so the uniform error is at most 3ε . This proves the claim. Substitution $d_n = n - \lfloor \theta n \rfloor$ in (82) gives (153) and the stated smaller-bound estimate. $\square$

42.4 The deficit measure

Second proof of the last assertion of theorem 38.2. For $0 < x_1 < x_2 \leq 1$, the mass of $[x_1, x_2]$ is the difference of the two tail probabilities, with the corresponding integer rounding. Hence theorem 38.1 gives convergence of these interval masses to $\Psi(x_1) - \Psi(x_2)$. On every $[x_1, 1]$, the total masses are uniformly bounded by corollary 40.8. Approximating continuous functions there by interval step functions proves weak convergence on compact subintervals whose endpoints are not atoms of the limit. This is vague convergence on $(0, 1]$.

Differentiation, by (210), gives

$$-\Psi'(x) = \frac{3}{2\sqrt{\pi}} [x(1-x)]^{-3/2} \quad (0 < x < 1/2).$$

There is no atom at $1/2$, because Ψ is continuous there and is zero to its right. These facts identify (158). Its density is asymptotic to $3x^{-3/2}/(2\sqrt{\pi})$ at zero, so its total mass is infinite, while its restriction to $[x_1, 1]$ is finite for every $x_1 > 0$. $\square$

43 The ends of the profile and the conditional geometry

43.1 Matching the boundary, and the edge at one half

As $u \downarrow 0$,

$$\Psi(u) = \frac{3}{\sqrt{\pi}} u^{-1/2} \left(1 - \frac{3}{2}u - \frac{5}{8}u^2 + O(u^3) \right). \quad (221)$$

(Manuscript 10 prints the expansion to $O(u^2)$, manuscript 11 to $O(u^3)$.) The leading term matches Part III's boundary equivalent $\mathbb{P}(D_n \geq d) \sim 3/\sqrt{\pi d}$ when $d \rightarrow \infty$ and $d = o(n)$ (theorem 22.2), and its limiting tail constant $3/\sqrt{\pi}$ ((88)): with $u = d/n$, $\Psi(u)/\sqrt{n} \approx 3/\sqrt{\pi d}$. This is a consistency check between two proven regimes, not a proof of a uniform two-parameter expansion down to arbitrary $u = u_n \rightarrow 0$, and not an illicit substitution of a moving $u = d/n$ into a fixed- u theorem. Theorem 38.1 gives compact *absolute* uniformity away from zero; it does not by itself give relative error estimates when $u \rightarrow 0$.

At the other endpoint, if $u = 1/2 - \varepsilon$ with $\varepsilon \downarrow 0$, then

$$\Psi(u) = \frac{12}{\sqrt{\pi}} \varepsilon + O(\varepsilon^3). \quad (222)$$

The profile vanishes linearly at one half. This signals the need for a smaller probability scale or a separate transition window, not an absence of bounded permutations on the other side:

a shrinking neighbourhood of $1/2$ may interact with a smaller finite-size scale that the first-order theorem does not determine. (Both manuscripts; the two readings are theirs.) [Added 30 September 2026, batch 68: Part VI determines both at the level of orders: $\mathbb{P}(M_n \leq \lfloor n/2 \rfloor) = \Theta(n^{-1})$, and $\mathbb{P}(M_n \leq m) \asymp n^{-1} + (2m - n)_+ n^{-3/2}$ uniformly near $m = n/2$, so the order changes on the scale $\sqrt{n}$ (corollary 68.4); no limiting transition function is proved.] [Added 1 October 2026, batch 71: Part VII proves the limit, $n \mathbb{P}(M_n \leq \lfloor n/2 + t\sqrt{n} \rfloor) \rightarrow 6 + (12/\sqrt{\pi})t_+$ locally uniformly in t (theorem 80.1); its ramp has the slope $12/\sqrt{\pi}$ of (222).]

43.2 Three equally weighted endpoint channels

Fix $0 < u < 1/2$ and condition on $D_n > un$. The split integral $I(u)$ of (220) satisfies

$$I(u) = \int_u^{1-u} [y(1-y)]^{-3/2} dy = \frac{4(1-2u)}{\sqrt{u(1-u)}}, \quad (223)$$

and $\Psi(u) = 3I(u)/(4\sqrt{\pi})$. The following is a sampling description of the limiting conditional law, not a claim that a unique finite tree split has already been identified (manuscript 10; but see theorem 43.2).

Corollary 43.1 (Three equally weighted endpoint channels; manuscript 10). *Let $U^{(u)}$ have density $I(u)^{-1}[y(1-y)]^{-3/2}$ on $(u, 1-u)$, and independently let ξ be uniform on {low, interior, high}. Define*

$$V = \min(U^{(u)}, 1 - U^{(u)}), \quad \zeta = \begin{cases} 0, & \xi = \text{low}, \\ 1 - U^{(u)}, & \xi = \text{interior}, \\ 1, & \xi = \text{high}. \end{cases}$$

Then

$$\mathcal{L}((D_n/n, L_n/n) \mid D_n > un) \implies \mathcal{L}(V, \zeta). \quad (224)$$

The limiting last-entry law has atoms of mass $1/3$ at each of 0 and 1 ; its remaining mass $1/3$ has the displayed interior density. Also $F_n/n \rightarrow 1$ in probability under the same conditioning.

Proof. Restrict theorem 38.2 to $x > u$ and divide by the positive limiting mass $\Psi(u) = 3I(u)/(4\sqrt{\pi})$. The three equal summands yield the sampling description. For the first value, lemma 39.2 makes $\mathbb{P}(J_n > \varepsilon n)$ exponentially small, whereas the conditioning event has probability asymptotic to $\Psi(u)n^{-1/2}$. Their ratio tends to zero. $\square$

43.3 Universal outer word and macroscopic split

The canonical parser of section 40.4 yields more than the scalar tail. Fix $u \in (0, 1/2)$, $d_n/n \rightarrow u$, and $m_n = n - d_n$. Condition on $\mathcal{E}_{n, m_n}$. Let W_n be the canonical outer word and U_n the terminal left size divided by n .

Theorem 43.2 (Universal outer word and macroscopic split; manuscript 11). *Conditionally on $\mathcal{E}_{n, m_n}$, the pair (W_n, U_n) converges in law to an independent pair $(W, U^{(u)})$, where*

$$\mathbb{P}(W = w) = \frac{\chi(w)4^{-\text{cost}(w)}}{3} \quad (225)$$

and $U^{(u)}$ has density

$$f_u(y) = \frac{\mathfrak{f}(y)}{I(u)} \mathbf{1}_{[u, 1-u]}(y). \quad (226)$$

The word space is the countable discrete space of individually decorated finite words. Its limiting law is independent of u .

Proof. For each fixed word, divide (213) by $\mathbb{P}(\mathcal{E}_{n,m_n}) \sim 3I(u)/(4\sqrt{\pi n})$. This gives the product law on that word and every continuous test function of U_n . The normalized version of (196), together with the positive denominator constant, gives tightness of the word costs. There are finitely many words at each bounded cost, so the word variables are tight on the discrete space. The split variable lies in a compact interval. Finite-word convergence plus this tightness proves joint convergence. Equation (219) verifies that (225) is normalized. $\square$

The split law $U^{(u)}$ is the same in corollary 43.1 and theorem 43.2: manuscript 10's split variable (its Y) and manuscript 11's (its U_x) have the same density.

Corollary 43.3 (A heavy-tailed cost but a short word; manuscript 11). *Let $T_W = \text{cost}(W)$. Then*

$$\mathbb{P}(T_W = t) = \frac{C_t + C_{t+1}}{6 \cdot 4^t} \quad (t \geq 0), \quad (227)$$

$$\mathbb{E} z^{T_W} = \frac{C(z/4) + C(z/4)^2}{6}, \quad (228)$$

$$\mathbb{P}(\text{len}(W) = k) = \frac{(1/2)^k + (3/4)^k}{6} \quad (k \geq 0), \quad (229)$$

$$\mathbb{E} \text{len}(W) = \frac{7}{3}, \quad \mathbb{P}(W \text{ has no } A) = \frac{2}{3}. \quad (230)$$

Furthermore,

$$\mathbb{P}(T_W > h) \sim \frac{5}{3\sqrt{\pi h}}, \quad \mathbb{E} T_W = \infty. \quad (231)$$

These expectation statements concern the limiting word law; convergence of finite- n word expectations is not asserted.

Proof. At cost t , there are C_t words without A and $C_{t+1} - C_t$ with at least one A . Sum (225) to obtain (227), and then use (191)–(192) for its probability generating function.

The critical total weight of one R operation is $1/2$, and that of one A is $1/4$. At length k , all words have weight $(3/4)^k$, whereas those without A have weight $(1/2)^k$. The accepted weight is half the sum of these two quantities. Division by three gives (229). The geometric-series identities give

$$\mathbb{E} \text{len}(W) = \frac{1}{6} \left\{ \frac{1/2}{(1 - 1/2)^2} + \frac{3/4}{(1 - 3/4)^2} \right\} = \frac{7}{3}.$$

The no- A probability is $\mathcal{W}_R(1/4)/3 = 2/3$. Finally, (181) and $C_{t+1}/C_t \rightarrow 4$ imply $\mathbb{P}(T_W = t) \sim 5/(6\sqrt{\pi} t^{3/2})$. Summing this regularly varying power tail gives (231) and infinite mean cost. $\square$

The word typically has only a few operations, but one decoration can be large. This explains why summability alone was insufficient for the rare-scale interchange: the limiting cost has no finite first moment. (The limiting word cost T_W is not Part III's skeleton cost T , whose tail is $\sim 4/\sqrt{\pi h}$ by lemma 24.1.)

Three equally weighted endpoint modes (manuscript 11). Call the terminal right block's last-entry mode *high* when $L(\tau) > b/2$ and *low* otherwise. In the interior of the split range, the low and high modes each have unconditioned probability tending to one half. The proof of proposition 42.1 therefore also yields the following table under $\mathcal{E}_{n,m_n}$:

Outer word	Terminal last-entry mode	Limiting probability
No append	Low	1/3
No append	High	1/3
At least one append	High	1/3
At least one append	Low	0

For a fixed no-append word, each mode has mass $4^{-t}/6$ in the conditional law. For a fixed word containing an append, only the high mode survives, also with mass $4^{-t}/6$. Summing over their respective critical word weights, both equal to two, proves the table. Boundary strips are removed exactly as in proposition 42.1. This is another interpretation of the factor three.

Proposition 43.4 (Geometric meaning of the rare-event measure; manuscript 11). *Conditionally on $\mathcal{E}_{n,m_n}$,*

$$\frac{D_n}{n} - \min(U_n, 1 - U_n) \longrightarrow 0 \quad \text{in probability.} \quad (232)$$

Consequently, for $y \geq u$,

$$\mathbb{P}(D_n/n \geq y \mid \mathcal{E}_{n,m_n}) \longrightarrow \frac{\Psi(y)}{\Psi(u)}, \quad (233)$$

with the right side zero for $y \geq 1/2$.

Proof. First restrict to a bounded outer cost and an interior split interval. By corollary 40.8, a uniform block of size a has maximum jump $a - o(n)$ in probability, uniformly when a is proportional to n ; the same is true for the right block of size b . Deleting a set of such terminal pairs of relative probability $o(1)$ remains harmless after imposing the endpoint acceptance event, whose limiting fraction is either one or one half.

The terminal downward jump is $a + 1 + J(\tau) = a + o(n)$ by lemma 39.2, and the upward jump into the terminal maximum is at most a . Every outer R jump is $o(n)$ on the same good endpoint set. If an append occurs, acceptance selects the high last-entry mode, and lemma 39.3 gives $L(\tau) = b - o(n)$ in conditional probability. Its first append jump is therefore $a + o(n)$; later append jumps are bounded by the fixed outer cost. Hence the global maximum jump is $\max(a, b) + o(n)$, and the deficit is $\min(a, b) + o(n)$. Since $(a + b)/n \rightarrow 1$, this proves (232) under the two restrictions. Remove the boundary strips and then the cost cutoff by theorem 43.2 and lemma 40.7.

The map $y \mapsto \min(y, 1 - y)$ sends (226) to density $2\mathfrak{f}(y)/I(u)$ on $[u, 1/2]$. Its upper tail is $\Psi(y)/\Psi(u)$ by (220), proving (233). The same tail statement also follows directly by taking the ratio in theorem 38.1. $\square$

43.4 How the two descriptions of the factor three match

The two manuscripts explain the coefficient 3 by three equally weighted alternatives, and the alternatives correspond. On the terminal split, with $y = a/n$ the left fraction, manuscript 11's modes and manuscript 10's channels match as follows. Without an append, the whole permutation ends with the terminal right block's last entry, since R operations preserve the last entry; the low mode puts L_n/n near 0 (manuscript 10's *low* channel), and the high mode puts L_n/n near $b/n \approx 1 - y$ (its *interior* channel, the mark $1 - U^{(u)}$). With an append, the last entry is a maximum at least $N = n - t$, so $L_n/n \rightarrow 1$ (its *high* channel). The deficit is $\min(y, 1 - y)$ in all three cases (proposition 43.4 and the proof of lemma 41.1). Manuscript 10's corollary does not identify a canonical split; manuscript 11's parser supplies one, and corollary 43.1's interior mark is then its high-mode, no-append terminal block. This correspondence is the merge's reading of the two proofs; neither manuscript states it, and no joint limit theorem for $(W_n, U_n, L_n/n)$ is claimed.

44 Moment constants and size-biased limits

44.1 Uniform integrability at exactly the required threshold

Proof of theorem 38.3 (both manuscripts). For $p > 0$, the tail formula for a nonnegative bounded random variable gives the layer-cake identity

$$n^{1/2-p} \mathbb{E} D_n^p = p \int_0^1 x^{p-1} \sqrt{n} \mathbb{P}(D_n > nx) dx. \quad (234)$$

Using a weak rather than a strict inequality changes only the integer threshold by at most one and has no effect on the pointwise limits. The bound (187) dominates the integrand by $4\sqrt{2} p x^{p-3/2}$ (manuscript 10). Equivalently (manuscript 11), by corollary 40.8 the scaled tail is at most $Kx^{-1/2}$ for $nx \geq 1$, and for $nx < 1$ the trivial bound $\sqrt{n} \leq x^{-1/2}$ gives the same envelope, so the integrand is at most $Kp x^{p-3/2}$. This is integrable at zero exactly when $p > 1/2$. Dominated convergence and theorem 38.1 therefore prove

$$n^{1/2-p} \mathbb{E} D_n^p \longrightarrow p \int_0^{1/2} x^{p-1} \Psi(x) dx. \quad (235)$$

Integration by parts is legitimate, because the boundary term at zero vanishes, $x^p \Psi(x) = O(x^{p-1/2}) \rightarrow 0$, and $\Psi(1/2) = 0$; with (210) it gives (160). The constant is finite and positive. For the alternative formula, substitute

$$x = \frac{z^2}{1+z^2}, \quad 0 < z < 1, \quad dx = \frac{2z}{(1+z^2)^2} dz.$$

The integrand $x^{p-3/2}(1-x)^{-3/2} dx$ becomes $2(z^2/(1+z^2))^{p-1} dz$, proving (161). At $p = 1$ the integral is one, so $m_1 = 3/\sqrt{\pi}$. At $p = 2$,

$$\int_0^1 \frac{z^2}{1+z^2} dz = [z - \arctan z]_0^1 = 1 - \frac{\pi}{4}.$$

Since $(\mathbb{E} D_n)^2 \sim m_1^2 n = o(n^{3/2})$, subtraction from the second moment gives $\text{Var}(D_n) \sim \mathbb{E} D_n^2 \sim m_2 n^{3/2}$, which is (163). The identity $M_n = n - D_n$ gives the mean formula (162) and equality of the variances. $\square$

The convergence proof is not a formal interchange of limits. The finite counting envelope supplies exactly the near-zero domination needed for the moment integral; vague convergence alone would not do so.

44.2 Closed forms

It is convenient to set

$$I(p) = \int_0^1 \left(\frac{z^2}{1+z^2} \right)^{p-1} dz, \quad m_p = \frac{3}{\sqrt{\pi}} I(p). \quad (236)$$

Direct integration gives (manuscript 11; manuscript 10 prints the orders 1, 2, 3):

p	$I(p)$	m_p
1	1	$3/\sqrt{\pi}$
3/2	$\sqrt{2} - 1$	$\frac{3}{\sqrt{\pi}}(\sqrt{2} - 1)$
2	$1 - \pi/4$	$\frac{3}{\sqrt{\pi}}(1 - \frac{\pi}{4})$
3	$5/4 - 3\pi/8$	$\frac{3}{\sqrt{\pi}}(\frac{5}{4} - \frac{3\pi}{8})$
4	$3/2 - 15\pi/32$	$\frac{3}{\sqrt{\pi}}(\frac{3}{2} - \frac{15\pi}{32})$

For example,

$$m_1 = 1.6925687506\dots, \quad m_2 = 0.3632283625\dots$$

For $p = 2$ the integral is $\int_0^1 t^2/(1+t^2) dt = 1 - \pi/4$; for $p = 3$, expand $t^4/(1+t^2)^2 = 1 - 2/(1+t^2) + 1/(1+t^2)^2$. For any positive integer p , partial fractions in (161) reduce the constant to rational numbers and π , followed by the common factor $3/\sqrt{\pi}$ (manuscript 10). For every $p > 1/2$ with $p \neq 1$, integration of

$$\frac{d}{dz} \frac{z^{2p-1}}{(1+z^2)^{p-1}} = (2p-1) \left(\frac{z^2}{1+z^2} \right)^{p-1} - 2(p-1) \left(\frac{z^2}{1+z^2} \right)^p$$

over $[0, 1]$ yields the recurrence (manuscript 11)

$$I(p+1) = \frac{(2p-1)I(p) - 2^{1-p}}{2(p-1)}. \quad (237)$$

At $p = 1$ this identity degenerates; one must not divide by zero or use it to obtain $I(2)$ without an additional calculation.

The integral in (160) can also be written $\frac{3}{2\sqrt{\pi}} B_{1/2}(p-1/2, -1/2)$, an incomplete beta function. The second parameter is negative, but the defining integral stops at $1/2$, not at its singular endpoint 1, so it is convergent for every $p > 1/2$ (manuscript 10).

44.3 The two limits of the constant function

Proposition 44.1 (Two limits of the constant function; manuscript 11, the first also manuscript 10). *As $p \downarrow 1/2$ and as $p \rightarrow \infty$, respectively,*

$$m_p \sim \frac{3}{\sqrt{\pi}(2p-1)} = \frac{3}{2\sqrt{\pi}} \frac{1}{p-1/2}, \quad (238)$$

$$m_p \sim \frac{6}{\sqrt{\pi} p} 2^{-p}. \quad (239)$$

These are statements about the constants after taking the fixed- p limit in n , not joint limits with $p = p(n)$.

Proof. Near zero the integrand of $I(p)$ is $z^{2p-2}(1+z^2)^{1-p}$. For p in a fixed right neighbourhood of $1/2$, subtracting z^{2p-2} leaves an integrable function with uniformly bounded integral: the factor difference is $O(z^2)$. Hence $I(p) = (2p-1)^{-1} + O(1)$, proving (238). (Manuscript 10's argument is the same: the only divergent contribution is the integral of t^{2p-2} near zero, and the remaining factor tends regularly to one there.)

For large p , write $\varphi(z) = \log(z^2/(1+z^2))$. It is strictly increasing on $(0, 1]$, with $\varphi(1) = -\log 2$ and $\varphi'(1) = 1$. For fixed $\eta > 0$, the integral over $[0, 1-\eta]$ is exponentially smaller than $2^{-(p-1)}$. On $[1-\eta, 1]$, substitute $v = (p-1)(1-z)$ and use $\varphi(1-v/(p-1)) = \varphi(1) - v/(p-1) + O(v^2/(p-1)^2)$. A one-sided derivative bound near one supplies an integrable exponential majorant. Dominated convergence yields $I(p) \sim 2^{-(p-1)}/(p-1)$, which is (239). $\square$

The pole in (238) is consistent with the logarithmic critical moment (92) of Part III. It does not prove convergence of its centered $O(1)$ remainder. The proof of theorem 38.3 deliberately stops at $p > 1/2$: its integrable envelope fails at the critical order, dominated convergence fails at $p = 1/2$, and vague convergence alone cannot repair that failure. That question remains separate (section 47.6). [Added 30 September 2026, batch 64: Part V answers it (theorem 53.1) and sharpens (238) to $m_p = \frac{3/(2\sqrt{\pi})}{p-1/2} + B_1 + O(p-1/2)$, with the finite part $B_1 \approx 0.382499$ of theorem 53.3 (lemma 55.2 and remark 55.3).]

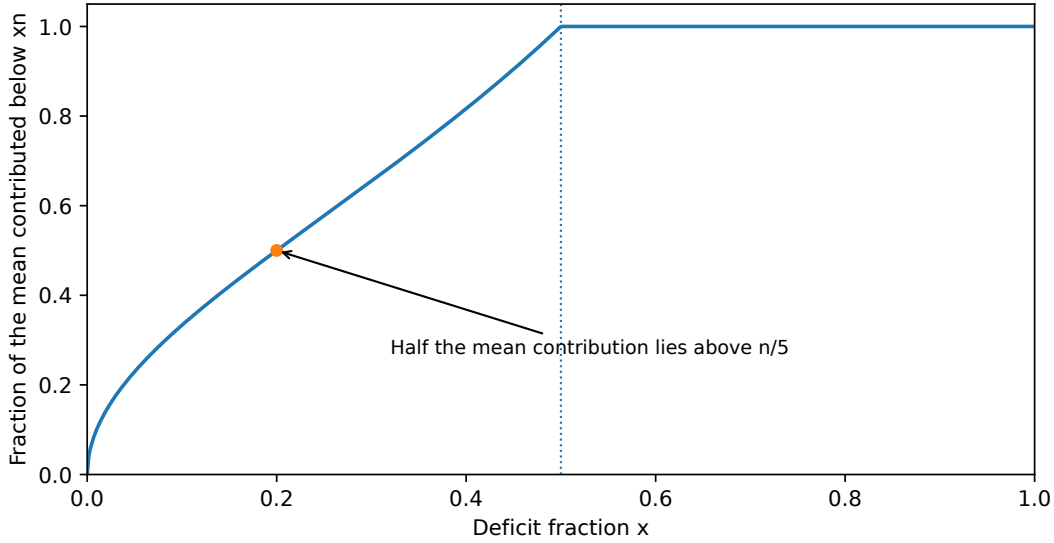


Figure 8: Manuscript 10: the limiting fraction $H^*(x)$ of the mean deficit contributed by permutations with $D_n \leq xn$. This is a size-biased distribution, not the ordinary distribution of D_n/n , which converges to a point mass at zero.

44.4 General power bias and the uniform transformation

For $p > 1/2$, define (manuscript 10)

$$\mathbb{P}_{n,p}^*(B) = \frac{\mathbb{E}[D_n^p \mathbf{1}_B]}{\mathbb{E}D_n^p}.$$

The same domination argument, first away from zero and then with the cutoff removed, proves

$$\mathcal{L}_{\mathbb{P}_{n,p}^*}(D_n/n) \Rightarrow \frac{x^p \rho(x)}{m_p} dx, \quad 0 < x < 1/2. \quad (240)$$

To be explicit about the cutoff removal, (187) and layer cake bound the contribution from $D_n/n \leq \varepsilon$ by a constant times $\varepsilon^{p-1/2}$, uniformly in n .

Proof of theorem 38.4. For $p = 1$, the limiting density simplifies to

$$\frac{1}{2}x^{-1/2}(1-x)^{-3/2} = \frac{d}{dx} \sqrt{\frac{x}{1-x}}.$$

Its integral over $(0, 1/2)$ is one. Thus its distribution function is (166); applying the increasing change of variable $t = \sqrt{x/(1-x)}$ proves (165). At $x = 1/5$, the value is $1/2$, giving (167). $\square$

45 Constructive lower bounds beyond the half-size cutoff

The vanishing first-order profile below $M_n/n = 1/2$ should not be mistaken for exponential suppression. A simple explicit family gives polynomial lower bounds in every fixed proportional-bound regime (manuscript 10).

Theorem 45.1 (A many-block lower bound; manuscript 10). *Fix $0 < \theta < 1$ and an integer $r \geq 1$ such that $(r+1)\theta > 1$. There is a constant $\underline{c}_{\theta,r} > 0$ such that, for all sufficiently large n ,*

$$\mathbb{P}(M_n \leq \lfloor \theta n \rfloor) \geq \underline{c}_{\theta,r} n^{-r/2}. \quad (241)$$

Equivalently, $a_n^{(\lfloor \theta n \rfloor)} \geq \underline{c}'_{\theta,r} 4^n n^{-(r+3)/2}$ for a positive constant $\underline{c}'_{\theta,r}$.

Proof. Put $\theta_r = 1/(r+1) < \theta$. Choose $\eta > 0$ sufficiently small that $\theta_r - r\eta > 0$ and $\theta_r + r\eta < \theta$. Choose r integers $a_1, \dots, a_r$ in intervals $|a_i/n - \theta_r| < \eta$, and put

$$b = n - r - \sum_{i=1}^r a_i.$$

After slightly shrinking the intervals if necessary, there are at least cn^r choices, every component size is bounded between two positive multiples of n , and all a_i, b are at most $(\theta - \varepsilon)n$ for a fixed $\varepsilon > 0$.

Take arbitrary left decorations $\beta_i \in \text{Av}_{a_i}(132)$ and a terminal core $\sigma \in \text{Av}_b(132)$. Form

$$\pi = R_{\beta_1} R_{\beta_2} \cdots R_{\beta_r}(\sigma).$$

The maximum decomposition along the right child recovers this fixed-length chain and every decoration uniquely. Restrict each component to have first deficiency at most $\varepsilon n/3$. By lemma 39.2, the excluded proportion in each component is exponentially small in n .

For an inner right subtree, (174) says that its first deficiency is either the deficiency of its left decoration plus one, or, at the final step, the deficiency of the core. The downward jump introduced by R_{β_i} is therefore at most $a_i + 1 + \varepsilon n/3 + 1 < \theta n - 1$ for large n . By (170), no internal jump in a left decoration can exceed that boundary jump. The core itself has maximum jump at most $b - 1 < \theta n$. Thus all these good reconstructions obey the required integer bound.

For each length vector, their number is $(1 - o(1))C_{a_1} \cdots C_{a_r} C_b$, uniformly on the chosen compact size intervals. Stirling's formula gives

$$\frac{C_{a_1} \cdots C_{a_r} C_b}{C_n} \geq c'' n^{-3r/2},$$

the fixed exponential factor is 4^{-r} because the component sizes sum to $n - r$. Summing over at least cn^r disjoint length vectors proves (241). $\square$

For $1/3 < \theta \leq 1/2$, this gives a lower bound of order n^{-1} , while theorem 38.1 gives $o(n^{-1/2})$. The exact order and profile between those bounds are not proved here. For nonreciprocal θ , the smallest admissible r is $\lceil 1/\theta \rceil - 1$; at a reciprocal endpoint the strict condition requires a larger r . (For $\theta > 1/2$, $r = 1$ gives the order $n^{-1/2}$ of (153); the merge's remark.) [Added 30 September 2026, batch 68: Part VI proves matching upper bounds. The smallest admissible r is $\lfloor 1/\theta \rfloor$ in every case, and with it (241) gives the exact polynomial order (theorem 68.1); for $1/3 < \theta \leq 1/2$ that order is n^{-1} . The profile, that is an equivalent, is still not proved.] [Added 1 October 2026, batch 71: for $1/3 < \theta \leq 1/2$ Part VII proves the equivalent $I_3(\theta)/(2\pi n)$ (theorem 80.2); for $\theta \leq 1/3$ it is still not proved.] [Added 1 October 2026, batch 72: for every fixed $\theta \leq 1/3$ Part VIII proves it (theorem 91.1).]

46 Exact checks, exact data and sampling

46.1 Exact finite checks of manuscript 10

The standard-library program code/04-macroscopic-jump-verify.py exhaustively checks all 82,499 avoiding permutations of sizes 1 through 11 generated by the Catalan decomposition. For each it compares the direct adjacent maximum with the exact root recurrence of proposition 39.1. It checks both endpoint counting formulas and the finite tail envelope at every threshold in those sizes. For sizes 1 through 8, a separate three-index test applied to all ordinary permutations

independently verifies the generated avoidance class. It additionally checks the exact stopping-cost tail (184) for $h = 0, \dots, 100$, and verifies the sampler's rotation multiplicities and full statistic histograms on all 2,352 degree words of sizes 1 through 6. The outcome is recorded in `data/04-macroscopic-jump-verification.json`; the exact finite deficit histograms for $n = 1, \dots, 11$ are retained as integer counts in `data/04-macroscopic-jump-exact_small_distributions.csv`. These tests are finite regression evidence, not a substitute for any asymptotic proof.

46.2 Exact finite and symbolic checks of manuscript 11

Manuscript 11's package contains a standard-library integer implementation of the endpoint recurrence, a separate exhaustive generator, a canonical parser and reconstructor, and exact-count data. The verification (`code/05-macroscopic-deficits-verify.py`) was executed with the following scope:

Check	Executed cases
Avoiders examined, sizes 2 through 9	6,916
Total counts versus exhaustive enumeration, every $1 \leq m < n$	36
Cumulative endpoint states versus enumeration, sizes through 7	644
Canonical parsing and exact reconstruction checks	18,738
Positive-cost coefficients of both word identities	60

The first- and last-entry formulas were also checked against the same exhaustive classes. The report is retained in `data/05-macroscopic-deficits-verification.json`. These are finite checks of the combinatorial objects and implementation. They do not establish the universal analytic limits; the written proofs do that work.

A separate SymPy script (`code/05-macroscopic-deficits-verify_symbolic.py`) checked eleven groups of exact symbolic identities, including the derivative of Ψ , the split-integral primitive, the moment substitution and recurrence, the displayed moment integrals, and the word generating functions. Its executed report is retained in `data/05-macroscopic-deficits-symbolic_verification.json`; these checks do not verify limit interchanges or analytic hypotheses.

46.3 The exact recurrence used for manuscript 11's data

Manuscript 11 computes $a_n^{(m)}$ exactly with Part I's endpoint counts $T_{u,v}^{(m)}(n)$ (proposition 2.1): the number of m -bounded avoiders of length n with $n - \pi_1 \leq u$ and $n - \pi_n \leq v$. A negative threshold gives zero, $T_{u,v}^{(m)}(1) = 1$ for nonnegative thresholds, and thresholds above $n - 1$ can be capped at $n - 1$. Let $\mathcal{N}_s(u) = \#\{\pi \in \text{Av}_s(132) : s - \pi_1 \leq u\}$; its values follow by summing (175), and $\mathcal{N}_0(u) = 1$ for $u \geq 0$ and zero for $u < 0$. The known endpoint recurrence of Mayama–Akita is

$$T_{u,v}^{(m)}(n) = \sum_{k=1}^{\min(m,n-1)} c_{k,u} T_{m-k,v-k}^{(m)}(n-k) + T_{u-1,m-1}^{(m)}(n-1), \quad (242)$$

where $c_{1,u} = 1$ and $c_{k,u} = \mathcal{N}_{k-1}(u-1)$ for $k \geq 2$: this is the coefficient form of Part I's (8), and $c_{k,u}$ is Part I's (5). Manuscript 11 rederives it from the maximum bijection, as Part I does, to make its computational certificates self-contained; the derivation is not repeated. When $n \leq m + 1$, the adjacency constraint is vacuous, and the last-entry formula gives the useful exact shortcut (manuscript 11)

$$T_{u,v}^{(m)}(n) = \mathcal{N}_{n-1}(u-1) + \sum_{q=1}^{\min(v,n-1)} C_{n-q-1} \mathcal{N}_q(u) \quad (n \geq 2, n \leq m+1). \quad (243)$$

For (243), consider a final value $\ell = n - q$. When $q \geq 1$, the last-entry decomposition has a high initial block of size q , which contains the first entry, and a lower block of size $n - q - 1$. The whole first deficiency is exactly the first deficiency in the high block. The count is therefore $C_{n-q-1}N_q(u)$, while the last threshold is equivalent to $q \leq v$. The separate case $q = 0$ appends the maximum and contributes $N_{n-1}(u - 1)$.

Memoization of (242), using (243) whenever possible, produces the retained counts. The implementation caps vacuous thresholds and memoizes smaller-size states. It uses Python arbitrary-precision integers throughout. It is intended as a transparent verifier and moderate-size data generator, not as a claim of a new optimal counting algorithm. Every count and every Catalan number in the CSV files is an exact integer. Floating point is used only for the displayed ratios and figures.

46.4 The bulk profile is approached slowly

Table 15: Manuscript 11: exact normalized counts $\sqrt{n} a_n^{(n-d)}/C_n$ and their proved limits $\Psi(d/n)$. Here d/n is exactly the displayed fraction. A dash means that case was not computed.

n	$d/n = 1/5$	$d/n = 3/10$	$d/n = 2/5$
40	2.794323	1.960484	1.291077
80	2.823905	1.897805	1.178743
160	2.806836	1.821163	1.071493
320	—	1.746102	—
Limit	2.538853	1.477395	0.690988

The data in table 15 do not show rapid convergence, and the first column is not even monotone at these sizes. No monotonicity or finite-size error rate is claimed. This makes it especially important that the factor three was obtained from the endpoint calculation, not fitted to numerical values. The full integer certificates for all ten cases are in `data/05-macroscopic-deficits-bulk_counts.csv`.

46.5 Moment data without a fitted constant

For integer-valued positive D_n ,

$$\mathbb{E}D_n = \sum_{d=1}^{n-1} \mathbb{P}(D_n \geq d),$$

$$\mathbb{E}D_n^2 = \sum_{d=1}^{n-1} (2d - 1) \mathbb{P}(D_n \geq d).$$

These identities allow the moments to be computed from exact bounded counts at every $m = 1, \dots, n - 1$. The retained results (manuscript 11) are:

n	$\mathbb{E}D_n/\sqrt{n}$	$\mathbb{E}D_n^2/n^{3/2}$
20	1.229950	0.498514
40	1.350574	0.495311
80	1.443085	0.479054
Limit	1.692569	0.363228

The second column of moments is a *second moment*, not a variance. Its leading constant equals that of the variance only asymptotically, because the squared mean is of smaller order. The complete

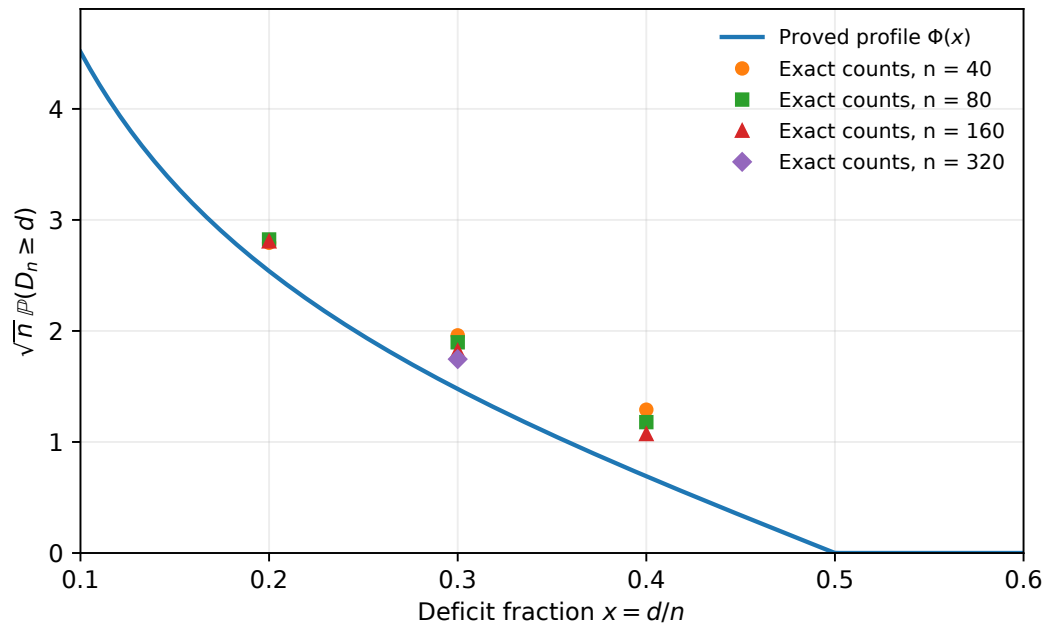


Figure 9: Manuscript 11: the proved limiting profile (labelled $\Phi(x)$ in the figure, $\Psi(u)$ here) and exact finite-size values $\sqrt{n} \mathbb{P}(D_n \geq d)$ against $x = d/n$. The visible separation is real finite-size behaviour, not a confidence interval or an error estimate for the theorem. Values at or beyond $1/2$ are zero only in the limiting profile.

rational values are retained as numerator and denominator integers in `data/05-macroscopic-deficits-moment_counts.csv`.

46.6 Uniform Catalan sampling without constructing permutations

Manuscript 10's C++ sampler (`code/04-macroscopic-jump-simulate.cpp`) uses a binary-tree preorder word with n symbols $+1$ (internal nodes) and $n + 1$ symbols -1 (leaves). A uniform shuffle of these symbols is rotated immediately after the first occurrence of its global minimum partial sum. The resulting word has nonnegative partial sums before its final -1 and encodes a full ordered binary tree.

The elementary cycle lemma says each tree word has exactly $2n + 1$ preimages under this rotation map. One can see uniqueness of the valid rotation from the integer partial sums and total sum -1 . A nontrivial period is impossible: it would make -1 an integer multiple of the sum of one shorter period. Thus all $2n + 1$ rotations are distinct. Uniformity of the shuffle, followed by the constant preimage count, proves uniformity of the Catalan tree. The maximum decomposition is a bijection between these trees and 132-avoiders.

The tree is then evaluated bottom-up using [proposition 39.1](#); no length- n permutation array is required. Each subtree stores only its size, first entry, last entry, and largest jump. One sample uses $O(n)$ time and $O(n)$ working memory. The implementation uses a fixed pseudorandom seed, so its retained output is reproducible with the stated implementation. Different standard-library implementations of `std::shuffle` may produce different sample streams even with the same engine seed.

46.7 What the numerics do and do not say

The retained simulation has 675,000 pseudorandom samples across five sizes. It illustrates approach to the profile and moment constants, but finite-size bias is visible, especially in the

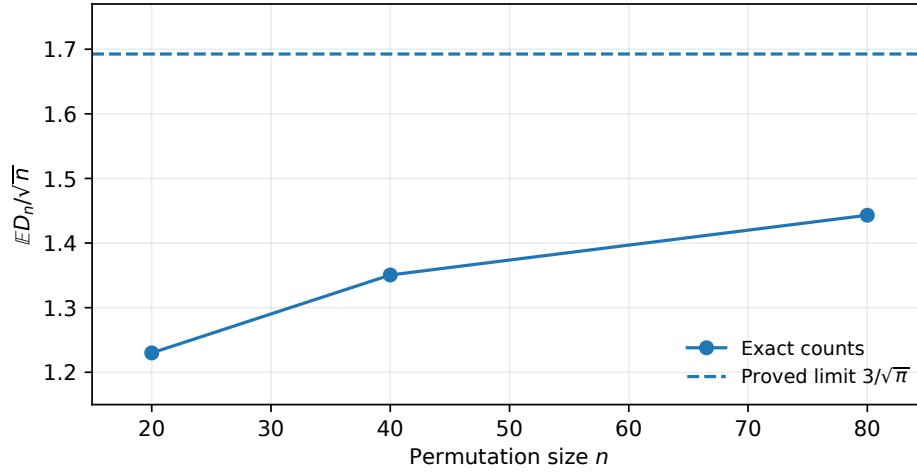


Figure 10: Manuscript 11: exact normalized mean deficits at three finite sizes. The dashed line is the analytically proved constant $3/\sqrt{\pi}$; it was not estimated from these three data points.

Table 16: Manuscript 10: numerical illustrations of the moment limits, from its shipped simulation data (section 46.8). Parenthesized values are one estimated Monte Carlo standard error for the mean column, not a bound on finite-size bias. The second-moment column is a sample estimate and is not presented as a certified interval.

n	Samples	$\widehat{\mathbb{E}D_n}/\sqrt{n}$	$\widehat{\mathbb{E}D_n^2}/n^{3/2}$
64	300,000	1.41333 (0.00250)	0.48329
256	200,000	1.55368 (0.00486)	0.44632
1,024	100,000	1.62981 (0.01027)	0.41261
4,096	50,000	1.69621 (0.02139)	0.40232
16,384	25,000	1.66394 (0.04260)	0.37599
Proved limit		$3/\sqrt{\pi} \approx 1.69257$	$3(4 - \pi)/(4\sqrt{\pi}) \approx 0.36323$

second moment and near the cutoff. The theorems are established by the proofs, not by a fit to table 16 or table 15. The data are neither outward-rounded interval certificates nor a claimed convergence-rate estimate.

46.8 Rebuilding and rerunning

Part IV's files are

code/04-macroscopic-jump-verify.py	Standard-library exact finite checks
code/04-macroscopic-jump-simulate.cpp	C++17 uniform Catalan-tree sampler
code/04-macroscopic-jump-plot_results.py	Its two figures (NumPy, Matplotlib)
code/04-macroscopic-jump-build.sh	Manuscript 10's three-pass build
data/04-macroscopic-jump-*	Recorded check, exact histograms n = 1..11, 675,000 samples, delivery build audit
04-macroscopic-jump-SOURCES.md	Manuscript 10's provenance ledger
code/05-macroscopic-deficits-model.py	Exact endpoint counts, generator
code/05-macroscopic-deficits-verify.py	Exhaustive combinatorial checks
code/05-macroscopic-deficits-verify_symbolic.py	SymPy identity checks
code/05-macroscopic-deficits-numeric.py	Exact bulk and moment counts
code/05-macroscopic-deficits-plot_results.py	Its two figures
code/05-macroscopic-deficits-build.sh	Manuscript 11's latexmk build
data/05-macroscopic-deficits-*	Recorded checks, exact counts, environment, package audit
05-macroscopic-deficits-SOURCES.md	Manuscript 11's source ledger
figures/04-macroscopic-jump-*, figures/05-macroscopic-deficits-*	The four figures of this part

The scripts are shipped byte-identical to the deliveries and still use the delivered names: they read and write data/verification.json, data/exact_small_distributions.csv, data/simulation.csv, data/symbolic_verification.json, data/bulk_counts.csv, data/moment_counts.csv and unprefix files in figures/. Manuscript 11's verifier and data generator execute from model import; run in place, that import would load Part I's different code/model.py and fail. The two build scripts change to their own directory and look for an article.tex there, so they fail in code/. The report's README gives a recipe that restores each delivered layout in a scratch copy; there, the manuscripts' instructions apply, and rerunning overwrites only the copy's recorded outputs. The exact verifications need Python 3.10 or later and no third-party package:

```
python code/verify.py          # either package
python code/numeric.py        # manuscript 11; exact counts
```

Do not use Python's -O option, which disables the verifiers' assertions. Manuscript 11's symbolic checks need SymPy, and both plot scripts need NumPy and Matplotlib:

```
python code/verify_symbolic.py # manuscript 11
python code/plot_results.py    # either package
g++ -O3 -std=c++17 code/simulate.cpp -o simulate # manuscript 10
./simulate > data/simulation.csv
```

Regenerating samples can change the numerical table unless the table is updated too; the article and data correspond to the retained run. The retained figures make Python unnecessary for rebuilding the article. The included count files record exact integers; rounded display columns and plots are not interval-arithmetic certificates.

47 Further research questions

The proved results remove the unknown leading constants from all supercritical moments and identify a positive part of the linear-bound regime. Several neighbouring problems remain substantial. The questions below separate statements proved above from plausible next steps. Manuscript 10 proposes ten directions and manuscript 11 ten questions; where both ask the same question it is printed once with both texts, so thirteen remain.

47.1 The smaller-bound regime and the hierarchy below half size

Manuscript 11. Determine an asymptotic equivalent for $a_n^{(\lfloor \theta n \rfloor)} / C_n$ for each fixed $0 < \theta < 1/2$.
Manuscript 10. For a nonreciprocal $0 < \theta < 1$, set $r = \lceil 1/\theta \rceil - 1$. Is there a positive function $\Omega_r(\theta)$ such that

$$\mathbb{P}(M_n \leq \theta n) \sim \Omega_r(\theta) n^{-r/2}?$$

This is a proposed conjectural hierarchy, not a theorem. The case $r = 1$ is now proved (theorem 38.1), and theorem 45.1 supplies the lower bound in every open interval. A proof of a matching upper bound would need control of repeated macroscopic splits and their endpoint marks. Manuscript 11 proves only $o(n^{-1/2})$ there, and adds: a single split into two blocks below the threshold is no longer available near the top. Several macroscopic pieces must be created before the threshold parser can stop. Thresholds at reciprocal integers $\theta = 1/k$ are natural candidates for changes of behaviour, but no hierarchy of exponents or constants is established by manuscript 11. An analysis must retain the last-entry constraints through all large pieces rather than multiply independent split probabilities. [Added 30 September 2026, batch 68: the exponent part is answered in Part VI: $\mathbb{P}(M_n \leq \lfloor \theta n \rfloor) = \Theta(n^{-\lceil 1/\theta \rceil/2})$ for every fixed $0 < \theta < 1$, so for nonreciprocal θ the exponent is $r/2$ with r as above, while at $\theta = 1/q$ it is $q/2$ (theorem 68.1). The existence and value of $\Omega_r(\theta)$, and an equivalent for $0 < \theta < 1/2$, stay open (section 74.1).] [Added 1 October 2026, batch 71: for $r = 2$ the amplitude exists: $\Omega_2(\theta) = I_3(\theta)/(2\pi)$ for $1/3 < \theta < 1/2$, and manuscript 11's equivalent holds there (theorem 80.2). For $r \geq 3$, that is $0 < \theta < 1/3$, both stay open (section 89.1).] [Added 1 October 2026, batch 72: answered in Part VIII. For every r the amplitude exists, $\Omega_r(\theta) = \kappa_{r+1} I_{r+1}(\theta)$ with the capped-simplex integral (377), and manuscript 11's equivalent holds for every fixed $0 < \theta \leq 1/2$, at $\theta = 1/q$ with the constant $\mathfrak{R}_q$ (theorems 91.1 and 91.2); the case $r = 1$ is theorem 38.1.]

47.2 Reciprocal thresholds and the critical window at $m = n/2$

Manuscript 10. Determine the order of $\mathbb{P}(M_n \leq n/2)$ and the transition when $M_n \leq n/2 + s_n$.
Manuscript 11. Find the probability scale and the transition function when $m = n/2 + s_n$ and $s_n = o(n)$. The linear vanishing in (222) is a constraint on any crossover formula, but does not determine the size of s_n , the width of the window, or the probability exactly at its centre. More generally, do the $\theta = 1/r$ mark distinct polynomial or logarithmic transitions? The discarded terminal boundary strips, negligible at a fixed bulk fraction, may become part of the main term here. Rounding effects, parity and the terminal maximum's unit cost must be retained at a lattice scale rather than replaced by a continuum approximation too early. [Added 30 September 2026, batch 68: answered at the level of orders in Part VI. $\mathbb{P}(M_n \leq \lfloor n/2 \rfloor) = \Theta(n^{-1})$, and uniformly near $m/n = 1/2$, $\mathbb{P}(M_n \leq m) \asymp n^{-1} + (2m - n)_+ n^{-3/2}$, so the order changes on the scale $s_n \asymp \sqrt{n}$ (corollary 68.4). Every $\theta = 1/q$ marks a distinct polynomial transition: the exponent jumps from $(q-1)/2$ to $q/2$, the endpoint belonging to the rarer phase, across the scale $n^{1-1/(2(q-1))}$ (theorem 68.1 and corollary 68.3). The limiting transition function stays open (section 74.3).] [Added 1 October 2026, batch 71: at $m = n/2$ the transition function is proved in Part VII: for $s_n = t\sqrt{n}$, $n \mathbb{P}(M_n \leq \lfloor n/2 + t\sqrt{n} \rfloor) \rightarrow 6 + (12/\sqrt{\pi})t_+$, and $\mathbb{P}(M_n \leq \lfloor n/2 \rfloor) \sim 6/n$ (theorem 80.1). At $1/q$ for $q \geq 3$ it stays open (section 89.2).] [Added 1 October 2026, batch 72: at every $1/q$ it is proved in Part VIII: $n^{q/2} \mathbb{P}(M_n \leq \lfloor n/q + t w_q(n) \rfloor) \rightarrow \mathfrak{R}_q + \mathfrak{L}_q t_+^{q-1}$ (theorem 91.2), so every $\theta = 1/q$ marks a distinct transition with an explicit limiting function.]

47.3 A uniform relative boundary-to-bulk approximation

Manuscript 11. Find a formula with relative error tending to zero in a range simultaneously covering $d \rightarrow \infty$, $d = o(n)$, and d/n bounded inside $(0, 1/2)$. A useful goal is an explicit error bound that exposes dependence on both d and $n - 2d$. The proof here truncates the word cost before taking the limit. A quantitative version would keep the slowly decaying word tail, the endpoint acceptance correction, and the two split-boundary strips in one estimate. Compact absolute uniformity is not enough when the main profile itself tends to zero or infinity.

47.4 A local macroscopic limit

Both manuscripts. For $d_n/n \rightarrow x \in (0, 1/2)$ (manuscript 11), or uniformly on compact subsets of $(0, 1/2)$ with $d_n = \lfloor xn \rfloor$ (manuscript 10), is it true that

$$n^{3/2} \mathbb{P}(D_n = d_n) \longrightarrow \frac{3}{2\sqrt{\pi}} [x(1-x)]^{-3/2}?$$

This is consistent with theorem 38.2, but it is not a consequence of it: the vague limit controls intervals, not individual lattice masses, and differencing two tail estimates whose errors are merely $o(n^{-1/2})$ does not control a probability of order $n^{-3/2}$. A proof requires a local count or substantially sharper error control. A local theorem would substantially strengthen coefficient asymptotics and numerical certification.

47.5 The exact total-variation constant

Both manuscripts. Part III proves a sharp order but leaves the constant in $d_{\text{TV}}(\mathcal{L}(D_n), \mathcal{L}(D))$ open (its Section 32.3). Determine whether $\sqrt{n} d_{\text{TV}}(\mathcal{L}(D_n), \mathcal{L}(D))$ converges, and compute the limit if it exists. The limiting boundary mass beyond $n/2$ suggests a possible contribution $3\sqrt{2}/\sqrt{\pi}$, but neither manuscript asserts that this is the full constant (manuscript 10). The bulk measure identifies one source of mass at large deficits. Total variation also depends on signed finite-size corrections at small and intermediate deficits, so the answer cannot be obtained just by integrating the unsigned bulk density; the distinction between a weak rare-event measure and a signed correction measure is essential (manuscript 11). Establishing it requires control of the signs and aggregate size of discrepancies inside the finite range, which the present weak rare-event limit does not provide.

47.6 Centered critical and subcritical corrections

Manuscript 10. Does

$$\mathbb{E}\sqrt{D_n} - \frac{3}{2\sqrt{\pi}} \log n$$

converge, and what is its limit? Can one identify the leading correction to $\mathbb{E}D_n^p$ for $0 < p < 1/2$? These questions require a quantitative matching of the fixed-deficit law and the macroscopic measure near the singular boundary $x = 0$. (The first is the critical part of Part III's Section 32.2, which Part IV leaves open.) [Added 30 September 2026, batch 64: both questions are answered in Part V, from Part III's uniform total-variation bound and Part IV's pointwise profile alone. The limit exists and equals an explicit constant $c_\star \approx -0.942121414$ (theorem 53.1), and $\mathbb{E}D_n^p = \mathbb{E}D^p + m(p)n^{p-1/2} + o(n^{p-1/2})$ for $0 < p < 1/2$, with an explicit negative coefficient that continues m_p through its pole (theorem 53.2 and remark 55.3). No finer matching near $x = 0$ was needed. A rate for the centered half-moment remains open (section 63).]

47.7 Second-order moment asymptotics

Both manuscripts. Determine the next term after $3\sqrt{n/\pi}$ in $\mathbb{E}D_n$, the next term in the second moment, and the next terms in $\mathbb{E}D_n^p$ for fixed $p > 1/2$. The numerical tables show significant

finite-size corrections, and the finite data do not establish whether the next mean term is constant, logarithmic, or of another order. The next contribution can mix a boundary correction with a bulk correction. An expansion for a fixed positive deficit fraction alone need not be integrable down to zero. Matched subtraction of the small- u singular terms of (221) is a possible route, but must come with a uniform remainder. A correct expansion must separate finite outer decorations, mesoscopic tails, and repeated macroscopic split events, rather than extrapolating a few simulated sizes.

47.8 A geometric conditional limit for the whole permutation

Manuscript 10. Corollary 43.1 describes two marked statistics. Can one construct a conditioned tree or permutation limit under $D_n > un$, with an actual canonical macroscopic split and a finite outer decoration process? Such a result could determine the position of the largest jump and the joint sizes of the largest ascent and descent. The sampling representation of the limiting pair is not, by itself, such a full geometric theorem. [Merge note: manuscript 11's parser supplies the canonical split and its theorem 43.2 the limiting law of the finite outer word jointly with the split; the question is re-scoped to the full conditioned permutation, the position of the largest jump and the joint largest ascent and descent, which remain open. Compare Part III's Section 32.4.]

47.9 Conditional cost moments and mesoscopic decorations

Manuscript 11. Under $D_n \geq d_n$ with $d_n/n \rightarrow u \in (0, 1/2)$, determine the growth of $\mathbb{E} \text{cost}(W_n)$ and the distribution of the largest outer decoration. The limiting word has finite mean length $7/3$ but infinite mean cost. Hence convergence in law from theorem 43.2 does not determine these expectations. A joint theory of the word cost and a second large split may describe how the infinite limiting expectation is truncated at finite n .

47.10 A uniform bridge to shrinking adjacency fractions

Manuscript 10. The current result fixes $\theta > 1/2$, while the fixed- m spectral program fixes m before letting n grow. Determine regimes with $m \rightarrow \infty, m/n \rightarrow 0$. The lower-bound hierarchy suggests that the required number of large components itself grows, so a fixed-depth renewal expansion may cease to be uniform. (Compare Part III's Section 32.10.) [Added 1 October 2026, batch 72: answered in Part IX on the ranges of table 32: for $m \rightarrow \infty, n/m \rightarrow \infty$ the number of large components does grow, like $(n-1)/m$, and explicit equivalents hold from $1 \ll n/m \ll \log m$ up to $n(\log m)^2/m^2 \rightarrow 0$, with a logarithmic bridge to the fixed- m rate beyond (section 101.2).]

47.11 Weighted refinements

Manuscript 11. Extend the rare-event profile to avoiders weighted by descents, or record additional statistics jointly with the terminal split. The constant three arose from three equally weighted endpoint modes. Under a nonuniform weight those modes may no longer be equal, and the critical operation masses will change. A correct extension should derive those weights from the refined generating functions instead of replacing Catalan numbers by a new sequence without rechecking the endpoint law. (Compare Part III's Section 32.6.)

47.12 Other classes, statistics and endpoint operators

Manuscript 10. Identify statistics on other pattern-avoiding classes whose root recursion has an endpoint projection like S . If the source is a Catalan split measure but the finite endpoint transition has a different matrix, which rare-event exponents and coefficients persist? The useful target is a proved uniform-remainder theorem, not merely a formal matrix inverse. *Manuscript 11.* Identify other recursively decomposable classes in which an extreme adjacency statistic has a tight ordinary limit but a nontrivial $n^{1/2}$ -weighted macroscopic measure. The proof suggests a mechanism: a summable critical outer-word series, a square-root Catalan split law, and a finite

endpoint acceptance matrix. These are hypotheses to verify in each class, not a universal theorem already proved. Symmetries preserving absolute adjacent differences give immediate equivalent formulations, but a genuinely new class needs a separate decomposition.

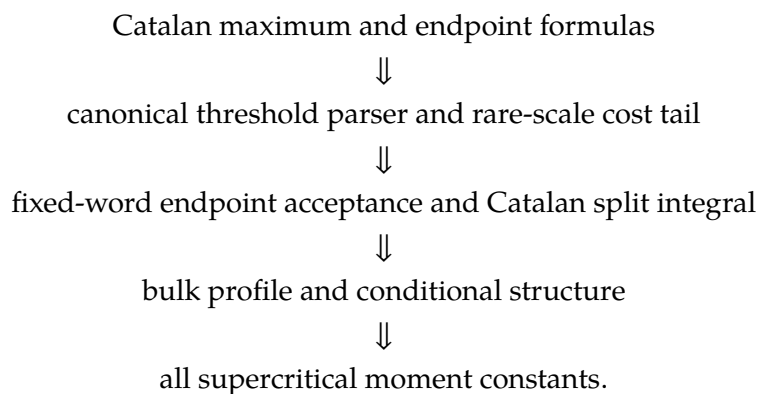
47.13 Formalization, certified and efficient computation

Manuscript 10. A Lean development could first formalize the finite decomposition, proposition 39.1, the endpoint counts, and the stopping-word injection. The analytic layer would then add the uniform tail envelope, the Riemann-sum source theorem, and the bounded-sequence contraction argument. The last two steps deserve separate reusable interfaces. No claim is made that these declarations already exist in the repository or that the current Python checks constitute a formal proof. *Manuscript 11.* Formalize the threshold parser and the rare-scale tail argument, and improve the exact endpoint implementation for large linear bounds. Natural proof-assistant milestones are the reconstruction bijection, the word generating-function identities, the convolution tail estimate, and the fixed-word endpoint acceptance lemma. The dominated-convergence step should then expose its precise $p > 1/2$ hypothesis. For computation, a batched treatment of all bounds could avoid rebuilding overlapping endpoint states. Neither a new formal library nor an optimal complexity claim is supplied.

48 Dependency audit, scope and conclusion

48.1 The two dependency chains

Manuscript 11's proof dependency is



Manuscript 10's replaces the second and third lines by the exact deficit recursion with the stopping-word envelope, and the truncated renewal equation with its contraction proof. Neither depends on a fixed- m exponential growth approximation, an unproved component ordering, a presumed uniform spectral gap, or the asymptotic theorems of Parts I–III. The finite-state recurrence is used for computations and is explicitly attributed to its prior source.

48.2 What is resolved

The advances claimed are deliberately specific. The positive linear-bound range $m/n \in (1/2, 1)$ now has an explicit relative equivalent. Every fixed moment order above the threshold $1/2$ now has a proved leading constant, including the previously undetermined mean and variance constants. The conditional rare event has an explicit two-block geometry and a universal outer-word law. The entire linear-bound problem is not claimed solved, because smaller bounds are only shown to have lower-order probability, with polynomial lower bounds (manuscript 11's summary, extended by manuscript 10's lower bounds). [Added 30 September 2026, batch 68: Part VI proves matching upper bounds, so the smaller bounds have exact polynomial orders (theorem 68.1); their equivalents remain open.] [Added 1 October 2026, batch 71: for

Table 17: Claim-by-claim status of Part IV (manuscripts 10 and 11).

Claim	Basis and limitation
Exact root identity (10)	Direct finite comparison of all possible adjacent jumps; independently checked on all avoiders through size 11.
Uniform tail envelope (both)	An injective stopping-word reconstruction and exact Catalan coefficient identities (10); the canonical parser and a convolution bound (11).
Marked rare-event measure (10)	Endpoint limits, an explicit mesoscopic remainder bound, and a contraction proof of convergence.
Bulk profile for $0 < d/n < 1/2$ (both)	Two written proofs, each including rare-scale tail control.
Scaled zero limit at and above $d/n = 1/2$ (both)	Written monotonicity proof; no finer equivalent.
All fixed- $p > 1/2$ moment constants (both)	Evaluation of that measure and a proved uniform integrable envelope for moment integrals.
Conditional word and split laws (11); three channels (10)	Written joint convergence proof (11); a sampling description of the limiting marked pair (10).
Deficit-size-biased uniform law (10)	A change of variable in the limiting weighted density; not the unweighted permutation law.
Sub-half-size regimes (10, 11)	Constructive polynomial lower bounds only (10), $o(n^{-1/2})$ (both); matching upper bounds and equivalents remain open. [Matching upper bounds, hence exact orders: Part VI, added 30 September 2026; equivalents still open.] [Equivalents for $1/3 < \theta \leq 1/2$: Part VII, added 1 October 2026.] [Every fixed $\theta \leq 1/3$: Part VIII, added 1 October 2026.]
Computations (both)	Exact integer finite tests and exact symbolic checks (11); explicitly labelled Monte Carlo illustrations (10); rounded displays of exact counts (11). No asymptotic claim rests on computation.
New Lean or Rocq declarations	None supplied or claimed.
Novelty, review and formal status	New relative to the inspected report's stated scope; not an exhaustive priority assessment, external review or a proof-assistant verification.
Local limit, critical window, smaller-bound bulk, centered half-moment, exact TV constant	Open. [Centered half-moment: answered in Part V, added 30 September 2026.] [Critical window and smaller-bound bulk: orders and order-crossover scales in Part VI, added 30 September 2026; limiting functions open.] [Limiting function at $m = n/2$, and the smaller-bound bulk for $1/3 < \theta \leq 1/2$: Part VII, added 1 October 2026.] [Limiting functions at every $m = n/q$ and the smaller-bound bulk for every fixed θ : Part VIII, added 1 October 2026.]

$1/3 < \theta \leq 1/2$ Part VII proves the equivalent (theorem 80.2).] [Added 1 October 2026, batch 72: for every fixed $\theta \leq 1/3$ Part VIII proves it (theorem 91.1).]

48.3 Conclusion

Manuscript 10. The main obstruction in passing from a boundary law to exact moments is not integration but missing probability mass at macroscopic distances. The present work identifies that mass. A two-block Catalan source, modified by a finite endpoint renewal, produces the explicit measure (156); its deficit marginal gives the complete first-order profile and all supercritical constants. The proof also isolates a useful methodological lesson. A finite exact identity can remove an apparently necessary subtree statistic. Once this identity is combined with a uniform counting envelope, the mesoscopic remainder becomes summable and a heuristic renewal equation becomes a convergent one. The equal three-channel endpoint mixture and the size-biased uniform law then emerge from an elementary explicit calculation.

Manuscript 11. The fixed-deficit law and the macroscopic deficit measure describe different parts of the same random statistic. The former governs ordinary probability and low moments; the latter accounts for rare configurations large enough to determine high moments. A canonical threshold parser makes that second regime accessible without importing a nonuniform spectral estimate. Its summable but heavy-tailed outer structure contributes an endpoint-weighted mass of three, and the remaining Catalan split integral can be evaluated exactly. This turns an order-of-magnitude moment statement into a complete family of explicit leading constants while leaving clear, nontrivial questions at the critical window and beyond it.

49 Small exact examples and normalization checks

This section was manuscript 10's Appendix B. For $n = 3$, the five avoiders are 123, 213, 231, 312, 321. Their deficits are respectively 2, 1, 1, 1, 2. For example, $\pi = 213$ is an append operation on 21:

$$D(213) = \min\{D(21) + 1, L(21)\} = \min\{2, 1\} = 1.$$

For $\pi = 231$, the nonempty right block is 1 and the left size is one:

$$D(231) = \min\{F(1), 2 + D(1)\} = 1.$$

These examples illustrate why the append state needs the last value while a nonempty right block removes the left block's internal maximum.

For the marked source, the total tail integral is

$$\int_u^{1-u} [y(1-y)]^{-3/2} dy = 4 \frac{1-2u}{\sqrt{u(1-u)}}.$$

Multiplying by the source coefficient $1/(4\sqrt{\pi})$ gives the unmarked two-block source. The full renewal contributes a factor three, so its tail is exactly (150). The folding factor two and the three endpoint channels are distinct factors; conflating them would produce an incorrect constant.

Finally, the size-biased density integrates to one because

$$\frac{1}{2} \int_0^{1/2} x^{-1/2} (1-x)^{-3/2} dx = \left[\sqrt{\frac{x}{1-x}} \right]_0^{1/2} = 1.$$

This independently checks the normalization linking the marked source, the mean constant, and the uniform-law transformation.

50 Repository pins, literature scope and merge

50.1 Manuscript 10's pin and literature scope

This subsection was manuscript 10's Appendix A. The inspected repository was pinned at commit

1ee53d57de253d16cdaad79d1f54bbd95d76d682

(root tree c772efcde0cc2db7902005598190606da6cb338c). The relevant directory is

```
SetTheory/Cardinals/docs/reports/
enumerative-combinatorics/adjacency-bounded-132-avoiders/.
```

Its README at that pin describes the merger of three reports. The third, *The Largest Jump Is Almost Maximal*, was placed with the 03-largest-jump- prefix and appears as Part III of the combined article. The README explicitly excludes a bulk joint limit and exact leading supercritical moment constants. The original standalone source, `article(20260929-161704).tex`, was also consulted through Vladimir's file library to inspect the questions and proofs in contiguous form; its own reference pin is 8d936ee23, not manuscript 10's.

Manuscript 10 uses the same notation for the largest-jump deficit, credits the stopping-word construction, and answers the designated bulk and moment questions. Its proof does not require the earlier algebraic probability generating function, a total-variation theorem, or any fixed-bound spectral asymptotic. Those statements are contextual results, not unverified premises imported into the present argument.

A targeted literature check used the arXiv records of Nadler and Mayama–Akita, the latter's available HTML, and searches for largest adjacent differences in 132-avoiding permutations. No matching macroscopic formula was established from the retrieved material. This is not an exhaustive search or evidence sufficient to certify priority. External papers and earlier repository manuscripts are not redistributed. The full ledger is 04-macroscopic-jump-SOURCES.md.

50.2 Manuscript 11's pin and literature scope

Manuscript 11 inspected the repository snapshot

1ee53d57de253d16cdaad79d1f54bbd95d76d682,

the report at the repository-relative path

```
SetTheory/Cardinals/docs/reports/
enumerative-combinatorics/
adjacency-bounded-132-avoiders/article.tex,
```

with Git blob aeadaf364b083f7e25a475bd42b777cf926769b9, including the report's statements about its unresolved bulk and moment questions. It builds on the report's subject and elementary decompositions, but does not assume its probabilistic or spectral theorems. The primary external records inspected were the arXiv version-one records and HTML texts of [1, 2], accessed September 29, 2026. Targeted searches for maximum adjacent differences and largest jumps in 132-avoiders were also made. These checks locate the contribution relative to the inspected materials; they do not certify first-in-literature priority or exclude independent unpublished work. The full ledger is 05-macroscopic-deficits-SOURCES.md.

50.3 Where the merge had to choose

- *Placement.* Part IV follows Part III and precedes Part I's appendices, which contain no numbered equations, tables or figures; no number of Parts I–III changed. Manuscript 10's two appendices are printed as section 49 and the first subsection of section 50; manuscript 11's

Appendix A (the recurrence) is section 46.3 and its Appendix B (reproduction) is replaced by section 46.8. Both title pages and abstracts are replaced by the opening note of Part IV, the status box of section 37.1 (which prints both research-status boxes) and a paragraph of the report's abstract; every claim of both abstracts is stated in section 38.

- *Base and order.* Manuscript 10 is the base. Its sections are printed in its order, with manuscript 11's parser and tightness sections beside its envelope (section 40), manuscript 11's proof of the profile after manuscript 10's (section 42), and manuscript 11's conditional structure, closed forms and exact data beside the corresponding material of manuscript 10.
- *Duplicates between the manuscripts.* Printed once, credited to both: the profile theorem (manuscript 10's Theorem 1.1 with manuscript 11's sequence form, Theorem 1.1 there), the deficit measure (manuscript 11's Theorem 1.2 as the last assertion of theorem 38.2), the moment theorem (both Theorems 1.3; one proof, with both domination arguments), the first-entry bound (lemma 39.2, both constants), the small- u and half-edge expansions, the critical pole (proposition 44.1), and seven research questions. Kept as second routes: manuscript 11's uniform last-entry estimate (second proof of lemma 39.3), its all-deficit envelope (corollary 40.8, a second proof of proposition 40.3 up to the constant), its proof of the profile (section 42) and of the deficit measure (section 42.4). The constants checked equal: the mean constant $3/\sqrt{\pi}$; the variance constants $3(4 - \pi)/(4\sqrt{\pi})$ and $(3/\sqrt{\pi})(1 - \pi/4)$; manuscript 10's size-biased form and manuscript 11's moment formula (the size-biased law has total mass $\mathcal{I}(1) = 1$, and manuscript 10's integral forms equal manuscript 11's m_p at $p = 3/2$ and $p = 3$ numerically and in general by the substitution of theorem 38.3); and the small- u tail, which is Part III's $3/\sqrt{\pi d}$.
- *Duplicates with Parts I–III.* The maximum decomposition and positional restriction (Part I, section 2), the first- and last-entry counts (Part III, (98) and (101)), Part III's stopping words with their cost series, exact cost tail and disjointness (section 24, lemmas 24.1 and 25.2), and the endpoint recurrence (Part I, proposition 2.1) are printed once, in the earlier parts, and credited; both manuscripts rederive them by the same arguments, which are therefore not reprinted. What the manuscripts add to them (the uniform bounds of lemmas 39.2 and 39.3, the deficit bound (183), the bound in (184), the shortcut (243)) is printed with its proof.
- *Notation.* Every rename is listed in table 14: manuscript 11's $\Phi \rightarrow \Psi$, $x \rightarrow u$ (threshold), $u \rightarrow y$ (split), $\kappa_p \rightarrow m_p$ (both), $I(p) \rightarrow \mathcal{I}(p)$, $I_x \rightarrow I(u)$, $k(u) \rightarrow \mathfrak{k}(y)$, $W, W_R \rightarrow \mathcal{W}, \mathcal{W}_R$, $r_t \rightarrow w_t^R$, $T \rightarrow T_W$, $K \rightarrow \text{len}(W)$, $\alpha(w) \rightarrow \chi(w)$, $r_n(w) \rightarrow \ell_n(w)$, $E_{n,m}, E_n(w) \rightarrow \mathcal{E}_{n,m}, \mathcal{E}_n(w)$, $U_x \rightarrow U^{(u)}$, $H_s(u) \rightarrow \mathcal{N}_s(u)$, $\mu \rightarrow \mu_D$, its core ρ in the operation definitions (not printed), its mode indicator H_n (written out) and its $T_n^{(m)}(u, v) \rightarrow T_{u,v}^{(m)}(n)$ (Part I's form); manuscript 10's $E \rightarrow \mathcal{X}$, $Y_n \rightarrow \Xi_n$, $P \rightarrow S^*$, $R_H \rightarrow \bar{r}_H$, $h (= Pf) \rightarrow g$, $c_h, c_f \rightarrow \bar{Z}_g, \bar{Z}_f$, $E_h, E_f \rightarrow e_g, e_f$, the resolvent weights $r, a \rightarrow w_{\text{id}}, w_S$, $H \rightarrow H^*$, $\mathbb{P}^* \rightarrow \mathbb{P}^*$, its corollary's $Y, B, W \rightarrow U^{(u)}, \xi, \zeta$, $q \rightarrow \theta_r$ in theorem 45.1, $c_{\theta,r} \rightarrow \underline{c}_{\theta,r}$, $A_r(\theta) \rightarrow \Omega_r(\theta)$, and the left and right blocks L, R (written out). No normalization changed. Manuscript 10's $\mathbb{P}(D_n > un)$ and manuscript 11's $\mathbb{P}(D_n \geq \lceil xn \rceil)$ are kept as each stated them.
- *References to this report.* Both manuscripts cite Parts I–III as a repository report (their references “Repo” and “ProveIt”); here those citations are internal cross-references, including Part III's Sections 32.1 and 32.2, whose headings received labels in the merge. Their entries for Nadler and Mayama–Akita are Part I's.
- *Research questions.* Manuscript 10's ten directions and manuscript 11's ten questions are printed as thirteen subsections of section 47; seven pairs asked the same question and are printed once with both texts. Manuscript 10's question on a geometric conditional limit is re-scoped by a merge note, because manuscript 11's theorem 43.2 supplies part of it.
- *The merge's own text.* Section 37, section 43.4, the comparison with $3/\sqrt{\pi(1 - \theta)n}$ in section 37.3, the m_p column of the table of closed forms, the sentences and parentheses explicitly marked as the merge's, the attributions “(manuscript 10)”, “(manuscript 11)”, “(both)”, and this list are the merge's.

- *Reproduction.* The manuscripts' package listings and commands are replaced by the shipped names and the copy recipes (section 46.8 and the README); their `article.tex`, `article.pdf`, `README.md` and `SHA256SUMS` are not shipped.
- *Figures.* The four figure PDFs were regenerated in the write phase by the delivered plot scripts, run unchanged on copies of the delivered layouts with Matplotlib 3.10.8 and `pdf.fonttype 42`, so that they embed TrueType rather than Type 3 fonts; the delivered PDFs had Type 3 DejaVu and STIX fonts. The plotted data are the shipped data. The PNG previews in `figures/` are the delivered files.
- *Stale sentences elsewhere.* The abstract, Part II's research direction on joint growth and its scope ledger (table 10), Part III's scope paragraph (section 21.3), its remark 29.3, the remark after corollary 30.1, its audit table (table 13), its research questions on the bulk profile and the moment constants, and Part I's section B carry dated pointers to Part IV; their original wording is kept.

Part V. Critical moments of the largest-jump deficit

Part V was added on 30 September 2026 in batch 64 of ProveIt's incoming-report intake, from one manuscript written against the state of Parts I–IV printed above. Part IV found the macroscopic profile of the largest-jump deficit $D_n = n - M_n$ and the leading constant of every moment of order $p > 1/2$, and asked in section 47.6 whether $\mathbb{E}\sqrt{D_n} - \frac{3}{2\sqrt{\pi}} \log n$ converges and what the leading correction to $\mathbb{E}D_n^p$ is for $0 < p < 1/2$. Part V answers both questions, and with them the critical half of Part III's Section 32.2, from three proved results of Parts III and IV. The abstract transfer theorems behind the answer are independent of permutations.

51 Provenance, conventions and the answer

51.1 One manuscript, one addition

Part V is built from one AI-assisted research manuscript, *manuscript 06* of batch 64, *Every Logarithmic Scale Contributes: Critical moment crossover and endpoint corrections for the largest jump of a 132-avoiding permutation*, dated September 30, 2026 (27 pages, US Letter, as delivered). It arrived as the archive `ProveIt_Critical_Jump_Moments.zip` (inner directory `Critical_Jump_Moments/`, main file `article.tex`) in ProveIt commit `7747fcfd`, and was placed in commit `3025c15df`. It was written against ProveIt commit

`cb1646dc442724f8e298cf259195b6c308367d80`,

at which this report's `article.tex`, `README.md` and `code/05-macroscopic-deficits-model.py` had the blobs `c673f447a2`, `9b6865915b` and `f0248e0d61`, the ones the manuscript records. All three were unchanged at the placement commit, so every statement the manuscript makes about this report refers to Parts I–IV as printed above. Its code, data, figures and two provenance notes are shipped beside the earlier parts' files with the prefix `06-critical-moments-`, continuing the report's local sequence after Part IV's `05-macroscopic-deficits-`. Its manuscript, README and PDF are not shipped and survive in the arrival commit; its checksum ledger was verified at placement (21 of 21 files) and retired. Its three figures were regenerated in the write phase (section 66.2).

The manuscript contributes, in the order printed here: the centered critical half-moment with an exact convergent constant (theorem 53.1); the first correction to every subcritical moment, with its sign (theorem 53.2); a two-term crossover in the exponent window $p = 1/2 + \lambda/\log n$ (theorem 53.3); the logarithmic-scale law under square-root bias with a signed two-endpoint correction (theorem 53.4); the abstract uniform transfer theory behind all four (sections 54, 56 and 57), with tilted laws and logarithm-weighted moments (section 58), a counterexample and a stability theorem (section 59); a Laplace-integral formula for the microscopic constant (proposition 60.1); exact finite checks through $n = 64$; ten research questions; a derivation of the exact endpoint recurrence; and a proof audit. Every result, proof, example, remark, limitation and question of the manuscript is printed.

Status (manuscript 06 of batch 64). The theorems of Part V have written mathematical proofs. They are not Lean- or Rocq-verified, have not been externally peer reviewed, and no publication priority is claimed. The abstract transfer theorems are proved from explicitly stated hypotheses. The permutation application uses three results of Parts III and IV as proved inputs; it therefore has the same status as those unrefereed and unformalized parts, and a correction to an input would affect the application but not the abstract theorems.

The printed decimals of the constants are high-precision numerical approximations, not interval certificates; the finite data and the figures are regression checks and illustrations, not evidence for the asymptotic statements.

51.2 The question of Part IV, and what answers it

Part III proves the fixed-deficit limiting law of D_n , the sharp order $n^{-1/2}$ of its total-variation error and the moment threshold at $p = 1/2$; at that threshold it shows only that $\mathbb{E}\sqrt{D_n} - \frac{3}{2\sqrt{\pi}} \log n$ is bounded (theorem 22.3). Part IV proves the macroscopic rare-event profile and every supercritical moment constant, but not the centered half-moment or corrections below the threshold: those are its question section 47.6, and the README’s list of what Parts III and IV do not claim named both.

Part V answers the two questions *relative to the proved results of Parts III and IV*. It does not require a new enumeration of large permutations or a local limit theorem. The decisive observation is that a uniform distribution-function error can be integrated after the microscopic power tail has been subtracted. This makes the critical exponent accessible without trying to insert $p = 1/2$ into a divergent supercritical integral. (Part IV expected that the question would “require a quantitative matching of the fixed-deficit law and the macroscopic measure near the singular boundary”; the matching needed turns out to be supplied by Part III’s uniform total-variation bound together with Part IV’s pointwise profile.)

There is a second, more structural conclusion. The critical moment is the normalizing constant of a probability measure that weights a permutation by $\sqrt{D_n}$. Under that measure, the logarithm of the deficit explores every scale between order one and order n . A general theorem identifies not only the leading uniform law on logarithmic scales, but also its first signed correction. That correction is concentrated at the two ends of the scale interval.

51.3 Inherited status

Parts III and IV are AI-assisted, unrefereed research proofs, and their probabilistic theorems are not formalized in Lean or Rocq. Part V uses their limiting law, total-variation estimate and bulk profile as previously proved mathematical inputs, not as machine-checked facts. Every new transfer statement below has a complete proof from explicitly stated hypotheses. This separation permits an independent audit: a correction to an input from Parts III or IV would affect the application, but not the abstract transfer theorem. The manuscript’s pin and its record of the inspected files are in section 66.

51.4 Relation to public work

Nadler [1] initiated the adjacency-constrained enumeration problem and analyzed the case of bound two. Mayama and Akita [2] developed an endpoint-state approach for arbitrary fixed bounds, including rational generating functions. Their maximum decomposition also supplies the finite recurrence used for the exact checks of Part V. The limit here takes the permutation size to infinity while the statistic itself ranges through multiple scales; it is not an additional fixed-bound rationality result.

The literature check used the primary arXiv records of those papers and targeted searches for the largest-jump and critical-moment questions on September 30, 2026. It was not an exhaustive priority investigation. Accordingly, “new” in the contribution ledger (table 18) means proved here beyond the explicit scope of Parts III and IV, not certified absent from all other literature.

Table 18: What Part V inherits from Parts III and IV, and what it adds.

Inherited from Parts III and IV	Proved in Part V
Algebraic limiting law and power tail	Exact centered critical moment, including a convergent constant formula
$O(n^{-1/2})$ total-variation estimate	First correction for every $0 < p < 1/2$, with its sign
Macroscopic deficit-tail profile	Locally uniform moment transfer and the $1/\log n$ exponent window
Leading moments for fixed $p > 1/2$	Critical logarithmic-scale law and two-endpoint signed correction
Exact finite endpoint recurrence	Independent finite regression checks; no new formalization

51.5 Notation of Part V

The symbols $a_n^{(m)}$, C_n , C , $\text{Av}_n(132)$, Π_n , M_n , D_n , D (Part III's limit law), G , s , r (the square roots of (83)), $N_{N,f}$, Ψ , ρ , u (a threshold fraction) and $\mathcal{N}_s(u)$ keep their meanings from Parts III and IV. Several of the manuscript's letters are used in Parts I–IV for other objects, and one of its functions is Part IV's moment constant under another name. Part V fixes one name for each notion (table 19). In particular:

- the coefficient function that the manuscript writes $\mathcal{K}(p)$ is $m(p)$ here, because for $p > 1/2$ it is Part IV's moment constant m_p (remark 55.3). The tempting false reading is that $m(p)$ is a leading constant for every p : for $0 < p < 1/2$ it is the coefficient of the *correction* $n^{p-1/2}$, and it is negative, while the leading term is the limiting moment $\mathbb{E}D^p$. The calligraphic $\mathcal{K}$ of the manuscript would also clash with Part III's clipping series $\mathcal{K}(z)$, and a Greek κ with Part III's quantile $\kappa_{1-\epsilon}$, which is why Part IV already renamed its moment constants from κ_p ;
- the manuscript's Lebesgue measure m on $[0, 1]$ is Leb , since m is the adjacency bound throughout the report and m_p is Part IV's constant;
- the microscopic and macroscopic constants h and j of the manuscript are c_{mic} and c_{mac} , since h is Part III's cutoff $\lfloor n/4 \rfloor$ and Part IV's cutoff, and Part III's $c_j = C_j/4^j$ is always indexed by a letter;
- the tail coefficient a of the manuscript is α , since $a_n^{(m)}$ counts avoiders and a, b are block sizes (as in Part IV and in section 64).

No normalization changed: every renamed symbol denotes exactly the manuscript's quantity. Probability, expectation, total variation and law are written $\mathbb{P}$, $\mathbb{E}$, d_{TV} , $\mathcal{L}$ as in Parts III and IV; the manuscript's Lip and BL are kept. The survival function of the manuscript's application, $\mathcal{S}(x)$, is Part III's $\mathbb{P}(D > x)$. The letters σ , τ , p^* , f , v , δ_n , ε_n and v are local to single proofs and declared where used. Equations are numbered consecutively with those of Parts I–IV.

52 The permutation model and the three inputs

52.1 Definitions and conventions

Recall from Part III that a permutation $\pi \in S_n$ avoids 132 when there are no indices $i < j < k$ with $\pi_i < \pi_k < \pi_j$, that $\text{Av}_n(132)$ is this set and has $C_n = \frac{1}{n+1} \binom{2n}{n}$ elements, and that Π_n is uniform on $\text{Av}_n(132)$. Put

$$M_n = \max_{1 \leq i < n} |\Pi_n(i+1) - \Pi_n(i)|, \quad D_n = n - M_n. \quad (244)$$

For $n = 1$ the empty maximum is zero, so $D_1 = 1$, as in Part IV. Thus $1 \leq D_n \leq n$. Since $a_n^{(m)}$ counts avoiders whose adjacent differences are all at most m , Part III's (82) reads

$$\frac{a_n^{(m)}}{C_n} = \mathbb{P}(M_n \leq m) = \mathbb{P}(D_n \geq n - m). \quad (245)$$

Table 19: Symbols of Part V. “Source” gives manuscript 06’s symbol; the last column prints the tempting false reading.

Symbol	Meaning in Part V	Source	Not to be confused with
α	tail coefficient; $3/\sqrt{\pi}$ in the application	a	the counts $a_n^{(m)}$; Part III’s a_2, a_3 ; block sizes a, b .
α, β	tail and remainder exponents (abstract)	α, β	Part I’s growth constants α_m , Part II’s β_m .
$c_{\text{mic}}, c_{\text{mac}}, c_\star$	microscopic, macroscopic and total critical constants	$h, j, c_\star$	Part III’s cutoff $h = \lfloor n/4 \rfloor$; $c_j = C_j/4^j$.
B_0, B_1	endpoint constants	B_0, B_1	Part III’s $B(x)$; Part II’s $\mathcal{B}_N$.
$m(p)$	regularized coefficient function	$\mathcal{K}(p)$	a leading constant (only for $p > 1/2$, where it is m_p); Part III’s $\mathcal{K}$ and quantile $\kappa_{1-\epsilon}$.
$\mathcal{M}(\lambda)$	$\int_0^1 e^{\lambda t} dt$	$\Phi(\lambda)$	Part I’s shift map Φ ; Part IV’s profile Ψ .
ϑ_n	logarithmic scale $\log D_n / \log n$ ($\log X_n / \log n$)	T_n	Part III’s cost T ; Part IV’s $T_{n,m}, T_W$; Part I’s $T_{p,q}^{(m)}(n)$.
η_n	critical weighted measure on $[0, 1]$	η_n	the local η of Part IV’s proofs.
$\mathbb{P}_n^\lambda, \mathbb{E}_n^\lambda$	power-tilted probability and expectation	Q_n^λ	Part III’s Q ; Part IV’s $Q_{N,m}$ and size bias $\mathbb{P}_n^\star$ ($p = 1$).
ω_λ	limiting tilted law on $[0, 1]$	μ_λ	Part IV’s marked measure μ .
Leb	Lebesgue measure on $[0, 1]$	m	the adjacency bound m ; m_p .
S, S_n, S_Y	survival functions of D (or X), D_n (or X_n), Y	S, S_n, S_Y	Part IV’s endpoint map S .
$\varrho(x)$	remainder in the limiting tail	$r(x)$	r of (83) (kept); Part IV’s r_a .
$b(u), g_n(u)$	cancellation function; scaled tail discrepancy	$b(u), g_n(u)$	Part III’s $b_t, B(x)$; Part IV’s test function g .
$\mathcal{H}(p), \mathcal{J}(p), \mathcal{J}_n(p)$	microscopic and macroscopic regularized functions	H, J, J_n	Part III’s series H, J ; Part IV’s first deficiency J_n and $H^\star$.
$\mathcal{R}, \mathcal{R}(p); c(p); \ell(t)$	regularized Laplace integral; $p/\Gamma(1-p); 1 - G(e^{-t})$	$\mathcal{A}, \mathcal{A}(p); c(p); \ell(t)$	Part I’s class $\mathcal{A}$; c_j ; Part IV’s $\ell_n(w)$.
$\mathcal{D}; K, K_{\mathcal{D}}$	compact set of exponents; constants	$K; C, C_K$	Catalan numbers C_n ; Part III’s $\mathcal{K}$.
$N_n(u), c_{k,u}, T_{u,v}^{(m)}(n), \mathcal{G}_j(x)$	the counts of section 64	$h(n, u), c_k(u), T_m(n, u, v), A_j(x)$	Part IV’s notation for the same counts (242); Part I’s $A_m(x)$.
θ, λ	exponent of an added atom; exponent-window parameter	θ, λ	Part IV’s fraction $\theta = m/n$; Part I’s eigenvalues λ_U, λ_V .

All logarithms are natural. For a positive real x and complex p , x^p means $\exp(p \log x)$.

For a positive random variable Y define its strict survival function by $\mathcal{S}_Y(x) = \mathbb{P}(Y > x)$. Abbreviate

$$\mathcal{S}_n(x) = \mathbb{P}(D_n > x), \quad \mathcal{S}(x) = \mathbb{P}(D > x), \quad \alpha = \frac{3}{\sqrt{\pi}}.$$

Here D is the positive-integer random variable of the next input. The strict/non-strict distinction matters for exact lattice formulas but not for the Lebesgue integrals used in the proofs. The manuscript also writes $A_0 = \alpha/2 = \frac{3}{2\sqrt{\pi}}$ for the critical logarithmic coefficient; Part V writes the fraction out.

The three inputs below are printed as the manuscript states them, although each is a result of Parts III or IV: they are the entire interface between the abstract theory and the permutations, and printing them here makes that interface auditable in one place.

52.2 Input I: the microscopic law

Part III (theorem 22.1, (84)) gives the probability generating function

$$G(z) = \mathbb{E}z^D = \frac{(1-s)(4+8s+3s^2-s^2r)}{4(1+2s)(1+s)^2}, \quad s = \sqrt{1-z}, \quad r = \sqrt{\frac{1+s}{2}}, \quad (246)$$

with the branches positive on $0 \leq z < 1$. In particular, $\mathbb{P}(D = 1) = 7/48$. Its tail satisfies (theorem 22.2, (88))

$$\mathcal{S}(x) = \alpha x^{-1/2} + O(x^{-3/2}), \quad x \geq 1. \quad (247)$$

The integer version stated in Part III implies the real version: $\mathcal{S}(x) = \mathcal{S}(\lfloor x \rfloor)$, and changing x by at most one changes $x^{-1/2}$ by $O(x^{-3/2})$.

The leading coefficient is consistent with a direct algebraic expansion:

$$1 - G(z) = 3s + \left(-\frac{23}{4} + \frac{\sqrt{2}}{8}\right)s^2 + \left(\frac{43}{4} - \frac{9\sqrt{2}}{16}\right)s^3 + O(s^4). \quad (248)$$

The manuscript verifies this identity symbolically. This finite algebraic check does not replace Part III's limiting-law and tail proofs. [Merge note: this is Part III's expansion in the proof of theorem 22.2, $3(1-z)^{-1/2} - a_2 - a_3(1-z)^{1/2} + O(1-z)$ for $\sum_k \mathbb{P}(D > k)z^k = (1-G)/s^2$ ((133)), with $a_2 = 23/4 - \sqrt{2}/8$ and $a_3 = -43/4 + 9\sqrt{2}/16$; the two agree.]

52.3 Input II: a uniform finite-size error

The total-variation theorem of Part III ((85), with the explicit bound (86)) implies

$$\sup_{x \geq 0} |\mathcal{S}_n(x) - \mathcal{S}(x)| \leq Kn^{-1/2} \quad (249)$$

for an absolute constant K . No value of the best constant is required. Indeed a half-line is one of the measurable sets controlled by total variation. The proofs below use only (249); the full strength of total variation is unnecessary.

52.4 Input III: the macroscopic profile

Part IV's theorem 38.1 gives

$$\sqrt{n} \mathcal{S}_n(nu) \longrightarrow \Psi(u), \quad 0 < u \leq 1, \quad (250)$$

where, as in (150),

$$\Psi(u) = \begin{cases} \alpha \frac{1-2u}{\sqrt{u(1-u)}}, & 0 < u < 1/2, \\ 0, & 1/2 \leq u \leq 1. \end{cases} \quad (251)$$

Part V needs only pointwise convergence, not a convergence rate or a local limit. Part IV supplies a stronger uniformity away from zero.

The formal leading density on $(0, 1/2)$ is

$$-\Psi'(u) = \rho(u) = \frac{\alpha}{2[u(1-u)]^{3/2}}. \quad (252)$$

This density has infinite total mass at zero. It is a rare-event limit measure, not a probability density on the entire interval. It is the deficit density of Part IV (theorem 38.2).

53 Main results for the largest-jump deficit

53.1 The centered square-root moment

Define the microscopic finite part

$$c_{\text{mic}} = 1 + \frac{1}{2} \int_1^\infty x^{-1/2} \left(\mathcal{S}(x) - \alpha x^{-1/2} \right) dx \quad (253)$$

and the elementary macroscopic correction

$$c_{\text{mac}} = \alpha \left(\log 2 - \log(1 + \sqrt{2}) + \sqrt{2} - 2 \right). \quad (254)$$

The integral in (253) is absolutely convergent by (247).

Theorem 53.1 (Resolution of the centered half-moment question). *Using the three inputs of section 52,*

$$\mathbb{E}\sqrt{D_n} = \frac{3}{2\sqrt{\pi}} \log n + c_\star + o(1), \quad c_\star = c_{\text{mic}} + c_{\text{mac}}. \quad (255)$$

The exact definitions (253)–(254), together with the algebraic function (246), specify $c_\star$ without a limiting finite-size fit. Numerical quadrature gives

$$c_\star \approx -0.942121413999402.$$

This answers the first question of section 47.6 and the critical question of Part III's Section 32.2, and sharpens Part III's (92) from $O(1)$ to $c_\star + o(1)$.

53.2 Every subcritical moment has an explicit first correction

For $\Re p > -1/2$, $p \neq 1/2$, define the regularized expression

$$m(p) = \frac{\alpha}{2} \left\{ \frac{2^{1/2-p}}{p-1/2} + \int_0^{1/2} u^{p-3/2} ((1-u)^{-3/2} - 1) du \right\}. \quad (256)$$

The integral converges in this domain because its bracket is $O(u)$ at zero. Equivalently,

$$m(p) = \frac{\alpha}{2} B_{1/2}(p-1/2, -1/2), \quad (257)$$

where the incomplete beta function in (257) is *analytically continued* when $p < 1/2$. The convergent expression (256), not a divergent unregularized integral, is the definition used here.

Theorem 53.2 (Subcritical correction). *For each fixed real $0 < p < 1/2$,*

$$\mathbb{E}D_n^p = \mathbb{E}D^p + m(p)n^{p-1/2} + o(n^{p-1/2}). \quad (258)$$

Moreover $m(p) < 0$ throughout this interval. Consequently $\mathbb{E}D_n^p < \mathbb{E}D^p$ for all sufficiently large n , for each fixed such p . This last conclusion is eventual one-sided convergence, not monotonicity in n .

This answers the second question of section 47.6, and sharpens Part III's (91) from $O(n^{p-1/2})$ to an exact first term with its sign.

For comparison, the supercritical leading coefficient of Part IV is the same function: for fixed $p > 1/2$, $\mathbb{E}D_n^p \sim m(p)n^{p-1/2}$, that is, $m(p) = m_p$ (theorem 38.3; remark 55.3 proves the identity). Thus one regularized analytic function governs both sides of the moment threshold, although its role is an error coefficient below the threshold and a leading coefficient above it. Its pole at the threshold must be canceled before one takes a limit.

53.3 The moving critical window

Let

$$\mathcal{M}(\lambda) = \int_0^1 e^{\lambda t} dt = \begin{cases} (e^\lambda - 1)/\lambda, & \lambda \neq 0, \\ 1, & \lambda = 0, \end{cases} \quad (259)$$

which is an entire function. Set

$$B_0 = c_{\text{mic}} - \alpha, \quad B_1 = \alpha + c_{\text{mac}}. \quad (260)$$

Theorem 53.3 (Two-term moment crossover). *For $p_n(\lambda) = 1/2 + \lambda/\log n$,*

$$\mathbb{E}D_n^{p_n(\lambda)} = \frac{3}{2\sqrt{\pi}} \log n \mathcal{M}(\lambda) + B_0 + B_1 e^\lambda + o(1), \quad (261)$$

locally uniformly for complex λ . In particular, the error is uniform on each fixed bounded real interval. The endpoint constants have the numerical values

$$\begin{aligned} B_0 &\approx -1.324620212145336, \\ B_1 &\approx 0.382498798145934. \end{aligned}$$

At $\lambda = 0$, their sum is $c_\star$.

53.4 The logarithmic-scale law and its signed correction

Write $\vartheta_n = \log D_n / \log n$ for $n \geq 2$. Define a finite measure on $[0, 1]$ by

$$\eta_n(\varphi) = \mathbb{E}[\sqrt{D_n} \varphi(\vartheta_n)]. \quad (262)$$

For a Lipschitz function φ , let $\|\varphi\|_{\text{BL}} = \|\varphi\|_\infty + \text{Lip}(\varphi)$.

Theorem 53.4 (Two endpoint corrections). *Uniformly over complex-valued φ with $\|\varphi\|_{\text{BL}} \leq 1$,*

$$\eta_n(\varphi) = \frac{3}{2\sqrt{\pi}} \log n \int_0^1 \varphi(t) dt + B_0 \varphi(0) + B_1 \varphi(1) + o(1). \quad (263)$$

Equivalently, in the dual bounded-Lipschitz norm,

$$\eta_n - \frac{3}{2\sqrt{\pi}} \log n \text{Leb} \longrightarrow B_0 \delta_0 + B_1 \delta_1. \quad (264)$$

This is convergence of a signed residual, not a decomposition of a positive measure into three positive pieces.

Under the probability measure that weights Π_n proportionally to $\sqrt{D_n}$, ϑ_n therefore converges weakly to $\text{Unif}[0, 1]$. For real λ , weighting instead by $D_n^{p_n(\lambda)}$ produces the limiting density

$$\frac{e^{\lambda t}}{\mathcal{M}(\lambda)}, \quad 0 \leq t \leq 1. \quad (265)$$

The normalization and the first correction to this probability law are proved in section 58. Theorem 53.4 is the critical analogue of Part IV's power-biased law (240) and its size-biased limit (theorem 38.4), which live on the scale D_n/n for $p > 1/2$; at $p = 1/2$ the natural scale is logarithmic.

54 A general uniform moment-transfer theorem

This section contains the common argument behind all the application results. It is independent of permutations and of the rest of the report.

54.1 Hypotheses and regularized functions

Let X, X_n be real random variables with $X \geq 1$ and $1 \leq X_n \leq n$. Write $\mathcal{S}(x) = \mathbb{P}(X > x)$ and $\mathcal{S}_n(x) = \mathbb{P}(X_n > x)$. Suppose that constants $\alpha, \alpha, \beta > 0$ satisfy the following three conditions:

$$\mathcal{S}(x) = \alpha x^{-\alpha} + \varrho(x), \quad |\varrho(x)| \leq Kx^{-\alpha-\beta} \quad (x \geq 1), \quad (266)$$

$$\sup_{x \geq 0} |\mathcal{S}_n(x) - \mathcal{S}(x)| \leq Kn^{-\alpha}, \quad (267)$$

$$n^\alpha \mathcal{S}_n(nu) \longrightarrow \Psi(u) \quad (0 < u \leq 1). \quad (268)$$

Constants denoted by K may be enlarged. Set

$$\mathfrak{b}(u) = \Psi(u) - \alpha u^{-\alpha}, \quad 0 < u \leq 1, \quad (269)$$

and

$$\mathcal{H}(p) = 1 + p \int_1^\infty x^{p-1} \varrho(x) dx, \quad \Re p < \alpha + \beta, \quad (270)$$

$$\mathcal{J}(p) = p \int_0^1 u^{p-1} \mathfrak{b}(u) du, \quad \Re p > 0. \quad (271)$$

In this section Ψ is any function satisfying (268); in the application it is Part IV's profile (251).

Lemma 54.1 (The cancellation is uniformly bounded). *The function $\mathfrak{b}$ is bounded on $(0, 1]$. The integrals defining $\mathcal{H}$ and $\mathcal{J}$ converge absolutely and define holomorphic functions in their indicated half-planes.*

Proof. For fixed $u > 0$, (266) implies $n^\alpha \mathcal{S}(nu) \rightarrow \alpha u^{-\alpha}$. Subtracting this limit from (268) and applying (267) gives $|\mathfrak{b}(u)| \leq K$. For p in a compact subset of $\Re p < \alpha + \beta$, the integrand for $\mathcal{H}$ is bounded by a constant times $x^{-1-\varepsilon}$ for some $\varepsilon > 0$. For p in a compact subset of $\Re p > 0$, the integrand for $\mathcal{J}$ is bounded by a constant times $u^{-1+\varepsilon}$. Additional powers of $|\log x|$ or $|\log u|$ remain integrable, so differentiation under the integral is justified locally. This proves holomorphy. $\square$

Define a scaled tail discrepancy

$$g_n(u) = n^\alpha (\mathcal{S}_n(nu) - \mathcal{S}(nu)), \quad 0 < u \leq 1. \quad (272)$$

It is bounded by K , vanishes when $u < 1/n$, and converges pointwise to $\mathfrak{b}(u)$. The vanishing follows because both survival functions are one on $0 \leq x < 1$.

Lemma 54.2 (Uniform integral convergence). *Put $\mathcal{J}_n(p) = p \int_0^1 u^{p-1} g_n(u) du$. On every compact subset of $\Re p > 0$, $\mathcal{J}_n \rightarrow \mathcal{J}$ uniformly.*

Proof. If $\sigma = \inf \Re p > 0$ and $p^* = \sup |p|$ on the compact set, then

$$\sup_p |\mathcal{J}_n(p) - \mathcal{J}(p)| \leq p^* \int_0^1 u^{\sigma-1} |g_n(u) - \mathfrak{b}(u)| du.$$

The integrand tends to zero pointwise and is bounded by $2Ku^{\sigma-1}$. Dominated convergence proves the assertion. $\square$

54.2 An exact identity before any asymptotics

For $\Re p > 0$, finite support and the fundamental theorem of calculus give

$$\mathbb{E}X_n^p = 1 + p \int_1^n x^{p-1} \mathcal{S}_n(x) dx. \quad (273)$$

This is also valid for complex p , by applying Fubini to the absolutely integrable identity $y^p - 1 = p \int_1^y x^{p-1} dx$.

Lemma 54.3 (Exact decomposition). *For $0 < \Re p < \alpha + \beta$,*

$$\begin{aligned} \mathbb{E}X_n^p &= \mathcal{H}(p) + \alpha p \frac{n^{p-\alpha} - 1}{p - \alpha} + n^{p-\alpha} \mathcal{J}_n(p) \\ &\quad - p \int_n^\infty x^{p-1} \varrho(x) dx. \end{aligned} \quad (274)$$

At $p = \alpha$, the quotient is interpreted as $\log n$.

Proof. In (273), split $\mathcal{S}_n = \alpha x^{-\alpha} + \varrho + (\mathcal{S}_n - \mathcal{S})$. Integrating the first term yields the quotient. Complete the second integral to infinity and subtract its tail. In the last term set $x = nu$. The resulting lower limit $1/n$ can be replaced by zero because g_n vanishes there. These are exact operations, not formal asymptotic manipulations. $\square$

Theorem 54.4 (Locally uniform complex-moment transfer). *Under (266)–(268), for every compact set $\mathcal{D} \subset \{p : 0 < \Re p < \alpha + \beta\}$,*

$$\sup_{p \in \mathcal{D}} \left| n^{\alpha-p} \left(\mathbb{E}X_n^p - \mathcal{H}(p) - \alpha p \frac{n^{p-\alpha} - 1}{p - \alpha} \right) - \mathcal{J}(p) \right| \rightarrow 0. \quad (275)$$

The singularity at $p = \alpha$ is removable in the displayed expression.

Proof. By lemma 54.3, the expression inside the absolute value is

$$\mathcal{J}_n(p) - \mathcal{J}(p) - p n^{\alpha-p} \int_n^\infty x^{p-1} \varrho(x) dx.$$

The first term tends uniformly to zero by lemma 54.2. For the second, let $\tau = \sup_{p \in \mathcal{D}} \Re p < \alpha + \beta$ and $p^* = \sup_{p \in \mathcal{D}} |p|$. Then its absolute value is at most

$$p^* K n^{\alpha-\Re p} \frac{n^{\Re p-\alpha-\beta}}{\alpha + \beta - \Re p} \leq \frac{p^* K}{\alpha + \beta - \tau} n^{-\beta}.$$

This bound is uniform. The quotient in (275) is an entire function of p , after filling its value at $p = \alpha$, which verifies the last assertion. $\square$

54.3 What uniformity does and does not buy

For a fixed p in the strip, theorem 54.4 gives an error $o(n^{\Re p-\alpha})$ in the unscaled moment expression. At $p = \alpha$, this is $o(1)$. At fixed real $p > \alpha$, it need not be $o(1)$. Consequently, extracting the constant-looking term from the formula for a fixed supercritical exponent does *not* establish a second-order constant. A quantitative rate for $\mathcal{J}_n(p) - \mathcal{J}(p)$ would be required.

The moving window $p - \alpha = O(1/\log n)$ is different: $n^{p-\alpha}$ stays bounded on compact windows. Precisely there, the uniform theorem upgrades the unscaled error to $o(1)$ and yields a genuine two-term expansion.

55 Fixed exponents, the critical constant, and beta regularization

55.1 The general subcritical and critical formulas

For real $0 < p < \alpha$, the limiting moment exists and

$$\mathbb{E}X^p = \mathcal{H}(p) + \frac{\alpha p}{\alpha - p}. \quad (276)$$

This follows from the same tail integration as before, now on $[1, \infty)$. Define

$$\mathfrak{m}(p) = \frac{\alpha p}{p - \alpha} + \mathcal{J}(p). \quad (277)$$

(In the application, $\alpha = 1/2$, this is the function (256), by lemma 55.2.) Then theorem 54.4 gives

$$\mathbb{E}X_n^p = \mathbb{E}X^p + \mathfrak{m}(p)n^{p-\alpha} + o(n^{p-\alpha}), \quad 0 < p < \alpha. \quad (278)$$

At the critical exponent itself it gives

$$\mathbb{E}X_n^\alpha = \alpha \log n + \mathcal{H}(\alpha) + \mathcal{J}(\alpha) + o(1). \quad (279)$$

Thus the finite constant splits into a microscopic term determined by the entire limiting law and a macroscopic term determined by the bulk profile. Neither term alone supplies the answer.

55.2 Applying the theorem to the permutation inputs

Equations (247)–(250) satisfy the general hypotheses with

$$\alpha = \frac{1}{2}, \quad \beta = 1, \quad \alpha = \frac{3}{\sqrt{\pi}}.$$

In particular, the abstract uniform strip includes $0 < \Re p < 3/2$. Formula (253) is exactly $\mathcal{H}(1/2)$. It remains to evaluate $\mathcal{J}(1/2)$ and identify (277) with the regularized beta expression.

Lemma 55.1 (Evaluation of the macroscopic critical contribution). *For the profile (251), $\mathcal{J}(1/2) = c_{\text{mac}}$ as defined in (254).*

Proof. By direct substitution,

$$\mathcal{J}(1/2) = \frac{\alpha}{2} \left\{ \int_0^{1/2} \left(\frac{1-2u}{u\sqrt{1-u}} - \frac{1}{u} \right) du - \int_{1/2}^1 \frac{du}{u} \right\}. \quad (280)$$

The first integrand is bounded at zero; its two displayed summands must not be integrated separately there. An antiderivative of their difference is

$$\mathcal{F}(u) = -2 \operatorname{artanh} \sqrt{1-u} + 4\sqrt{1-u} - \log u.$$

As $u \downarrow 0$, $\mathcal{F}(u) \rightarrow 4 - \log 4$. At $u = 1/2$, $\operatorname{artanh}(1/\sqrt{2}) = \log(1 + \sqrt{2})$, so

$$\mathcal{F}(1/2) = -2 \log(1 + \sqrt{2}) + 2\sqrt{2} + \log 2.$$

Subtract $\mathcal{F}(0+)$ and then the last integral $\log 2$ in (280). The result is

$$\frac{\alpha}{2} (\log 4 - 2 \log(1 + \sqrt{2}) + 2\sqrt{2} - 4) = c_{\text{mac}}.$$

□

Combining (279) with lemma 55.1 proves theorem 53.1.

Lemma 55.2 (A single coefficient function on both sides of the threshold). *For $\Re p > 0$, $p \neq 1/2$, the expression $\alpha p/(p - 1/2) + \mathcal{J}(p)$ equals (256). In particular its residue at $p = 1/2$ is $\alpha/2$, and its finite part there is $\alpha + c_{\text{mac}} = B_1$.*

Proof. First take real $p > 1/2$. The singular integrals are then separately convergent, and

$$\frac{\alpha p}{p - 1/2} + \mathcal{J}(p) = p \int_0^{1/2} u^{p-1} \Psi(u) \, du.$$

Since $\Psi(1/2) = 0$, $u^p \Psi(u) \rightarrow 0$ at zero, and $-\Psi' = \rho$, integration by parts gives

$$p \int_0^{1/2} u^{p-1} \Psi(u) \, du = \frac{\alpha}{2} \int_0^{1/2} u^{p-3/2} (1-u)^{-3/2} \, du.$$

Subtract one from $(1-u)^{-3/2}$ and integrate the subtracted power explicitly. This gives (256). Both sides are holomorphic on the connected domain $\{\Re p > 0\} \setminus \{1/2\}$, so the identity extends throughout that domain by analytic continuation. Finally,

$$\frac{\alpha p}{p - 1/2} = \frac{\alpha/2}{p - 1/2} + \alpha, \quad \mathcal{J}(p) = c_{\text{mac}} + O(p - 1/2),$$

which gives the residue and the finite part. $\square$

Remark 55.3 (Merge note: $m(p)$ is Part IV's m_p for $p > 1/2$). The second display of the proof of lemma 55.2 shows, for real $p > 1/2$,

$$m(p) = \frac{3}{2\sqrt{\pi}} \int_0^{1/2} u^{p-3/2} (1-u)^{-3/2} \, du,$$

which is Part IV's first formula (160) for m_p . Hence $m(p) = m_p$ for every $p > 1/2$, and the manuscript's sentence that the supercritical leading coefficient “is the same function” is an identity, not only a numerical coincidence (the write phase also checked it numerically at $p = 3/4, 1, 3/2, 2, 3$, to twenty digits against (161)). For $1/2 < p < 3/2$ it is also what theorem 54.4 says directly, since there the transfer theorem gives the limit of theorem 38.3:

$$n^{1/2-p} \mathbb{E} D_n^p \longrightarrow m(p).$$

Part IV's proposition 44.1 gives the pole (238) as $p \downarrow 1/2$; lemma 55.2 sharpens it to

$$m_p = \frac{3}{2\sqrt{\pi}} \frac{1}{p - 1/2} + B_1 + O(p - 1/2),$$

so the finite part of Part IV's pole is $B_1 \approx 0.382498798$, the constant of theorem 53.3. Below the threshold, the coefficient $m(p)$, $0 < p < 1/2$, of the correction in (258) is the analytic continuation of Part IV's constants through that pole.

Proof of theorem 53.2. The expansion follows from (278) and lemma 55.2. To prove the sign, observe that

$$\frac{1-2u}{\sqrt{1-u}} < 1 \quad (0 < u < 1/2).$$

Indeed both sides are positive, and squaring reduces the claim to $u(4u - 3) < 0$. For $1/2 \leq u \leq 1$ the bulk profile is zero. Therefore $b(u) = \Psi(u) - \alpha u^{-1/2} < 0$ on $(0, 1]$, so $\mathcal{J}(p) < 0$ for every real $p > 0$. If $p < 1/2$, the term $\alpha p/(p - 1/2)$ is also negative. Hence $m(p) < 0$. Dividing the remainder in (258) by its strictly negative leading coefficient proves eventual one-sidedness. $\square$

56 The $1/\log n$ critical exponent window

Theorem 56.1 (General two-term crossover). *Under the hypotheses of theorem 54.4, let $p_n(\lambda) = \alpha + \lambda/\log n$. Define*

$$B_0 = \mathcal{H}(\alpha) - \alpha, \quad B_1 = \alpha + \mathcal{J}(\alpha).$$

Then

$$\mathbb{E}X_n^{p_n(\lambda)} = \alpha \log n \mathcal{M}(\lambda) + B_0 + B_1 e^\lambda + o(1) \quad (281)$$

locally uniformly for complex λ .

Proof. Fix a compact set of λ values. For large n , all $p_n(\lambda)$ lie in a fixed compact subset of the strip in theorem 54.4. In addition $n^{p_n-\alpha} = e^\lambda$ is uniformly bounded. The theorem therefore gives, uniformly on the chosen compact set,

$$\mathbb{E}X_n^{p_n} = \mathcal{H}(p_n) + \alpha p_n \log n \mathcal{M}(\lambda) + e^\lambda \mathcal{J}(p_n) + o(1).$$

Continuity gives $\mathcal{H}(p_n) = \mathcal{H}(\alpha) + o(1)$ and $\mathcal{J}(p_n) = \mathcal{J}(\alpha) + o(1)$. The remaining term is exactly

$$\alpha p_n \log n \mathcal{M}(\lambda) = \alpha \log n \mathcal{M}(\lambda) + \alpha \lambda \mathcal{M}(\lambda) = \alpha \log n \mathcal{M}(\lambda) + \alpha(e^\lambda - 1).$$

Collecting constants proves (281), including at $\lambda = 0$ by continuity. $\square$

Theorem 53.3 is the specialization $\alpha = 1/2$ with the constants already computed.

56.1 Why the width is logarithmic

The crossover width is forced by the factor $n^{p-\alpha}$:

$$n^{p-\alpha} = \exp((p - \alpha) \log n).$$

It changes by a constant factor when $p - \alpha$ changes by order $1/\log n$. At that scale, the pole cancellation between the microscopic limiting moment and the macroscopic contribution is visible in the entire function $\mathcal{M}$.

For real λ one has $\mathcal{M}(\lambda) > 0$. The asymptotic expression never asks for a moment of the limiting variable X above its convergence threshold; it concerns the finite variable X_n throughout. In particular, using $\mathbb{E}X^{p_n}$ for positive λ would be invalid.

56.2 An exact model that checks the normalization

Example 56.2 (A truncated Pareto variable). Let $\mathbb{P}(X > x) = x^{-\alpha}$ for $x \geq 1$, and put $X_n = \min(X, n)$. Here $\alpha = 1$, $\rho = 0$, $\mathcal{H} = 1$, and $\mathfrak{b} = 0$ almost everywhere on $(0, 1)$. The value of $\mathfrak{b}$ at $u = 1$ is immaterial to all integrals. Thus $\mathcal{J} = 0$, $B_0 = 0$, and $B_1 = 1$. Direct calculation gives the exact identities

$$\mathbb{E}X_n^p = 1 + p \frac{n^{p-\alpha} - 1}{p - \alpha}, \quad (282)$$

$$\mathbb{E}X_n^{\alpha+\lambda/\log n} = \alpha \log n \mathcal{M}(\lambda) + e^\lambda. \quad (283)$$

The atom at $X_n = n$ contributes exactly one to the critical weighted mass. It is the endpoint term B_1 , not an extra piece of the logarithmic coefficient. This elementary example checks both the factor α in the leading term and the distinction between $\mathcal{J}(\alpha)$ and B_1 .

57 The logarithmic-scale measure and endpoint theorem

The moment crossover probes the test functions $e^{\lambda t}$. A direct argument treats the entire bounded-Lipschitz class at once and gives a stronger conclusion.

57.1 Statement in the general setting

Put $\vartheta_n = \log X_n / \log n$ and define

$$\eta_n(\varphi) = \mathbb{E}[X_n^\alpha \varphi(\vartheta_n)].$$

Let Leb be Lebesgue measure on $[0, 1]$ and define, for a finite signed or complex measure ν ,

$$\|\nu\|_{\text{BL}^*} = \sup_{\|\varphi\|_\infty + \text{Lip}(\varphi) \leq 1} |\nu(\varphi)|.$$

Theorem 57.1 (General two-endpoint correction). *Under (266)–(268),*

$$\|\eta_n - \alpha \log n \text{ Leb} - B_0 \delta_0 - B_1 \delta_1\|_{\text{BL}^*} \longrightarrow 0, \quad (284)$$

where $B_0 = \mathcal{H}(\alpha) - \alpha$ and $B_1 = \alpha + \mathcal{J}(\alpha)$. More precisely, the norm is at most

$$|\mathcal{J}_n(\alpha) - \mathcal{J}(\alpha)| + O((\log n)^{-1} + n^{-\beta}), \quad (285)$$

with a constant depending only on the hypotheses.

57.2 Proof by three integrals

Proof. For a Lipschitz function φ on $[0, 1]$, let

$$f(x) = x^\alpha \varphi(\log x / \log n), \quad 1 \leq x \leq n.$$

This is absolutely continuous, with almost-everywhere derivative

$$f'(x) = x^{\alpha-1} \left\{ \alpha \varphi(\log x / \log n) + \frac{1}{\log n} \varphi'(\log x / \log n) \right\}. \quad (286)$$

The derivative is integrable, so Fubini gives

$$\eta_n(\varphi) = \varphi(0) + \int_1^n f'(x) \mathcal{S}_n(x) dx. \quad (287)$$

Split the survival function into its power tail, remainder and finite-size discrepancy, exactly as in lemma 54.3.

The power tail. The part with $\alpha x^{-\alpha}$ equals

$$\alpha \int_1^n \frac{\alpha \varphi(\log x / \log n) + (\log n)^{-1} \varphi'(\log x / \log n)}{x} dx = \alpha \log n \int_0^1 \varphi(t) dt + \alpha(\varphi(1) - \varphi(0)). \quad (288)$$

This identity explains why two endpoint terms occur even before the microscopic and macroscopic corrections are considered.

The microscopic remainder. Combine the initial term $\varphi(0)$ in (287) with the integral involving $\varrho(x)$. Uniformly for $\|\varphi\|_{\text{BL}} \leq 1$, their sum is

$$\mathcal{H}(\alpha) \varphi(0) + O((\log n)^{-1} + n^{-\beta}). \quad (289)$$

Indeed $|\varphi(\log x / \log n) - \varphi(0)| \leq \log x / \log n$, and $|\varphi'| \leq 1$ almost everywhere. The error from replacing φ by $\varphi(0)$ and dropping the derivative term is bounded by

$$\frac{1}{\log n} \int_1^\infty x^{\alpha-1} |\varrho(x)| (\alpha \log x + 1) dx,$$

which is finite by (266). Completing the remaining integral to infinity costs $O(n^{-\beta})$.

The finite-size discrepancy. After $x = nu$, this contribution is

$$\int_{1/n}^1 u^{\alpha-1} g_n(u) \left\{ \alpha \varphi \left(1 + \frac{\log u}{\log n} \right) + \frac{1}{\log n} \varphi' \left(1 + \frac{\log u}{\log n} \right) \right\} du. \quad (290)$$

The bound $|g_n| \leq K$ and the Lipschitz property show that replacing the function by $\varphi(1)$ and deleting its derivative costs at most

$$\frac{K}{\log n} \int_0^1 u^{\alpha-1} (\alpha |\log u| + 1) du = O((\log n)^{-1}).$$

The remaining integral is exactly $\mathcal{J}_n(\alpha)\varphi(1)$, since $g_n = 0$ on $(0, 1/n)$. Replacing $\mathcal{J}_n(\alpha)$ by $\mathcal{J}(\alpha)$ costs at most $|\mathcal{J}_n(\alpha) - \mathcal{J}(\alpha)|$.

Adding (288)–(290), the coefficient of $\varphi(0)$ is $\mathcal{H}(\alpha) - \alpha$, and that of $\varphi(1)$ is $\alpha + \mathcal{J}(\alpha)$. The errors are uniform on the unit ball, proving (285). Finally lemma 54.2 proves convergence. $\square$

Applying this theorem with $\alpha = 1/2$ proves theorem 53.4. The same proof also shows exactly what quantitative information is missing for a rate: it is the single weighted integral discrepancy $\mathcal{J}_n(\alpha) - \mathcal{J}(\alpha)$, not a new pointwise local limit.

57.3 The topology is essential

For the permutation application, every η_n is supported on finitely many points $\log d/\log n$. Its critical normalization therefore has total-variation distance one from the continuous uniform distribution, with the convention $d_{TV}(\mu, \nu) = \sup_A |\mu(A) - \nu(A)|$ (Part III's). Weak or bounded-Lipschitz convergence is the correct conclusion.

Moreover, $B_0 < 0$ in this application. There is no contradiction with positivity of η_n : the signed residual after removing $\frac{3}{2\sqrt{\pi}} \log n$ Leb need not be positive. Nor does an atom in the correction mean that the original discrete measure has a limiting atom at that endpoint under probability normalization.

58 Tilted laws, logarithmic moments, and interpretation

58.1 A full first correction under power bias

For real λ define the probability measure $\mathbb{P}_n^\lambda$, with expectation $\mathbb{E}_n^\lambda$, by

$$\mathbb{E}_n^\lambda \varphi(\vartheta_n) = \frac{\mathbb{E}[X_n^{\alpha+\lambda/\log n} \varphi(\vartheta_n)]}{\mathbb{E}X_n^{\alpha+\lambda/\log n}}. \quad (291)$$

Let

$$\omega_\lambda(\varphi) = \frac{\int_0^1 e^{\lambda t} \varphi(t) dt}{\mathcal{M}(\lambda)}.$$

Corollary 58.1 (Tilted logarithmic-scale laws). *Uniformly for real λ in a compact interval and $\|\varphi\|_{BL} \leq 1$,*

$$\mathbb{E}_n^\lambda \varphi(\vartheta_n) = \omega_\lambda(\varphi) + \frac{B_0[\varphi(0) - \omega_\lambda(\varphi)] + B_1 e^\lambda [\varphi(1) - \omega_\lambda(\varphi)]}{\alpha \log n \mathcal{M}(\lambda)} + o((\log n)^{-1}). \quad (292)$$

In particular ϑ_n converges weakly under $\mathbb{P}_n^\lambda$ to the density $e^{\lambda t}/\mathcal{M}(\lambda)$.

Proof. Apply theorem 57.1 to $\varphi_\lambda(t) = e^{\lambda t} \varphi(t)$. Its bounded-Lipschitz norm is uniformly bounded for the stated range of λ and φ . This gives

$$\mathbb{E}[X_n^{\alpha+\lambda/\log n} \varphi(\mathfrak{D}_n)] = \alpha \log n \int_0^1 e^{\lambda t} \varphi(t) dt + B_0 \varphi(0) + B_1 e^\lambda \varphi(1) + o(1).$$

The denominator has the expansion (281) and has a positive leading coefficient uniformly bounded away from zero after division by $\log n$. Expanding the ratio to order $1/\log n$ gives (292). Bounded-Lipschitz convergence implies weak convergence on the compact interval $[0, 1]$. $\square$

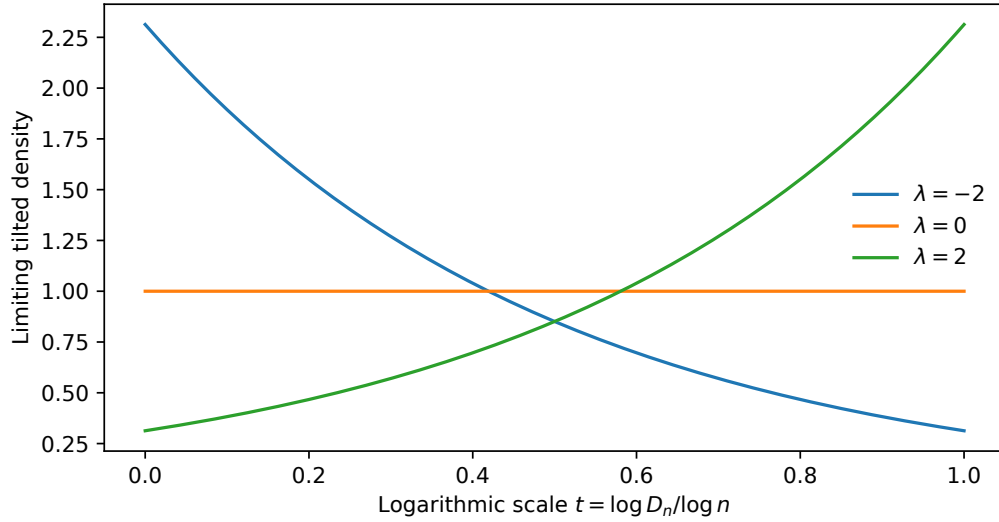


Figure 11: The limiting densities $e^{\lambda t} / \mathcal{M}(\lambda)$ (manuscript 06's $e^{\lambda t} / \Phi(\lambda)$), for $\lambda = -2, 0, 2$. At critical square-root bias, $\lambda = 0$ and the density is uniform. These curves are the exact functions in the theorem, not fitted data.

At critical bias, each fixed interval $(s, t) \subset (0, 1)$ receives limiting probability $t - s$. Equivalently, for $0 < s < t < 1$,

$$\mathbb{E}[X_n^\alpha; n^s < X_n < n^t] \sim \alpha \alpha(t - s) \log n. \quad (293)$$

This leading statement follows from weak convergence because interval boundaries have zero limiting mass. The second-order formula (292) is asserted for Lipschitz test functions, not for arbitrary discontinuous indicators.

58.2 Logarithm-weighted critical moments

Corollary 58.2 (All fixed logarithmic powers). *For each fixed integer $k \geq 1$,*

$$\mathbb{E}[X_n^\alpha (\log X_n)^k] = \frac{\alpha \alpha}{k+1} (\log n)^{k+1} + B_1 (\log n)^k + o((\log n)^k). \quad (294)$$

For $k = 0$ the corresponding constant is $B_0 + B_1$, as in (279).

Proof. Use $\varphi(t) = t^k$ in theorem 57.1 and multiply by $(\log n)^k$. Its integral is $1/(k+1)$, its value at zero is zero, and its value at one is one. No differentiation of an asymptotic expansion is needed. $\square$

For the permutation model the case $k = 1$ reads

$$\mathbb{E}[\sqrt{D_n} \log D_n] = \frac{3}{4\sqrt{\pi}} (\log n)^2 + B_1 \log n + o(\log n). \quad (295)$$

It also follows that the critical-biased mean scale satisfies

$$\mathbb{E}_n^0 \vartheta_n = \frac{1}{2} + \frac{B_1 - B_0}{\alpha \log n} + o((\log n)^{-1}). \quad (296)$$

Here $B_1 - B_0 > 0$. The slow positive correction helps explain the finite-size illustrations in section 61; the theorem does not predict monotonicity at moderate sizes.

58.3 Two different notions of typical behavior

Under the original uniform distribution on avoiders, D_n converges to a proper finite-valued variable D . It is therefore tight, and $\log D_n / \log n \rightarrow 0$ in probability. Under square-root bias, the same logarithmic ratio instead tends to a uniform distribution. These statements refer to different sampling measures and do not conflict.

The critical bias compensates exactly for the power-law decay: a scale increment $d(\log x)$ contributes asymptotically the same amount to $\mathbb{E}X_n^\alpha$. Positive λ in the moving exponent window favors larger logarithmic scales; negative λ favors smaller ones. Both still retain a nondegenerate distribution over the entire interval for fixed finite λ .

59 Sharpness and stability of the transfer mechanism

59.1 The uniform tail error cannot be discarded

The limiting law and the bulk profile alone do not determine the centered critical moment. An intermediate-scale perturbation can be invisible to both while leaving an order-one critical contribution.

Proposition 59.1 (A surviving intermediate-scale correction). *Fix $\alpha > 0$, $0 < \theta < 1$, and $c > 0$. There are random variables $Y_n \in [1, n]$ having the same distributional limit and the same pointwise bulk profile as the truncated Pareto variables of example 56.2, but*

$$\mathbb{E}Y_n^{\alpha+\lambda/\log n} = \alpha \log n \mathcal{M}(\lambda) + e^\lambda + ce^{\lambda\theta} + o(1) \quad (297)$$

locally uniformly for complex λ . Their critical logarithmic weighted residual converges in BL^ to $\delta_1 + c\delta_\theta$, not just the two-endpoint measure. They do not satisfy (267) with order $n^{-\alpha}$.*

Proof. For sufficiently large n , put $\varepsilon_n = cn^{-\alpha\theta} \leq 1$. With probability $1 - \varepsilon_n$, let Y_n have the law of $X_n = \min(X, n)$ from example 56.2; with probability ε_n , set $Y_n = n^\theta$. Define the finitely many smaller n arbitrarily in $[1, n]$.

Because $\varepsilon_n \rightarrow 0$, Y_n has the same limiting law as X_n . For fixed $0 < u < 1$, the added atom lies below nu for large n , so

$$n^\alpha \mathbb{P}(Y_n > nu) = (1 - \varepsilon_n)u^{-\alpha} \longrightarrow u^{-\alpha}.$$

At $u = 1$ both survival functions vanish. Thus the bulk profile is also unchanged. On the other hand, the new atom contributes

$$\varepsilon_n (n^\theta)^{\alpha+\lambda/\log n} = ce^{\lambda\theta}.$$

Removing the fraction ε_n of the original law changes its moment by $O(\varepsilon_n \log n) = o(1)$ uniformly on compact λ sets, by (283). This proves (297).

For the weighted-measure assertion, the critical contribution of the new atom is exactly $c\varphi(\theta)$, and deleting the corresponding fraction of the Pareto weighted measure costs $O(\varepsilon_n \log n) \|\varphi\|_\infty = o(1)$ uniformly. Finally, at $x = n^\theta/2$, for large n , the survival discrepancy from the untruncated Pareto law equals

$$\varepsilon_n (1 - x^{-\alpha}) \sim cn^{-\alpha\theta},$$

which is not $O(n^{-\alpha})$. □

This example explains the role of Part III's total-variation theorem. It excludes persistent critical weighted mass at intermediate logarithmic scales. The bulk profile by itself would not exclude such mass and would not resolve the centered half-moment question.

59.2 Robustness under smaller perturbations

Proposition 59.2 (Stability at the relevant error scale). *Suppose X_n satisfies the general hypotheses, $1 \leq Y_n \leq n$, and*

$$\delta_n := \sup_x |\mathbb{P}(Y_n > x) - \mathbb{P}(X_n > x)| = o(n^{-\alpha}).$$

Then the uniform transfer expansion, critical crossover, and signed bounded-Lipschitz endpoint correction hold for Y_n with the same functions $\mathcal{H}, \mathcal{J}$ and the same constants B_0, B_1 .

Proof. For p in a compact subset $\mathcal{D}$ of $\mathbb{R}^p > 0$, tail integration bounds

$$|\mathbb{E}Y_n^p - \mathbb{E}X_n^p| \leq |p|\delta_n \int_1^n x^{\mathbb{R}^p-1} dx \leq K_{\mathcal{D}}\delta_n n^{\mathbb{R}^p}.$$

After multiplication by $n^{\alpha-\mathbb{R}^p}$ this is $o(1)$ uniformly. For a test function in the unit bounded-Lipschitz ball, use (286) to bound the corresponding weighted expectation difference by $K\delta_n n^{\alpha} = o(1)$. These estimates transfer all the assertions directly. $\square$

The sufficient condition in this proposition is an error bound, not a claim that every natural modification of a permutation statistic satisfies it. Verifying the condition for a particular modification remains a separate problem.

60 Computing the constant without fitting finite permutations

60.1 A convergent discrete formula

Because D is positive-integer-valued, (253) also gives

$$c_{\text{mic}} = 1 + \sum_{k=1}^{\infty} \left[(\sqrt{k+1} - \sqrt{k})\mathbb{P}(D > k) - \frac{\alpha}{2} \log \frac{k+1}{k} \right]. \quad (298)$$

The summands are $O(k^{-2})$, by (247). Hence this is an absolutely convergent series, with truncation error $O(N^{-1})$ after N terms. Its probabilities are rational coefficients of the algebraic function G . It is an exact specification, although not the fastest numerical route. (The summand is the one Part III's proof of (92) estimates by $\frac{3}{2\sqrt{\pi}}k^{-1} + O(k^{-2})$.)

Equivalently,

$$c_{\text{mic}} = \lim_{N \rightarrow \infty} \left(\mathbb{E}\sqrt{\min(D, N)} - \frac{\alpha}{2} \log N \right). \quad (299)$$

This limit concerns a truncation of the *limiting* variable D , not a finite permutation variable D_n . Their centered constants differ by the macroscopic contribution c_{mac} .

60.2 A regularized Laplace integral

Set

$$\ell(t) = 1 - G(e^{-t}), \quad t > 0,$$

and define

$$\mathcal{R} = \int_0^1 \frac{\ell(t) - 3\sqrt{t}}{t^{3/2}} dt + \int_1^{\infty} \frac{\ell(t)}{t^{3/2}} dt. \quad (300)$$

Both integrals converge absolutely: (248) gives $\ell(t) = 3\sqrt{t} + O(t)$ at zero, and $0 \leq \ell(t) \leq 1$ at infinity.

Proposition 60.1 (A computable integral for the microscopic constant). *With Euler's constant γ ,*

$$c_{\text{mic}} = \frac{\mathcal{R}}{2\sqrt{\pi}} + \frac{\alpha}{2}(\gamma + 2 \log 2). \quad (301)$$

Proof. For $0 < p < 1$, integrating by parts after the substitution $u = tx$ gives

$$x^p = \frac{p}{\Gamma(1-p)} \int_0^\infty (1 - e^{-tx}) t^{-p-1} dt.$$

For $0 < p < 1/2$, Tonelli's theorem therefore yields

$$\mathbb{E}D^p = \frac{p}{\Gamma(1-p)} \int_0^\infty \ell(t) t^{-p-1} dt. \quad (302)$$

Subtract $3\sqrt{t}$ on $(0, 1)$. If $c(p) = p/\Gamma(1-p)$ and

$$\mathcal{R}(p) = \int_0^1 [\ell(t) - 3\sqrt{t}] t^{-p-1} dt + \int_1^\infty \ell(t) t^{-p-1} dt,$$

then $\mathcal{R}(p)$ is holomorphic for $0 < \Re p < 1$, with value $\mathcal{R}$ at $p = 1/2$, and

$$\mathbb{E}D^p = c(p)\mathcal{R}(p) + \frac{3c(p)}{1/2 - p}.$$

Compare this with (276):

$$\mathcal{H}(p) = c(p)\mathcal{R}(p) + \frac{3c(p) - \alpha p}{1/2 - p}.$$

The numerator in the quotient vanishes at $p = 1/2$. Its derivative there is $\frac{\alpha}{2}\psi(1/2)$, where $\psi = \Gamma'/\Gamma$, since $c'(p) = \Gamma(1-p)^{-1}[1 + p\psi(1-p)]$. Consequently the quotient tends to $-\frac{\alpha}{2}\psi(1/2)$. The duplication formula for Γ , differentiated at $1/2$, gives $\psi(1/2) = -\gamma - 2 \log 2$. Finally $c(1/2) = 1/(2\sqrt{\pi})$, proving (301). $\square$

60.3 Stable evaluation near zero

A naive evaluation of $1 - G(e^{-t}) - 3\sqrt{t}$ loses digits near zero. The verification code instead uses the algebraically equivalent form

$$\ell(t) = \frac{s[12 + (25+r)s + (11-r)s^2]}{4(1+2s)(1+s)^2}, \quad s = \sqrt{-\expm1(-t)}, \quad r = \sqrt{(1+s)/2}. \quad (303)$$

Writing $t = v^2$, it subtracts $3v$ by the identity

$$\ell(v^2) - 3v = 3(s - v) + \frac{s^2[(r - 23) - (49 + r)s - 24s^2]}{4(1+2s)(1+s)^2}. \quad (304)$$

The factor $s - v$ is rationalized; the remaining exponential cancellation can be removed by

$$e^{-x} - 1 + x = \frac{x^2}{2} {}_1F_1(1; 3; -x),$$

whose identity follows term by term from the exponential series. The transformed integrand $2[\ell(v^2) - 3v]/v^2$ has finite endpoint limit $(1/\sqrt{2} - 23)/2$.

For the second integral in (300), the code uses

$$\int_1^\infty \ell(t) t^{-3/2} dt = 2 - \int_1^\infty G(e^{-t}) t^{-3/2} dt.$$

The last integrand decays exponentially. These rearrangements make the reported numerical agreement meaningful; they are not an interval-based error proof.

Table 20: Constants evaluated from convergent integrals. The displayed decimals are numerical approximations, not certified enclosures.

Quantity	Approximation
c_{mic}	0.367948538497933
c_{mac}	-1.310069952497335
$c_{\star} = c_{\text{mic}} + c_{\text{mac}}$	-0.942121413999402
$B_0 = c_{\text{mic}} - \alpha$	-1.324620212145336
$B_1 = \alpha + c_{\text{mac}}$	0.382498798145934
$m(1/4)$	-2.575500874573932

61 Exact finite checks and numerical illustrations

61.1 Counting rather than sampling

The manuscript’s code computes exact integer distributions of D_n through $n = 64$. It uses the endpoint-threshold recurrence derived in section 64, not Monte Carlo sampling. Independent recursive generation of every 132-avoider through size ten agrees with the resulting deficit histograms. For sizes through seven, a separate test of the defining 132 condition on all permutations verifies the recursive generator.

The verification run records:

Check	Count or range
Definition-based avoiders checked	625 (sizes 1–7)
Generated avoiders checked against histograms	23 713 (sizes 1–10)
Endpoint-state equalities checked	784
Exact deficit distributions	sizes 1–64
Exact PGF coefficients checked	129 (degrees 0–128)
Symbolic algebraic identities	5

The integer totals equal the Catalan numbers at every computed size. The PGF coefficients are exact rational numbers; multiplication by the algebraic denominator is checked coefficient by coefficient, and the coefficient of z is exactly $7/48$. (Part III’s data/03-largest-jump-limit_probabilities.csv records the same probabilities $\mathbb{P}(D = k)$ for $k \leq 256$, computed there independently.)

61.2 Independent numerical regressions

The constant calculation was repeated at 40- and 70-decimal working precision. The two computations agreed to more than 29 decimal places in the recorded regression check. The elementary formula for c_{mac} was checked against a separately evaluated convergent integral. The value of c_{mic} was also compared with the symmetric removable limit of the regularized fractional-moment expression at $p = 1/2 \pm 10^{-6}$; its observed difference was about 2.68×10^{-12} .

A different, slower check uses the first 128 exact PGF coefficients in (299). It gives 0.377930026226763 for the truncated microscopic constant, about 0.00998149 above c_{mic} , consistent with its $O(1/N)$ truncation scale. No interval enclosure is inferred from that comparison. The pole finite part of m was also checked numerically against B_1 , and $m(1) = \alpha$ was verified at high precision as a normalization check. (That is Part IV’s mean constant, $m_1 = 3/\sqrt{\pi}$.)

61.3 What the finite data actually show

These sizes are not close enough to identify the limiting constant by inspection. In particular, the critical-biased mean scale is moving away from $1/2$ in the displayed range even though the

Table 21: Finite-size moments from exact integer distributions, rounded for display. The last row lists the proved limits, not fitted estimates.

n	$\mathbb{E}\sqrt{D_n} - \frac{3}{2\sqrt{\pi}} \log n$	$n^{1/4}(\mathbb{E}D_n^{1/4} - \mathbb{E}D^{1/4})$	$\mathbb{E}_n^0 \vartheta_n$
8	-0.1296061114	-2.0053474893	0.5061596109
16	-0.2961230527	-2.0986386677	0.5651776479
32	-0.4402451849	-2.1871612273	0.585553734
64	-0.5577086463	-2.2648950027	0.5922137801
∞	-0.9421214140	-2.5755008746	0.5

theorem proves its convergence to $1/2$. The data test the implementations and illustrate slow finite-size behavior. They are not evidence sufficient to prove the asymptotic formulas, and no coefficient in a theorem was fitted to them.

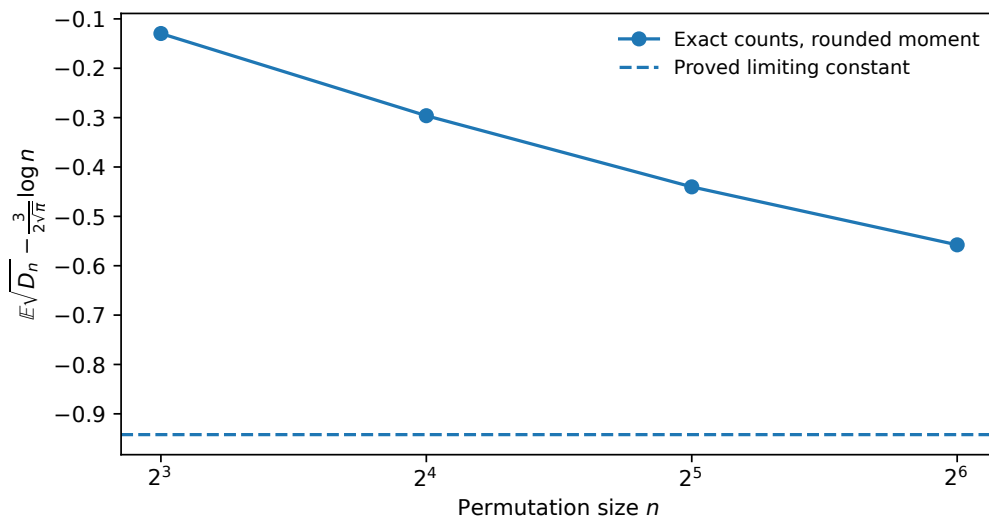


Figure 12: The centered square-root moment at the four computed sizes and its proved limiting constant. The substantial remaining gap illustrates why numerical fitting was not used to derive $c_\star$.

61.4 Rebuilding and rerunning

Part V's files are

code/06-critical-moments-exact_counts.py	Exact distributions, exhaustive comparisons, rational PGF coefficients
code/06-critical-moments-verify_symbolic.py	Five exact algebraic identity checks
code/06-critical-moments-numerics.py	Stable quadrature, independent numerical regressions, finite moments
code/06-critical-moments-figures.py	The three figures of this part
code/06-critical-moments-build.sh	The manuscript's three-pass build
data/06-critical-moments-*	Recorded checks, exact counts and coefficients, numerical output, requirements, delivery PDF audit
figures/06-critical-moments-*.pdf	The three figures of this part
06-critical-moments-SOURCES.md	Pinned source paths, literature scope
06-critical-moments-PROOF_STATUS.md	Trust boundary and statement ledger

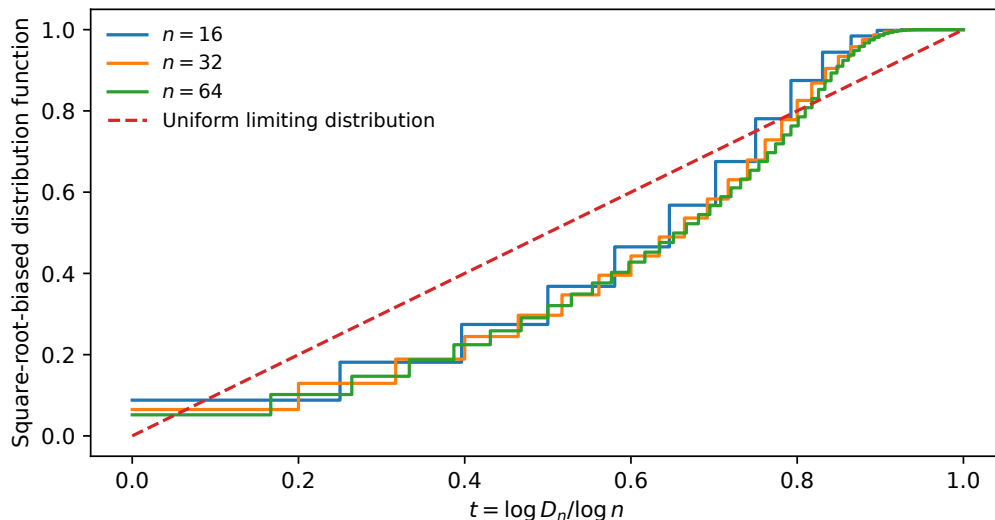


Figure 13: Critical-biased logarithmic distribution functions of ϑ_n (the axis label's $t = \log D_n / \log n$) from exact counts, for $n = 16, 32, 64$. The limiting distribution is uniform, but finite distributions are atomic and their convergence is not in total variation.

The scripts are shipped byte-identical to the delivery and still use the delivered names: they read and write `data/exact_distributions.json`, `data/pgf_coefficients.json`, `data/exact_checks.json`, `data/symbolic_checks.json`, `data/constants.json`, `data/finite_moments.csv` and unpre-fixed files in `figures/`, and the build script looks for an `article.tex` beside itself. The report's README gives a recipe that restores the delivered layout in a scratch copy; there, the manuscript's instructions apply,

```
python code/exact_counts.py --max-n 64 --pgf-terms 128
python code/verify_symbolic.py
python code/numerics.py
python code/figures.py
```

and rerunning overwrites only the copy's recorded outputs. The counting program uses the Python standard library. The symbolic, quadrature and figure programs additionally use SymPy, mpmath, NumPy and Matplotlib; the versions used are recorded in `data/06-critical-moments-requirements.txt`. None of these computations is connected to a proof-assistant checker. Their output certifies only that the indicated finite assertions passed in the execution environment. The retained figures make Python unnecessary for rebuilding the article.

62 What has been settled, and what has not

The two explicit questions answered here are convergence of the centered critical half-moment and the first correction to every fixed positive subcritical moment (section 47.6). Their proofs are consequences of exactly the three inputs of section 52. The new logarithmic-scale statements organize the same information into a distributional picture, and the general theorem makes that mechanism reusable outside the permutation problem.

Several stronger assertions would exceed the proof. There is no new local limit for individual lattice masses, no exact leading constant in the original total-variation error, and no description of the adjacency-bound transition at $m = n/2$ [added 30 September 2026, batch 68: Part VI describes it at the level of orders, corollary 68.4; added 1 October 2026, batch 71: Part VII proves its limiting function, theorem 80.1]. The coefficient function $\mathfrak{m}(p)$ is analytically continued below $1/2$, but

the moment theorem is not extended to $p \leq 0$ on that basis. A continuation of a coefficient formula does not by itself continue an asymptotic theorem.

The complex crossover is locally uniform for fixed compact sets of λ , not uniform as $|\lambda| \rightarrow \infty$. For fixed supercritical p , the present transfer theorem does not supply an $o(1)$ -accurate second-order constant. The numerical constants are exact only through their convergent definitions; their printed decimal approximations have not been enclosed by interval arithmetic. Discrete biased laws do not converge to their continuous limit in total variation, and the negative endpoint coefficient B_0 has no positive-measure interpretation. No elementary closed form, rationality or transcendence result is claimed for the microscopic constant.

Finally, no proof of Part V has been checked by Lean or Rocq, and neither the advance relative to Parts III and IV nor a bounded literature search constitutes external validation of publication priority. These limits do not weaken the stated abstract implications; they identify precisely the boundary of the contribution.

63 Further research questions and proposed routes

The manuscript proposes ten questions. Four overlap with questions of Part IV; they are printed here with pointers to Part IV's text, which is not repeated.

63.1 A rate for the centered half-moment

The exact decomposition reduces the rate problem to $\mathcal{J}_n(1/2) - \mathcal{J}(1/2)$, up to an $O(n^{-1})$ microscopic tail error. Can one prove an explicit bound, or an asymptotic equivalent, for that weighted macroscopic discrepancy? A promising route is to quantify Part III's stopping-word or Part IV's maximum-decomposition argument in a weighted integral norm rather than demand a uniform relative tail estimate at every scale. Such a result would immediately sharpen both (255) and (285). (This refines, at the critical exponent, Part IV's question on second-order moment asymptotics, section 47.7.)

63.2 A local limit and the exact total-variation constant

The derivative $\rho(u)$ suggests a lattice approximation for $\mathbb{P}(D_n = d)$ when d/n stays inside $(0, 1/2)$. Proving it requires more than differentiation of the existing tail limit. Exact endpoint-state counts may identify a useful discrete difference equation. Even a local bulk theorem would not alone determine the total-variation constant; one must control all remaining scales and the signs of probability-mass errors. Part V does not identify that constant. (These are Part IV's questions section 47.4 and section 47.5, asked there by both of its manuscripts.)

63.3 The half-size threshold

The leading bulk profile vanishes at $u = 1/2$, corresponding to adjacency bound $m = n/2$. What scale and limiting function describe the transition there? The endpoint expansion in (251) is linear from below, but gives no order on or beyond the threshold. A refined terminal split analysis, retaining lattice offsets and two-large-block events, seems more suitable than the present integrated method. (This is Part IV's question section 47.2.) [Added 30 September 2026, batch 68: Part VI answers the scale. The order on the threshold is n^{-1} , the order changes on the scale $\sqrt{n}$, and $\mathbb{P}(M_n \leq \lfloor n/2 + t\sqrt{n} \rfloor) \asymp n^{-1}(1 + t_+)$ for bounded t (corollary 68.4); the limiting function stays open (section 74.3).] [Added 1 October 2026, batch 71: Part VII proves the limiting function, $n \mathbb{P}(M_n \leq \lfloor n/2 + t\sqrt{n} \rfloor) \rightarrow 6 + (12/\sqrt{\pi})t_+$ locally uniformly in t (theorem 80.1).]

63.4 Wider moving-exponent windows

Can (261) be made uniform for $|\lambda| \leq L_n$ with $L_n \rightarrow \infty$? Such a theorem would match the critical window to fixed subcritical or supercritical regimes. The general proof exposes two

issues: controlling the range of p in $\mathcal{J}_n(p) - \mathcal{J}(p)$, and ensuring that its error remains small after multiplication by e^λ . A quantitative estimate from the first question is an essential ingredient.

63.5 Negative exponents and logarithms without critical bias

For $p \leq 0$, the integral involving $b(u)$ is no longer controlled merely by boundedness at zero. The explicit limiting profile has a better small- u expansion, but the finite discrepancies need a matching uniform estimate. Can one prove that stronger estimate, and thereby determine corrections to $\mathbb{E} \log D_n$ or $\mathbb{E} D_n^p$ for negative p ? This is a separate boundary-layer problem, not an automatic consequence of (256).

63.6 Geometry of a permutation under critical bias

The theorem identifies the scale of the deficit but not the conditioned permutation. Under square-root bias, condition further on $\log D_n / \log n \in (s, t)$. What happens to the location of the largest jump, the relative ascent/descent contribution, or the sizes along the maximum-decomposition path? A joint extension of Part IV's marked macroscopic measure (theorem 38.2) could lead to a scale-indexed geometric limit. (This is the critical-bias counterpart of Part IV's question section 47.8, which conditions on a macroscopic deficit instead.)

63.7 Other Catalan models and other tail exponents

The abstract theorem permits arbitrary $\alpha > 0$. Which statistics on Catalan trees, other pattern classes, or conditioned branching structures satisfy its three hypotheses? The crucial target is a uniform $O(n^{-\alpha})$ survival error, not merely convergence in distribution. The counterexample in proposition 59.1 provides a diagnostic for proposed generalizations: hidden intermediate-scale mass must be excluded.

63.8 Higher-order endpoint corrections

The first signed correction consists of two point masses. A next term might involve derivatives of endpoint evaluations, a regular measure on $[0, 1]$, or slower scales, depending on the rate and shape of $g_n - b$. Determine the appropriate test-function topology and the next expansion before guessing its support. The direct three-integral proof provides a natural starting point, with its $1/\log n$ terms still visible.

63.9 Certified constants and their analytic nature

The formulas for c_{mic} and $c_\star$ are absolutely convergent and amenable to interval quadrature. A useful computational continuation would supply a rigorous decimal enclosure, with explicit small- t and large- t error bounds. A more theoretical question asks whether c_{mic} admits a simpler special-function expression. No elementary closed form, rationality or transcendence claim is suggested by the present decimal computation.

63.10 Formalization in the repository

A manageable first Lean target is the abstract theorem for real exponents on a fixed compact interval, followed by the critical half-moment and the bounded-Lipschitz estimate. The proof uses tail integration, dominated convergence, elementary power integrals, and uniform bounds; it need not initially formalize complex holomorphy. The application then requires formal counterparts of the three probabilistic inputs from Parts III and IV. Keeping these interfaces separate would make proof status explicit and permit independent replacement of any input argument. (Part IV's formalization targets are in section 47.13; no Lean or Rocq declaration for any part of this report exists.)

64 The exact endpoint recurrence used by the checks

This section was the manuscript's Appendix A. It gives the finite combinatorics used by the verification code. The recurrence is inherited from the endpoint-state approach of Mayama–Akita [2] and is Part I's proposition 2.1, in the coefficient form Part IV prints as (242); the first-entry count is Part III's (98). The manuscript derives both again, by the same maximum decomposition as Parts I and III, to make its checks reproducible independently of the repository's code, and its derivation is kept here in Part IV's notation (table 19).

64.1 Maximum decomposition and first entries

Every nonempty 132-avoider has a unique decomposition

$$\pi = (\sigma + b) n \tau, \quad \sigma \in \text{Av}_a(132), \quad \tau \in \text{Av}_b(132), \quad a + b = n - 1. \quad (305)$$

All left entries must exceed all right entries: otherwise a left entry, the maximum, and a larger right entry form a 132 pattern. Conversely, this condition and avoidance in the two blocks ensure avoidance globally. It follows that the recursive generator is exhaustive and has no duplicates, and yields the Catalan recurrence. (This is Part I's maximum decomposition, section 2.)

Let $\mathcal{N}_n(u)$ count unrestricted size- n avoiders with first deficiency $n - \pi_1 \leq u$. Set $\mathcal{N}_n(u) = 0$ for $u < 0$ and $\mathcal{N}_0(u) = 1$ for $u \geq 0$. For $1 \leq f \leq n$, the number with first entry f is

$$N_{n,f} = \frac{n - f + 1}{n} \binom{n + f - 2}{n - 1}, \quad (306)$$

Part III's (98). Here is the manuscript's generating-function proof, which also checks the endpoint convention. Write $C(x) = \sum_{n \geq 0} C_n x^n$, so $C(x) = 1 + xC(x)^2$, and for $j \geq 0$ put

$$\mathcal{G}_j(x) = \sum_{n \geq j+1} N_{n,n-j} x^n.$$

When the first deficiency is zero, the maximum is first and the rest is an arbitrary avoider, so $\mathcal{G}_0 = xC(x)$. When the first deficiency is $j \geq 1$, the left block in (305) is nonempty. Its first entry has deficiency $j - 1$ inside that block, while the right block is arbitrary. Thus $\mathcal{G}_j = xC(x)\mathcal{G}_{j-1}$, giving

$$\mathcal{G}_j(x) = (xC(x))^{j+1}.$$

Lagrange inversion applied to $C - 1 = xC^2$ gives, for $k \geq 0$,

$$[x^k]C(x)^{j+1} = \frac{j+1}{2k+j+1} \binom{2k+j+1}{k}.$$

Substitute $k = n - j - 1$ and simplify the binomial coefficient. The result is (306) with $f = n - j$. (Part III reaches $(xC)^{j+1}$ through the bivariate series (95) and (97); the manuscript's one-line recursion in j is the same decomposition read one deficiency at a time.) Thus

$$\mathcal{N}_n(u) = \sum_{j=0}^{\min(u, n-1)} \frac{j+1}{n} \binom{2n-j-2}{n-1}, \quad n \geq 1. \quad (307)$$

The program checks that each complete row sums to C_n .

64.2 Two endpoint thresholds

Let $T_{u,v}^{(m)}(n)$ count size- n avoiders with every adjacent difference at most m , first deficiency at most u , and last deficiency at most v (Part I's $T_{p,q}^{(m)}(n)$ with finite thresholds). Negative thresholds give zero, $T_{u,v}^{(m)}(1) = 1$ for nonnegative thresholds, and thresholds above $n - 1$ are clipped to $n - 1$.

For $n \geq 2$ one has

$$\begin{aligned} T_{u,v}^{(m)}(n) &= T_{u-1,m-1}^{(m)}(n-1) \\ &\quad + \sum_{k=1}^{\min(m,n-1)} c_{k,u} T_{m-k,v-k}^{(m)}(n-k), \\ c_{k,u} &= \begin{cases} 1, & k = 1, \\ \mathcal{N}_{k-1}(u-1), & k \geq 2, \end{cases} \end{aligned} \quad (308)$$

which is (242). The unrestricted adjacency count is $a_n^{(m)} = T_{n-1,n-1}^{(m)}(n)$.

Proof of (308). If the maximum is last, deleting it changes the first deficiency bound to $u - 1$. The final jump has size $n - \pi_{n-1}$, so the old last deficiency must be at most $m - 1$. The new last deficiency is zero, explaining the first term.

Otherwise let the maximum be at position k , so the right block has size $b = n - k \geq 1$. The jump from the maximum to its following entry is at least k , hence $k \leq m$. The left block has size $k - 1 \leq m - 1$, so its internal jumps and its jump to the maximum automatically meet the adjacency bound. When nonempty, its first deficiency must be at most $u - 1$; it contributes $\mathcal{N}_{k-1}(u - 1)$. If it is empty, the first entry is the maximum, and the coefficient is one.

The jump from the maximum to the first entry f of the standardized right block has size $n - f = k + (b - f)$, so its first deficiency is at most $m - k$. The final deficiency of the whole permutation is k plus the right-block last deficiency, so that bound is $v - k$. This gives the summand. Uniqueness of (305) makes all cases disjoint. $\square$

The manuscript's code uses this recurrence for every size, without the faster unrestricted-state shortcut (243) of Part IV's code/05-macroscopic-deficits-model.py, which it inspected. It is therefore intended as a finite verification implementation, not a claim of optimal enumeration complexity. It explicitly clears the memoization cache when the adjacency bound changes.

65 Proof and computation audit

This section was the manuscript's Appendix B.

Statement	Dependency	Verification status
Algebraic limiting law and tail	Part III, theorems 22.1 and 22.2	Adopted theorem of Part III; not formalized
Uniform survival error	Part III's total-variation theorem, (85)	Immediate half-line consequence; Part III not formalized
Bulk profile	Part IV, theorem 38.1	Adopted theorem of Part IV; not formalized
Uniform moment transfer	Three abstract hypotheses	Complete proof in section 54
Centered half-moment	Transfer theorem and explicit profile	Complete deduction; convergent exact constant

Statement	Dependency	Verification status
Subcritical correction	Transfer theorem and regularization	Complete deduction; negative sign proved
Complex critical window	Locally uniform transfer	Complete proof; compact windows only
Signed endpoint measure	Three abstract hypotheses	Complete proof in dual bounded-Lipschitz norm
Tilted limits and logarithmic powers	Endpoint theorem	Complete proofs by normalization and test functions
Intermediate-scale counterexample	Truncated Pareto mixture	Explicit construction and proof
Exact finite counts	Endpoint recurrence and independent generation	Integer regression checks through the stated ranges
Constant decimals	Convergent quadrature formulas	Precision and alternate-form checks; not interval certified
Lean/Rocq endpoint	Not supplied	No formal verification claimed

An audit should check in particular the following possible failure points. The leading tail coefficient is $\alpha = 3/\sqrt{\pi}$, but the critical logarithm coefficient is $\alpha/2$. The finite part of $m(p)$ at its pole is $B_1 = \alpha + c_{\text{mac}}$, not c_{mac} . The microscopic constant c_{mic} and the finite-permutation constant $c_\star$ differ. The negative endpoint coefficient is a signed correction, not a negative probability. The complex moment theorem has an explicit strip, and its fixed-supercritical error cannot be silently strengthened. Finally, the finite numerical checks have no bearing on the validity of dominated convergence beyond their stated regression role. The incomplete beta expression is regularized below $p = 1/2$; no divergent integral is treated as an ordinary integral, and complex powers always use the real logarithm of a positive base (06-critical-moments-PROOF_STATUS.md).

66 Repository pin, literature scope and merge

66.1 Manuscript 06's pin and literature scope

The manuscript records the ProveIt commit `cb1646dc442724f8e298cf259195b6c308367d80`, resolved through GitHub's main-branch reference on 30 September 2026, and the report directory

```
SetTheory/Cardinals/docs/reports/
enumerative-combinatorics/
adjacency-bounded-132-avoiders/.
```

It inspected three files there, with the Git blobs

```
README.md          9b6865915b549427272025beaf2826d59131b6d1
article.tex        c673f447a22396f5c66b36b4ce8141acc0ceee3d
code/05-macroscopic-deficits-model.py f0248e0d6181200257e2949c1a8fa8f9fecf5617
```

The README's list of what Parts III and IV do not claim names the centered half-moment and the subcritical corrections; the article supplied Parts III and IV; the model's endpoint-state recurrence and first-entry formula informed the manuscript's independent verification code. The interfaces it names are Part III's theorem 22.1, section 27, section 28, section 26 and corollary 28.2 (it does not assume the exact total-variation constant) and Part IV's theorem 38.1; the open problems it cites are section 47.6, Part IV's scope discussion (section 37.3) and the README's

list. It requires no result about the fixed- m growth constants of Parts I and II. The full ledger is 06-critical-moments-SOURCES.md.

The primary arXiv records and relevant HTML of [1, 2] were checked on September 30, 2026. They provide problem context and the endpoint-state enumeration method; the largest-jump critical-moment results are not attributed to them. The investigation compared the report's explicit list of open questions with the new deductions and made targeted public searches for the relevant permutation and critical-moment terms. This does not establish that the abstract transfer principle, the constants, or related conclusions have never appeared elsewhere. There is no claim of exhaustive bibliographic coverage, external peer review, or established publication priority; claims of progress are relative to the pinned report. No complete report or external paper is redistributed.

66.2 Where the merge had to choose

- *Placement.* Part V follows Part IV and precedes Part I's appendices, which contain no numbered equations, tables or figures; no number of Parts I–IV changed. The manuscript's Appendices A and B are printed as sections 64 and 65; its Appendix C (package map) and its reproduction subsection are replaced by section 61.4. Its title page and abstract are replaced by the opening note of Part V, the status box of section 51.1 and a paragraph of the report's abstract; every claim of its abstract is stated in section 53. The delivery's title-page and metadata wording naming the assistant's addressee is not printed; the manuscript is described here only as AI-assisted.
- *Introduction.* The manuscript's Section 1.1 ("A specific gap, rather than a new encoding of an old theorem") and 1.2 ("Version boundary and inherited status") describe this report from outside, by pin and README lines. They are rewritten as the second and third subsections of section 51, citing Parts III and IV by label; the pin and file record moved to the previous subsection. Its contribution table is table 18.
- *Duplicates with Parts III and IV.* The three inputs of section 52 are results of Parts III and IV and are printed again as the interface of Part V, each with its source; the manuscript's "Part IV, Theorem 38.1 of the pinned report" is theorem 38.1. The expansion (248) is Part III's, the density (252) Part IV's. The maximum decomposition, the first-entry count and the endpoint recurrence of section 64 are Part I's, Part III's and Part I's (as Part IV's (242)); the manuscript's derivations are kept there, marked, because its checks were built on them.
- *What Part V sharpens.* Theorem 53.1 sharpens Part III's (92), theorem 53.2 its (91), and lemma 55.2 Part IV's pole (238). The manuscript's sentence that its coefficient function is the supercritical leading coefficient is proved as an identity with Part IV's m_p in remark 55.3 (the merge's). Nothing in Parts I–IV is refuted.
- *Notation.* Every rename is listed in table 19: $\mathcal{K}(p) \rightarrow m(p)$, $h \rightarrow c_{\text{mic}}$, $j \rightarrow c_{\text{mac}}$, $a \rightarrow \alpha$ (the tail coefficient; the block sizes a, b keep their letters), $A_0 = a/2 \rightarrow \frac{3}{2\sqrt{\pi}}$ (written out), $\Phi \rightarrow \mathcal{M}$, $T_n \rightarrow \mathfrak{S}_n$, $Q_n^\lambda \rightarrow \mathbb{P}_n^\lambda$, $\mathbb{E}_n^\lambda \rightarrow \mu_\lambda$, $\mu_\lambda \rightarrow \omega_\lambda$, $m \rightarrow \text{Leb}$, $S, S_n, S_Y \rightarrow \mathcal{S}, \mathcal{S}_n, \mathcal{S}_Y$, $r(x) \rightarrow \varrho(x)$, $b \rightarrow \mathfrak{b}$, $H, J, J_n \rightarrow \mathcal{H}, \mathcal{J}, \mathcal{J}_n$, $\mathcal{A}, \mathcal{A}(p) \rightarrow \mathcal{R}, \mathcal{R}(p)$, $c(p) \rightarrow \mathfrak{c}(p)$, the compact set $K \rightarrow \mathcal{D}$, the constants $C, C_K \rightarrow K, K_{\mathcal{D}}$, $P = \sup |p| \rightarrow p^*$, $L = \log n$ (written out), $\psi \rightarrow \varphi_\lambda$ in the proof of corollary 58.1 (the manuscript's ψ is also the digamma function), $F(u) \rightarrow \mathcal{F}(u)$ in the proof of lemma 55.1, $C(x) \rightarrow \mathcal{C}(x)$, $h(n, u) \rightarrow N_n(u)$, $A_j \rightarrow \mathcal{G}_j$, $c_k(u) \rightarrow c_{k,u}$, $T_m(n, u, v) \rightarrow T_{u,v}^{(m)}(n)$; the macros $\mathbb{E}, \mathbb{P}, d_{\text{TV}}, \mathcal{L}$ are Part III's. No normalization changed.
- *Labels.* The manuscript's 102 labels are kept with the prefix `cjm:`; its `eq:h` and `sec:computation` would otherwise collide with Part I's. The merge added labels to eight new headings, tables and remarks of Part V and to six unlabelled question headings of Part IV (sections 47.2, 47.4, 47.5, 47.7, 47.8 and 47.13) so that Part V can cite them.
- *Research questions.* All ten are printed. Four overlap with Part IV: the local limit and total-variation constant (Part IV's sections 47.4 and 47.5), the half-size threshold (section 47.2), and

formalization (section 47.13); two bear on Part IV's questions on second-order asymptotics and the conditioned geometry. They keep the manuscript's text with a pointer. Part IV's section 47.6, and the critical half of Part III's Section 32.2, carry dated notes that Part V answers them.

- *The merge's own text.* The opening note, section 51.1, the notation subsection and table, the sentences after theorems 53.1, 53.2 and 53.4, remark 55.3, this section's first subsection (condensed from the manuscript's 06-critical-moments-SOURCES.md and its Section 1.2), the bracketed or parenthesized merge notes, and this list are the merge's.
- *Reproduction and figures.* The manuscript's package listing and commands are replaced by the shipped names and the copy recipe (section 61.4 and the README). The three figure PDFs were regenerated in the write phase by the delivered figures.py, run unchanged on a copy of the delivered layout with Matplotlib 3.10.8, NumPy 2.3.5 and pdf.fonttype 42, so that they embed TrueType rather than Type 3 fonts; they plot the shipped data. Their axis labels keep the manuscript's $t = \log D_n / \log n$; the captions say that this is $\mathfrak{g}_n$.
- *Stale sentences elsewhere.* The abstract gained a paragraph; Part III's Section 32.2, Part IV's scope (section 37.3), its account of what neither manuscript claims, its remark after proposition 44.1, its question section 47.6 and its scope table, and Part I's section B, carry dated pointers to Part V; their original wording is kept.

Part VI. A staircase of polynomial rarity

Part VI was added on 30 September 2026 in batch 68 of ProveIt's incoming-report intake, from one manuscript written against the state of Parts I–V printed above. Part IV found the first-order profile of $\mathbb{P}(M_n \leq \lfloor \theta n \rfloor)$ for $1/2 < \theta < 1$; below half size it proved only $o(n^{-1/2})$ and polynomial lower bounds (theorems 38.1 and 45.1). It asked for the hierarchy below half size (section 47.1) and for the order and the window at $m = n/2$ (section 47.2), and Part V asked the second question again (section 63.3). Part VI answers the exponent part of the first and the order-level part of the second: the probability has the exact polynomial order $n^{-\lfloor 1/\theta \rfloor/2}$ for every fixed $0 < \theta < 1$, and the order changes at every reciprocal threshold $1/q$ across an explicit order-crossover scale. Its proofs use no asymptotic theorem of Parts I–V.

67 Provenance, conventions and the answer

67.1 One manuscript, one addition

Part VI is built from one AI-assisted research manuscript, *manuscript 05* of batch 68, *A Staircase of Polynomial Rarity: Bulk exponents and reciprocal-threshold crossover scales for adjacency-bounded 132-avoiding permutations*, dated September 30, 2026 (25 pages, US Letter, as delivered). It arrived as the archive `ProveIt_Polynomial_Rarity.zip` (inner directory `ProveIt_Polynomial_Rarity/`, main file `article.tex`) in ProveIt commit `a866ff9a2`, and was placed in commit `29e52fcdf`. It was written against ProveIt commit

`ffddaa8b9c89e7bf027e1442cc6216bb010906d0`,

at which this report's `README.md`, `code/model.py` and `code/05-macroscopic-deficits-model.py` had the blobs `a40d20342f`, `ae5528c72` and `f0248e0d61`, the ones the manuscript records; the manuscript did not read this report's `article.tex`, whose blob at the pin was `0b7cb2f2e6`. All four files were unchanged at the placement commit, so every statement the manuscript makes about this report refers to Parts I–V as printed above. Its code, data, figures and source ledger are shipped beside the earlier parts' files with the prefix `07-polynomial-rarity-`, continuing the report's local sequence after Part V's `06-critical-moments-`. Its manuscript, README and PDF are not shipped and survive in the arrival commit; its checksum ledger was verified at placement (15 of 15 files) and retired. Its two figure PDFs were regenerated in the write phase (section 78.2).

The manuscript contributes, in the order printed here: the complete fixed-fraction polynomial-order staircase, reciprocal endpoints included (theorem 68.1); a uniform two-term comparison near every reciprocal threshold (theorem 68.2), with its order-crossover scales (corollary 68.3), in particular at half size (corollary 68.4), and a continuum of logarithmic exponents in shrinking neighbourhoods (corollary 71.2); a local theorem for truncated subcritical renewal sequences for every power-tail index $0 < \alpha < 1$ (theorem 70.1) with its two lemmas, and a renewal-sandwich transfer principle (proposition 72.1); the two counting envelopes it transfers through, which are Part I's, reproved in the normalization it needs (section 69); a discussion of robustness and limitations; exact finite checks through $n = 128$; ten research questions; a derivation of the endpoint recurrence of its checks; and a proof audit. Every result, proof, example, remark, limitation and question of the manuscript is printed.

Status (manuscript 05 of batch 68). The theorems of Part VI have written mathematical proofs. They are not Lean- or Rocq-verified, have not been independently refereed, and no literature priority is claimed. Unlike Part V, Part VI uses no asymptotic theorem of Parts I–V as an input: it needs only the maximum decomposition, an explicitly verified injection (both Part I's, reproved here), elementary power-sum bounds and Catalan asymptotics. Its con-

clusions are matching two-sided *orders*, not asymptotic equivalents: no amplitude, endpoint constant, limiting crossover function or conditioned-permutation geometry is determined. The finite data and the figures are regression checks and illustrations, not evidence for the asymptotic statements. [Added 1 October 2026, batch 71: Part VII determines the amplitude for $1/3 < \theta < 1/2$, the endpoint constant $\mathfrak{K}_2 = 6$ and the crossover function $\mathfrak{S}_2$ at half size (theorems 80.1 and 80.2).] [Added 1 October 2026, batch 72: Part VIII determines every amplitude, endpoint constant and crossover function for fixed indices, and the conditioned law of the parser's macroscopic pieces (theorems 91.1 and 91.2 and corollary 98.1).]

67.2 The questions of Parts IV and V, and what answers them

For fixed $1/2 < \theta < 1$, Part IV proves the equivalent (153), $\mathbb{P}(M_n \leq \lfloor \theta n \rfloor) \sim \frac{3(2\theta-1)}{\sqrt{\pi\theta(1-\theta)}} n^{-1/2}$.

For $0 < \theta \leq 1/2$ it proves only $o(n^{-1/2})$ (theorem 38.1) and the lower bound $c n^{-r/2}$ whenever $(r+1)\theta > 1$ (theorem 45.1). Its question section 47.1 asks whether, for nonreciprocal θ and $r = \lceil 1/\theta \rceil - 1$, there is a positive $\Omega_r(\theta)$ with $\mathbb{P}(M_n \leq \theta n) \sim \Omega_r(\theta) n^{-r/2}$; its question section 47.2 asks for the order of $\mathbb{P}(M_n \leq n/2)$, for the transition when $M_n \leq n/2 + s_n$, and whether the thresholds $\theta = 1/r$ mark distinct polynomial transitions. Part V repeats the half-size question (section 63.3), and the README's list of what Part IV does not claim names the hierarchy and the order and window at $m = n/2$; that list is the manuscript's starting point.

Part VI answers these questions at the level of orders:

- *The exponents.* For every fixed $0 < \theta < 1$, $\mathbb{P}(M_n \leq \lfloor \theta n \rfloor) = \Theta(n^{-\lfloor 1/\theta \rfloor/2})$ (theorem 68.1). This answers the exponent part of section 47.1; the existence and value of $\Omega_r(\theta)$ stay open (section 74.1). The lower bounds are Part IV's theorem 45.1, obtained again by a different route; the matching upper bounds are new. For $\theta > 1/2$ the theorem gives only $\Theta(n^{-1/2})$, which is weaker than (153). [Added 1 October 2026, batch 72: $\Omega_r(\theta)$ exists for every r and is found in Part VIII (theorem 91.1).]
- *The half-size threshold.* $\mathbb{P}(M_n \leq \lfloor n/2 \rfloor) = \Theta(n^{-1})$; the order-crossover scale is $\sqrt{n}$, and $\mathbb{P}(M_n \leq \lfloor n/2 + t\sqrt{n} \rfloor) \asymp n^{-1}(1 + t_+)$ uniformly for bounded t (corollary 68.4). A limiting transition function is not proved (section 74.3). [Added 1 October 2026, batch 71: Part VII proves it, $6 + (12/\sqrt{\pi})t_+$ (theorem 80.1).]
- *Every reciprocal threshold.* Each $\theta = 1/q$, $q \geq 2$, is a jump of the logarithmic exponent from $(q-1)/2$ to $q/2$, the endpoint belonging to the rarer phase, and the order changes across the scale $n^{1-1/(2(q-1))}$ above n/q (theorem 68.2 and corollary 68.3). [Added 1 October 2026, batch 72: the limiting crossover function at every $1/q$ is proved in Part VIII (theorem 91.2).]

Two indexings of the exponent meet here. Part VI writes $r = \lfloor 1/\theta \rfloor$, Part IV's question $r = \lceil 1/\theta \rceil - 1$. They agree whenever $1/\theta$ is not an integer. At $\theta = 1/q$ the first is q and the second $q-1$: Part IV states its question for nonreciprocal θ only, and the endpoint exponent is $q/2$, not $(q-1)/2$. In theorem 45.1 the condition $(r+1)\theta > 1$ admits exactly the integers $r \geq \lfloor 1/\theta \rfloor$, so its best bound is the staircase order in every case, at a reciprocal endpoint as well (as the remark after theorem 45.1 says).

67.3 Independence and inherited status

The manuscript read this report's README, Part I's code/model.py and Part IV's code/05-macroscopic-deficits-model.py (section 78). The README's status statement calls the earlier parts AI-assisted and unrefereed, and the manuscript therefore avoids treating their unverified asymptotic assertions as axioms. Its proof needs only the direct maximum decomposition, a directly verified injection, elementary power-sum bounds, and Catalan asymptotics. It uses neither (153) nor any other asymptotic theorem of Parts I–V. The elementary upper majorant and the skew-block lower construction are Part I's (proposition 5.1, and proposition 5.2 at $d = 1$);

the manuscript reproves precisely the versions it needs. Part VI's theorems therefore do not inherit the status of Parts III–V, in contrast with Part V's application.

The delivered code tests finite counting identities, not infinite assertions. It contains no claimed Lean or Rocq formalization. Neither finite agreement nor an attractive plot is offered as a substitute for the proofs in sections 70 and 71.

67.4 Relation to public work

Nadler [1] initiated the adjacency-constrained enumeration and identified the restriction on the position of the maximum. Mayama and Akita [2] developed the two-sided endpoint-state recurrence, rational generating functions for fixed m , and the corresponding spectral growth theory. The question of Part VI is different: both m and n grow, with m a substantial fraction of n . The main advance is a complete polynomial-order phase diagram, including all reciprocal boundaries, rather than a new fixed- m enumeration; a uniform two-term comparison near each boundary explains how the exponent changes. The analytic contribution is the local truncated-renewal comparison and its application to the two envelopes.

No claim of exhaustive literature priority is made. The mechanism belongs to the established family of “fewest big jumps” ideas. Kerriou and Mörters [6] prove a different, precise large-deviation theorem for a growing number of truncated heavy-tailed summands. The setting here is a subcritical renewal sum, with an exact lattice target, a power-tail index below one, and estimates that remain valid at the reciprocal threshold itself. The necessary result is proved from first principles. The literature check (30 September 2026) used the arXiv records of [1, 2] and the record and an author-hosted PDF of [6], with targeted searches; it was not an exhaustive priority search (section 78). Accordingly, “new” in the contribution ledger (table 23) means proved here beyond the explicit scope of Parts I–V, not certified absent from all other literature.

Table 23: What Part VI takes from Parts I–V, and what it adds.

In Parts I–V	Proved in Part VI
Maximum decomposition and the Catalan majorant (Part I, proposition 5.1)	Reproved; also $\bar{A}_m \leq C$
Injective skew blocks (Part I, proposition 5.2, $d = 1$)	Reproved; both envelopes normalized as subcritical renewal kernels
Polynomial lower bounds below half size (Part IV, theorem 45.1)	Matching upper bounds for every $0 < \theta < 1$, reciprocal endpoints included
First-order profile for $\theta > 1/2$ (Part IV, theorem 38.1)	Only the order $n^{-1/2}$; weaker, and not used
Hierarchy and half-size window asked (Parts IV, V)	Uniform two-term comparison at every $1/q$; order-crossover scales
—	Local truncated-renewal theorem for $0 < \alpha < 1$; a transfer principle
Exact endpoint recurrence (Parts I, IV, V)	Independent finite checks through $n = 128$; no formalization

67.5 Notation of Part VI

The symbols $a_n^{(m)}$, C_n , C , $\text{Av}_n(132)$, M_n , the generating function $A_m(x) = \sum_{n \geq 0} a_n^{(m)} x^n$ of Part I, the Catalan polynomial $K_m(x) = \sum_{k=1}^m C_{k-1} x^k$ of Part II (with Part I's majorant $R_m = x + K_m$ of (16)), Part I's block polynomial $K_{m,1}$ (20), Part IV's fixed fraction θ and conjectured amplitude $\Omega_r(\theta)$, and the endpoint counts $T_{u,v}^{(m)}(n)$, $N_n(u)$, $c_{k,u}$ and $N_{n,f}$ of Parts III–V keep their meanings. Several of the manuscript's letters are used in Parts I–V for other objects. Part VI fixes one name for each notion (table 24). In particular:

- $p_n(m) = \mathbb{P}(M_n \leq m)$ is a probability; it is not a moment exponent p as in Parts III–V, which occurs in Part VI only in the moment bound (338);
- the two envelopes are $\underline{A}_m \leq A_m \leq \bar{A}_m$ (the manuscript's L_m and B_m): $\bar{A}_m = 1 + x/(1 - R_m)$ is Part I's majorant (17), and $\underline{A}_m = (1 - K_{m,1})^{-1}$ is Part I's block lower bound (19) at $d = 1$; the manuscript's $H_{m-1}(x)$ is Part I's $K_{m,1}(x)$. The tempting false reading of the manuscript's B_1 is Part V's endpoint constant B_1 or Part I's mass B_d at $d = 1$;
- the ratio interval is $\theta_- \leq m/n \leq \theta_+$ (the manuscript's $a \leq m/n \leq b$), since $a_n^{(m)}$ counts avoiders and a, b are block sizes in Parts IV and V;
- the order function is $O_q(n, m)$ (the manuscript's F_q), since $F_{p,q}^{(m)}$ are Part I's endpoint series; the order-crossover scale is $w_q(n)$ (the manuscript's w_q), since w_k are renewal weights;
- the renewal kernels and weights W, w_k keep the manuscript's letters; their total masses are $\bar{w}, \bar{w}_\pm$ (the manuscript's $\rho, \rho_\pm$), since $\rho = -\Psi'$ is Part IV's deficit density; the resolvents are Z (the manuscript's R), since R_m is Part I's majorant polynomial;
- the conjectural endpoint constants, slopes and limiting crossover functions are $\mathfrak{R}_q, \mathfrak{Q}_q, \mathfrak{H}_q(t)$ (the manuscript's K_q, L_q, H_q), since $K_m, K_{m,d}, L_m$ and H are polynomials, a logarithm and a series of Parts I–III.

No normalization changed: every renamed symbol denotes exactly the manuscript's quantity. Probability is written $\mathbb{P}$ as in Parts III–V, the Catalan generating function C as in Part I. Throughout Part VI,

$$f_n \asymp g_n \iff 0 < c \leq f_n/g_n \leq C < \infty \quad \text{for all sufficiently large } n,$$

and Θ has the same meaning. The constants may depend on fixed parameters stated in the theorem, but not on varying n or m . This notation does *not* mean that the ratio converges; in particular, the scale functions printed here are not numerical approximations to probabilities. The letters $S, T, V, J, L, A, h, \ell, \gamma, \delta, \varepsilon, \lambda_n, \Delta, d_i, r_s, s, e, f, c_w, C_w$ and k_0 are local to section 70 and to single proofs, and are declared where used; in particular $S_n(z), T_n(z)$ and $V_{n,m}(z)$ are not Part V's survival functions, Part III's cost T or Part I's component V_m . Equations, tables and figures are numbered consecutively with those of Parts I–V.

68 Main results

68.1 Objects and probabilities

A permutation $\pi = (\pi_1, \dots, \pi_n)$ avoids 132 when no indices $i < j < k$ satisfy $\pi_i < \pi_k < \pi_j$. Write $\text{Av}_n(132)$ for this class and

$$C_n = \frac{1}{n+1} \binom{2n}{n}, \quad M(\pi) = \max_{1 \leq i < n} |\pi_{i+1} - \pi_i|, \quad a_n^{(m)} = \#\{\pi \in \text{Av}_n(132) : M(\pi) \leq m\}. \quad (309)$$

For $n \leq 1$ the maximum of the empty set of jumps is defined to be zero. A uniformly chosen avoider has largest jump M_n , so

$$p_n(m) := \mathbb{P}(M_n \leq m) = \frac{a_n^{(m)}}{C_n}. \quad (310)$$

The Catalan enumeration is recalled in section 69.

68.2 The staircase and the reciprocal thresholds

Theorem 68.1 (Complete polynomial-order staircase). *For every fixed real $0 < \theta < 1$, put $r = \lfloor 1/\theta \rfloor$. Then*

$$p_n(\lfloor \theta n \rfloor) = \Theta_\theta(n^{-r/2}), \quad a_n^{(\lfloor \theta n \rfloor)} = \Theta_\theta(4^n n^{-(r+3)/2}). \quad (311)$$

Table 24: Symbols of Part VI. “Source” gives manuscript 05’s symbol; the last column prints the tempting false reading.

Symbol	Meaning in Part VI	Source	Not to be confused with
$p_n(m)$ $\theta; r = \lfloor 1/\theta \rfloor$	$\mathbb{P}(M_n \leq m) = a_n^{(m)}/C_n$ fixed fraction m/n ; staircase index	$p_n(m)$ $\theta; r$	a moment exponent p . Part IV’s $r = \lceil 1/\theta \rceil - 1$ (equal unless $1/\theta \in \mathbb{Z}$); Part III’s square root r ; Part V’s exponent θ .
q	reciprocal index of the threshold $1/q$	q	Part III’s $q = (1 - \sqrt{1-z})/2$; Part I’s root q_m and threshold q .
θ_-, θ_+	ends of a fixed ratio interval	a, b	the counts $a_n^{(m)}$; block sizes a, b .
$O_q(n, m)$	order function of theorem 68.2	$F_q(n, m)$	Part I’s series $F_{p,q}^{(m)}$; Part III’s first entry F .
$w_q(n)$	order-crossover scale $n^{1-1/(2(q-1))}$	$w_q(n)$	the renewal weights w_k .
$\bar{A}_m, \underline{A}_m$	upper and lower envelopes of A_m	B_m, L_m	Part V’s B_0, B_1 ; Part I’s B_d , L_m .
$K_{m,1}$	lower block polynomial	H_{m-1}	Part III’s series H ; Part V’s $\mathcal{H}$.
$\mathfrak{B}_k$	set of lower blocks of size k	$\mathcal{B}_k$	Part II’s $\mathcal{B}_N$.
$w_k^\pm, W_M^\pm; w_k, W_m$	renewal weights and kernels	same	Part I’s endpoint matrix $W_m(x)$; Part I’s weights b_k . Part IV’s density $\rho = -\Psi'$.
$\bar{w}_\pm, \bar{w}$ $Z_M^\pm, Z_m, Z$	total kernel masses renewal resolvents $(1 - W)^{-1}$	$\rho_\pm, \rho$ $R_M^\pm, R_m, R$	Part I’s majorant $R_m = x + K_m$; Part IV’s $Z_n(f)$. Part IV’s threshold fraction u .
$u_M^\pm(n), u_m(n)$ α	renewal coefficients power-tail index, $0 < \alpha < 1$	same α	Part I’s growth constants α_m (it is Part V’s abstract tail exponent).
$\eta_q(\beta), \beta$	exponent of corollary 71.2	$\eta_q(\beta), \beta$	Part V’s measure η_n ; Part II’s β_m .
$\lambda_\star, \tilde{a}_n^{(m)}$	growth base and counts of proposition 72.1	$\Lambda, b_n^{(m)}$	Part II’s $\Lambda = \frac{1}{2} \log m$; Part I’s b_k .
$\mathfrak{R}_q, \mathfrak{Q}_q, \mathfrak{S}_q(t)$	conjectural endpoint constant, slope, crossover function	K_q, L_q, H_q	Part II’s K_m ; Part I’s $K_{m,d}, L_m$; Part III’s H ; Part IV’s $\mathfrak{f}$.
$N_{n,n-d}, N_n(u),$ $T_{u,v}^{\text{free}}(n), T_{u,v}^{(m)}(n),$ $c_{k,u}$	the counts of section 76	$f(n, d),$ $h(n, u), T_{\text{free}},$ $T, c(k, u)$	Parts IV and V’s names for the same counts.

Consequently

$$\lim_{n \rightarrow \infty} \frac{-\log p_n(\lfloor \theta n \rfloor)}{\log n} = \frac{1}{2} \left\lfloor \frac{1}{\theta} \right\rfloor. \quad (312)$$

The constants are uniform on compact subintervals of each open phase. At every reciprocal endpoint $\theta = 1/q$, $q \geq 2$, the exponent is $q/2$, not $(q-1)/2$.

Theorem 68.2 (Uniform reciprocal-threshold comparison). *Fix an integer $q \geq 2$ and real constants θ_-, θ_+ satisfying*

$$\frac{1}{q+1} < \theta_- < \frac{1}{q} < \theta_+ < \frac{1}{q-1}. \quad (313)$$

There exist $c, C > 0$ and N such that for every integer $n \geq N$ and every integer m with $\theta_- \leq m/n \leq \theta_+$,

$$c O_q(n, m) \leq p_n(m) \leq C O_q(n, m), \quad O_q(n, m) := n^{-q/2} + n^{-3(q-1)/2} (qm - n)_+^{q-1}. \quad (314)$$

The constants depend only on q, θ_-, θ_+ .

The first term is the rarity of a configuration with $q+1$ macroscopic pieces. The second is the contribution that becomes available when q almost-maximal pieces can accommodate the total size. This explanation concerns the counting envelopes; it is not yet a theorem identifying the typical pieces of the conditioned permutation.

Corollary 68.3 (The order-crossover scales). *Define*

$$w_q(n) := n^{1-1/(2(q-1))}. \quad (315)$$

Under the assumptions of theorem 68.2,

$$p_n(m) \asymp n^{-q/2} \left[1 + \left(\frac{(qm - n)_+}{w_q(n)} \right)^{q-1} \right]. \quad (316)$$

In particular, $m = \lfloor n/q \rfloor + O(w_q(n))$ gives $p_n(m) = \Theta(n^{-q/2})$. If $s_n = m - n/q > 0$, $s_n/w_q(n) \rightarrow \infty$, and $s_n = o(n)$, then

$$p_n(m) \asymp n^{-3(q-1)/2} s_n^{q-1}. \quad (317)$$

On the lower side of the reciprocal threshold, throughout any fixed neighbourhood allowed by (313), the order remains $n^{-q/2}$.

Corollary 68.4 (The first unresolved threshold). *Uniformly for m/n in a fixed sufficiently small neighbourhood of $1/2$,*

$$p_n(m) \asymp \frac{1}{n} + \frac{(2m - n)_+}{n^{3/2}}. \quad (318)$$

Thus

$$p_n(\lfloor n/2 \rfloor) = \Theta(n^{-1}), \quad p_n(\lfloor n/2 + t\sqrt{n} \rfloor) \asymp n^{-1}(1 + t_+)$$

uniformly for t in any fixed bounded interval. There is no assertion here that $n p_n(\lfloor n/2 + t\sqrt{n} \rfloor)$ has a limit.

[Added 1 October 2026, batch 71: it has one. Part VII proves $n p_n(\lfloor n/2 + t\sqrt{n} \rfloor) \rightarrow 6 + (12/\sqrt{\pi})t_+$ uniformly for bounded t (theorem 80.1).]

(The title is the manuscript's: $m = n/2$ is the first threshold that Parts IV and V left unresolved.)

69 Two elementary counting envelopes

This section reproves, in the form Part VI needs, Part I's scalar majorant (proposition 5.1) and the $d = 1$ case of its injective block lower bound (proposition 5.2); the manuscript's proofs follow Part I's arguments and are kept because its normalization and checks are built on them.

Fixed adjacency fraction	Probability order	Count order
$1/2 < \theta < 1$	$n^{-1/2}$	$4^n n^{-2}$
$1/3 < \theta \leq 1/2$	n^{-1}	$4^n n^{-5/2}$
$1/4 < \theta \leq 1/3$	$n^{-3/2}$	$4^n n^{-3}$
$1/5 < \theta \leq 1/4$	n^{-2}	$4^n n^{-7/2}$
$1/(r+1) < \theta \leq 1/r$	$n^{-r/2}$	$4^n n^{-(r+3)/2}$

Table 25: All entries mean Θ for fixed θ . The right-closed endpoint convention is essential. The first row is weaker than Part IV's equivalent (153); the remaining orders are the new conclusion.

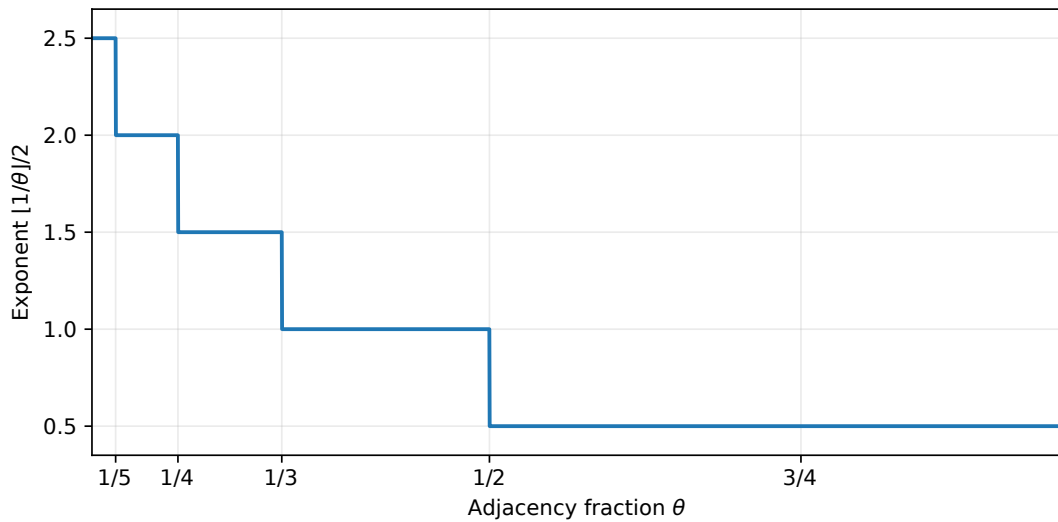


Figure 14: The proven logarithmic rarity exponent $[1/\theta]/2$ of (312). At a reciprocal discontinuity the value is the higher of the two neighbouring levels, as specified exactly in table 25. This is a graph of the theorem, not a fit to data.

69.1 Maximum decomposition and Catalan enumeration

If the maximum n occurs at position k in a 132-avoiding permutation, then every entry to its left is larger than every entry to its right. Indeed, a left entry smaller than a right entry, together with n , would be a 132 pattern. Conversely, choosing any avoiding left block and any avoiding right block, with all left values above all right values, produces an avoider (Part I's maximum decomposition, section 2). Hence

$$C_0 = 1, \quad C_n = \sum_{k=1}^n C_{k-1} C_{n-k}. \quad (319)$$

The generating function $C(x) = \sum_{n \geq 0} C_n x^n$ therefore satisfies

$$C(x) = 1 + xC(x)^2 = \frac{1 - \sqrt{1 - 4x}}{2x}. \quad (320)$$

The binomial expansion gives the formula in (309). Stirling's formula gives

$$C_n \sim \frac{4^n}{\sqrt{\pi} n^{3/2}}, \quad c_{\text{Cat}}(n+1)^{-3/2} \leq 4^{-n} C_n \leq C_{\text{Cat}}(n+1)^{-3/2} \quad (321)$$

for some absolute positive constants $c_{\text{Cat}}, C_{\text{Cat}}$ and all $n \geq 0$.

If $k < n$, the right block has size $n - k$ and values $1, \dots, n - k$. The jump immediately after the maximum is at least k . Thus an adjacency-bounded avoider has its maximum either at the end or in one of the first m positions. This is the structural observation of Nadler [1], reproduced here because it is the only input needed for the upper envelope.

69.2 The valid scalar upper majorant

With $K_m(x) = \sum_{k=1}^m C_{k-1} x^k$, let

$$\bar{A}_m(x) := 1 + \frac{x}{1 - x - K_m(x)} = 1 + \frac{x}{1 - R_m(x)}, \quad (322)$$

where $R_m = x + K_m$ is Part I's (16). For formal series, $F \leq G$ means coefficientwise inequality.

Lemma 69.1 (Upper envelope). *For every $m \geq 1$,*

$$A_m(x) = \sum_{n \geq 0} a_n^{(m)} x^n \leq \bar{A}_m(x). \quad (323)$$

Moreover $\bar{A}_m \leq C$.

The first assertion is Part I's (17).

The manuscript's proof. For $n \geq 2$, deleting a final maximum leaves an adjacency-bounded avoider of size $n - 1$. If the maximum is at position $k < n$, its left block can be chosen in at most C_{k-1} ways and its right block in at most $a_{n-k}^{(m)}$ ways. We ignore the endpoint compatibility conditions, which can only increase the count. Consequently

$$a_n^{(m)} \leq a_{n-1}^{(m)} + \sum_{k=1}^{\min(m, n-1)} C_{k-1} a_{n-k}^{(m)}. \quad (324)$$

The sequence with initial values $b_0 = b_1 = 1$ and equality in this recurrence has generating function $\bar{A}_m$. Induction proves the first assertion.

For the second assertion, increase m to infinity coefficientwise. Since $K_\infty = xC(x)$, the identity $C = 1 + xC^2$ implies

$$\frac{1}{1 - x - xC(x)} = C(x)^2, \quad \bar{A}_\infty(x) = 1 + xC(x)^2 = C(x).$$

All intermediate series have nonnegative coefficients. □

Remark 69.2 (A tempting but invalid shortcut). The upper bound is *not* $A_m \leq (1 - K_m)^{-1}$. At $m = 2$, the latter series has coefficient 3 at x^3 , whereas $a_3^{(2)} = C_3 = 5$. The extra x in the denominator of (322) accounts for a final maximum. Part II explicitly warns against confusing its stronger spectral comparison with a coefficientwise upper bound (remark 12.4, with the same counterexample in section 18).

69.3 An injective skew-block lower envelope

For a permutation σ of size u and τ of size v , their skew sum is

$$\sigma \ominus \tau = (\sigma_1 + v, \dots, \sigma_u + v, \tau_1, \dots, \tau_v).$$

Skew sums preserve 132-avoidance: a triple spanning distinct blocks cannot have its first entry smaller than a later entry from a lower-valued block.

For each $k \geq 1$, use the following blocks:

$$\mathfrak{B}_1 = \{(1)\}, \quad \mathfrak{B}_k = \{(k-1, \sigma_1, \dots, \sigma_{k-2}, k) : \sigma \in \text{Av}_{k-2}(132)\} \quad (k \geq 2). \quad (325)$$

Every such block avoids 132, starts either at its maximum ($k = 1$) or one below its maximum ($k \geq 2$), and ends at its maximum. There are C_{k-2} blocks of size $k \geq 2$. These are Part I's admissible blocks with $d = 1$.

Lemma 69.3 (Lower envelope). *For $m \geq 2$, define*

$$K_{m,1}(x) = x + \sum_{k=2}^{m-1} C_{k-2}x^k, \quad \underline{A}_m(x) := \frac{1}{1 - K_{m,1}(x)}. \quad (326)$$

Then $\underline{A}_m \leq A_m$.

This is Part I's (19) at $d = 1$, with $K_{m,1}$ as in (20).

The manuscript's proof. Take any finite ordered list of blocks from (325), each of size at most $m-1$, and form their skew sum. A jump within a block is at most its size minus one. If a block of size k is followed by a block whose first deficiency is $d \in \{0, 1\}$, the jump across their boundary is $k + d \leq m$. Thus the resulting permutation has all jumps at most m and avoids 132.

It remains to prove injectivity. A nonempty block ending at its maximum cannot be split as a nontrivial skew sum, because in such a split every value in the first part would have to exceed the final maximum. Each block is therefore skew-indecomposable. Every permutation has a unique decomposition into skew-indecomposable blocks: the possible cut positions are exactly those t for which all first t values exceed all later values, and cutting at all these positions gives the unique decomposition. Our block list is recovered by those cuts, and each internal σ is recovered by deleting the first and last entries. The sequence construction has generating function $\underline{A}_m$. □

This is precisely the simplest, first-deficiency-one specialization of Part I's skew-block lower construction (proposition 5.2; the function `block_counts` of `code/model.py`). Its role here is important: one need not approximate the whole class accurately in order to recover its polynomial exponents. A small, explicit subclass suffices when it has the correct power tail.

69.4 Normalizing at the Catalan radius

Replace x by $z/4$. The upper renewal kernel is

$$W_m^+(z) = \sum_{k=1}^m w_k^+ z^k, \quad w_1^+ = \frac{1}{2}, \quad w_k^+ = 4^{-k} C_{k-1} \quad (k \geq 2). \quad (327)$$

The doubled coefficient at $k = 1$ is the sum of the append term and the $k = 1$ term of K_m , as in Part I's R_m , whose coefficient of x is two ((16)). Its total mass is

$$\bar{w}_+ := \sum_{k \geq 1} w_k^+ = \frac{1}{4} + \frac{1}{4} C(1/4) = \frac{3}{4}. \quad (328)$$

The lower kernel is

$$W_{m-1}^-(z) = \sum_{k=1}^{m-1} w_k^- z^k, \quad w_1^- = \frac{1}{4}, \quad w_k^- = 4^{-k} C_{k-2} \quad (k \geq 2), \quad (329)$$

with total mass

$$\bar{w}_- := \sum_{k \geq 1} w_k^- = \frac{1}{4} + \frac{1}{16} C(1/4) = \frac{3}{8}, \quad (330)$$

which is Part I's B_d (23) at $d = 1$. Both kernels are strictly subcritical and have power tails

$$w_k^+ \sim \frac{1}{4\sqrt{\pi}} k^{-3/2}, \quad w_k^- \sim \frac{1}{16\sqrt{\pi}} k^{-3/2} \quad (331)$$

(the second is Part I's (25) at $d = 1$). Write

$$Z_M^\pm(z) = \frac{1}{1 - W_M^\pm(z)}, \quad u_M^\pm(n) = [z^n] Z_M^\pm(z).$$

For $n \geq 1$, the exact coefficientwise sandwich is

$$u_{m-1}^-(n) \leq 4^{-n} a_n^{(m)} \leq \frac{1}{4} u_m^+(n-1). \quad (332)$$

The finite shifts $m-1$ and $n-1$ must be retained until the final comparison. Dropping them before checking the reciprocal boundary would obscure the endpoint issue.

70 A local theorem for truncated subcritical renewals

70.1 Abstract hypotheses and theorem

Let $0 < \alpha < 1$ and let $(w_k)_{k \geq 1}$ be a nonnegative sequence satisfying

$$\bar{w} := \sum_{k \geq 1} w_k < 1, \quad w_k \leq C_w k^{-1-\alpha} \quad (k \geq 1), \quad w_k \geq c_w k^{-1-\alpha} \quad (k \geq k_0), \quad (333)$$

where $C_w, c_w > 0$ and $k_0 \geq 1$ are fixed. Put

$$W_m(z) = \sum_{k=1}^m w_k z^k, \quad Z_m(z) = \frac{1}{1 - W_m(z)}, \quad u_m(n) = [z^n] Z_m(z). \quad (334)$$

These coefficients count weighted ordered compositions whose parts are at most m . One can also view them as a geometric-sum model, with an appropriate normalizing factor; no probabilistic normalization is needed below.

Theorem 70.1 (Uniform local truncated-renewal estimate). *Assume (333). Fix $q \geq 2$ and*

$$\frac{1}{q+1} < \theta_- < \theta_+ < \frac{1}{q-1}. \quad (335)$$

The interval need not straddle $1/q$. Uniformly for integer m, n with n sufficiently large and $\theta_- \leq m/n \leq \theta_+$,

$$u_m(n) \asymp n^{-1-(q+1)\alpha} + n^{-q(1+\alpha)}(qm - n)_+^{q-1}. \quad (336)$$

The constants depend only on α , the kernel in (333), and q, θ_-, θ_+ .

The theorem uses only upper and lower power bounds, not a precise tail equivalent or regular variation with a limiting constant. It therefore determines orders robustly, but cannot by itself identify exact amplitudes.

70.2 A uniform local tail bound for unrestricted renewals

Let $W(z) = \sum_{k \geq 1} w_k z^k$ and $Z(z) = (1 - W(z))^{-1}$.

Lemma 70.2 (Local subcritical convolution bound). *For every fixed integer $h \geq 1$, there is $C_h < \infty$ such that*

$$[z^s]Z(z)^h \leq C_h(s+1)^{-1-\alpha} \quad (s \geq 0). \quad (337)$$

The same bound holds for $(1 - S(z))^{-h}$ whenever S is a coefficientwise subseries of W . Also

$$\sum_{s=0}^n s^p [z^s](1 - S(z))^{-h} \leq C_{h,p} n^{p-\alpha} \quad (p > \alpha, n \geq 1). \quad (338)$$

Proof. For $s \geq 1$ and $\ell \geq 1$, every ordered composition $k_1 + \dots + k_\ell = s$ has an index i with $k_i \geq s/\ell$. Designate one such index; allowing all possibilities only overcounts. Once the other $\ell - 1$ parts are fixed, k_i is determined. Its weight is at most $C_w \ell^{1+\alpha} s^{-1-\alpha}$. Summing the other weights without the sum constraint gives at most $\bar{w}^{\ell-1}$. Hence

$$[z^s]W(z)^\ell \leq C_w \ell^{2+\alpha} \bar{w}^{\ell-1} s^{-1-\alpha}. \quad (339)$$

Now expand

$$Z(z)^h = \sum_{\ell \geq 0} \binom{\ell + h - 1}{h - 1} W(z)^\ell.$$

The sum of the bounds (339) converges, because $\bar{w} < 1$ and the remaining factors grow polynomially in ℓ . The coefficient at $s = 0$ is one. This proves (337). Coefficientwise domination proves the assertion for S . Finally, sum $s^{p-1-\alpha}$ over $1 \leq s \leq n$; this is $O(n^{p-\alpha})$ because $p > \alpha$. $\square$

70.3 Suppressing a macroscopic sum made only of small parts

For a fixed small $\varepsilon > 0$, put

$$S_n(z) = \sum_{k \leq \lfloor \varepsilon n \rfloor} w_k z^k, \quad T_n(z) = (1 - S_n(z))^{-1}. \quad (340)$$

Lemma 70.3 (Arbitrarily strong polynomial suppression). *Given $c > 0, A > 0$ and a fixed integer $h \geq 1$, one can choose $\varepsilon > 0$ so that, for all sufficiently large n ,*

$$[z^s]T_n(z)^h \leq C n^{-A} \quad (s \geq cn). \quad (341)$$

The same ε works for any prescribed finite set of values of h .

Proof. Choose $L > A/c$ and then $\varepsilon > 0$ so small that $L\varepsilon < \alpha$. Set $\lambda_n = L \log n / n$. Since $e^t - 1 \leq te^t$ for $t \geq 0$,

$$\begin{aligned} S_n(e^{\lambda_n}) - S_n(1) &\leq \lambda_n e^{\lambda_n \varepsilon n} \sum_{k \leq \varepsilon n} k w_k \\ &\leq C \frac{\log n}{n} n^{L\varepsilon} n^{1-\alpha} = O((\log n) n^{L\varepsilon-\alpha}) = o(1). \end{aligned}$$

Here $\sum_{k \leq N} k w_k = O(N^{1-\alpha})$ follows directly from (333). Because $S_n(1) \leq \bar{w} < 1$, for large n we have

$$T_n(e^{\lambda_n}) \leq \frac{2}{1 - \bar{w}}.$$

All coefficients are nonnegative, so

$$[z^s] T_n(z)^h \leq e^{-\lambda_n s} T_n(e^{\lambda_n})^h \leq \left(\frac{2}{1 - \bar{w}} \right)^h n^{-Lc} \leq C n^{-A}$$

for $s \geq cn$. Only the final constant depends on h . $\square$

The cutoff is a fixed fraction εn , not $n/\log n$. That distinction prevents artificial logarithmic losses. The parameter ε can depend on the desired power A , but remains fixed while n tends to infinity.

70.4 Proof of the upper estimate

Upper bound in theorem 70.1. Set $\gamma = 1 - (q-1)\theta_+ > 0$. Choose $A > 1 + (q+1)\alpha$. Apply lemma 70.3 with $c = \gamma$ and choose $\varepsilon < \theta_-/2$ sufficiently small for the resulting estimate. Decompose

$$W_m = S_n + V_{n,m}, \quad V_{n,m}(z) = \sum_{\lfloor \varepsilon n \rfloor < k \leq m} w_k z^k.$$

Abbreviate these series by S, V and put $T = (1 - S)^{-1}$. The exact formal identity

$$Z_m(z) = \sum_{j \geq 0} V(z)^j T(z)^{j+1} \tag{342}$$

marks the j large parts and the $j+1$ intervening sequences of small parts. Because every part counted by V exceeds εn , only $j \leq J := \lceil 1/\varepsilon \rceil$ can contribute to the coefficient of z^n . Thus the number of cases is fixed independently of n .

Fewer than q large parts. For $j \leq q-1$, the total size supplied by the large parts is at most $j m \leq (q-1)\theta_+ n$. The small-part sequences must therefore supply at least γn . By lemma 70.3, every relevant coefficient of T^{j+1} is $O(n^{-A})$. Convolving with V^j and using $V(1)^j \leq \bar{w}^j$ gives

$$[z^n] V^j T^{j+1} = O(n^{-A}) = O(n^{-1-(q+1)\alpha}). \tag{343}$$

This argument includes $j = 0$.

At least $q+1$ large parts. For each fixed $j \geq q+1$, an individual large-part weight is $O(n^{-1-\alpha})$. For a specified sum $n-s$, the number of ordered j -tuples is $O(n^{j-1})$. Hence

$$[z^{n-s}] V(z)^j = O(n^{-1-j\alpha})$$

uniformly in s . Since $T(1)^{j+1} \leq (1 - \bar{w})^{-j-1}$,

$$[z^n] V^j T^{j+1} = O(n^{-1-j\alpha}) = O(n^{-1-(q+1)\alpha}). \tag{344}$$

There are at most J such terms.

Exactly q large parts. Put $\Delta = qm - n$. If the small-part sequences have total size s , write the q large parts as $k_i = m - d_i$. Their deficits satisfy

$$d_i \geq 0, \quad d_1 + \cdots + d_q = \Delta + s.$$

Ignoring upper bounds on the d_i gives at most

$$\binom{\Delta + s + q - 1}{q - 1} \leq C(\Delta + s + 1)^{q-1}$$

choices when $\Delta + s \geq 0$, and no choices otherwise. Every large-part product is $O(n^{-q(1+\alpha)})$. Writing $r_s = [z^s]T^{q+1}$, we obtain

$$\begin{aligned} [z^n]V^q T^{q+1} &\leq Cn^{-q(1+\alpha)} \sum_{s=0}^n (\Delta + s + 1)^{q-1} r_s \\ &\leq Cn^{-q(1+\alpha)} ((\Delta + 1)^{q-1} + n^{q-1-\alpha}). \end{aligned} \quad (345)$$

The last step uses $\sum r_s \leq (1 - \bar{w})^{-q-1}$, the elementary bound $(x + y)^{q-1} \leq C_q(x^{q-1} + y^{q-1})$, and lemma 70.2 with $p = q - 1 > \alpha$.

The second term in (345) is exactly of order $n^{-1-(q+1)\alpha}$. The harmless constant in $(\Delta + 1)^{q-1}$ is absorbed by that term because

$$q(1 + \alpha) - (1 + (q + 1)\alpha) = q - 1 - \alpha > 0.$$

Combining the three cases proves the upper half of (336). $\square$

70.5 Proof of the lower estimate

Lower bound in theorem 70.1. All terms of $Z_m = \sum_{j \geq 0} W_m^j$ are nonnegative, so it suffices to exhibit two families of compositions.

The background family with $q + 1$ macroscopic parts. Because $\theta_- > 1/(q + 1)$, choose a fixed $\delta > 0$ so small that

$$\frac{1}{q + 1} - q\delta > 0, \quad \frac{1}{q + 1} + q\delta < \theta_-.$$

Choose the first q parts independently in integer intervals of width proportional to n about $n/(q + 1)$, with deviation at most δn . Let the final part make the sum equal to n . Every part then lies between fixed positive multiples of n and is at most m . There are at least cn^q choices for sufficiently large n . The lower tail bound in (333) gives a weight of at least $cn^{-(q+1)(1+\alpha)}$ for each choice. Consequently

$$u_m(n) \geq cn^{-1-(q+1)\alpha}. \quad (346)$$

The near-cap family with q macroscopic parts. Suppose $\Delta = qm - n \geq 0$. Every weak composition

$$d_1 + \cdots + d_q = \Delta, \quad d_i \geq 0$$

gives $k_i = m - d_i$ with total size n . Moreover

$$k_i \geq m - \Delta = n - (q - 1)m \geq \gamma n > 0, \quad k_i \leq m.$$

Thus all these choices are admissible, all parts are macroscopic, and there are exactly $\binom{\Delta + q - 1}{q - 1}$ of them. Each weight is at least $cn^{-q(1+\alpha)}$, so

$$u_m(n) \geq cn^{-q(1+\alpha)}(\Delta + 1)^{q-1} \quad (\Delta \geq 0). \quad (347)$$

When $\Delta < 0$, the second term requested in (336) is zero. The two lower families belong to different powers of W_m and may be added; alternatively, the maximum of their lower bounds controls half their sum. Together with (346), this proves the theorem. $\square$

(Here $\gamma = 1 - (q - 1)\theta_+$ as in the proof of the upper bound.)

70.6 Why the reciprocal endpoint changes the exponent

Away from a boundary, q macroscopic parts have $q - 1$ free size coordinates. Their local weighted count is

$$n^{q-1} (n^{-1-\alpha})^q = n^{-1-q\alpha}.$$

At $n = qm$, those q parts cannot vary macroscopically: they would all have to sit at the cap. Allowing a small-part remainder of size s restores only s^{q-1} choices. The unrestricted renewal coefficients obey the upper bound $O(s^{-1-\alpha})$, so the relevant truncated moment is $O(n^{q-1-\alpha})$. This bounds the contribution by $O(n^{-1-(q+1)\alpha})$, an order that is actually attained by the separate lower family with $q + 1$ genuinely macroscopic parts.

The crossover occurs when

$$n^{-q(1+\alpha)} \Delta^{q-1} \quad \text{and} \quad n^{-1-(q+1)\alpha}$$

have the same order, equivalently

$$\Delta \asymp n^{1-\alpha/(q-1)}. \quad (348)$$

This algebra is not the proof of the estimate; it is an interpretation of the proved uniform bounds. The inequalities $q - 1 > \alpha$ and $\bar{w} < 1$ are exactly the two points that make the argument work without an extra logarithm or a critical-renewal correction.

71 Transferring the theorem to permutations

71.1 Finite shifts do not alter the order function

The lower and upper envelopes in (332) have slightly different cutoffs and coefficient indices. We first remove that bookkeeping issue.

Lemma 71.1 (Bounded-shift stability). *Fix $q \geq 2$ and a positive bounded interval of ratios m/n . For any fixed integers e, f , the two expressions*

$$O_q(n, m), \quad O_q(n + e, m + f)$$

are comparable by constants independent of n, m , whenever n is sufficiently large. The same is true of the renewal order expression in (336), for $0 < \alpha < 1$.

Proof. The powers of n change only by bounded factors. The excess $qm - n$ changes by the fixed amount $qf - e$. For every fixed $p \geq 1$ and fixed real c ,

$$(x + c)_+^p \leq C_{p,c}(x_+^p + 1), \quad x_+^p \leq C_{p,c}((x + c)_+^p + 1).$$

For O_q , the extra constant is multiplied by $n^{-3(q-1)/2}$, which is at most a constant times $n^{-q/2}$ because $q \geq 2$. In the renewal expression, it is absorbed because $q - 1 - \alpha > 0$, as in the proof of theorem 70.1. $\square$

71.2 Proof of the uniform permutation estimate

Proof of theorem 68.2. By (331), both normalized kernels satisfy (333) with $\alpha = 1/2$. Their masses are $3/4$ and $3/8$, strictly below one. Apply theorem 70.1 to $u_{m-1}^-(n)$ and $u_m^+(n-1)$, using slightly enlarged fixed ratio intervals if necessary to accommodate the shifts. The resulting common order, after lemma 71.1, is

$$4^{-n} a_n^{(m)} \asymp n^{-(q+3)/2} + n^{-3q/2} (qm - n)_+^{q-1}. \quad (349)$$

The lower inequality follows from the lower envelope, and the upper inequality from the upper envelope. Finally divide by $4^{-n} C_n \asymp n^{-3/2}$ from (321). This gives exactly (314). $\square$

71.3 Proof of the staircase and its endpoints

Proof of theorem 68.1. First suppose $1/2 < \theta < 1$. Use theorem 68.2 with $q = 2$ and a ratio interval containing both $1/2$ and θ . For $m = \lfloor \theta n \rfloor$, the positive excess $2m - n$ is of order n . The second term in (314) is therefore of order $n^{-1/2}$ and dominates n^{-1} .

Now suppose $0 < \theta \leq 1/2$ and put $r = \lfloor 1/\theta \rfloor \geq 2$. Then

$$\frac{1}{r+1} < \theta \leq \frac{1}{r}.$$

Choose $\theta_- < \theta$ and $\theta_+ > 1/r$ satisfying (313) with $q = r$. If $m = \lfloor \theta n \rfloor$, then $rm - n \leq 0$. The second term of (314) vanishes, leaving the order $n^{-r/2}$. This also handles $\theta = 1/r$ without a limiting argument.

The count statement follows from Catalan asymptotics. Taking logarithms gives (312), since the logarithms of the two comparison constants are $O(1)$. The same choices of $\theta_{\pm}$ work for θ in a fixed compact subinterval of a phase, proving the asserted uniformity. $\square$

Proof of corollaries 68.3 and 68.4. Factoring $n^{-q/2}$ from (314) leaves

$$1 + n^{-(2q-3)/2} (qm - n)_+^{q-1}.$$

The exponent identity

$$(q-1) \left(1 - \frac{1}{2(q-1)} \right) = \frac{2q-3}{2}$$

gives (316). All stated regimes follow by comparing the bracketed terms. For $q = 2$, the excess is $2m - n$ and the scale is $\sqrt{n}$, proving corollary 68.4. Integer rounding changes the excess by only $O(1)$ and is covered by lemma 71.1. $\square$

71.4 A continuum of logarithmic exponents inside a shrinking neighbourhood

Corollary 71.2 (Power-scale approach from above). *Fix $q \geq 2$, $c > 0$, and $0 < \beta < 1$. Set*

$$m_n = \left\lfloor \frac{n}{q} + cn^{\beta} \right\rfloor.$$

Then

$$p_n(m_n) = \Theta \left(n^{-\eta_q(\beta)} \right), \quad \eta_q(\beta) = \min \left\{ \frac{q}{2}, (q-1) \left(\frac{3}{2} - \beta \right) \right\}. \quad (350)$$

With $m_n = \lfloor n/q - cn^{\beta} \rfloor$, the order is always $n^{-q/2}$.

Proof. The ratio m_n/n tends to $1/q$, and $qm_n - n = qcn^{\beta} + O(1)$ in the first case. Substitute into (314). The sum of two positive powers of n has the order of the larger term, which gives the minimum exponent. In the second case the excess is eventually negative. $\square$

For example, $m = n/2 + n^{3/4} + O(1)$ gives rarity $\Theta(n^{-3/4})$, while $m = n/3 + n^{7/8} + O(1)$ gives $\Theta(n^{-5/4})$. These are genuine intermediate logarithmic exponents, even though the fixed-fraction diagram is a staircase.

71.5 The exact sense of “window”

A window often means that a rescaled probability converges to a nonconstant function of the scaled parameter. That convergence is not proved here. Here the phrase *order-crossover scale* has the following precise meaning: below a positive excess of order $w_q(n)$ the first term controls the order, and above it the second term eventually dominates by an unbounded factor. The uniform comparison (316) proves this statement directly. (Part IV’s question section 47.2 asks for a window in the stronger sense as well; that part stays open.) [Added 1 October 2026, batch 72: it is proved at $q = 2$ in Part VII and for every fixed $q \geq 2$ in Part VIII (theorems 80.1 and 91.2).]

Likewise (312) is a pointwise logarithmic rarity law, not a full large-deviation principle asserted without qualification. At a reciprocal boundary the pointwise cumulative exponent jumps, and ordinary large-deviation terminology would require care with lower semicontinuous rate functions and the distinction between open and closed events.

72 Interpretation, robustness, and limitations

72.1 Why a coarse sandwich recovers an exact exponent

The masses $3/8$ and $3/4$ and the tail constants in (331) differ substantially. Consequently the envelopes cannot be expected to agree to first relative order. Nevertheless, both are subcritical and both assign a macroscopic piece of size k a normalized weight of order $k^{-3/2}$. The deficit simplex has the same dimension in both models. Those three properties determine the powers of n , while the details discarded by the upper bound and restricted by the lower construction affect constants.

This is why the method reaches the entire lower-fraction hierarchy without resolving the complicated endpoint-state eigenvectors. It is also why the proof stops at a two-sided comparison rather than silently promoting it to an equivalent.

72.2 A general transfer principle

The same argument applies outside this permutation problem.

Proposition 72.1 (Renewal-sandwich universality). *Suppose a nonnegative counting sequence $\tilde{a}_n^{(m)}$ is normalized by $\lambda_\star^{-n}$, and its normalized coefficients lie between two truncated renewal sequences satisfying (333) with the same $0 < \alpha < 1$, up to fixed shifts of m, n and positive constant factors. Suppose the unrestricted normalization satisfies*

$$\lambda_\star^{-n} \tilde{a}_n \asymp n^{-1-\alpha}.$$

Then, uniformly near each fixed reciprocal threshold $m/n = 1/q$, $q \geq 2$,

$$\frac{\tilde{a}_n^{(m)}}{\tilde{a}_n} \asymp n^{-q\alpha} + n^{-(q-1)(1+\alpha)}(qm - n)_+^{q-1}. \quad (351)$$

The associated order-crossover scale is $n^{1-\alpha/(q-1)}$.

Proof. Apply theorem 70.1 to both bounds, use the bounded-shift argument, and divide by $n^{-1-\alpha}$. Equating the two terms gives the scale. $\square$

The hypotheses are substantive. A generic Catalan class does not automatically admit the required upper and lower cutoff encodings. The proposition isolates exactly what must be proved before importing the phase diagram into another class.

72.3 Finite changes, subcriticality, and the tail-index restriction

Changing finitely many kernel weights does not change the theorem's powers as long as the total mass stays below one and the two-sided tail bounds remain valid. More generally, the proof tolerates bounded multiplicative changes of all sufficiently large weights. This robustness is built into (333).

Subcriticality cannot simply be removed. For example,

$$W(z) = 1 - \sqrt{1 - z}$$

has nonnegative coefficients of order $k^{-3/2}$ and total mass one, but $(1 - W(z))^{-1} = (1 - z)^{-1/2}$ has coefficients of order $n^{-1/2}$, not $n^{-3/2}$. Thus lemma 70.2, the basic input to the proof, fails at criticality.

The condition $\alpha < 1$ ensures $q - 1 > \alpha$ at every $q \geq 2$. At $\alpha = 1$ and $q = 2$ the truncated moment used in (345) becomes logarithmic. The exact endpoint theorem in that regime requires a separate statement. It is not justified by replacing α by one in the displayed formulas.

72.4 Comparison with the known upper-half profile

When $q = 2$ and $\theta > 1/2$ is fixed, theorem 68.1 gives $\Theta(n^{-1/2})$, consistent with but weaker than Part IV's (153). Near $\theta = 1/2$ from above, that previously reported prefactor has the expansion

$$\frac{3(2\theta - 1)}{\sqrt{\pi\theta(1 - \theta)}} = \frac{12}{\sqrt{\pi}} \left(\theta - \frac{1}{2} \right) + O((\theta - \frac{1}{2})^3)$$

(this is Part IV's (222) with $u = 1 - \theta$). Substituting $\theta - 1/2 = s/n$ heuristically suggests the order $sn^{-3/2}$. The present proof establishes that order uniformly once $s \gg \sqrt{n}$, without assuming that the fixed- θ equivalent can be extrapolated into a shrinking interval. It also proves the missing n^{-1} background at and below the threshold.

72.5 Scope of the advance

The results resolve the exponent part of Part IV's proposed hierarchy (section 47.1) and the polynomial order and order-crossover scale at $n/2$ (sections 47.2 and 63.3). They extend the same analysis to every n/q . They do not establish the proposed functions $\Omega_r(\theta)$, do not determine the exact constants at reciprocal endpoints, and do not prove a limiting distribution for the macroscopic block sizes in the actual conditioned permutation. They also leave the fixed- m component-ordering problem untouched.

The constants in the proofs depend on the fixed integer q . No uniformity is claimed as $q \rightarrow \infty$, equivalently as the adjacency fraction tends to zero. The theorem therefore does not yet connect the linear-bound regime to fixed m or to an arbitrary mesoscopic sequence $m = o(n)$.

73 Exact checks and reproducibility

73.1 What the program checks

The delivered standard-library Python program `code/07-polynomial-rarity-verify.py` performs several independent finite checks. First, for $n \leq 7$, it compares a direct 132-pattern predicate on all permutations with the recursive Catalan generator. Second, it compares 54 endpoint-recurrence counts with exhaustive avoider counts for $n \leq 9$. Third, it generates 13,431 lower-construction objects across 60 parameter pairs, checking avoidance, the jump bound, absence of duplicate outputs, and agreement with the lower generating function. Fourth, it checks

$$[x^n] \underline{A}_m \leq a_n^{(m)} \leq [x^n] \overline{A}_m \leq C_n$$

for 276 pairs with $2 \leq m \leq n \leq 24$. Finally it reproduces the counterexample to the invalid upper shortcut in remark 69.2.

All those assertions passed in the delivered run. The finite checks test the implementation and the combinatorial interfaces. They do not certify theorem 70.1 or any asymptotic statement; those rest on the proofs above.

73.2 Illustrative exact counts

The program also computes exact integer counts for 20 fixed-fraction samples through $n = 128$ and 18 crossover samples. The CSV files retain the integers, so every displayed probability can be reconstructed as an exact rational number. The decimals below are rounded and are not interval certificates.

n	$\theta = 3/5$	$\theta = 1/2$	$\theta = 2/5$	$\theta = 1/3$	$\theta = 1/4$
32	1.287330	4.131442	1.135882	2.317767	2.862041
64	1.173974	4.629846	1.210151	3.094373	2.714275
96	1.107349	4.868996	1.196177	3.419091	2.535235
128	1.061964	5.016273	1.169257	2.937328	2.391952

Table 26: The scaled quantity $n^{1/\theta/2} p_n(\lfloor \theta n \rfloor)$, from exact counts. Neither monotonicity nor convergence of these columns is asserted. In particular, a small table cannot distinguish a genuine limit from bounded nonconvergence. [Added 1 October 2026, batch 71: Part VII proves the limits of two columns, 6 for $\theta = 1/2$ and $I_3(2/5)/(2\pi) \approx 0.4693$ for $\theta = 2/5$ (theorems 80.1 and 80.2); the $\theta = 2/5$ entries are about 2.5 times their limit.] [Added 1 October 2026, batch 72: Part VIII proves the limits of the $\theta = 1/3$ and $\theta = 1/4$ columns, $\mathfrak{R}_3 = 5/\sqrt{2\pi} \approx 1.9947$ and $\mathfrak{R}_4 = \kappa_5 I_5(1/4) \approx 0.677$ (theorem 91.2 at $t = 0$, corollary 97.1 and section 97); their entries are 1.2–1.7 and 3.5–4.2 times these limits.]

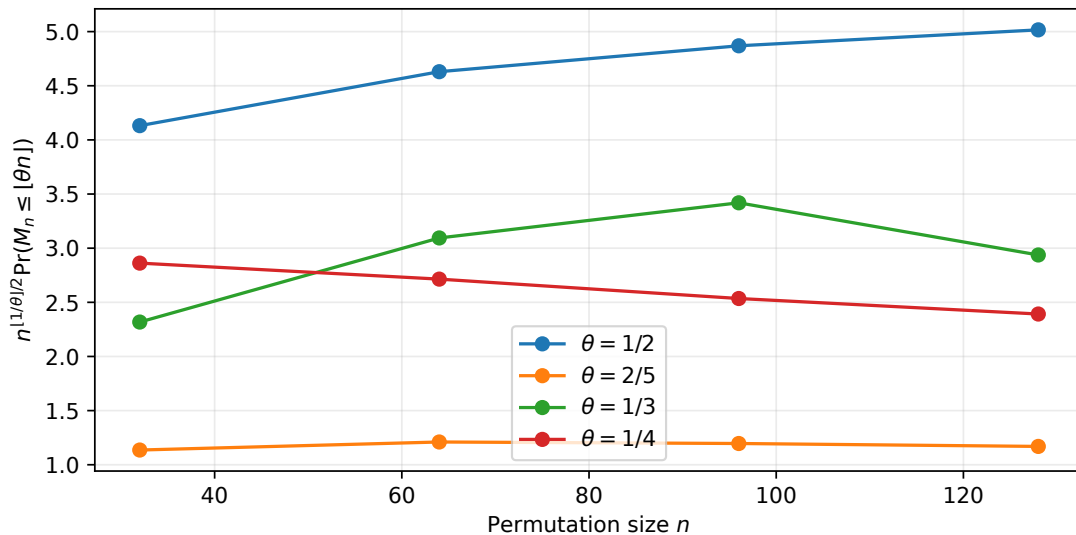


Figure 15: The same exact finite data for four lower-fraction or threshold cases. The theorem proves boundedness above and away from zero for large n ; this plot does not prove it and does not estimate limiting constants. (The axis label writes $\Pr$ for $\mathbb{P}$.)

73.3 A concrete lower-construction example

For $m = 5$, take the size-four block $(3, 2, 1, 4)$ and the size-three block $(2, 1, 3)$. Their skew sum is

$$(6, 5, 4, 7, 2, 1, 3).$$

Its largest jump is the boundary jump $7 - 2 = 5$. Each block ends at its maximum and starts one below it, exactly as required in lemma 69.3. The skew-indecomposable cuts recover the two chosen blocks. This tiny example exhibits the encoding used to obtain the lower bound at every scale.

73.4 Rebuilding and rerunning

Part VI's files are

<code>code/07-polynomial-rarity-model.py</code>	Exact counters, envelopes, lower-construction generator, order proxy
<code>code/07-polynomial-rarity-verify.py</code>	Finite tests and exact-count data (standard library)
<code>code/07-polynomial-rarity-figures.py</code>	The two figures of this part (NumPy, Matplotlib)
<code>code/07-polynomial-rarity-Makefile</code>	The manuscript's convenience targets
<code>data/07-polynomial-rarity-*</code>	The two exact-count CSV files and the recorded run log
<code>figures/07-polynomial-rarity-*</code>	The two figures (PDF and PNG)
<code>07-polynomial-rarity-SOURCES.md</code>	Pinned source files, literature ledger and status

The programs are shipped byte-identical to the delivery and still use the delivered names: `verify.py` imports `model`, which in this directory would load Part I's different `code/model.py`, and writes `data/phase_counts.csv`, `data/crossover_counts.csv` and `data/verification_log.txt`; `figures.py` reads `data/phase_counts.csv` and writes unprefixes files in `figures/`; the Makefile builds an `article.tex` beside itself. The report's README gives a recipe that restores the delivered layout in a scratch copy; there, the manuscript's instructions apply,

```
python code/verify.py
python code/figures.py
```

and rerunning overwrites only the copy's recorded outputs. The exact checks require only Python's standard library (Python 3.10 or later); regenerating the figures additionally requires NumPy and Matplotlib. The endpoint recurrence used in the counting program is explained in section 76. The source ledger `07-polynomial-rarity-SOURCES.md` identifies the exact repository files inspected; on the identifier returned by its initial tree query it is mistaken (section 78). The retained figures make Python unnecessary for rebuilding the article.

74 Further research questions and proposed approaches

The order theorem separates several previously entangled questions. The following agenda is ordered roughly from the closest analytic refinement to more structural extensions. None of the questions in this section is asserted to have been solved here. Several refine questions of Parts IV and V; each says which.

74.1 Exact amplitudes inside the lower phases

For fixed $r \geq 2$ and $1/(r+1) < \theta < 1/r$, does there exist a positive function $\Omega_r(\theta)$ such that

$$p_n(\lfloor \theta n \rfloor) \sim \Omega_r(\theta) n^{-r/2} \quad (352)$$

Theorem 68.1 proves that the normalized sequence is bounded above and away from zero, but that is not convergence. The two renewal envelopes have different masses and different tail constants, so squeezing between them cannot identify Ω_r . A promising next step is an exact decomposition into a finite macroscopic skeleton decorated by microscopic avoiding blocks. One would need both unique encoding and a summable bound on all decorations before interchanging a scaling limit with the sum over skeletons. The known upper-half formula (153) is a useful compatibility test, not a license to extrapolate. (This is the amplitude half of Part IV's section 47.1, whose exponent half theorem 68.1 answers.) [Added 1 October 2026, batch 71: answered for $r = 2$ in Part VII: $\Omega_2(\theta) = I_3(\theta)/(2\pi)$ for $1/3 < \theta < 1/2$, with I_3 the three-piece integral (359) (theorem 80.2). It is obtained from an exact decomposition of the kind proposed here, the canonical threshold parser of lemma 82.1. For $r \geq 3$ the question stays open (section 89.1).] [Added 1 October 2026, batch 72: answered for every $r \geq 2$ in Part VIII: $\Omega_r(\theta) = \kappa_{r+1}I_{r+1}(\theta)$ for $1/(r+1) < \theta < 1/r$, uniformly on compact subintervals, with the capped-simplex integral (377) and $\kappa_p = (p+1)/(2^p \pi^{(p-1)/2})$ (theorem 91.1). The skeleton is again the canonical parser; the new ingredient is the endpoint-prefix count (389).]

74.2 Constants at the reciprocal endpoints

For each $q \geq 2$, determine whether

$$\mathfrak{R}_q := \lim_{n \rightarrow \infty} n^{q/2} p_n(\lfloor n/q \rfloor)$$

exists and compute it. The theorem gives $0 < \liminf \leq \limsup < \infty$. It does not exclude residue-class oscillations caused by the lattice. An exact skeleton expansion should first establish whether changing $qm - n$ by a bounded integer changes the normalized limit. The bounded-shift lemma (lemma 71.1) proves only comparability, so an eventual proof must genuinely sharpen that step. [Added 1 October 2026, batch 71: answered for $q = 2$ in Part VII: $\mathfrak{R}_2 = 6$, along all n , so there is no residue-class oscillation in the limit (theorem 80.1). For $q \geq 3$ the question stays open (section 89.1).] [Added 1 October 2026, batch 72: answered for every $q \geq 2$ in Part VIII: $\mathfrak{R}_q = \kappa_{q+1}I_{q+1}(1/q)$ along all n , so no residue-class oscillation survives in the limit (theorem 91.2 at $t = 0$); for example $\mathfrak{R}_3 = 5/\sqrt{2\pi}$ (corollary 97.1).]

74.3 Limiting crossover functions

A stronger version of theorem 68.2 would establish, locally uniformly in real t ,

$$n^{q/2} p_n\left(\left\lfloor \frac{n}{q} + t \mathfrak{w}_q(n) \right\rfloor\right) \longrightarrow \mathfrak{H}_q(t). \quad (353)$$

The current estimates predict only $\mathfrak{H}_q(t) \asymp 1 + t_+^{q-1}$, assuming such a limit exists. A particularly simple candidate is

$$\mathfrak{H}_q(t) = \mathfrak{R}_q + \mathfrak{Q}_q t_+^{q-1}, \quad \mathfrak{R}_q, \mathfrak{Q}_q > 0.$$

This is a proposed hypothesis, not a conclusion of the proof. It reflects the two mechanisms in the renewal estimate: a background family and a newly available near-cap simplex. The actual permutation decorations could require a more complicated formula. For $q = 2$, matching this candidate to (153) would suggest $\mathfrak{Q}_2 = 12/\sqrt{\pi}$, but only after proving a suitable uniform overlap regime. Fixed- θ asymptotics alone do not determine that slope. (For $q = 2$ this is the transition function asked for in Part IV's section 47.2 and Part V's section 63.3.) [Added 1 October 2026, batch 71: answered for $q = 2$ in Part VII. The candidate holds: $\mathfrak{H}_2(t) = 6 + (12/\sqrt{\pi})t_+$, so $\mathfrak{R}_2 = 6$ and $\mathfrak{Q}_2 = 12/\sqrt{\pi}$, the suggested slope, proved directly rather than by matching (theorem 80.1). For $q \geq 3$ the question stays open (section 89.2).] [Added 1 October 2026, batch 72: answered for every $q \geq 2$ in Part VIII. The candidate holds, $\mathfrak{H}_q(t) = \mathfrak{R}_q + \mathfrak{Q}_q t_+^{q-1}$ with $\mathfrak{Q}_q = (q+1)q^{5q/2-1}/(2^q \pi^{(q-1)/2}(q-1)!)$ (theorem 91.2); at $q = 3$ it is $5/\sqrt{2\pi} + (729\sqrt{3}/(4\pi))t_+^2$ (corollary 97.1).]

74.4 The geometry of a conditioned permutation

What does a uniform avoider conditioned on $M_n \leq m$ look like when m/n lies in a lower phase? The proof marks large parts in two auxiliary renewal encodings, not in a canonical decomposition of every valid permutation. It therefore does not prove that the conditioned permutation has a prescribed number of macroscopic blocks. A useful objective is a canonical skeleton map for the actual class, followed by a limit theorem for its ordered size vector. One should distinguish the number of macroscopic blocks, their values, their positions, and the location of the largest jump. The reciprocal transition might then become a geometric degeneration of a simplex of allowed block sizes. (Part IV's theorem 43.2 gives the two-block geometry for $\theta > 1/2$, and its question section 47.8 asks for the whole permutation there.) [Added 1 October 2026, batch 71: for $1/3 < \theta \leq 1/2$ Part VII's canonical parser and its three-piece endpoint table (371) identify the leading families, but it proves no conditioned limit theorem; see section 89.4.] [Added 1 October 2026, batch 72: partly answered in Part VIII. In every fixed phase the conditioned permutation has exactly p macroscopic parser pieces with probability tending to one; their ordered sizes converge to the density $I_p(\theta)^{-1} \prod x_i^{-3/2}$, independently of $p + 1$ uniform endpoint/gap channels (corollary 98.1), and in each reciprocal window the law is a two-family mixture (corollary 98.2). The positions and values of the blocks in the permutation and the location of the largest jump stay open (section 100.4).]

74.5 Uniform relative errors and matching of scales

Can (352) be made uniform as θ approaches a reciprocal boundary, with an error that overlaps the scale in (353)? This requires more than adding a little- o term to the present bounds. The small-part suppression lemma deliberately chooses a very small fixed cutoff and does not optimize the resulting constants. A sharper method would track the cutoff through a dominated-convergence argument and isolate the contribution of intermediate-sized parts. A successful theorem should also specify the effect of rounding m and give a genuinely relative, rather than merely additive, error. [Added 1 October 2026, batch 71: at $q = 2$ Part VII's limits come without an error rate; see section 89.3.] [Added 1 October 2026, batch 72: so do Part VIII's for every q ; see section 100.1.]

74.6 Growing reciprocal index and the spectral regime

The theorem treats fixed q , so it does not apply uniformly when $m/n \rightarrow 0$. A next scale is $q = q(n) \rightarrow \infty$ slowly, or equivalently an adjacency bound tending to infinity but sublinear in n . Which growth rates of q preserve a few-macroscopic-pieces description? When does a saddle point or the fixed- m spectral radius become the correct language? The constants hidden in the proof depend strongly on q , on the available ratio interval, and on the chosen small-part cutoff. Keeping those dependencies explicit is a prerequisite for connecting the present theorem with the report's fixed-bound growth-constant theory (Parts I and II). (This is Part IV's question section 47.10, and Part III's section 32.10 from the other end.) [Added 1 October 2026, batch 72: answered in Part IX for $q \approx n/m \rightarrow \infty$. A few-pieces description with one or two Gamma terms survives while $q = o(\log m)$, outside a phase window near integers (theorems 109.3 and 109.4), and with a bounded number of Gamma terms at $q \asymp \log m$ (theorem 109.1); from $n \geq c_0 m \log m$ on, a saddle sum over the cluster count is the right language (theorem 106.1), with a theta modulation above $m(\log m)^{3/2}$ (theorem 108.2); and the fixed- m spectral rate governs at logarithmic precision for all $n \geq c_0 m \log m$ (theorem 105.1). Bounded q with m/n not converging stays open (section 113.1).]

74.7 Effective bounds and improved lower subclasses

The existence proof can, in principle, be made effective after supplying explicit Catalan bounds and explicit choices of $A, L, \varepsilon, \delta$. Such constants would probably be far from sharp. It would be useful to produce certified, practical inequalities for $p_n(m)$ over finite ranges. The lower

construction used here fixes the first deficiency at at most one. Allowing a bounded but larger deficiency enlarges the subclass, while still admitting exact block counts (Part I's $K_{m,d}$, (18)). How much can the constants improve before the cutoff loss starts to compete with the crossover scale? This is an optimization problem about rigorous inequalities, distinct from proving an exact asymptotic amplitude.

74.8 Other pattern classes and cutoff models

Proposition 72.1 supplies a reusable criterion, but not an automatic result for every Catalan class. For another permutation class, tree family, or weighted composition model, identify a genuine cutoff-respecting injection and a coefficientwise upper encoding with the same tail index. Merely having unrestricted growth $\lambda_\star^n n^{-3/2}$ is not sufficient. The central question is whether the local statistic being bounded can be translated into a cap on every part with only bounded shifts. Establishing that translation would turn the present analytic theorem into a phase-diagram theorem for another concrete model. (Compare Part V's section 63.7.)

74.9 Other tail indices and critical renewal mass

Two adjacent analytic problems are especially natural. At $\alpha = 1$, the moment at the first reciprocal threshold is borderline and may introduce logarithmic factors. At $\bar{w} = 1$, the renewal sequence is critical and the local tail lemma used here fails. Determine the correct two-term estimates in those regimes, including their endpoint orders and crossover scales. The two changes should initially be studied separately: altering the tail index affects moment integrability, whereas altering the total mass changes the number of microscopic parts. There is no reason for a single formal substitution to handle both.

74.10 A staged formalization plan

A feasible formalization should separate three interfaces. The finite combinatorial layer consists of maximum decomposition, skew sums, unique skew-indecomposable factorization, and the injection behind $\underline{A}_m$. The formal-series layer consists of nonnegative coefficientwise inequalities and the identity (342). The analytic layer consists of the geometric-polynomial summation in lemma 70.2, the exponential tilt in lemma 70.3, and the explicit composition bounds in theorem 70.1. Only after those layers are connected should an audit assert the main asymptotic theorem. The endpoint dynamic program and its test cases would be useful executable specifications, but they are not substitutes for any of these proof obligations. (Parts IV and V propose formalization targets in sections 47.13 and 63.10; no Lean or Rocq declaration for any part of this report exists.)

75 Conclusion

The lower-fraction regime has a simple logarithmic structure: each time the adjacency fraction crosses a reciprocal threshold from above, the rarity gains another factor $n^{-1/2}$. At the threshold itself the rarer exponent already applies. The transition from the threshold background to the neighbouring larger probability occurs at a fractional-power scale that depends on the reciprocal index.

The proof is economical in its dependence on previous work. It does not require the full endpoint-state spectrum, a conjectured exact asymptotic profile, or any unverified assertion elsewhere in the report. Two elementary encodings place the counting sequence between subcritical renewal coefficients with the same tail exponent. A local composition argument then transfers a robust phase diagram through that sandwich. What remains difficult is no longer identifying the powers: it is recovering the exact amplitudes and the structure of the conditioned objects.

76 The exact endpoint recurrence used in the checks

This section was the manuscript's Appendix A. It explains the implemented dynamic program. The recurrence is due to Mayama–Akita [2] and is Part I's proposition 2.1, in the coefficient form Part IV prints as (242), with Part IV's shortcut (243); Part V derives it again in section 64. The delivered implementation is a separately packaged adaptation of Part IV's counter code/05-macroscopic-deficits-model.py. The manuscript's derivation is kept here, in Parts IV and V's notation (table 24), because its checks were built on it; its first-entry argument differs from those of Parts I, III and V. No correctness claim for the main asymptotic theorem is delegated to this program.

76.1 First-entry counts

For a permutation of size n , call $n - \pi_1$ its first deficiency. For $0 \leq d < n$, the number of unrestricted avoiders with first deficiency exactly d is

$$N_{n,n-d} = \frac{d+1}{n} \binom{2n-d-2}{n-1}, \quad (354)$$

Part III's (98) with first entry $f = n - d$. Here is a direct derivation. Fix the first entry $f = n - d$. The d entries larger than f must occur in increasing order: a decreasing pair of them together with the first entry would form 132. These d entries divide the remaining $f - 1$ smaller entries into $d + 1$ gaps. All entries in an earlier gap must exceed those in a later gap, since otherwise the separating larger entry completes a 132 pattern. Conversely, these conditions, with an avoiding permutation inside each gap, are sufficient. Thus

$$N_{n,n-d} = [x^{n-d-1}]C(x)^{d+1}.$$

The generalized Catalan coefficient identity

$$[x^r]C(x)^k = \frac{k}{2r+k} \binom{2r+k}{r} \quad (k \geq 1, r \geq 0)$$

follows from $Y = x(1+Y)^2$, $Y = C - 1$, by formal coefficient inversion (or by the standard ballot decomposition); here r and k are local indices. Substitution of $r = n - d - 1$, $k = d + 1$ and cancellation of factorials gives (354). This also proves its integrality.

Write

$$\mathcal{N}_n(u) = \sum_{d=0}^{\min(u,n-1)} N_{n,n-d} \quad (n \geq 1, u \geq 0), \quad (355)$$

with $\mathcal{N}_n(u) = 0$ for $u < 0$, and $\mathcal{N}_0(u) = 1$ for $u \geq 0$ (Part V's (307)). The implementation precomputes these cumulative counts using integers and checks $\mathcal{N}_n(n-1) = C_n$ for every row it constructs.

76.2 Unrestricted two-endpoint counts

Let $T_{u,v}^{\text{free}}(n)$ count unrestricted avoiders with first deficiency at most u and last deficiency at most v , where $u, v \geq 0$. For $n \geq 2$,

$$T_{u,v}^{\text{free}}(n) = \mathcal{N}_{n-1}(u-1) + \sum_{j=1}^{\min(v,n-1)} C_{n-j-1} \mathcal{N}_j(u). \quad (356)$$

(This is Part IV's shortcut (243), which holds for $T_{u,v}^{(m)}(n)$ when $n \leq m + 1$.) To prove this, first consider a final maximum. Removing it leaves an arbitrary avoider of size $n - 1$, whose first deficiency is one less than the original one. This gives the first term.

Otherwise let the final value be $n - j$, with $j \geq 1$. Every larger value must occur before every smaller value: a smaller value followed by a larger one and then by $n - j$ would be a 132 pattern. Hence the permutation consists of an avoiding block of the j larger values, an avoiding block of the $n - j - 1$ smaller values, and the last entry. The first deficiency is the first deficiency of the larger block; the last deficiency is j . The factors in the sum follow. When $n = 1$, the unrestricted count is one for all nonnegative thresholds.

76.3 The adjacency-restricted recurrence

Fix m . Let $T_{u,v}^{(m)}(n)$ count the size- n avoiders with jump bound m and first and last deficiencies at most u, v . Negative thresholds give zero; $T_{u,v}^{(m)}(1) = 1$ for nonnegative thresholds. Thresholds larger than $n - 1$ can be clipped to $n - 1$.

For $n \geq 2$ the exact recurrence is

$$T_{u,v}^{(m)}(n) = T_{u-1,m-1}^{(m)}(n-1) + \sum_{k=1}^{\min(m,n-1)} c_{k,u} T_{m-k,v-k}^{(m)}(n-k), \quad (357)$$

for $u, v \geq 0$, where

$$c_{1,u} = 1, \quad c_{k,u} = \mathcal{N}_{k-1}(u-1) \quad (k \geq 2); \quad (358)$$

this is (242). The first term places the maximum at the end. The last jump is one plus the last deficiency in the shorter permutation, so that deficiency must be at most $m - 1$. The first deficiency drops by one when the maximum is removed.

For a maximum at position $k < n$, its left block has size $k - 1$, and $k \leq m$. All internal jumps and the jump from the left block to the maximum are automatically at most m . The first-deficiency restriction on that block is $u - 1$, giving $c_{k,u}$. The first deficiency of the right block must be at most $m - k$, because the jump after the maximum is k plus that deficiency. The original last deficiency is k plus the right block's last deficiency. These are exactly the two thresholds in the summand. The case $k = 1$ has an empty left block and original first deficiency zero.

If $n \leq m + 1$, the adjacency restriction is vacuous, and (356) provides a useful shortcut. The desired count is

$$a_n^{(m)} = T_{n-1,n-1}^{(m)}(n).$$

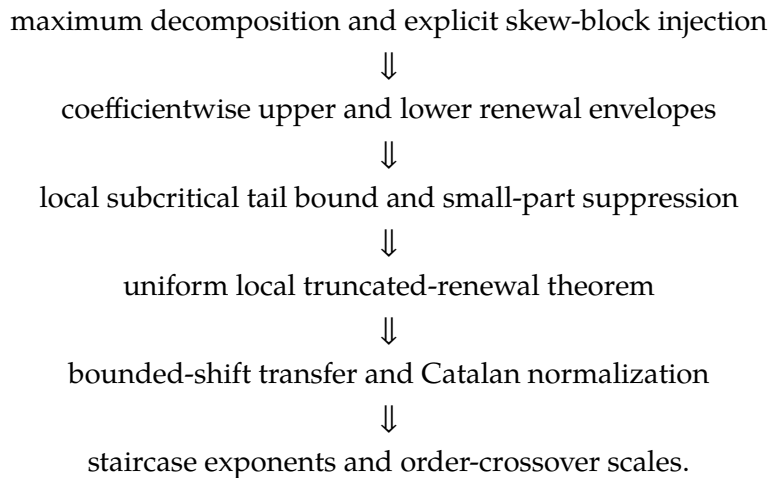
The recursion always reduces n , so memoization terminates. All operations are sums, products, and exact integer divisions in (354); no floating-point number affects a count.

77 Proof audit and reproducibility boundary

This section was the manuscript's Appendix B.

77.1 Dependency chain

The logical chain of the main result is



The endpoint dynamic program is not on this chain. It independently tests the finite combinatorial interfaces and generates the illustrations.

77.2 Quantifiers that must not be weakened

The constants in [theorem 70.1](#) are uniform over all integer m in one fixed ratio interval. That uniformity, not a collection of fixed- θ estimates, is what permits the crossover conclusions. The cutoff ε is chosen after the desired suppression exponent and then held fixed as n grows. Consequently the number of macroscopic-part cases is finite and does not introduce a hidden power of $\log n$.

The lower power bound on w_k is needed for every sufficiently large integer k , not just along a subsequence. This ensures that every composition in the lower families has positive weight and prevents an unexamined lattice-periodicity obstruction. The shifts of the cutoff and coefficient index in (332) are checked explicitly. Finally, q is fixed throughout the theorem; letting it grow would require new estimates.

77.3 What the delivered tests establish

The checks establish exact equalities or inequalities only for the listed finite ranges. They provide evidence that the endpoint recurrence, block generator, and coefficient recursions have been implemented consistently. They do not establish eventual bounds, uniform asymptotics, the existence of limits, independent peer review, or a machine-checked proof. The manuscript was prepared with AI assistance and should be subjected to independent mathematical review before any stronger status is assigned.

The delivery contained the $\text{\LaTeX}$ source, compiled PDF, vector figures, plotting source, exact-count CSV files, verification program and its run log, source ledger, and a SHA-256 manifest; the source, the PDF and the manifest are not shipped ([section 67.1](#)). A rebuild does not require network access once the usual $\text{\LaTeX}$ packages are installed. Regenerating the figures additionally requires NumPy and Matplotlib; rerunning the exact checks does not.

78 Repository pin, literature scope and merge

78.1 Manuscript 05's pin and literature scope

The manuscript's source ledger, shipped as 07-polynomial-rarity-SOURCES.md, records the inspection date 30 September 2026, the ProveIt commit

ffddaa8b9c89e7bf027e1442cc6216bb010906d0

(commit time 2026-09-30T23:38:21Z, root tree cb9c68b025), and the report directory

```
SetTheory/Cardinals/docs/reports/
enumerative-combinatorics/
adjacency-bounded-132-avoiders/.
```

It read three files there, with the Git blobs

README.md	a40d20342fcaf1a97c18ea0064b9c0c9bd9d694e
code/model.py	aee5528c72552036b5e630bd49afe2715f440da4
code/05-macroscopic-deficits-model.py	f0248e0d6181200257e2949c1a8fa8f9fecf5617

The README was read to identify the report's results, status and unresolved regimes: its list of what is not claimed leaves the lower-half equivalents and hierarchy and the order and window at $n/2$ open, it records the upper-half profile used here only for comparison, and its status statement describes the parts as AI-assisted and unrefereed, without Lean or Rocq formalization. Part I's model supplies the endpoint recurrence, the majorant counts and the first-deficiency skew-block construction (its `majorant_counts` and `block_counts`), from which the two envelopes are rederived and are not claimed as new. Part IV's counter supplies the exact endpoint recursion and unrestricted shortcut that the delivered EndpointCounter adapts. The manuscript did not read or independently audit the whole report (then about 140 pages), does not use its asymptotic theorems, and makes no claim that all incoming, unmerged or simultaneous reports were checked for duplication.

The ledger also says that an initial tree query on the moving main branch returned the identifier `f4457a1d19701d32e8f324240c497653d235b53e`, that this is a root-tree identifier and "NOT a commit identifier", that the initial file reads used it, and that the three blobs were then confirmed at the pin. The middle claim is mistaken: in this repository `f4457a1d1` is a commit (the arrival commit of batch 67, committed 16:35:42 on 30 September 2026, Pacific time, with root tree `32b9d20237`). The error does not affect the mathematics or the pin: the three blobs are the same at `f4457a1d1`, at the pin and at the placement commit. The ledger is shipped byte-identical, and the README records the discrepancy.

The primary arXiv records of [1, 2] were checked for the definition, the maximum-position restriction, the original fixed-bound problem and the endpoint-state method; the exact recurrence was read from Parts I and IV's implementations and rederived in section 76. The record and an author-hosted PDF of [6] were inspected, including the theorem assumptions; it is related context, not an invoked theorem, and the fewest-big-jumps mechanism is not claimed as new. Targeted searches covered adjacency-constrained 132-avoiders, macroscopic bounded adjacent jumps, and truncated heavy-tail and fewest-big-jumps asymptotics; they were not an exhaustive priority search. The plotted and tabulated decimals are rounded, not certified intervals. Exact amplitudes, limiting crossover functions, conditioned-permutation geometry and uniformity as q grows are left as research questions, and independent mathematical review and formal verification remain outstanding.

78.2 Where the merge had to choose

- *Placement.* Part VI follows Part V and precedes Part I's appendices, which contain no numbered equations, tables or figures; no number of Parts I–V changed. The manuscript's Appendices A and B are printed as sections 76 and 77; its "Commands and files" subsection is replaced by section 73.4. Its title page and abstract are replaced by the opening note of Part VI, the status box of section 67.1 and a paragraph of the report's abstract; every claim of its abstract is stated in section 68. The delivery's title-page, running-head and metadata wording naming the manuscript's addressee is not printed; the manuscript is described here only as AI-assisted.

- *Introduction.* The manuscript's Section 1 describes this report from outside, by README. Its definitions are section 68's first subsection; its account of the gap, its trust boundary and its literature paragraph are rewritten as sections 67.2 and 67.3 and the subsection on public work, citing Parts I–V by label; its asymptotic convention is in section 67.5. Its display of the upper-half equivalent is Part IV's (153) and is cited, not reprinted (the one manuscript label not kept); its citations of the repository's README, code/model.py and code/05-macroscopic-deficits-model.py are replaced by references to Parts I, II and IV.
- *Duplicates with Parts I, II, IV and V.* The two envelope lemmas are Part I's proposition 5.1 and proposition 5.2 at $d = 1$ (plus $\bar{A}_m \leq C$); the manuscript's proofs, which follow Part I's, are kept and marked because its renewal normalization and its checks are built on them. Remark 69.2 is Part II's warning with its counterexample. The lower bounds of theorem 68.1 are Part IV's theorem 45.1, credited; the new content is the matching upper bounds and the endpoints. The endpoint recurrence of section 76 is Part I's, as printed by Parts IV and V; its derivation is kept, marked. The masses and tails of (330) and (331) agree with Part I's (23) and (25) at $d = 1$; this identification is the merge's.
- *What Part VI sharpens.* Theorem 68.1 supplies the upper bounds missing beside theorem 45.1, and for $0 < \theta \leq 1/2$ improves theorem 38.1's $o(n^{-1/2})$ to an exact order. It is weaker than theorem 38.1 for $\theta > 1/2$, which it does not use. Nothing in Parts I–V is refuted.
- *Notation.* Every rename is listed in table 24: $F_q \rightarrow O_q$, $a, b \rightarrow \theta_-, \theta_+$, $w_q \rightarrow w_q$, $B_m \rightarrow \bar{A}_m$, $L_m \rightarrow \underline{A}_m$, $H_{m-1} \rightarrow K_{m,1}$, $\mathcal{B}_k \rightarrow \mathfrak{B}_k$, $\rho, \rho_\pm \rightarrow \bar{w}, \bar{w}_\pm$, $R, R_m, R_M^\pm \rightarrow Z, Z_m, Z_M^\pm$, $\Lambda \rightarrow \lambda_\star$, $b_n^{(m)} \rightarrow \tilde{a}_n^{(m)}$, $K_q, L_q, H_q \rightarrow \mathfrak{K}_q, \mathfrak{L}_q, \mathfrak{H}_q$, $C(x) \rightarrow C(x)$, $f(n, d) \rightarrow N_{n,n-d}$, $h(n, u) \rightarrow \mathcal{N}_n(u)$, $T_{\text{free}}(n, u, v) \rightarrow T_{u,v}^{\text{free}}(n)$, $T(n, u, v) \rightarrow T_{u,v}^{(m)}(n)$, $c(k, u) \rightarrow c_{k,u}$; the manuscript's $\mathbb{P}$ is Part III's $\mathbb{P}$. No normalization changed. The manuscript's $r = \lfloor 1/\theta \rfloor$ is kept, with its relation to Part IV's r stated in section 67.2.
- *Labels.* The manuscript's labels are kept with the prefix pr:, except eq:prior (above); unprefixed, its eq:upper and lem:shift would collide with Part I's. The merge added labels to new headings and tables of Part VI, to the ten research questions, and to four unlabelled question headings of Parts IV and V (sections 47.10, 63.3, 63.7 and 63.10) so that Part VI can cite them.
- *Research questions.* All ten are printed. Three refine questions of Parts IV and V (amplitudes, crossover functions, growing reciprocal index) and three bear on others (geometry, other classes, formalization); they keep the manuscript's text with a pointer. Part IV's sections 47.1 and 47.2 and Part V's section 63.3 carry dated notes that Part VI answers them at the level of orders.
- *The merge's own text.* The opening note, sections 67.1 and 67.2, most of section 67.3, the notation subsection and tables, the sentences attributing results to Parts I, II and IV, the parenthesized pointers in the research questions, this section's first subsection (condensed from 07-polynomial-rarity-SOURCES.md) and this list are the merge's.
- *Reproduction and figures.* The manuscript's package listing and commands are replaced by the shipped names and the copy recipe (section 73.4 and the README). The two figure PDFs were regenerated in the write phase by the delivered figures.py, run unchanged on a copy of the delivered layout with Matplotlib 3.10.8, NumPy 2.3.5 and pdf.fonttype 42, so that they embed TrueType rather than Type 3 fonts; they plot the shipped data. Their axis labels keep the manuscript's Pr; the caption of figure 15 says so.
- *Stale sentences elsewhere.* The abstract gained a paragraph; Part II's research direction on joint growth, Part IV's scope list (section 37.3), its scope paragraph on what neither manuscript claims, its edge at one half (section 43.1), the remark after theorem 45.1, its questions section 47.1 and section 47.2, its scope table and its "What is resolved", Part V's scope (section 62) and its half-size question, and Part I's section B, carry dated pointers to Part VI; their original wording is kept.

Part VII. The exact half-size crossover

Part VII was added on 1 October 2026 in batch 71 of ProveIt's incoming-report intake, from one manuscript. Part VI proved that $\mathbb{P}(M_n \leq \lfloor n/2 + t\sqrt{n} \rfloor) \asymp n^{-1}(1 + t_+)$ for bounded t (corollary 68.4) and asked for the limiting crossover functions, the endpoint constants and the amplitudes inside the lower phases (sections 74.1 to 74.3); Parts IV and V asked for the transition function at $m = n/2$ (sections 47.2 and 63.3). Part VII proves, locally uniformly in real t ,

$$n \mathbb{P}(M_n \leq \lfloor n/2 + t\sqrt{n} \rfloor) \longrightarrow 6 + \frac{12}{\sqrt{\pi}} t_+,$$

and the equivalent $\mathbb{P}(M_n \leq \lfloor \theta n \rfloor) \sim I_3(\theta)/(2\pi n)$ throughout the first sub-half-size phase $1/3 < \theta \leq 1/2$, with an explicit two-dimensional integral I_3 . This answers Part VI's questions on crossover functions and endpoint constants at $q = 2$ and on amplitudes at $r = 2$, and the transition questions of Parts IV and V. Its proofs re-derive every finite interface they use and assume no asymptotic theorem of Parts I–VI.

79 Provenance, conventions and the answer

79.1 One manuscript, one addition

Part VII is built from one AI-assisted research manuscript, *manuscript 03* of batch 71, *Exact half size crossover for adjacency bounded 132 avoiders*, dated 1 October 2026 (8 pages, US Letter, as delivered; its byline reads “AI-assisted mathematical research”). It arrived as the archive `132_Avoider_Crossover_Package.zip` (inner directory `Half_Size_Crossover/`, main file `half_size_crossover.tex`) in ProveIt commit `ae32ad45`, and was placed in commit `427bca743`. Its source ledger pins ProveIt commit `29e52fcd`, the placement commit of batch 68, at which this report's `article.tex` had the blob `0b7cb2f2e6` and ended at Part V; and it read the incoming batch-68 archive `ProveIt_Polynomial_Rarity.zip` (blob `cf18b47515`, arrival commit `a866ff9a2`), whose manuscript is printed above as Part VI. Part VI was written into this report after the pin, in commit `cfdefe6e3`, whose parent still had the article blob `0b7cb2f2e6`. The manuscript therefore cites Part VI's results from the incoming manuscript, which Part VI prints in full, and its statements about Parts III–V refer to the text printed above (apart from the dated batch-68 notes that Part VI added there).

Its code, data and source ledger are shipped beside the earlier parts' files with the prefix `08-half-size-crossover-`, continuing the report's local sequence after Part VI's `07-polynomial-rarity-`. Its manuscript, README and PDF are not shipped and survive in the arrival commit; its checksum ledger was verified at placement (11 of 11 files) and retired. Its `model.py` is byte-identical to Part VI's `code/07-polynomial-rarity-model.py` (Git blob `a4fdee3761`): the manuscript copied it from the batch-68 package and credits it as such. Both copies are shipped, the second as `code/08-half-size-crossover-model.py`, because the delivered verifier imports it by its delivered name (section 88.2).

The manuscript contributes, in the order printed here: the limiting half-size crossover (theorem 80.1) and the first sub-half-size equivalent (theorem 80.2); uniform endpoint estimates for a uniform avoiding block (lemma 81.1); a canonical threshold parser valid for every adjacency bound below the size (lemma 82.1) with its word masses; small-word estimates (lemma 83.1) and their use to discard large small-word costs; the two-piece contribution, with a uniform tail at the rare-event scale (lemma 84.1); the exact three-piece endpoint table and its mass eight; the exclusion of every other parser family at the scale n^{-1} ; the evaluation $I_3(1/2) = 12\pi$; finite regression checks; an exact statement of its repository boundary; and four further questions. Every result, proof, remark, limitation and question of the manuscript is printed.

Status (manuscript 03 of batch 71). The theorems of Part VII have written mathematical proofs, which the manuscript reports were checked by direct derivation. They are ordinary mathematical arguments, not proof-assistant formalizations: nothing is Lean- or Rocq-verified, nothing has been externally refereed, and no exhaustive publication priority is claimed. Part VII re-derives the finite interfaces it uses and does not assume Part IV's macroscopic equivalent (153); its small-word estimates are Part VI's lemmas, re-derived. Its conclusions are limits and asymptotic equivalents with no finite-size error rate (section 88). Its finite data are regression checks, not evidence for the limits.

79.2 The questions of Parts IV–VI, and what answers them

With $p_n(m) = \mathbb{P}(M_n \leq m)$ as in (310), Part VII answers the following questions in their first case.

- *The crossover function at $q = 2$* (Part VI's section 74.3). There $w_2(n) = \sqrt{n}$, and theorem 80.1 proves (353) at $q = 2$ with

$$\mathfrak{S}_2(t) = \mathfrak{R}_2 + \mathfrak{Q}_2 t_+, \quad \mathfrak{R}_2 = 6, \quad \mathfrak{Q}_2 = \frac{12}{\sqrt{\pi}}.$$

This is exactly Part VI's "particularly simple candidate" at $q = 2$, with the slope that matching with (153) "would suggest". The slope is proved directly, by the parser, not by that matching. For $q \geq 3$ the question stays open (section 89.2).

- *The endpoint constant at $q = 2$* (section 74.2). $\mathfrak{R}_2 = \lim_n n p_n(\lfloor n/2 \rfloor)$ exists and equals 6. Theorem 80.1 holds along all n , odd and even, so the limit has no residue-class oscillation; $\mathfrak{R}_q$ for $q \geq 3$ stays open.
- *The amplitude at $r = 2$* (section 74.1). For $1/3 < \theta < 1/2$, (352) holds with

$$\Omega_2(\theta) = \frac{I_3(\theta)}{2\pi},$$

I_3 as in (359), and at $\theta = 1/2$ the same formula gives 6. This is also the $r = 2$ case of the amplitude half of Part IV's section 47.1; Part IV's manuscript 11 asked there for an equivalent for every $0 < \theta < 1/2$, which is now known only for $1/3 < \theta < 1/2$. The amplitudes for $r \geq 3$ stay open (section 89.1).

- *The transition at $m = n/2$* (Part IV's section 47.2, Part V's section 63.3). For $s_n = t\sqrt{n}$, the probability scale is n^{-1} (Part VI) and the limiting transition function is $6 + (12/\sqrt{\pi})t_+$, continuous and piecewise linear, constant on the lower side.

Theorem 80.1 sharpens Part VI's corollary 68.4 from a two-sided order to a limit, and theorem 80.2 sharpens theorem 68.1 at $r = 2$ from an order to an equivalent; nothing in Parts I–VI is refuted. Two consistency checks connect the new constants with the earlier parts. The ramp slope $12/\sqrt{\pi}$ is the slope of Part IV's profile at its upper end, (222): substituting $\theta = 1/2 + t/\sqrt{n}$ into (153) and multiplying by n gives $12t/\sqrt{\pi} + o(1)$ for fixed $t > 0$, although no theorem licenses that substitution. And $I_3(\theta) \rightarrow 0$ as $\theta \downarrow 1/3$, because the integration domain of (359) shrinks to the single point $x = y = z = 1/3$, in agreement with the jump of the staircase exponent from 1 to $3/2$ at $\theta = 1/3$. Part VII does not answer Part VI's questions on the conditioned geometry, uniformity near reciprocal boundaries, growing reciprocal index, effective bounds, other classes, other tail indices or formalization (sections 74.4 to 74.10). [Added 1 October 2026, batch 72: Part VIII answers the cases $q \geq 3$ and $r \geq 3$ that the list above leaves open, and Part VI's question on the conditioned geometry at the level of the parser's macroscopic pieces (section 90.2).]

79.3 Notation of Part VII

The symbols C_n , C , $\text{Av}_n(132)$, $M(\pi)$, M_n , $p_n(m)$ of Part VI, $c_j = C_j/4^j$ and the constants $c_{\pm}$ of (181), the first entry F , first deficiency J and last entry L of a block (Parts III and IV), the

operations R_β and A with Part III's costs, the cost $\text{cost}(w)$ of an outer word, the word series $\mathcal{W}$, $\mathcal{W}_R$ of (191)–(192), the acceptance weight $\chi(w)$ of (211), Part IV's fixed fraction θ and amplitude $\Omega_r(\theta)$, and Part VI's w_q , $\mathfrak{R}_q$, $\mathfrak{Q}_q$, $\mathfrak{S}_q$ keep their meanings. Part VII fixes the remaining names as follows (table 27).

- The manuscript's word masses W , W_R , W_A are printed as $\mathcal{W}(1/4)$, $\mathcal{W}_R(1/4)$ and $\mathcal{W}_A(1/4)$, with $\mathcal{W}_A := \mathcal{W} - \mathcal{W}_R$ the series of outer words containing an A . A bare W is Part IV's random outer word and Part VI's renewal kernel; it does not occur in Part VII.
- The manuscript's small-word series $S(z)$, $T(z)$ are printed as $S_n(z)$, $T_n(z)$: they are Part VI's (340) for the upper kernel w^+ of (327). They are not Part V's survival functions, Part III's cost T or Part IV's terminal operation T_τ .
- x , y , z are both the three macroscopic size fractions of (359) and section 85 and, in z 's case, the formal variable of sections 82 and 83; the two uses never meet in one formula. The tempting false reading of z in (359) is the formal variable; there it is $1 - x - y$.
- K , K_h , K_B are constants, as in Part IV (as declared after (181)); they are not Part II's polynomial K_m , Part I's $K_{m,d}$ or Part VI's $\mathfrak{R}_q$. The cost cutoff H is not Part III's series H , and the bound B on $|t|$ is not Part I's B_d or Part V's B_0 , B_1 .
- The letter L is the last entry of a block, as in Part IV, except in the proof of lemma 83.1, where it is the tilt constant (as in Part VI's proof of lemma 70.3); ℓ is a value of the last entry, and l the number of letters of a composition.

Table 27: Symbols of Part VII. “Source” gives manuscript 03's symbol; the last column prints the tempting false reading.

Symbol	Meaning in Part VII	Source	Not to be confused with
$\mathcal{W}(1/4)$, $\mathcal{W}_R(1/4)$, $\mathcal{W}_A(1/4)$	normalized masses 4, 2, 2 of all outer words, of those without A , of those with an A	W , W_R , W_A	Part IV's random word W ; Part VI's kernel W .
$S_n(z)$, $T_n(z)$	small-word series and its resolvent	$S(z)$, $T(z)$	Part V's survival $\mathcal{S}$; Part III's cost T ; T_τ .
$I_3(\theta)$, $I_{3,\varepsilon}$	three-piece integral (359) and its cut-off version	same	Part IV's split integral $I(u)$.
$\chi(w)$; $\chi(w_1, w_2)$	two-piece weight (211); three-piece weight (371)	χ	—
x , y , z	size fractions k/n , a/n , b/n	same	the formal variable z ; Part IV's threshold u .
Δ	$2m - n$	Δ	Part VI's $\Delta = qm - n$ at $q = 2$ (equal).
K , K_h , K_B	constants	same	Part II's K_m ; $\mathfrak{R}_q$.
H ; h ; h_n	cost cutoff; a power of T_n ; deleted splits	same	Part III's series H ; Part III's $h = \lfloor n/4 \rfloor$.
L ; ℓ ; l	last entry (tilt constant in one proof); its value; number of letters	same	Part VI's $\mathfrak{Q}_q$; Part I's L_m .
k ; s ; j	cost of a large R ; small-word cost; number of large R	same	Part IV's word cost t (here t is the window variable).

No normalization changed: every renamed symbol denotes exactly the manuscript's quantity. Probability is written $\mathbb{P}$, positive parts $t_+ = (t)_+$, and cost cost , as in Parts III–VI. Within single proofs, α and β in the proof of lemma 81.1 are blocks (not Part I's α_m or Part II's β_m ; elsewhere β is the decoration of R_β), and λ , η are local parameters. Equations, tables and figures are numbered consecutively with those of Parts I–VI.

80 Results and research boundary

A permutation $\pi \in S_n$ avoids 132 if no $i < j < k$ satisfy $\pi_i < \pi_k < \pi_j$. The set $\text{Av}_n(132)$ has Catalan cardinality $C_n = (n+1)^{-1} \binom{2n}{n}$. Put

$$M(\pi) = \max_{1 \leq i < n} |\pi_{i+1} - \pi_i|, \quad p_n(m) = \frac{\#\{\pi \in \text{Av}_n(132) : M(\pi) \leq m\}}{C_n}.$$

An empty maximum is zero. (These are Part VI's (309) and (310).) Part VI, from the manuscript that this one inspected as an incoming archive (section 79.1), proves

$$p_n(m) \asymp n^{-1} + n^{-3/2}(2m - n)_+$$

uniformly near $m/n = 1/2$ (corollary 68.4), but explicitly leaves limiting crossover functions and exact amplitudes open (sections 74.1 and 74.3). The following strengthens that comparison to a limit.

Theorem 80.1 (Half-size crossover). *For every $B < \infty$,*

$$\sup_{|t| \leq B} \left| np_n\left(\lfloor n/2 + t\sqrt{n} \rfloor\right) - 6 - \frac{12}{\sqrt{\pi}}(t)_+ \right| \rightarrow 0.$$

In particular, $p_n(\lfloor n/2 \rfloor) \sim 6/n$.

Define

$$I_3(\theta) = \iint_{\substack{x+y+z=1 \\ 0 < x, y, z \leq \theta}} (xyz)^{-3/2} dx dy, \quad z = 1 - x - y. \quad (359)$$

The measure is the coordinate measure $dx dy$, not Euclidean surface area on the plane.

Theorem 80.2 (First sub-half-size equivalent). *For $1/3 < \theta \leq 1/2$,*

$$p_n(\lfloor \theta n \rfloor) \sim \frac{I_3(\theta)}{2\pi n}.$$

The convergence is uniform on compact subintervals of $(1/3, 1/2)$. The integral is finite and positive, and $I_3(1/2) = 12\pi$.

The fixed-bound enumeration was initiated by Nadler [1] and developed by Mayama and Akita [2]. The present argument uses elementary Catalan interfaces appearing in Parts III–IV of this report (sections 39.3, 39.4 and 40.4), and a small-letter estimate appearing in Part VI (lemmas 70.2 and 70.3). We rederive the needed estimates. We do not infer any infinite claim from numerical data, and do not assume the earlier macroscopic equivalent. Related fewest-big-jumps results, such as [6], provide context but are not invoked as theorems here.

81 Finite Catalan interfaces

Write $c_j = 4^{-j} C_j$. Stirling's formula and elementary bounds give

$$c_j \sim \frac{1}{\sqrt{\pi} j^{3/2}}, \quad c_-(j+1)^{-3/2} \leq c_j \leq c_+(j+1)^{-3/2}. \quad (360)$$

(This is Part IV's (181).) The maximum decomposition is unique:

$$\pi = (\sigma + b) N \tau, \quad \sigma \in \text{Av}_a(132), \quad \tau \in \text{Av}_b(132), \quad a + b + 1 = N.$$

Every left entry exceeds every right entry. Conversely, arbitrary avoiding children with these value sets give an avoider. If $b > 0$ and $M(\pi) \leq m$, then $a + 1 \leq m$, because the jump following N is at least $a + 1$. The left internal jumps and the jump into N are smaller than this downward boundary jump.

For a block of size N let F, L be its first and last entries, and $J = N - F$ its first deficiency.

Lemma 81.1 (Endpoint estimates). *For an unrestricted uniform avoiding block,*

$$\mathbb{P}(J \geq h) \leq (h + 2)2^{-h}, \quad (361)$$

$$\#\{\pi \in \text{Av}_N(132) : L(\pi) = \ell\} = C_{N-\ell}C_{\ell-1}. \quad (362)$$

For $1 \leq r \leq N$, uniformly,

$$\mathbb{P}(L \geq r) = \frac{1}{2} + O\left(r^{-1/2} + (N - r + 1)^{-1/2}\right). \quad (363)$$

In particular $L/N \Rightarrow \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1$.

Proof. The first-entry count at $F = N - j$ is

$$\frac{j+1}{N} \binom{2N-j-2}{N-1}.$$

This is the Catalan triangle: it follows, for example, from the first-entry generating function $(xC(x))^{j+1}$ and Lagrange inversion applied to $C = 1 + xC^2$. Division by C_N gives

$$(j+1) \frac{N+1}{2(2N-1)} \prod_{u=0}^{j-1} \frac{N-1-u}{2N-2-u} \leq (j+1)2^{-j-1}$$

for $N \geq 2$; summation proves (361), and $N = 1$ is immediate.

For the last entry ℓ , the entries larger than ℓ must precede all entries smaller than ℓ : otherwise a smaller entry, a larger entry and the final ℓ form a 132. Thus $\pi = (\alpha + \ell)\beta\ell$, with independent avoiding blocks of sizes $N - \ell$ and $\ell - 1$. This proves (362). The distribution is symmetric under $\ell \mapsto N + 1 - \ell$. The Catalan bounds give $\mathbb{P}(L = \ell) \leq K\ell^{-3/2}$ for $\ell \leq N/2$, and the reflected bound at the other endpoint. Summing from an endpoint cutoff to the middle yields (363). $\square$

Remark 81.2 (Merge note: the endpoint estimates in Parts III, IV and VI). The first-entry count is Part IV's (175) (Part III's (98), Part VI's (354)), and (361) with its proof is Part IV's lemma 39.2, (177), the same computation. Equation (362) is Part IV's (178) with the same decomposition. Equation (363) is uniform in $1 \leq r \leq N$, with an explicit error at both ends; Part IV's (180) states the same for r in $[\eta N, (1 - \eta)N]$ only, by the same symmetry argument, and its limit statement is (179). The manuscript's proofs are kept because they are short and self-contained.

82 A parser that remains valid below half size

Let R_β attach the left decoration β and the new maximum before a current avoiding block. Its cost is $k = |\beta| + 1$. Let A append a new maximum; its cost is one. Each operation preserves avoidance. If the old last deficiency is Q , then R_β changes it to $Q + k$. An append is accepted precisely when $Q + 1 \leq m$ and then sets last deficiency to zero.

Lemma 82.1 (Canonical threshold parsing). *Every m -bounded avoider of size $n > m$ has a unique representation by a decorated outer word in R, A applied to a terminal core*

$$(\sigma + b)N\tau, \quad a, b \leq m, \quad N = a + b + 1 > m.$$

Every outer R has cost at most m . Conversely, reconstruction from such a core and word is injective, and every accepted reconstruction has the prescribed parser.

Proof. At each maximum decomposition, follow a child of size $> m$ if there is one. With a nonempty right sibling, the left size plus one is at most m , so a large child is unique. A large right child records R ; a large left child necessarily has empty right sibling and records A . Stop when both children have size at most m . Sizes decrease strictly, and the size on stopping exceeds m . Reconstruction reverses these unique maximum decompositions. At every intermediate step the followed core exceeds m , while an R decoration has size below m , so the reconstructed parser follows exactly the specified core. $\square$

Remark 82.2 (Merge note: Part IV's parser). Lemma 82.1 is Part IV's lemma 40.4 (manuscript 11 of batch 52), with the same parsing rule, extended from the range $n/2 \leq m \leq n-1$ of (188) to every $m < n$. Above half size two children larger than m are excluded by their total size; below it, as the proof above shows, they are excluded by the positional restriction alone, so uniqueness survives. The price is that the terminal core no longer has fixed cost: its children may be small, and the outer word may contain macroscopic R letters, which section 86 controls.

The total normalized weight of a finite decorated word is obtained by giving a word of cost s weight 4^{-s} . Write $\mathcal{W}_R$ for words with no A , and $\mathcal{W}_A = \mathcal{W} - \mathcal{W}_R$ for words with at least one A . The Catalan identities give

$$\mathcal{W}(1/4) = \frac{1}{1 - zC(z) - z} \Big|_{z=1/4} = 4, \quad \mathcal{W}_R(1/4) = \frac{1}{1 - zC(z)} \Big|_{z=1/4} = 2, \quad \mathcal{W}_A(1/4) = 2. \quad (364)$$

The unnormalized cost generating function is $C(z)^2$, whose cost- s coefficient is C_{s+1} . In particular the normalized cost tail is $O((s+1)^{-3/2})$ locally. (These are Part IV's (191)–(192) at $z = 1/4$ and (193).)

83 Uniform bounds for small words

Fix $\varepsilon > 0$. An outer R is large if its cost exceeds εn . All remaining letters form small words, with normalized series

$$S_n(z) = \frac{z}{4} + \sum_{k \leq \lfloor \varepsilon n \rfloor} C_{k-1} 4^{-k} z^k, \quad T_n(z) = (1 - S_n(z))^{-1}.$$

For sufficiently large n , the first term accounts for A and the sum for all small R operations; in particular cost one has two letters. Since $S_n(1) \leq 3/4$, these are subcritical series. (They are Part VI's (340) for the upper kernel (327), whose mass is $\bar{w}_+ = 3/4$.)

Lemma 83.1 (Small-word estimates). *For every fixed positive integer h , uniformly in n and ε ,*

$$T_n(1)^h \leq 4^h, \quad (365)$$

$$[z^s]T_n(z)^h \leq K_h(s+1)^{-3/2}, \quad (366)$$

$$\sum_s s[z^s]T_n(z)^h \leq K_h \sqrt{\varepsilon n} \quad (\varepsilon n \geq 1). \quad (367)$$

Given $c, A_0 > 0$, choosing ε sufficiently small gives

$$[z^s]T_n(z)^h = O_h(n^{-A_0}) \quad (s \geq cn), \quad (368)$$

with the same choice of ε for any prescribed finite set of h .

Proof. The mass bound is immediate. In a composition of s with l letters, mark a largest letter, whose size is at least s/l . Its weight is at most $Kl^{3/2}s^{-3/2}$. There are at most l choices for the mark, and the other letters have total weight at most $(3/4)^{l-1}$. Expanding $T_n^h = \sum_{l \geq 0} \binom{l+h-1}{h-1} S_n^l$ proves

(366) by a convergent polynomial-geometric sum. Differentiate T_n^h at one and use $S'_n(1) = O(\sqrt{\varepsilon n})$ to obtain (367).

For (368), choose $L > A_0/c$ and then $L\varepsilon < 1/2$. With $\lambda = L \log n/n$,

$$S_n(e^\lambda) - S_n(1) \leq \lambda e^{\lambda \varepsilon n} \sum_{k \leq \varepsilon n} k[z^k]S_n = O\left((\log n)n^{L\varepsilon-1/2}\right) = o(1).$$

Thus $T_n(e^\lambda)$ stays bounded. Positivity bounds the coefficient at $s \geq cn$ by $e^{-\lambda s}T_n(e^\lambda)^h = O_h(n^{-A_0})$. $\square$

Remark 83.2 (Merge note: Part VI's lemmas). Equations (366) and (368) are Part VI's lemma 70.2 and lemma 70.3 for the upper kernel, with tail index $\alpha = 1/2$; the proofs are the same (Part VI's condition $L\varepsilon < \alpha$ is $L\varepsilon < 1/2$ here). Equation (367) is a first-moment bound for the truncated series, which grows like $\sqrt{\varepsilon n}$; Part VI's (338) bounds the moments of order $p > \alpha$ of a partial sum instead. The manuscript credits the local bound and the tilt to Part VI's manuscript and re-derives them.

Choose $\varepsilon < 1/16$ small enough that (368) holds with $c = 1/4$, $A_0 = 10$. A parser word has at most $\lceil 1/\varepsilon \rceil$ large letters. If it has j large letters, its $j+1$ intervening gaps are counted by T_n^{j+1} . For this fixed ε , large-gap estimates may depend on j and ε , but the polynomial exponent A_0 need not grow with j : after applying (368), sum every remaining normalized letter and core weight without a size constraint. Their total is bounded by $4(1/2)^j$. Division by c_n loses only $O(n^{3/2})$. Hence total small-gap cost at least $n/4$ contributes $o(n^{-1})$ in every case.

Remark 83.3 (Editorial note, 1 October 2026: the bound $4(1/2)^j$). The preceding paragraph does not say where the factor $(1/2)^j$ comes from, and one reading of “every remaining normalized letter” (all letters, including those of the gaps) makes it false. Here is a justification, added in the write phase and checked for this report. Fix $j \leq \lceil 1/\varepsilon \rceil$ and a total small-gap cost $s \geq n/4$. A parser word with j large letters splits uniquely as $g_0 R_{\beta_1} g_1 \cdots R_{\beta_j} g_j$ with small gaps g_i , and its terminal core is $(\sigma + b)N\tau$. Dropping every adjacency constraint only increases the count, and the total size is the sum of the costs and of the core size, so the normalized weight of all such configurations is at most the product of three factors:

- the gaps: $[z^s]T_n(z)^{j+1}$, which is $O_j(n^{-A_0})$ by (368) with $h = j+1$;
- the large letters: each carries at most $\sum_{k \geq 1} C_{k-1}4^{-k} = \frac{1}{4}C(1/4) = \frac{1}{2}$, the “critical total weight of one R operation” of Part IV's proof of corollary 43.3, so together at most $(1/2)^j$;
- the core: at most $\sum_{a,b \geq 0} C_a C_b 4^{-(a+b+1)} = \frac{1}{4}C(1/4)^2 = 1$.

Summing over the at most $n+1$ values of s gives the normalized weight $O_j(n^{1-A_0})(1/2)^j$; the paragraph's “ $4(1/2)^j$ ” is this bound for the letters and the core, with the core's mass 1 enlarged to 4, and the gap factor kept from (368). Division by c_n gives $O_j(n^{5/2-A_0}) = O_j(n^{-15/2})$, and the finitely many values of j give $o(n^{-1})$, as stated. The dependence on j is harmless: the proof of (368) chooses ε from c and A_0 alone, and $h = j+1$ enters only through $T_n(e^\lambda)^h \leq 5^h$ for large n . The tempting false reading sums the gap letters without a size constraint as well; that gives $T_n(1)^{j+1}(1/2)^j \leq 4^{j+1}2^{-j} = 4 \cdot 2^j$, not $4(1/2)^j$, and loses the suppression, so it cannot be the intended step. The conclusion of the paragraph stands. (The three masses $1/2$, 1 and $S_n(1) \leq 3/4$ were recomputed exactly for this note.)

84 The two-piece contribution in the window

Set

$$m = \lfloor n/2 + t\sqrt{n} \rfloor, \quad \Delta = 2m - n, \quad |t| \leq B.$$

First consider parser words with no large R . If their cost is $s \leq n/4$, the terminal children are both at least $n/4 - O(\sqrt{n})$. Their weight divided by C_n is $O(4^{-s}n^{-3/2})$, and there are at most $(\Delta + s + 2)_+$ possible splits.

Lemma 84.1 (Uniform tail at the rare-event scale). *For any cost cutoff H , the no-large- R contribution from $s > H$, multiplied by n , is at most*

$$K_B \left((H+1)^{-1/2} + \sqrt{\varepsilon} \right) + o(1). \quad (369)$$

Proof. Costs $s > n/4$ were suppressed above. For the rest, apply lemma 83.1 to

$$Kn^{-1/2} \sum_{s>H} (\Delta_+ + s + 2) [z^s] T_n.$$

The constant term is bounded by $K_B \sum_{s>H} (s+1)^{-3/2}$, and the first-moment term by $K\sqrt{\varepsilon}$. $\square$

For a fixed decorated word w of cost s , define

$$\chi(w) = \begin{cases} 1, & w \text{ has no } A, \\ 1/2, & w \text{ has an } A. \end{cases}$$

(This is Part IV's acceptance weight (211).) If $t < 0$, there are no terminal splits for all large n ; if $t = 0$, there are only $O(1)$, giving $o(n^{-1})$. If $t > 0$, their number is $2t\sqrt{n} + O(1)$. Delete h_n splits near each end, with $h_n \rightarrow \infty$ and $h_n = o(\sqrt{n})$. On the remaining splits, the slack below the adjacency cap grows, so (361) makes the downward constraints asymptotically automatic. If w contains an append, its first such append is accepted with probability $1/2 + o(1)$ by (363); subsequent appends have bounded jumps after the endpoint reset. No endpoint independence is needed: removing a first-deficiency event of probability $o(1)$ changes every other acceptance probability by at most $o(1)$.

Uniform Stirling at these near-half splits gives

$$\frac{C_a C_b}{C_n} \sim \frac{84^{-s-1}}{\sqrt{\pi} n^{3/2}}.$$

Therefore the contribution of w , multiplied by n , tends to $4t_+ \chi(w) 4^{-s} / \sqrt{\pi}$. Summing costs up to H , then using lemma 84.1, gives in the order $n \rightarrow \infty, H \rightarrow \infty, \varepsilon \downarrow 0$ the limit

$$\frac{4t_+}{\sqrt{\pi}} \sum_w \chi(w) 4^{-\text{cost}(w)} = \frac{4t_+}{\sqrt{\pi}} \left(2 + \frac{1}{2} \cdot 2 \right) = \frac{12t_+}{\sqrt{\pi}}. \quad (370)$$

Remark 84.2 (Merge note: the two-piece family is Part IV's). The sum in (370) is Part IV's word mass (219), 3, and the normalized weights $\chi(w) 4^{-\text{cost}(w)} / 3$ are Part IV's limiting outer-word law (225), whose cost marginal is $\mathbb{P}(T_W = t) = (C_t + C_{t+1}) / (6 \cdot 4^t)$ ((227)). The argument above is Part IV's fixed-word computation (proposition 42.1), moved from a fixed split fraction $u < 1/2$ to the moving window $|a/n - 1/2| \leq t/\sqrt{n}$, where the split kernel $\mathfrak{k}(y)$ of (212) is $\mathfrak{k}(1/2) = 8$ and the window has width $2t/\sqrt{n}$; that product, $16t/\sqrt{n}$, times $3/(4\sqrt{\pi})$ is the ramp $12t/(\sqrt{\pi}\sqrt{n})$ on Part IV's scale (the merge's consistency check, not a proof). New are the uniform tail lemma 84.1 at the rarer scale n^{-1} and the deletion of the large- R families, which do not exist above half size.

85 The three-piece contribution

Now require exactly one large R and both terminal children larger than εn . Write the word as $w_1 R_\beta w_2$, with w_1, w_2 the two small-word gaps. For fixed gaps of combined cost s , set

$$k = |\beta| + 1, \quad x = k/n, \quad y = a/n, \quad z = b/n.$$

The limiting domain is $x + y + z = 1, 0 < x, y, z \leq 1/2$. Restrict initially to a compact subset of its interior. The first deficiencies in β, σ, τ are uniformly tight by lemma 81.1, and all costs have linear slack below m , so every R join is accepted apart from a vanishing fraction.

The only remaining conditions come from appends. If τ has its low last-endpoint mode, the terminal last deficiency is asymptotic to $(y + z)n > m$; neither gap may contain an append. If it has its high mode, the terminal deficiency is asymptotic to $yn < m$. An append in w_2 resets it, so after the large R the deficiency is asymptotic to $xn < m$ and w_1 is arbitrary. If w_2 has no append, the large R instead makes the deficiency asymptotic to $(x + y)n > m$, so w_1 must have no append. Each endpoint mode has limiting probability one half. Hence

$$\chi(w_1, w_2) = \begin{cases} 1, & w_1, w_2 \text{ both have no } A, \\ 1/2, & w_2 \text{ has } A \text{ and } w_1 \text{ is arbitrary,} \\ 0, & \text{otherwise.} \end{cases} \quad (371)$$

Its summed normalized mass is

$$\mathcal{W}_R(1/4)^2 + \frac{1}{2}\mathcal{W}(1/4)\mathcal{W}_A(1/4) = 4 + 4 = 8. \quad (372)$$

There are three Catalan factors, with two maximum vertices outside them. Uniformly on the compact interior,

$$\frac{C_{k-1}C_aC_b}{C_n} \sim \frac{4^{-s-2}}{\pi n^3}(xyz)^{-3/2}. \quad (373)$$

The two free size coordinates give a Riemann sum with n^2 cells. Equations (371)–(373) therefore give a limiting contribution $I_{3,\varepsilon}(1/2)/(2\pi)$ after multiplication by n , where the integral initially restricts all three coordinates to be at least ε . Removing this final cutoff gives $I_3(1/2)/(2\pi)$.

Here is the precise justification for removing them. For fixed ε , the unrestricted weight of this family at any fixed total gap cost, divided by c_n , is $O_\varepsilon(n^{-1})$: every macroscopic variable is at least εn , there are $O(n^2)$ choices, and the three factors contribute $O_\varepsilon(n^{-9/2})$. By (366), removing gap costs above H costs $O_\varepsilon(H^{-1/2})$ after scaling by n . For fixed H , remove a strip of width η at every cap boundary; its integral and Riemann sum are $O_\varepsilon(\eta)$. On the remaining compact set all endpoint estimates are uniform. Take $n \rightarrow \infty$, then $\eta \downarrow 0$, then $H \rightarrow \infty$. The bounds in the next section permit $\varepsilon \downarrow 0$.

86 Excluding all other parser families

The stopping condition, rather than an unrestricted renewal majorant, is essential for small terminal children. If one terminal child has size $a \leq \varepsilon n$, its sibling obeys

$$m - a \leq b \leq m, \quad (374)$$

so there are at most $a + 1$ choices for b . This includes $a = 0$ and, after exchanging names, $b = 0$.

For exactly one large R and small-gap cost $s \leq n/4$, both its cost k and the large terminal child b are bounded below by a positive multiple of n . Their normalized product is $O(n^{-3})$. Once a, b, s are fixed, k is determined. Summing (374), either choice of small child, and the bounded gap mass gives

$$\mathbb{P}(\text{one large } R, \min(a, b) \leq \varepsilon n) \leq Kn^{-3/2} \sum_{a \leq \varepsilon n} (a + 1)c_a + o(n^{-1}) \leq K\sqrt{\varepsilon} n^{-1} + o(n^{-1}). \quad (375)$$

The large-gap part is suppressed by (368). If there is no large R and a terminal child is small, the gap cost is at least $n - m - \varepsilon n - 1 > n/4$, so it is suppressed as well.

Suppose there are $j \geq 2$ large outer R operations. If both terminal children exceed εn , there are at most $O(n^{j+1})$ choices at a fixed gap cost, and $j + 2$ macroscopic Catalan factors. Division by c_n gives

$$O_\varepsilon\left(n^{j+1}n^{-3(j+2)/2}n^{3/2}\right) = O_\varepsilon\left(n^{-(j+1)/2}\right) = O_\varepsilon(n^{-3/2}). \quad (376)$$

If one terminal child is small, (374) leaves only $O(n^{j-1}(a+1))$ choices for the $j+1$ large variables. Their weight is $O_\varepsilon(n^{-3(j+1)/2})$; summing $(a+1)c_a$ costs $O(\sqrt{\varepsilon n})$. Division by c_n gives the same order as (376). Gap masses are bounded by (365); the number of possible j is finite at fixed ε . All these families are $o(n^{-1})$.

These estimates exhaust the parser. They also directly explain the integrability at $\theta = 1/2$: when one coordinate x approaches zero, the allowed interval for a second coordinate has width x , giving local mass $O(x^{-1/2} dx)$ and therefore an $O(\sqrt{\varepsilon})$ omitted corner.

87 Evaluation and completion of the proofs

Integrating the y coordinate first gives

$$\begin{aligned} I_3(1/2) &= 8 \int_0^{1/2} x^{-1/2}(1-x)^{-2}(1-2x)^{-1/2} dx \\ &= 8\sqrt{2} \int_0^{\pi/2} (1 - \tfrac{1}{2} \sin^2 \phi)^{-2} d\phi = 12\pi. \end{aligned}$$

The substitution in the second line is $x = \frac{1}{2} \sin^2 \phi$. For the final evaluation, differentiating $\int_0^{\pi/2} (1 - a \sin^2 \phi)^{-1} d\phi = \pi/(2\sqrt{1-a})$ gives

$$\int_0^{\pi/2} (1 - a \sin^2 \phi)^{-2} d\phi = \frac{\pi(2-a)}{4(1-a)^{3/2}}.$$

The three-piece contribution is thus $12\pi/(2\pi) = 6$. Add (370); the remainder estimates prove theorem 80.1 pointwise. For every n , the scaled probability is nondecreasing in t , and its limit is continuous. A finite partition of any compact interval upgrades pointwise convergence to uniform convergence.

For fixed $1/3 < \theta < 1/2$, no-large- R words require a positive linear cost and are suppressed by the same tilt with $c < 1 - 2\theta$. With one large R , the domain is that of (359); every coordinate is at least $1 - 2\theta > 0$. Every pair of coordinates is at least $1 - \theta > \theta$, so the endpoint table (371) remains unchanged. The same Riemann sum gives theorem 80.2; all remainders are uniform on compact phase interiors. Its endpoint $\theta = 1/2$ follows from theorem 80.1.

88 Checks and provenance

88.1 Finite regression checks

The package includes an exact endpoint recurrence and a script comparing it to independent generation of every 132-avoider through size eight. The table records finite values at the central threshold. These are reproducible regression checks, not proof of a limit. Convergence can be slow, particularly on the negative side of the window, where the limiting constant arises from an integrable corner. No finite-size error rate is asserted here.

[Merge note. The “independent generation” is the generator avoiders of the credited model module, which builds the avoiders of each size by the maximum decomposition; the delivered verifier compares the endpoint counter with it for all $n \leq 8$, $1 \leq m \leq n$ (36 counts) and checks the parser of lemma 82.1 by reconstruction on every bounded avoider there (5,599 cases), but it does not test the generator against the 132 predicate. Part VI’s verifier did, through $n = 7$, for the same module.]

88.2 Rebuilding and rerunning

Part VII’s files are

n	$np_n(\lfloor n/2 \rfloor)$
64	4.629846221170498
96	4.868996036164201
128	5.016273257983964
192	5.194724686827719
256	5.302598836522250

Table 28: The scaled central probability, from exact counts; the limit is $\mathfrak{R}_2 = 6$ (theorem 80.1). The rows $n \leq 128$ are those of Part VII’s shipped exact-count CSV file (section 88.2) at $t = 0$ (and of Part VI’s table 26, $\theta = 1/2$); the rows 192 and 256 come from the delivered verifier’s longer run, whose output is not shipped.

<code>code/08-half-size-crossover-model.py</code>	The credited endpoint model (byte-identical to Part VI’s <code>code/07-polynomial-rarity-model.py</code>)
<code>code/08-half-size-crossover-verify.py</code>	Generator, parser and word checks; exact-count table (standard library)
<code>code/08-half-size-crossover-build.sh</code>	The manuscript’s three-pass build
<code>data/08-half-size-crossover-*</code>	The exact-count CSV, the recorded run (JSON, and its stdout capture) and the delivery validation record
<code>08-half-size-crossover-SOURCES.md</code>	Pinned sources, literature ledger and novelty boundary

The programs are shipped byte-identical to the delivery and still use the delivered names: `verify.py` imports `model`, which in this directory would load Part I’s different `code/model.py` (it has no `EndpointCounter`), and it writes `finite_checks.csv` and `verification.json` beside itself; `build.sh` builds a `half_size_crossover.tex` beside itself, which is not shipped (its text is printed here). The report’s README gives a recipe that restores the delivered layout in a scratch copy; there, the manuscript’s instructions apply,

```
python verify.py
python verify.py --max-n 256
```

and rerunning overwrites only the copy’s recorded outputs. The checks require only Python’s standard library; the longer run takes several minutes.

88.3 Exact repository boundary

The inspected combined report source is the Git blob

```
0b7cb2f2e63ad60760ddce0916d287e731d413f4;
```

it ends at Part V. The incoming source of Part VI was retrieved from the archive at arrival reference `a866ff9a2`, ZIP blob `cf18b47515f793c49d82cce5928967234e5a362f`. The ancillary addition07 files occur in batch68 commit `29e52fcdf8bfa99d52a76d3de380f58c24584b1e`; the combined article and README were not staged with that integration. Accordingly, this manuscript compares with the incoming Polynomial Rarity text and does not describe the older combined article as already containing Part VI. [Merge note: this was true at the pin. Part VI has since been printed above from that same archive (section 67.1), so in Part VII every reference to the incoming Polynomial Rarity manuscript is printed as a reference to Part VI; its order-level half-size result is corollary 68.4, its local bound lemma 70.2 and its small-letter suppression lemma 70.3.]

The inherited ingredients are Catalan maximum decomposition, endpoint counts, threshold parsing and word masses, plus the subcritical local bound and small-letter exponential tilt. The new contribution is the three-piece endpoint table, its exact mass eight, the corner exclusions at scale n^{-1} , the leading integral and the limiting ramp. The proof has been checked by direct derivation. It has not been externally refereed or machine-formalized.

The primary arXiv records [1, 2, 6] and targeted searches were checked on 1 October 2026. No matching exact half-window result was located in that limited search. This statement does not establish exhaustive priority. No third-party paper is redistributed in the package.

88.4 Where the merge had to choose

- *Placement.* Part VII follows Part VI and precedes Part I's appendices, which contain no numbered equations, tables or figures; no number of Parts I–VI changed. The manuscript's title page and abstract are replaced by the opening note of Part VII, the status box of section 79.1 and a paragraph of the report's abstract; every claim of its abstract is stated in sections 79.1 and 80. Its Sections 1–8 are sections 80 to 87; its Section 9 is this section, without its "Further questions", which open section 89. Its unnumbered table of finite values is table 28, with a caption that is the merge's. The subsection on rebuilding and rerunning is the merge's. The manuscript's metadata author line reads "AI assisted mathematical research"; it names no addressee.
- *Citations.* The manuscript cites its two repository inputs as bibliography entries: the report at the pin ("Parts III–V") and the incoming Polynomial Rarity archive. Both are replaced by references to Parts IV and VI by label (sections 80 and 88.3); its Kerriou–Mörsters entry is Part VI's [6], and its Nadler and Mayama–Akita entries are the report's. No bibliography entry was added.
- *Duplicates with Parts III, IV and VI.* Equation (360) is Part IV's (181); lemma 81.1 is Part IV's lemma 39.2 and (178), with a more uniform form of (180), and its first-entry count is also Part VI's (354); (364) is Part IV's (191)–(193); the two-piece weight and mass are Part IV's (211) and (219), and its normalized law is (225), whose cost marginal is (227); lemma 83.1 is a second route to Part VI's lemmas 70.2 and 70.3. They are kept, with merge notes, because Part VII's limits are built on these exact forms. Lemma 82.1 generalizes Part IV's lemma 40.4 to all $m < n$. New are theorems 80.1 and 80.2, the three-piece table (371) and its mass (372), the corner and many-letter exclusions of section 86, and the evaluation $I_3(1/2) = 12\pi$; the manuscript's own list of new content (section 88.3) agrees.
- *What Part VII sharpens.* Theorem 80.1 turns Part VI's corollary 68.4 into a limit and theorem 80.2 turns theorem 68.1 into an equivalent for $1/3 < \theta \leq 1/2$. Nothing in Parts I–VI is refuted.
- *An unjustified step.* The bound $4(1/2)^j$ after lemma 83.1 is printed as delivered, followed by the editorial remark 83.3, which gives the missing justification and names the false reading. No other step of the manuscript was changed.
- *Notation.* Every rename is listed in table 27: $W, W_R, W_A \rightarrow \mathcal{W}(1/4), \mathcal{W}_R(1/4), \mathcal{W}_A(1/4)$ and $S, T \rightarrow S_n, T_n$; the manuscript's $\mathbb{P}$, positive part and cost are Part III's $\mathbb{P}$, Part VI's $(\cdot)_+$ and Part III's cost, and its thin space before each differential is written out. No normalization changed.
- *Labels.* The manuscript's 25 labels are kept with the prefix hc.; unprefixed, its generic eq: 8, eq: J, eq: L and sec: remainders would name nothing specific in a report of this size. The merge added labels to the new sections, subsections, tables and the editorial remark.
- *Research questions.* The manuscript's one paragraph of further questions is printed verbatim and then split into four subsections, each pointing to the question of Part VI it continues.
- *Stale sentences elsewhere.* The abstract gained a paragraph; dated pointers to Part VII were added where Parts II, IV, V and VI say that an equivalent below half size, the half-size transition function, the endpoint constant at $q = 2$ or the amplitude at $r = 2$ is open, to the caption of table 26, and to Part I's section B; their original wording is kept.

89 Further research questions

The next concrete targets are exact constants in the lower phases, the analogous crossover functions at $1/q$ for $q \geq 3$, quantitative errors in theorem 80.1, and the conditioned distribution of the three macroscopic pieces. The same parser suggests finite endpoint-state calculations for those problems, but this manuscript claims only the results stated above.

The four subsections below split that paragraph; the pointers in parentheses are the merge's.

89.1 Exact constants in the lower phases

Determine $\Omega_r(\theta)$ for $r \geq 3$, $1/(r+1) < \theta < 1/r$, and the endpoint constants $\mathfrak{R}_q$ for $q \geq 3$. (Part VI's sections 74.1 and 74.2, now answered at $r = 2$ and $q = 2$; for $r \geq 3$ the parser has r macroscopic outer letters and the endpoint table of (371) has to be extended.) [Added 1 October 2026, batch 72: answered in Part VIII: $\Omega_r(\theta) = \kappa_{r+1}I_{r+1}(\theta)$ for every $r \geq 3$ (theorem 91.1) and $\mathfrak{R}_q = \kappa_{q+1}I_{q+1}(1/q)$ for every $q \geq 3$ (theorem 91.2). The endpoint table extends as proposed: the accepted channels are the low endpoint with no append and the high endpoint with an initial prefix of append-containing gaps, of total mass $2^{p-2}(p+1)$ (section 94 and remark 94.1).]

89.2 Crossover functions at the higher reciprocal thresholds

Find the analogue of theorem 80.1 at $1/q$ for $q \geq 3$, on Part VI's scale $w_q(n)$. (Part VI's section 74.3; at $q = 2$ its candidate $\mathfrak{R}_q + \mathfrak{L}_q t_+^{q-1}$ is now a theorem.) [Added 1 October 2026, batch 72: answered for every $q \geq 3$ in Part VIII (theorem 91.2); the candidate holds with $\mathfrak{L}_q$ of (380).]

89.3 Quantitative errors in the half-size window

Give a rate in theorem 80.1. Table 28 and section 88.1 show slow convergence, especially on the negative side of the window, where the constant comes from an integrable corner. (Part VI's section 74.5 asks for uniform relative errors near every reciprocal boundary.) [Added 1 October 2026, batch 72: still open, and Part VIII's limits for $q \geq 3$ have no rate either (section 100.1).]

89.4 The conditioned three-piece law

Find the limiting law of the three macroscopic pieces of a uniform avoider conditioned on $M_n \leq \lfloor \theta n \rfloor$, $1/3 < \theta \leq 1/2$. The endpoint table (371) and the density $(xyz)^{-3/2}$ of (373) identify the leading families of the parser, but no conditioned limit theorem is proved. (Part VI's section 74.4; Part IV's theorem 43.2 is the two-piece analogue above half size.) [Added 1 October 2026, batch 72: answered in Part VIII, in every phase (corollary 98.1): conditionally on $M_n \leq \lfloor \theta n \rfloor$, $1/3 < \theta \leq 1/2$, the parser's three macroscopic pieces converge to the density $I_3(\theta)^{-1}(x_1 x_2 x_3)^{-3/2}$ on $\mathcal{D}_3(\theta)$, independently of four uniform endpoint/gap channels.]

Part VIII. Every fixed phase amplitude and reciprocal crossover

Part VIII was added on 1 October 2026 in batch 72 of ProveIt's incoming-report intake, from one manuscript. Part VII proved the first sub-half-size equivalent and the half-size crossover (theorems 80.1 and 80.2) and asked for the higher constants, the higher crossover functions and the conditioned piece law (sections 89.1, 89.2 and 89.4). Part VIII proves them for every fixed index: in every phase $1/p < \theta < 1/(p-1)$,

$$\mathbb{P}(M_n \leq \lfloor \theta n \rfloor) \sim \kappa_p I_p(\theta) n^{-(p-1)/2}, \quad \kappa_p = \frac{p+1}{2^p \pi^{(p-1)/2}},$$

with a capped-simplex integral I_p , and at every reciprocal threshold $1/r$, on Part VI's scale $w_r(n)$,

$$n^{r/2} \mathbb{P}(M_n \leq \lfloor n/r + t w_r(n) \rfloor) \longrightarrow \mathfrak{R}_r + \mathfrak{L}_r t_+^{r-1},$$

locally uniformly in t ; at $1/3$ the limit is $5/\sqrt{2\pi} + (729\sqrt{3}/(4\pi))t_+^2$. Part VII's two theorems are the cases $p = 3$ and $r = 2$, and Part IV's profile is the case $p = 2$; they are printed once, above, and cited here as the first cases of the general theorems. The same proof gives the conditioned law of the parser's macroscopic pieces. This answers Part VII's questions on lower constants, higher crossovers and the conditioned three-piece law, Part VI's questions 74.1–74.3 and, at the level of the parser's pieces, 74.4, and the amplitude half of Part IV's question 47.1.

90 Provenance, conventions and the answer

90.1 One manuscript, one addition

Part VIII is built from one AI-assisted research manuscript, *manuscript 41* of batch 72, *Exact Phase Amplitudes and Reciprocal Crossovers for 132-Avoiders*, dated 1 October 2026 (8 pages, A4, as delivered; its byline reads “Research note prepared for Vladimir Reshetnikov with OpenAI”). It arrived as the archive `General_132_Avoider_Crossovers_Package.zip` (inner directory `General_132_Avoider_Crossovers/`, main file `general_avoider_crossovers.tex`) in ProveIt commit 1512ef835, and was placed in commit 9e6b04645. Its source ledger pins ProveIt commit 7421a4ca6, at which this report's `article.tex` had the blob b9697af6c8 and contained Parts I–VII exactly as printed here; the report did not change between the pin and the placement. So every statement of the manuscript about Parts III, IV, VI and VII refers to the text printed above.

Its code, data, source ledger and checks README are shipped beside the earlier parts' files with the prefix `09-general-crossovers-`, continuing the report's local sequence after Part VII's `08-half-size-crossover-`. Its manuscript, README and PDF are not shipped and survive in the arrival commit; its two checksum ledgers were verified at placement (15 of 15 and 8 of 8 files) and retired. Its `model.py` is byte-identical to Part VI's `code/07-polynomial-rarity-model.py` (Git blob a4fdee3761), which the manuscript copied and credits; it is not shipped a third time, and the rerun recipe of the report's README restores it under its delivered name (section 99.2).

The manuscript contributes, in the order printed here: the equivalent in every fixed phase (theorem 91.1) and the crossover limit at every fixed reciprocal threshold (theorem 91.2); uniform word bounds with a fixed gap count and all moment orders (lemma 93.1); counting bounds for families with many pieces or a microscopic terminal child; the endpoint-prefix count, whose accepted mass $2^{p-2}(p+1)$ generalizes the masses 3 and 8 of Parts IV and VII (section 94); the shrinking near-cap simplex at a reciprocal threshold and its slope $\mathfrak{L}_r$; the background family, its integrable corners and the microscopic terminal children; the evaluation $I_4(1/3) = 8\sqrt{2}\pi$ and the one-third window (corollary 97.1); the conditioned piece and channel law (corollary 98.1) and the window mixture (corollary 98.2); finite regression checks; and its scope. Every result, proof,

remark, limitation and question of the manuscript is printed. The Catalan, endpoint, parser and word-mass interfaces that it recalls from Part VII (and Parts IV and VI) are printed once, above, and cited by label (section 92).

Status (manuscript 41 of batch 72). The theorems of Part VIII have written mathematical proofs. They are ordinary mathematical arguments, not proof-assistant formalizations: nothing is Lean- or Rocq-verified, nothing has been externally refereed, and no worldwide priority is claimed. The indices p and r are fixed, and all constants and estimates may depend on them; no uniform assertion as the index grows, no effective convergence rate and no bridge to the fixed-bound spectral regime is made. The finite data are regression checks, not evidence replacing any proof step; at $n \leq 192$ they are still far from the one-third limit (section 99.1).

90.2 The questions of Parts IV, VI and VII, and what answers them

With $p_n(m) = \mathbb{P}(M_n \leq m)$ as in (310), Part VIII answers the following questions.

- *Amplitudes in the lower phases* (Part VI's section 74.1, Part VII's section 89.1, the amplitude half of Part IV's section 47.1). For every $r \geq 1$ and $1/(r+1) < \theta < 1/r$, (352) holds with

$$\Omega_r(\theta) = \kappa_{r+1} I_{r+1}(\theta)$$

(theorem 91.1 with $p = r + 1$). The index shifts by one: Part VIII's phase index p counts macroscopic pieces, and Part VI's $r = p - 1$ is the exponent index. The case $r = 1$ is Part IV's (153), the case $r = 2$ Part VII's theorem 80.2; $r \geq 3$ is new. Part IV's manuscript 11 asked for an equivalent for every fixed $0 < \theta < 1/2$; with the upper endpoints of the phases (theorem 91.1) one now exists for every fixed $0 < \theta \leq 1/2$.

- *Constants at the reciprocal endpoints* (section 74.2, section 89.1). For every $q \geq 2$, $\mathfrak{R}_q = \lim_n n^{q/2} p_n(\lfloor n/q \rfloor)$ exists along all n and equals $\kappa_{q+1} I_{q+1}(1/q)$ (theorem 91.2 at $t = 0$, and the endpoint clause of theorem 91.1); there is no residue-class oscillation in the limit. $\mathfrak{R}_2 = 6$ is Part VII's; $\mathfrak{R}_3 = 5/\sqrt{2\pi}$ (corollary 97.1); for $q \geq 4$ the constant is a q -dimensional integral, not evaluated in closed form.
- *Limiting crossover functions* (section 74.3, section 89.2). For every $q \geq 2$, (353) holds with Part VI's candidate

$$\mathfrak{H}_q(t) = \mathfrak{R}_q + \mathfrak{L}_q t_+^{q-1}, \quad \mathfrak{L}_q = \frac{(q+1)q^{5q/2-1}}{2^q \pi^{(q-1)/2} (q-1)!}.$$

Part VI called it “a proposed hypothesis” that “could require a more complicated formula”; it is the theorem. At $q = 2$ it is Part VII's $6 + (12/\sqrt{\pi})t_+$.

- *The conditioned geometry* (section 74.4, section 89.4). Conditionally on $M_n \leq \lfloor \theta n \rfloor$ in a fixed phase, the ordered macroscopic piece vector of the canonical parser converges to the density $I_p(\theta)^{-1} \prod x_i^{-3/2}$ on $\mathcal{D}_p(\theta)$, independently of an endpoint/gap channel uniform over $p+1$ values, with independent gap words (corollary 98.1); in a reciprocal window the law is a two-family mixture with uniform-simplex deficits (corollary 98.2). This answers Part VII's question as posed and Part VI's question *partly*: the conditioned permutation has exactly p macroscopic parser pieces with probability tending to one, but their positions in the permutation, their value sets beyond the parser's ordering and the location of the largest jump are not described.
- *Transitions at every $\theta = 1/r$* (Part IV's section 47.2, Part V's section 63.3). Every reciprocal threshold carries a window in the strong sense, a rescaled limit that is a nonconstant function of the scaled parameter, constant on the lower side.

The theorems sharpen Part VI's theorems 68.1 and 68.2 from orders to equivalents and limits for every fixed index; nothing in Parts I–VII is refuted. Part VIII does not answer Part VII's question

on quantitative errors (section 89.3), or Part VI's questions on uniform relative errors, a growing reciprocal index, effective bounds, other classes, other tail indices or formalization (sections 74.5 to 74.10).

90.3 Notation of Part VIII

Part VI's $C_n, C, \text{Av}_n(132), M_n, p_n(m), w_q, \mathfrak{R}_q, \mathfrak{Q}_q, \mathfrak{S}_q$ and Ω_r , Part IV's fixed fraction θ , and Part VII's $c_j = C_j/4^j$, endpoint letters $F, L, J = N - F$, operations R_β and A , word series $\mathcal{W}, \mathcal{W}_R, \mathcal{W}_A$, small-word series S_n, T_n , constants K, K_B , cost cutoff H , bound B on $|t|$, cutoff ε and tilt constant L keep their meanings. Part VIII fixes the remaining names as follows (table 29).

- The manuscript's endpoint factor $C_p^\star$ is printed κ_p ; a bare C with an index is a Catalan number. The manuscript's window constants B_r, A_r are printed with Part VI's names $\mathfrak{R}_r, \mathfrak{Q}_r$, and the limit function is $\mathfrak{S}_r$: they are not Part I's B_d , Part VII's bound B , the append operation A or the generating function A_m . The manuscript's integration domain $D_p(\theta)$ is printed $\mathcal{D}_p(\theta)$; it is not Part III's limit variable D , the deficit D_n , or Part V's compact set $\mathcal{D}$ of exponents.
- The manuscript's coefficient $t_s = [z^s]T(z)^g$ is printed $T_s^{(g)} := [z^s]T_n(z)^g$, because t is the window variable. Its S, T are Part VII's S_n, T_n .
- The phase index p is a number of macroscopic pieces, not Part IV's or Part V's moment order p ; Part VI's amplitude index is $r = p - 1$, and its threshold index q is the window index r of theorem 91.2.
- The letter ℓ is a number of genuine macroscopic pieces in (386)–(387) and a moment order in (384); it is not Part VII's value of the last entry.
- In section 97 the manuscript's pair convolution $F_a(s)$ and auxiliary integral $J(A)$ are printed $\mathcal{P}_a(s)$ and $\mathcal{J}(\gamma)$, to keep them apart from the first entry F , the first deficiency J and the operation A .

Table 29: Symbols of Part VIII. “Source” gives manuscript 41's symbol; the last column prints the tempting false reading.

Symbol	Meaning in Part VIII	Source	Not to be confused with
κ_p	endpoint factor $(p+1)/(2^p \pi^{(p-1)/2})$	$C_p^\star$	the Catalan number C_p ; Part IV's sources' $\kappa_p (= m_p)$.
$I_p(\theta); \mathcal{D}_p(\theta)$	capped-simplex integral (377); its domain	$I_p; D_p$	Part IV's $I(u) (= I_2(1-u))$; Part III's D ; Part V's $\mathcal{D}$.
$\mathfrak{R}_r, \mathfrak{Q}_r, \mathfrak{S}_r$	window background, slope and limit function	$B_r, A_r, —$	Part I's B_d ; the bound B ; the operation A ; $A_m(x)$.
$\Omega_r(\theta)$	Part VI's amplitude, $= \kappa_{r+1} I_{r+1}(\theta)$	$C_{r+1}^\star I_{r+1}$	the index shift: $\Omega_r \neq \kappa_r I_r$.
p	number of macroscopic pieces in a phase	p	a moment order (Parts IV, V).
$S_n, T_n; T_s^{(g)}$	small-word series, resolvent; $[z^s]T_n^g$	$S, T; t_s$	the window variable t ; Part III's cost T .
$g; \ell$	number of gaps; number of genuine pieces, or a moment order	same	Part VII's last-entry value ℓ .
$x_i; \Delta$	scaled piece sizes; $rm - n$	same	the formal variable z ; Part VII's $\Delta = 2m - n$ (the case $r = 2$).
$\varepsilon, \varepsilon', c; h_n$	macroscopic-letter cutoff, word-cost fraction, cost fraction; strip width	same	—
$\mathcal{P}_a(s); \mathcal{J}(\gamma)$	pair convolution; auxiliary integral	$F_a(s); J(A)$	the first entry F ; the first deficiency J .

No normalization changed: every renamed symbol denotes exactly the manuscript's quantity. Probability is written $\mathbb{P}$, positive parts $t_+ = (t)_+$, and the manuscript's thin space before each differential is written out. Equations, tables and figures are numbered consecutively with those of Parts I–VII.

91 Statements

Let $C_n = (n+1)^{-1} \binom{2n}{n}$ and let π be uniform in $\text{Av}_n(132)$. Set

$$M_n = \max_{1 \leq i < n} |\pi_{i+1} - \pi_i|, \quad p_n(m) = \mathbb{P}(M_n \leq m),$$

with an empty maximum equal to zero (Part VI's (309) and (310)). Part VI proves the polynomial rarity orders and the reciprocal window scales (theorems 68.1 and 68.2 and corollary 68.3), and Part VII the exact half-size crossover and the first sub-half-size amplitude (theorems 80.1 and 80.2). Part VII's sections 89.1, 89.2 and 89.4 leave the higher amplitudes, crossover functions and conditioned piece laws open; Part VIII addresses those questions.

For $p \geq 2$ define

$$I_p(\theta) = \int_{\substack{x_1 + \dots + x_p = 1 \\ 0 < x_i \leq \theta}} \prod_{i=1}^p x_i^{-3/2} dx_1 \cdots dx_{p-1}, \quad \kappa_p = \frac{p+1}{2^p \pi^{(p-1)/2}}. \quad (377)$$

Write $\mathcal{D}_p(\theta)$ for the integration domain. The measure is coordinate measure, not Euclidean surface area. (I_3 is Part VII's (359).)

Theorem 91.1 (Every fixed phase amplitude). *For fixed $p \geq 2$ and $1/p < \theta < 1/(p-1)$,*

$$p_n(\lfloor \theta n \rfloor) \sim \kappa_p I_p(\theta) n^{-(p-1)/2}. \quad (378)$$

The convergence is uniform on compact subintervals of that open phase. For $p \geq 3$, the same equivalent holds at $\theta = 1/(p-1)$.

Theorem 91.2 (Every fixed reciprocal crossover). *Fix an integer $r \geq 2$, and put*

$$\mathfrak{R}_r = \kappa_{r+1} I_{r+1}(1/r), \quad (379)$$

$$\mathfrak{Q}_r = \frac{(r+1)r^{5r/2-1}}{2^r \pi^{(r-1)/2} (r-1)!}. \quad (380)$$

Then, locally uniformly for $t \in \mathbb{R}$,

$$n^{r/2} p_n\left(\left\lfloor \frac{n}{r} + t n^{1-1/[2(r-1)]} \right\rfloor\right) \longrightarrow \mathfrak{R}_r + \mathfrak{Q}_r(t)_+^{r-1}. \quad (381)$$

The integral defining $\mathfrak{R}_r$ is finite and positive.

The scale $n^{1-1/[2(r-1)]}$ is Part VI's $w_r(n)$ of (315), so (381) is (353) with $\mathfrak{H}_r(t) = \mathfrak{R}_r + \mathfrak{Q}_r t_+^{r-1}$.

The $p = 2$ phase recovers Part IV, since

$$I_2(\theta) = \frac{4(2\theta-1)}{\sqrt{\theta(1-\theta)}}, \quad \kappa_2 I_2(\theta) = \frac{3(2\theta-1)}{\sqrt{\pi\theta(1-\theta)}}.$$

The $p = 3$ phase and $r = 2$ window are Part VII's results. The new amplitude cases begin at $p = 4$ and the new windows at $r = 3$; the common formula includes the known cases consistently. All constants and estimates may depend on the fixed index. Growing-index and quantitative convergence questions remain outside the theorem.

Remark 91.3 (Merge note: the first cases are Parts IV and VII). At $p = 2$, (378) is Part IV's (153) (theorem 38.1), whose split integral is $I(u) = I_2(1 - u)$ by (223). At $p = 3$, $\kappa_3 = 4/(8\pi) = 1/(2\pi)$, so (378) is Part VII's theorem 80.2, including its endpoint $\theta = 1/2$. At $r = 2$, $\mathfrak{R}_2 = \kappa_3 I_3(1/2) = 12\pi/(2\pi) = 6$ and $\mathfrak{L}_2 = 3 \cdot 2^4/(4\sqrt{\pi}) = 12/\sqrt{\pi}$, so (381) is Part VII's theorem 80.1. These three theorems are printed once, in their own Parts, and are cited here as the first cases; their proofs are the explicit first-case routes of the general proof below. The manuscript proves the general theorems without importing them. The merge checked the three reductions symbolically and, at $\theta = 0.6, 0.75, 0.8$ and 0.9 , numerically.

Remark 91.4 (Merge note: the slope is the edge of the next phase). The slope $\mathfrak{L}_r$ is what the r -piece amplitude predicts at its lower edge. Just above $1/r$ the domain $\mathcal{D}_r(\theta)$ is a simplex of deficits $\theta - x_i \geq 0$ with sum $r\theta - 1$, on which the integrand is $r^{3r/2} + o(1)$, so

$$I_r(1/r + \delta) \sim \frac{r^{3r/2}(r\delta)^{r-1}}{(r-1)!} \quad (\delta \downarrow 0).$$

Put $\delta = t n^{-1/[2(r-1)]}$ in (378) at $p = r$ and multiply by $n^{r/2}$: the result is $\kappa_r r^{3r/2} r^{r-1} t^{r-1}/(r-1)!$, which is $\mathfrak{L}_r t^{r-1}$; this is the manuscript's calculation in section 95. No theorem licenses the substitution, so this is a consistency check, not a proof, exactly as for Part VII's slope at $q = 2$ (section 79.2). A ProveIt check made for this write-up confirmed the displayed asymptotic of I_r numerically for $r = 2, 3, 4$ (ratios $1 + O(\delta)$ at $\delta = 10^{-3}, 10^{-4}, 10^{-5}$).

92 Catalan interfaces and the exact parser

The manuscript recalls the interfaces of Part VII “to specify the counting and normalization completely”, and says itself that “these parser and endpoint interfaces are already in the source”. They are printed once, in Part VII, and used here by label:

- the Catalan bounds (360) for $c_j = 4^{-j}C_j$ (Part IV's (181));
- the unique maximum decomposition $\pi = (\sigma + b)N\tau$, $a = |\sigma|$, $b = |\tau|$, $a + b + 1 = N$, and its consequence $a + 1 \leq m$ when $b > 0$ and all jumps are at most m (section 81);
- the endpoint facts for an unrestricted avoiding block of size N with first and last entries F, L : $\mathbb{P}(N - F \geq u) \leq (u + 2)2^{-u}$, the last-entry count $C_{N-\ell}C_{\ell-1}$ and the equal limiting atoms of L/N at 0 and 1, that is (361), (362) and (363), uniform in their displayed ranges (lemma 81.1);
- the operations R_β (cost $k = |\beta| + 1$; it adds k to the last deficiency) and A (cost one; allowed exactly when the previous last deficiency plus one is at most m , after which it resets it to zero), and the canonical threshold parser lemma 82.1: an m -bounded avoider of size $n > m$ is uniquely an outer decorated word in R, A applied to a terminal core with children of sizes

$$a, b \leq m, \quad a + b + 1 > m, \quad (382)$$

every outer R has cost at most m , and accepted reconstruction is injective;

- the normalized word masses $\mathcal{W}(1/4) = 4$, $\mathcal{W}_R(1/4) = \mathcal{W}_A(1/4) = 2$ of (364), and the full word series $C(z)^2$, whose cost- s coefficient is C_{s+1} .

Remark 92.1 (Merge note: what the manuscript recalls). The manuscript's equations for these interfaces (its eq:catalan, eq:first, eq:last, eq:lastlimit, lem:parser and eq:wordmass) are condensed restatements, partly near-verbatim, of (360), (361), (362), (363), lemma 82.1 and (364), with the same proofs (the Catalan-triangle first-entry count $\frac{j+1}{N} \binom{2N-j-2}{N-1}$ and its bound $(j+1)2^{-j-1}$, the 132-argument for the last entry, and the parser's rule of following a child larger than m); it writes v for Part VII's r in (363) and W, W_R, W_A for the masses. Only the terminal condition (382) is displayed again, because the later sections use it.

93 Small-word estimates and counting bounds

Fix a small cutoff $\varepsilon > 0$. An outer R is called macroscopic when its cost exceeds εn . The other letters form small-word gaps with normalized generating function

$$S_n(z) = \frac{z}{4} + \sum_{1 \leq k \leq \lfloor \varepsilon n \rfloor} C_{k-1} 4^{-k} z^k, \quad T_n(z) = (1 - S_n(z))^{-1}.$$

There are at most $1/\varepsilon$ macroscopic outer letters. For each fixed gap count g , write $T_s^{(g)} = [z^s]T_n(z)^g$.

Lemma 93.1 (Uniform word bounds). *For fixed g and integers $\ell \geq 1$,*

$$T_n(1)^g \leq 4^g, \quad T_s^{(g)} \leq K_g (s+1)^{-3/2}, \quad (383)$$

$$\sum_{s>H} T_s^{(g)} = O_g((H+1)^{-1/2}), \quad \sum_{s \leq x} s^\ell T_s^{(g)} = O_{g,\ell}((x+1)^{\ell-1/2}). \quad (384)$$

For prescribed $c, A_0 > 0$, choosing ε sufficiently small gives

$$T_s^{(g)} = O_g(n^{-A_0}) \quad (s \geq cn). \quad (385)$$

Proof. The letter mass is at most $3/4$. In a word of length l and cost s , a largest letter has size at least s/l and weight at most $Kl^{3/2}s^{-3/2}$. Marking it gives at most l choices, while the other letters have total mass at most $(3/4)^{l-1}$. Summing with the polynomial multiplicity $\binom{l+g-1}{g-1}$ proves (383); summation gives (384).

For (385), choose $L > A_0/c$ and $L\varepsilon < 1/2$. At $\lambda = L \log n/n$,

$$S_n(e^\lambda) - S_n(1) \leq \lambda e^{\lambda \varepsilon n} S'_n(1) = O((\log n)n^{L\varepsilon-1/2}) = o(1).$$

Thus $T_n(e^\lambda)^g$ stays bounded, and coefficient positivity yields the claim. The choice of ε works for every fixed finite set of gap counts. $\square$

Remark 93.2 (Merge note: the word bounds of Parts VI and VII). The first three bounds and (385) are Part VII's (365), (366) and (368) with $h = g$ (Part VI's lemmas 70.2 and 70.3 for the upper kernel, tail index $1/2$), with the same proofs, here condensed; the tail sum is a summation of (366). New is the moment bound in (384) for every order $\ell \geq 1$, the analogue for T_n^g of Part VI's partial-sum moment bound (338); at $\ell = 1$ and $x = \varepsilon n$ it is a truncated form of Part VII's (367). The higher orders are what the shrinking simplex at $1/r$ needs (section 95).

When applying (385), sum all remaining normalized decoration and terminal weights without the total-size constraint. With j macroscopic outer letters that mass is at most a fixed constant times $(1/2)^j$. The word cost is then determined by the remaining sizes. Division by c_n loses only $O(n^{3/2})$. A fixed exponent A_0 larger than the desired rare-event scaling plus $3/2$ therefore suffices; it need not grow with j . Constants may depend on the fixed cutoff, and n tends to infinity before that cutoff is removed.

[Merge note. The bound “a fixed constant times $(1/2)^j$ ” is asserted without proof, as in Part VII; Part VII's editorial remark 83.3 supplies it: each macroscopic R letter carries total normalized mass at most $\frac{1}{4}C(1/4) = \frac{1}{2}$ and the terminal core at most $\frac{1}{4}C(1/4)^2 = 1$. The manuscript's wording, “decoration and terminal weights”, is the correct reading that the remark reconstructs, and because the word cost is then determined by the remaining sizes, it needs no summation over the gap cost; that is why the exponent condition here, A_0 larger than the rare-event scaling plus $3/2$, is sharper than the n^{1-A_0} of remark 83.3.]

Two elementary family bounds will be used repeatedly. If there are ℓ genuine macroscopic pieces and both terminal children are large, their Catalan factors and the $\ell - 1$ free sizes give

$$O_\varepsilon(n^{-3\ell/2}n^{\ell-1}n^{3/2}) = O_\varepsilon(n^{-(\ell-1)/2}) \quad (386)$$

after summing the bounded gap mass. If a terminal child has size $a \leq \varepsilon n$, its partner satisfies Part VII's (374), $m - a \leq b \leq m$, so it has at most $a + 1$ choices. With ℓ genuine large pieces, the remaining count is $O(n^{\ell-2}(a+1))$. Since $\sum_{a \leq \varepsilon n} (a+1)c_a = O(\sqrt{\varepsilon n})$, the corresponding probability is

$$O_\varepsilon(n^{-\ell/2}). \quad (387)$$

This count has $\ell + 1$ nominal pieces including the small terminal child; it must not be confused with (386). For sufficiently small ε , both terminal children cannot be small by (382).

[Merge note. With $\ell = j + 2$ pieces (j large outer letters and two large terminal children), (386) is Part VII's (376), and (387) its small-child variant; the manuscript's eq:partner is (374) and is not displayed again.]

94 The endpoint-prefix count and phase amplitudes

Suppose p macroscopic pieces carry the total size. There are $p - 2$ outer R decorations, two terminal children, and $p - 1$ intervening small-word gaps, ordered from the core outward. In the minimal p -piece phase,

$$x_i + x_j \geq 1 - (p - 2)\theta > \theta. \quad (388)$$

In particular the terminal pair exceeds the cap.

For fixed gap words and pieces in the domain interior, all first-deficiency constraints are negligible by (361); cap-boundary strips with growing slack remove the remaining joins. Only the terminal-right last-endpoint mode matters, classified as low or high according as $L(\tau) \leq b/2$ or $L(\tau) > b/2$. Conditional on these modes, respectively $L(\tau)$ or $b - L(\tau)$ is tight by (363). The following acceptance rules therefore hold with probability $1 - o(1)$, uniformly on the retained interior. In the low mode the initial last deficiency exceeds the cap, and no append is possible in any gap. In the high mode the initial deficiency is the left terminal size up to a tight error. An append in the first gap resets it. Each next macroscopic R adds its own cost, which is below the cap. If an append gap is skipped, any later first append would encounter two macroscopic sizes, forbidden by (388). Once the first append within a fixed gap is accepted, its remaining bounded costs introduce no new macroscopic constraint. Therefore append-containing gaps form an initial contiguous prefix of length $0, \dots, p - 1$.

There is one low-endpoint channel and p high-endpoint channels. Each endpoint mode has limiting mass $1/2$, and each gap type has mass two by (364). The exact limiting accepted normalized gap mass is

$$\frac{1}{2}(1 + p)2^{p-1} = 2^{p-2}(p + 1). \quad (389)$$

No permutation factor is present: the parser orders the outer decorations and distinguishes the two terminal children.

For fixed words of total cost s , the Catalan product divided by C_n is

$$4^{-s-(p-1)}\pi^{-(p-1)/2}n^{-3(p-1)/2} \prod_{i=1}^p x_i^{-3/2}(1 + o(1)). \quad (390)$$

There are n^{p-1} cells. Multiplying the Riemann integral by (389) yields $\kappa_p I_p(\theta)$.

Here are the exclusions making this a full proof in the phase interior. Fewer than p genuine large pieces leave positive linear word cost, because $\theta < 1/(p - 1)$; choose the cutoff below

that phase margin and use (385). Exactly p pieces with both terminal children large give the preceding calculation. They are bounded below by a fixed positive fraction of n when the word cost is sufficiently small, since $1 - (p - 1)\theta > 0$. A fixed word-cost cutoff can be removed by (384). Families with more pieces are smaller by at least $n^{-1/2}$ using (386); those with a microscopic terminal child are handled by (387), or require linear word cost if fewer than p genuine pieces remain. These cases exhaust the parser. The same margins hold uniformly on compact phase subintervals, proving (378) there.

Remark 94.1 (Merge note: the masses 3 and 8). At $p = 2$ the accepted mass (389) is 3, Part IV's word mass (219) and the sum $2 + \frac{1}{2} \cdot 2$ of Part VII's (370); the three channels (low; high with an empty or a one-gap prefix) are Part IV's three equally weighted endpoint channels (corollary 43.1). At $p = 3$ it is 8, Part VII's (372): with w_2 the gap next to the core, the low channel and the high channel with the empty prefix both require that neither gap contain an A (weight $\frac{1}{2} + \frac{1}{2} = 1$), and the prefixes of lengths one and two together require an A in w_2 with w_1 arbitrary (weight $\frac{1}{2}$), which is Part VII's table (371). Equation (390) at $p = 3$ is Part VII's (373). The prefix rule, and hence the mass $2^{p-2}(p + 1)$, is new for $p \geq 4$.

95 The shrinking contribution at a reciprocal threshold

Fix $r \geq 2$, $|t| \leq B$, and

$$m = \left\lfloor n/r + t n^{1-1/[2(r-1)]} \right\rfloor, \quad \Delta = rm - n = rt n^{1-1/[2(r-1)]} + O(1).$$

For r genuine pieces and total small-word cost s , cap deficits have sum $\Delta + s + 1$. Their number is bounded by $K_r(\Delta + s + 1)^{r-1}$. For $s \leq cn$ with c sufficiently small, every piece is bounded below by a fixed positive multiple of n . Thus, after multiplication by $n^{r/2}$, the contribution has prefactor $O(n^{-r+3/2})$.

Choose a small word-cost fraction $\varepsilon' > 0$, then choose the macroscopic-letter cutoff ε sufficiently smaller so that (385) applies for costs at least $\varepsilon'n$. For a fixed cutoff H , the intermediate word-cost tail is at most

$$\begin{aligned} & K n^{-r+3/2} \sum_{H < s \leq \varepsilon'n} T_s^{(g)} (\Delta + s + 1)^{r-1} \\ & \leq K_B (H + 1)^{-1/2} + K_r (\varepsilon')^{r-3/2} + o(1). \end{aligned} \quad (391)$$

This uses $(u + v)^{r-1} \leq K_r(u^{r-1} + v^{r-1})$, $\Delta_+^{r-1} = O_B(n^{r-3/2})$, and (384). Larger costs are negligible by (385), with $A_0 > r/2 + 3/2$. There is no circular dependence on the number of macroscopic letters.

For fixed words and $t > 0$, the deficit simplex has $\Delta^{r-1}/(r-1)!(1 + o(1))$ lattice points. Delete strips of width h_n at each cap face, where $h_n \rightarrow \infty$ and $h_n = o(\Delta)$. Their relative count tends to zero. The endpoint-prefix argument applies on the remainder, and all scaled pieces tend to $1/r$. Hence the limit of this family's contribution is

$$\kappa_r r^{3r/2} \frac{(rt)^{r-1}}{(r-1)!} = \mathfrak{L}_r t^{r-1}.$$

For $t < 0$ no fixed-word configuration remains; at $t = 0$ at most a bounded number of compositions remain for each fixed word. Both contributions vanish after scaling. Bound (391) removes nonfixed words and proves the contribution $\mathfrak{L}_r(t)_+^{r-1}$ in all cases.

[Merge note. At $r = 2$ this is Part VII's two-piece ramp (section 84): the deficit simplex is the interval of $2t\sqrt{n} + O(1)$ splits, the accepted mass is 3, and $\kappa_2 2^3 \cdot 2t = 12t/\sqrt{\pi}$ is (370); (391) at $r = 2$ is the tail bound lemma 84.1.]

96 The background and microscopic terminal children

With $r + 1$ genuine pieces, both terminal children large, fixed words give the Riemann integral on

$$\mathcal{D}_{r+1}(1/r) = \left\{x_i > 0 : \sum x_i = 1, x_i \leq 1/r\right\}.$$

Every pair sum exceeds $1/r$ except on cap-boundary faces. Delete thin boundary strips first; the endpoint-prefix count then gives accepted mass $2^{r-1}(r+2)$ and contribution $\mathfrak{R}_r$ after scaling by $n^{r/2}$. At a fixed macroscopic cutoff, word-cost tails are removed by (384).

The only singular integral corners have one coordinate tending to zero. Two vanishing coordinates would leave at most $r - 1$ cap-sized pieces, which cannot sum to one. If $x_1 = x$ is small, the remaining r coordinates have deficits from $1/r$ summing to x . Their cross-sectional volume is $O(x^{r-1})$ and they stay bounded away from zero. Therefore the corner mass is

$$O\left(\int_0^\varepsilon x^{r-5/2} dx\right) = O(\varepsilon^{r-3/2}). \quad (392)$$

This proves finiteness and justifies restoring the integral corners.

The discrete family with one microscopic terminal child needs a separate bound, because the parser imposes (374). Let that child have size $a \leq \varepsilon n$, with r genuine large pieces remaining. At fixed a, s , the number of choices is at most

$$K_r(a+1)(\Delta_+ + a + s + 1)^{r-2}.$$

Choose $\varepsilon, c < 1/(4r)$. For $s \leq cn$, all r large sizes are at least $n/(3r)$ for large n . Their Catalan bound is therefore independent of ε . The entire scaled contribution is at most

$$Kn^{-r+3/2} \sum_{a \leq \varepsilon n} (a+1)c_a \sum_{s \leq cn} T_s^{(g)} (\Delta_+ + a + s + 1)^{r-2}. \quad (393)$$

For $r = 2$, this is $O(\sqrt{\varepsilon})$. For $r \geq 3$, split the power into the Δ , a , and s terms. Respectively they give

$$O_B\left(\sqrt{\varepsilon} n^{-(r-2)/[2(r-1)]}\right), \quad O(\varepsilon^{r-3/2}), \quad O(\sqrt{\varepsilon} n^{-1/2}).$$

For the last bound use $\sum_{s \leq cn} s^{r-2} T_s^{(g)} = O(n^{r-5/2})$. The terms $s > cn$ are negligible by (385), summing the remaining weights before dividing by c_n . This proves that the small-terminal family vanishes as the cutoffs are removed, including the degenerate child sizes zero.

For clarity, the exhaustive classification at a fixed cutoff is as follows. If both terminal children are large, fewer than r pieces force linear word cost; r pieces give the shrinking simplex; $r + 1$ give the background; and at least $r + 2$ vanish by (386). If one terminal child is small, fewer than r genuine pieces again force linear word cost, r are covered by (393), and at least $r + 1$ vanish by (387). Both terminal children cannot be small. Small outer decorations already belong to the small-word gaps, so there is no extra mesoscopic family to add or double-count.

The limit order is explicit: first fix ε' , then choose ε small enough for the required tilts and terminal bounds; at those cutoffs let $n \rightarrow \infty$ and then remove the fixed word cutoff H . Finally let $\varepsilon' \downarrow 0$ with its corresponding $\varepsilon \downarrow 0$. The shrinking-word error is controlled by (391); background corners and small terminal children by (392)–(393). This proves pointwise (381). Since $p_n(m)$ is nondecreasing in m and the limiting function is continuous, the finite-grid monotonicity argument makes convergence locally uniform in t . Taking $t = 0$ also proves the endpoint assertion in theorem 91.1.

[Merge note. At $r = 2$ the background is Part VII's three-piece contribution (section 85), its accepted mass $2^{r-1}(r+2) = 8$ is (372), the corner bound (392) is the $O(x^{-1/2} dx)$ corner of section 86, and (393) is (375); the monotone finite-grid argument is that of section 87.]

97 The first two reciprocal constants

The known integral $I_3(1/2) = 12\pi$ (section 87) gives $\mathfrak{R}_2 = 6$ and $\mathfrak{Q}_2 = 12/\sqrt{\pi}$. For the next threshold we give an ordinary-integral evaluation.

For $a < s < 2a$, pair convolution gives

$$\mathcal{P}_a(s) = \int_{s-a}^a \frac{dx}{[x(s-x)]^{3/2}} = \frac{4(2a-s)}{s^2\sqrt{a(s-a)}}.$$

Taking $a = 1/3$ and pairing the four coordinates,

$$\begin{aligned} I_4(1/3) &= 48 \int_{1/3}^{2/3} \frac{\sqrt{(2/3-s)(s-1/3)}}{s^2(1-s)^2} ds \\ &= 1728 \int_{-1}^1 \frac{\sqrt{1-x^2}}{(9-x^2)^2} dx. \end{aligned}$$

For $\gamma > 1$, the substitution $x = \sin u$ yields

$$\mathcal{J}(\gamma) = \int_{-1}^1 \frac{\sqrt{1-x^2}}{\gamma-x^2} dx = \pi \left(1 - \sqrt{1 - \frac{1}{\gamma}} \right).$$

Differentiation under this bounded integral gives $-\mathcal{J}'(9) = \pi/(108\sqrt{2})$. Hence

$$I_4(1/3) = 8\sqrt{2}\pi, \quad \mathfrak{R}_3 = \frac{5}{\sqrt{2\pi}}, \quad \mathfrak{Q}_3 = \frac{729\sqrt{3}}{4\pi}.$$

Corollary 97.1 (The one-third window). *Locally uniformly for real t ,*

$$n^{3/2}p_n\left(\lfloor n/3 + tn^{3/4} \rfloor\right) \longrightarrow \frac{5}{\sqrt{2\pi}} + \frac{729\sqrt{3}}{4\pi}(t)_+^2.$$

[Merge note. Numerically $\mathfrak{R}_3 = 1.99471\dots$ and $\mathfrak{Q}_3 = 100.4797\dots$. The delivered SymPy check (section 99.1) verifies the pair antiderivative, its endpoint value and $-1728\mathcal{J}'(9) = 8\sqrt{2}\pi$, taking $\mathcal{J}$ as an input; an independent mpmath quadrature made for this write-up confirms $\mathcal{J}(9)$, both integral reductions and $I_4(1/3)$ to $4 \cdot 10^{-14}$. The next constant, $\mathfrak{R}_4 = \kappa_5 I_5(1/4)$, is not evaluated in closed form; two quadratures by different nested pair convolutions, made for this write-up, agree on $I_5(1/4) \approx 35.6357$, so $\mathfrak{R}_4 \approx 0.6770$ (a numerical value, not a certified enclosure).]

98 Conditioned geometry

The proof also gives marked limits. The macroscopic vector is ordered as the two terminal children followed by the large outer decorations from the core outward; outer entries are their costs, divided by n . Statements about its dimension are interpreted by first taking $n \rightarrow \infty$ at a fixed cutoff and then letting the cutoff tend to zero. The probability of excluded microscopic terminal cases vanishes by the proved bounds.

Corollary 98.1 (Piece and channel law in a phase). *Condition on $M_n \leq \lfloor \theta n \rfloor$, in a fixed phase of theorem 91.1, including its allowed upper endpoint. The ordered macroscopic-size vector converges to the probability density*

$$I_p(\theta)^{-1} \prod_{i=1}^p x_i^{-3/2} \quad \text{on } \mathcal{D}_p(\theta).$$

Independently of this vector, the endpoint/gap channel is uniform over $p + 1$ possibilities: the low terminal-right endpoint with no append in any gap, or the high endpoint with append-containing prefix length $0, \dots, p - 1$. Given a channel, the gap words are independent, and a decorated word w of the prescribed type has probability $4^{-|w|}/2$.

Proof. Insert a bounded continuous test function of the scaled pieces into the fixed-word Riemann sums. The same absolute bounds control its tails. Each allowed endpoint channel has limiting mass $\frac{1}{2} 2^{p-1}$, independent of the piece sizes, giving the uniform channel law after (389). Fixed gap words contribute only their factors $4^{-|w|}$; the tail bound makes their total cost tight. Dividing by the proved probability equivalent gives the result, including independence. $\square$

Corollary 98.2 (The reciprocal-window mixture). *Condition on the event in (381). For fixed $t > 0$, the r -piece family has limiting weight*

$$\frac{\mathfrak{L}_r t^{r-1}}{\mathfrak{R}_r + \mathfrak{L}_r t^{r-1}},$$

and all its scaled sizes tend to $1/r$. Its cap deficits, divided by $\Delta = rm - n$, converge to the uniform simplex law Dirichlet(1, ..., 1) on r coordinates. The complementary $(r + 1)$ -piece family has the density from corollary 98.1 at $\theta = 1/r$. For fixed $t \leq 0$, only the latter family survives.

Proof. The unconditioned family masses were proved separately. In the shrinking family, fixed words give deficits summing to $\Delta + s + 1$, with s fixed under truncation and then tight. Since $\Delta \rightarrow \infty$ for fixed $t > 0$, this shift disappears after normalization. The Catalan density is asymptotically constant on the deficit simplex, so marked lattice sums give the uniform Dirichlet law. The existing tail estimates justify removal of the word cutoff and division by the total limiting mass. The background marked limit is the preceding Riemann argument. $\square$

Remark 98.3 (Merge note: the case $p = 2$ is Part IV's conditional law). At $p = 2$ and $1/2 < \theta < 1$, put $u = 1 - \theta$. The density of corollary 98.1 is $I_2(\theta)^{-1}[x(1-x)]^{-3/2}$ on $[u, 1-u]$, the split law f_u of Part IV's (226) (with $I(u) = I_2(\theta)$ by (223)). The three uniform channels give an A -free word probability $\frac{1}{3} \cdot 4^{-|w|}/2 + \frac{1}{3} \cdot 4^{-|w|}/2 = 4^{-|w|}/3$ and a word with an A probability $\frac{1}{3} \cdot 4^{-|w|}/2 = 4^{-|w|}/6$, which is Part IV's universal word law (225), and independence of word and split is Part IV's theorem 43.2 (manuscript 11 of batch 52). The channel itself refines that theorem; it is the three equally weighted endpoint channels of Part IV's corollary 43.1 (manuscript 10 of batch 52), there a sampling description and here part of a proved joint limit for the parser's data. At $p = 3$, $1/3 < \theta \leq 1/2$, the corollary is the conditioned three-piece law that Part VII asked for (section 89.4). The merge checked the case $p = 2$ against Part IV's statements.

99 Finite checks, attribution and limits

99.1 Finite regression checks

The package includes the credited exact endpoint recurrence, checked against independently generated avoiders through size eight, and exact finite endpoint-prefix checks of (389). Exact integer counts at the one-third threshold give the rounded scaled probabilities of table 30. At $n = 192$ the value is 3.2587832927, while the limit is $5/\sqrt{2\pi} \approx 1.9947114020$. These are slow-convergence regression values, not error certificates or evidence replacing any proof step. Exact count numerators and Catalan denominators are retained in the package.

[Merge note on the finite checks. (i) The “exact finite endpoint-prefix checks” are a finite automaton check: for $p = 2, \dots, 14$ every binary mask of the $p - 1$ gaps is passed through the endpoint channel logic under the pair-sum geometry (388), and exactly $p + 1$ mode/mask channels are accepted, giving (389); it is not a check against permutations, as the shipped 09-general-crossovers-CHECKS.md says. The “independent” generation is the generator avoiders

n	48	72	96	144
$n^{3/2}p_n(n/3)$	3.4610742065	3.4548874596	3.4190907163	3.3352928324

Table 30: The scaled one-third probability, from exact counts; the limit is $\mathfrak{R}_3 = 5/\sqrt{2\pi} \approx 1.9947$ (corollary 97.1). With the row $n = 192$ of the text these are the five rows of the shipped boundary-count CSV file (section 99.2).

of the credited model module, here, unlike in Part VII (section 88.1), compared with a literal enumeration of all $n!$ permutations under the 132 predicate for $n \leq 8$, together with 44 total counts and 1,500 endpoint-state counts of the recurrence. The shipped data also record four window rows at $n = 48, 96$ and $t = \pm 1/8$. (ii) Convergence. The tabulated values alone, $n \leq 192$, do not determine the limit: three-term fits $a + bn^{-1/4} + cn^{-1/2}$ to them extrapolate to about 0.5, and fits $a + bn^{-1/2} + cn^{-1}$ to about 2.3–2.6. A ProveIt check made for this write-up goes further. An endpoint recurrence written independently of the credited model reproduces all five exact counts digit for digit and agrees with literal enumeration for $1 \leq m \leq n \leq 8$. Its floating-point evaluation up to $n = 1050$ shows that $n^{3/2}p_n(n/3)$ peaks near $n = 60$ and then decreases steadily (2.7343 at $n = 1050$), and fits with corrections $n^{-1/2}, n^{-1}, n^{-3/2}$ give the limits 2.0015, 1.9971 and 1.9951 from the rows $n \geq 100, 190$ and 290, against $5/\sqrt{2\pi} = 1.99471$; other correction ladders approach 1.99–2.00 from both sides. A wrong channel mass would change $\mathfrak{R}_3$ by a factor such as $4/5$ or $6/5$, to about 1.60 or 2.39, and these data exclude that. The proved Part VII constant at $\theta = 0.4$ is approached equally slowly from above (still 1.72 times its limit at $n = 800$). The proof’s own error terms at $t = 0$ are $O(\sqrt{\varepsilon} n^{-1/2})$ plus cutoff terms (section 96), consistent with the $n^{-1/2}$ ladder. This is numerical support, not a proof; the theorem rests on the proof alone, and none of these computations is shipped. The window slope $\mathfrak{Q}_3$ cannot be read off finite data at these sizes; it is pinned by the matching of remark 91.4.]

99.2 Rebuilding and rerunning

Part VIII’s files are

code/09-general-crossovers-verify.py	Channel masks, literal enumeration, endpoint states, exact counts (standard library)
code/09-general-crossovers-verify_integral.py	Symbolic check of the $I_4(1/3)$ algebra (SymPy)
code/09-general-crossovers-build.sh	The manuscript’s three-pass build
data/09-general-crossovers-*	Recorded run (JSON), the two exact-count CSV files, the symbolic record and the delivery validation
09-general-crossovers-CHECKS.md	The checks README: scope, run, model lineage
09-general-crossovers-SOURCES.md	Pinned source, credited results, literature ledger

The programs are shipped byte-identical to the delivery and still use the delivered names: `verify.py` imports `model`, which in this directory would load Part I’s `different/code/model.py`, checks the SHA-256 of a `model.py` beside itself, and writes `results.json` and the two CSV files beside itself; `verify_integral.py` writes `integral_symbolic.json` beside itself; `build.sh` builds a `general_avoider_crossovers.tex` beside itself, which is not shipped (its text is printed here). The report’s README gives a recipe that restores the delivered layout in a scratch copy, with Part VI’s `code/07-polynomial-rarity-model.py` as `model.py`; there, the manuscript’s instructions apply,

`python verify.py` (or: `python verify.py --quick`)

`python verify_integral.py`

and rerunning overwrites only the copy's recorded outputs. The main checks require only Python's standard library; the symbolic check needs SymPy.

99.3 Attribution and repository boundary

The public fixed-bound work of Nadler [1] and Mayama–Akita [2] provides the enumeration context. The fewest-big-jumps principle [6] is related heavy-tail context, not an invoked theorem: the present parser and endpoint constraints are handled directly. The source pin is the current combined article containing Parts VI and VII, not its earlier incomplete merge. No external paper is redistributed.

[Merge note. The manuscript cites this report as an external item, “combined article at commit 7421a4ca60fd, Git blob b9697af6c8489fb1e0b3670037a50aa70fdc6f80”, for Parts III–IV, VI and VII; every such citation is printed as a reference by label. “Its earlier incomplete merge” refers to the pin of Part VII, which preceded the writing of Part VI (section 88.3); at Part VIII's pin the report contained Parts I–VII as printed. Its source ledger records limited targeted searches on 1 October 2026 of the arXiv records of Nadler, Mayama–Akita and Kerriou–Mörsters, which “did not identify the same general cap-window formulas; this is not a certification of worldwide novelty”.]

99.4 Limits

The manuscript gives fixed-index equivalents, locally uniform fixed-index crossover limits, and the stated conditioned laws. It does not give an effective convergence rate, uniformity as p or r grows, or a bridge to the fixed-bound spectral regime. It is an ordinary mathematical proof, not formally verified or externally refereed, and makes no worldwide priority claim.

99.5 Where the merge had to choose

- *Placement.* Part VIII follows Part VII and precedes Part I's appendices; no number of Parts I–VII changed. The manuscript's title page and abstract are replaced by the opening note of Part VIII, the status box of section 90.1 and a paragraph of the report's abstract; every claim of its abstract is stated in sections 91 and 98. Its Sections 1–8 are sections 91 to 98; its Section 9 is this section, split into subsections, and its closing scope paragraph is section 99.4 and, as questions, section 100. Its unnumbered table is table 30 (one row moved to the text, as in the manuscript), with a caption that is the merge's; its unnumbered corollary is corollary 97.1. The subsection on rebuilding and rerunning is the merge's.
- *The first cases are not reprinted.* Theorem 91.1 at $p = 2$ and $p = 3$ and theorem 91.2 at $r = 2$ are Part IV's and Part VII's theorems; they stay in their Parts and remark 91.3 names them. The general theorems are the statements with the right generality, and the earlier proofs are the explicit first-case routes.
- *Recalled interfaces printed by reference.* The manuscript's Section 2 is a condensed restatement of Part VII's interfaces; it is printed as a list of references (section 92), with only (382) displayed. Its eq:partner is (374). Lemma 93.1 is printed in full because its moment bound is new; a merge note names the earlier parts.
- *Citations.* The manuscript's citation of this report is replaced by references by label (section 99.3); its Nadler, Mayama–Akita and Kerriou–Mörsters entries are the report's [1], [2] and [6]. No bibliography entry was added.
- *Additions of the merge.* The merge notes after lemma 93.1 (the bound $(1/2)^j$, citing remark 83.3), on the family bounds, after sections 95 and 96, after corollary 97.1, and in section 99.1 (the scope of the channel check, the convergence evidence); remarks 91.3, 91.4, 94.1 and 98.3. No step of the manuscript was changed.

- *Notation.* Every rename is listed in table 29: $C_p^\star \rightarrow \kappa_p, B_r, A_r \rightarrow \mathfrak{R}_r, \mathfrak{L}_r, D_p \rightarrow \mathcal{D}_p, S, T \rightarrow S_n, T_n, t_s \rightarrow T_s^{(g)}, F_a, J(A) \rightarrow \mathcal{P}_a, \mathcal{J}(\gamma)$; the manuscript's $\mathbb{P}$, positive part and thin-spaced differential are Part III's $\mathbb{P}$, Part VI's $(\cdot)_+$ and d . No normalization changed.
- *Labels.* The manuscript's 29 labels were all bare generic names (thm:phase, eq:A, eq:many, ...). The 22 that name printed material are kept with the prefix gc.; the six that name the recalled interfaces (eq:catalan, eq:first, eq:last, eq:lastlimit, lem:parser, eq:wordmass) and eq:partner are replaced by Part VII's labels. The merge added labels to the new sections, subsections, the two tables, the corollary and the merge remarks.
- *Research questions.* The manuscript states its limits in one paragraph; section 100 splits them into questions, each pointing to the question of Parts VI and VII it continues.
- *Stale sentences elsewhere.* The abstract gained a paragraph; dated pointers to Part VIII were added where Parts II, IV, VI and VII say that the amplitudes for $\theta \leq 1/3$, the endpoint constants or crossover functions for $q \geq 3$, or the conditioned geometry are open, to the caption of table 26, and to Part I's section B; their original wording is kept.

100 Further research questions

The manuscript states that it does not give an effective convergence rate, uniformity as p or r grows, or a bridge to the fixed-bound spectral regime (section 99.4). The subsections below turn that sentence into questions; the pointers in parentheses are the merge's.

100.1 Effective rates

Give a rate in theorems 91.1 and 91.2, and a uniform relative error near every reciprocal boundary that overlaps the window scale. The one-third table (table 30) converges slowly, and the proof's error terms at $t = 0$ are of order $n^{-1/2}$ only up to cutoff terms. (Part VII's section 89.3 at $q = 2$; Part VI's section 74.5.)

100.2 Uniformity in the index

Make the constants of theorems 91.1 and 91.2 explicit in p and r , and find the range of growing indices on which the few-pieces description survives. The endpoint-prefix mass $2^{p-2}(p+1)$ and the factor κ_p are explicit, but the cutoffs $\varepsilon, \varepsilon'$ and the tilt depend on the index. (Part VI's section 74.6.) [Added 1 October 2026, batch 72: for a growing index Part IX proves equivalents when $\lceil (n-1)/m \rceil \rightarrow \infty$ (theorems 109.1, 109.3 and 109.4); a bounded index with m/n not converging stays open (section 113.1).]

100.3 The bridge to the fixed-bound spectral regime

Connect the fixed-fraction theorems with the fixed- m growth constants of Parts I and II as $m/n \rightarrow 0$. (Part VI's section 74.6, Part IV's section 47.10, Part III's section 32.10.) [Added 1 October 2026, batch 72: answered in Part IX as $m/n \rightarrow 0$: at logarithmic precision for all $n \geq c_0 m \log m$ (theorem 105.1), and by relative equivalents while $n(\log m)^2/m^2 \rightarrow 0$ (section 101.2).]

100.4 The whole conditioned permutation

Corollary 98.1 describes the parser's macroscopic pieces, channels and gap words. Describe the positions and value sets of the macroscopic blocks in the permutation and the location of the largest jump, as Part VI's section 74.4 and Part IV's section 47.8 ask. (This question is the merge's, from the part of Part VI's question that the manuscript leaves unanswered.)

100.5 Closed forms of the constants

Evaluate $I_p(\theta)$ and $\mathfrak{R}_r$ in closed form beyond $p \leq 4$ and $r \leq 3$, or show that the pair-convolution method stops there. (The merge's question: the manuscript evaluates only $I_3(1/2)$ and $I_4(1/3)$.)

Part IX. Growing adjacency caps

Part IX was added on 1 October 2026 in batch 72 of ProveIt's incoming-report intake, merged from six manuscripts written the same day. Part II asked for the joint growth of length and adjacency bound and warned that its fixed- m constants are not uniform (section 19.5); Parts III–VIII answered the side where m/n stays bounded below. Part IX treats the other side, $m \rightarrow \infty$ with $n/m \rightarrow \infty$. For every fixed $c_0 > 0$ it proves, uniformly for $n \geq c_0 m \log m$, the logarithmic bridge $\log \mathbb{P}(M_n \leq m) = -(n/m)t_m + O(n(\log m)^2/m^2 + \log n)$ to Part II's scalar rate t_m , with no upper restriction on n (theorem 105.1); one positive saddle sum for the squared-renewal envelope on the same range, and for $a_n^{(m)}$ itself when $n(\log m)^2/m^2 \rightarrow 0$ (theorem 106.1); a theta-function modulation of the equivalent that is needed down to $n \asymp m(\log m)^{3/2}$, with a cubic correction there (theorem 108.2); and explicit Gamma-sum equivalents on $n \asymp m \log m$ (theorem 109.1) and on $1 \ll n/m \ll \log m$ (theorems 109.3 and 109.4). The six main theorems nest or continue one another, and four tools are proved in several of them; each is printed once, with its sources named, and the genuinely different proofs are kept as marked second routes.

101 Provenance, conventions and the answer

101.1 Six manuscripts, one addition

Part IX is built from six AI-assisted research manuscripts of batch 72, all dated 1 October 2026, all with the byline “Research note prepared for Vladimir Reshetnikov with OpenAI”, and all pinned to ProveIt commit 7421a4ca6, at which this report's article.tex had the blob b9697af6c8 and contained Parts I–VII as printed (the report did not change between the pin and the placement). They arrived in commit 1512ef835 and were placed in commit 9e6b04645. Table 31 lists them in the order in which they were written (the times of their archive entries), which is also their order of dependence: each cites only earlier ones. Their code, data, source ledgers and checks READMEs are shipped with the prefixes of the table, continuing the report's local sequence after Part VIII's 09-general-crossovers-. Their manuscripts, PDFs, delivery READMEs, checksum ledgers (all verified at placement) and TeX build transcripts are not shipped and survive in the arrival commit; two of them, manuscripts 36 and 33, also delivered a model.py that is byte-identical to Part VI's code/07-polynomial-rarity-model.py (Git blob a4fdee3761), which is not shipped again (section 110).

What each contributes, and where it is printed:

- *Manuscript 36* (the bridge): the squared-renewal coefficient majorant (proposition 102.1), a negative-tilt cluster law on Part I's block kernel (section 103), the uniform logarithmic bridge and its sharp constant (theorem 105.1 and corollary 105.2), and a Fourier-free two-term version (proposition 105.3).
- *Manuscript 33* (the relative equivalent): the relaxed class whose counts are the squared envelope (section 102), the cluster scales and the summable pointwise deviation bound (lemma 103.2, (412)), the cluster-count summation (proposition 108.5), the good-subclass inequality (lemma 104.1), the nearby-root comparison and the unmodulated equivalent $p_n(m) \sim \sqrt{\pi} n^{5/2} m^{-2} e^{-nt_m/m}$ for $m(\log m)^2 \ll n \ll m^2/(\log m)^2$ (part (iii) of theorem 108.2).
- *Manuscript 30* (the theta boundary): the exact finite cluster parameters, the periodic Gaussian ϑ and its bounds (lemma 108.1), the theta-modulated equivalent for $n \geq c_0 m(\log m)^2$ and its lattice-sum proof (parts (i)–(ii) of theorem 108.2, section 108.1), phase stability under a shortened cap (lemma 108.6), and the boundary limits in which every phase occurs (corollary 108.3).
- *Manuscript 27* (integer counts): compact tilts with a small cutoff, in the general form printed here (lemma 103.1), a finite saddle sum with controlled overflow, the Gamma equivalent at $n \asymp m \log m$ (theorem 109.1), the phase-dependent rate and the integer-count transition (corollary 109.2), and the endpoint-layer route to cap shortening.

Table 31: The six sources of Part IX. “Ms.” is the manuscript number in batch 72; “Src.” the report’s local source number (Part VIII is source 09). Pages are those of the delivered A4 PDFs.

Src.	Ms.	Title (pages; written, UTC)	Archive; main file	Prefix; labels
10	36	<i>A Uniform Logarithmic Bridge for Adjacency-Bounded 132-Avoiders</i> (8; 00:46–01:07)	Uniform_132_Avoider_Spectral_Bridge_Package.zip; uniform_avoider_spectral_bridge.tex	10-logarithmic-bridge-; jg:lb:
11	33	<i>A Relative Asymptotic for Adjacency-Bounded 132-Avoiders</i> (7; 01:21–01:32)	Relative_132_Avoider_Asymptotic_Package.zip; relative_avoider_asymptotic.tex	11-relative-asymptotic-; jg:ra:
12	30	<i>Theta Modulation at the 132-Avoider Renewal Boundary</i> (7; 01:37–01:48)	Theta_132_Avoider_Boundary_Package.zip; theta_avoider_boundary.tex	12-theta-boundary-; jg:tb:
13	27	<i>Integer-Count Transitions for 132-Avoiders at the $m \log m$ Scale</i> (8; 02:06–02:20)	Integer_Count_132_Avoider_Asymptotic_Package.zip; integer_count_avoider_asymptotic.tex	13-integer-count-; jg:ic:
14	24	<i>Uniform Saddle Sums for Bounded Gap 132 Avoiders</i> (9; 02:42–02:52)	Uniform_132_Avoider_Saddle_Sum_Package.zip (no inner directory); uniform_avoider_saddle_sum.tex	14-saddle-sum-; jg:ss:
15	22	<i>Growing Macro Counts for Bounded Gap 132 Avoiders</i> (8; 03:05–03:19)	Growing_Macro_132_Avoider_Counts_Package.zip (no inner directory); sublogarithmic_avoider_counts.tex	15-growing-macro-counts-; jg:gm:

- *Manuscript 24* (saddle sums): one positive saddle sum for all $n \geq c_0 m \log m$ (theorem 106.1), the two exact actions in the intermediate band (theorem 107.1), and the sharp Gaussian threshold $m(\log m)^{3/2}$ with its cubic boundary (theorem 108.2).
- *Manuscript 22* (growing macro counts): the Catalan-power small-run law (lemma 109.6), the general composition estimate (lemma 109.7), the Gamma equivalent for $1 \ll n/m \ll \log m$ and the shrinking transition below an integer (theorems 109.3 and 109.4), an endpoint-scale maximal-atom bound, a three-range uniform-integrability argument, and the Gamma-term route to cap shortening.

Every result, proof, example, remark, limitation and question of the six manuscripts is printed. Where several prove the same statement it is printed once, in the most general form proved, with every source named; genuinely different proofs are kept as marked second routes, and restatements with proof sketches are replaced by references (section 112).

Status (manuscripts 36, 33, 30, 27, 24 and 22 of batch 72). The theorems of Part IX have written mathematical proofs. They are ordinary mathematical arguments, not proof-assistant formalizations: nothing is Lean- or Rocq-verified, nothing has been externally refereed, and no worldwide priority is claimed (each manuscript calls its claims “source-relative”). All statements are asymptotic as $m \rightarrow \infty$, for fixed constants c_0, c_1, δ ; none gives an effective threshold, an error rate for a relative equivalent, or uniformity as these constants tend to their limits. Every statement about the actual class $a_n^{(m)}$, rather than the scalar envelope, needs $n(\log m)^2/m^2 \rightarrow 0$; only the logarithmic bridge of manuscript 36 holds without an upper restriction on n . The finite checks are regressions and diagnostics, several far from the asymptotic regime (section 110).

101.2 The questions answered, and the map of ranges

Part II's research direction "Joint growth of length and adjacency bound" (section 19.5) asks for $a_n^{(m)}/C_n$ when m grows with n , with "an explicit range of uniformity before proposing a transition law", and warns that substituting $m(n)$ into α_m^n is not justified. Part III's section 32.10, Part IV's section 47.10, Part VI's section 74.6 and Part VIII's sections 100.2 and 100.3 ask the same question from the boundary and bulk sides: which growth of n/m keeps a few-pieces description, and when do a saddle point or the fixed- m spectral rate become the right language. Part IX answers on the following ranges, all with $m \rightarrow \infty$ (table 32); each theorem states its own range of uniformity.

Table 32: Ranges of Part IX, from small to large n/m . Every statement holds for the actual count $a_n^{(m)}$ under the extra condition $n(\log m)^2/m^2 \rightarrow 0$, except where the envelope or the absence of an upper bound is named.

Range	Statement	Source; where
$1 \ll n/m \ll \log m$, phase of $(n-1)/m$ away from integers same, just below an integer $(n-1)/(m \log m) \in [c_0, c_1]$	one Gamma term two Gamma terms, logistic shares finite Gamma sum; phase-dependent rate	22; theorem 109.3 22; theorem 109.4 27; theorem 109.1 and corollary 109.2
$n \geq c_0 m \log m$	one positive saddle sum (envelope: no upper bound)	24; theorem 106.1
$n \geq c_0 m \log m$, no upper bound	logarithmic bridge to t_m and α_m , actual class	36; theorem 105.1
$m \log m \ll n \ll m(\log m)^2$	the two neighbouring counts with exact actions	24; theorem 107.1
$n \asymp m(\log m)^{3/2}$ $n \gg m(\log m)^{3/2}$	cubic correction, factor $\cosh(1/(192c^2))$ theta-modulated equivalent	24; theorem 108.2 24 ($\supset 30, 33$); theorem 108.2
$n \asymp m(\log m)^2$ $n \gg m(\log m)^2$	every phase of $\vartheta(2\sqrt{c}, \cdot)$ occurs unmodulated equivalent; $(\alpha_m/4)^n$ form	30; corollary 108.3 33 (= the $H \rightarrow \infty$ case); theorem 108.2

The answer to Part II's question is therefore: on $n \geq c_0 m \log m$ the fixed-cap rate is the right exponential language, at logarithmic precision with no upper restriction on n (manuscript 36), and with a relative equivalent whenever also $n(\log m)^2/m^2 \rightarrow 0$ (manuscripts 24, 30, 33); the number of macroscopic clusters is $\approx n/\mu$ and its lattice structure survives as a theta modulation down to $m(\log m)^{3/2}$; below $m \log m$ the answer is a few-pieces description with explicit Gamma terms (manuscripts 27, 22). Part IV's remark that "the number of macroscopic components grows" in this regime (section 47.10) is borne out: it is $\lceil (n-1)/m \rceil$ or one more in the Gamma regimes and about n/μ above. What stays open is listed in section 113: in particular bounded counts $\lceil (n-1)/m \rceil$ with $m \rightarrow \infty$ outside Part VIII's fixed- θ statements, phases near integers outside the specified window, uniformity as $\delta, c_0 \rightarrow 0$, the actual class when $n(\log m)^2/m^2 \not\rightarrow 0$ (the hand-off to the fixed- m Perron regime, where only the logarithmic bridge holds), and error rates. Manuscript 36 is thus not superseded by manuscript 24: its statement is weaker in precision but has no upper restriction on n .

101.3 Notation and conventions of Part IX

The report's $C_n, C, \text{Av}_n(132), a_n^{(m)}, p_n(m) = a_n^{(m)}/C_n = \mathbb{P}(M_n \leq m), \alpha_m$, Part I's endpoint recurrence (8), block counts $c_{k,p}$ (5), block polynomial $K_{m,d}$ (18) and its root $s_{m,d}$, and Part II's $K_m(x) = \sum_{k=1}^m C_{k-1}x^k$, scalar root r_m with $K_m(r_m) = 1, \beta_m = r_m^{-1}, t_m = m \log(4r_m)$ (58), partial sums H_q (49), weights $b_k = C_{k-1}4^{-k}$ (57) and tail $\varepsilon_d = (d+2)2^{-d}$ (54) keep their meanings. All

logarithms are natural. Part IX fixes the following conventions (table 33).

- *The envelope.* All six manuscripts write $E_m(n) = [x^n]x/(1 - K_m(x))^2$. It is printed $\mathfrak{E}_m(n)$. The tempting false reading is Parts I–II’s centered remainder E_m , defined by $m(4 - \alpha_m) = 2 \log m + 4 \log \log m + E_m$ and tending to $2 \log \pi$; the two are unrelated, and E_m has no argument. $\mathfrak{E}_m(n)$ is also not Part IV’s event $\mathcal{E}_{n,m}$.
- *The periodic Gaussian.* Manuscripts 30 and 24 write $\Theta(H, u)$; it is printed $\vartheta(H, u)$, defined in (460). In Part IX, as in Parts VI–VIII, $\Theta(\cdot)$ is always order notation (bounded ratios); ϑ with two arguments is not Part V’s logarithmic scale ϑ_n .
- *Clusters.* The tilted weights p_k are split at a cutoff into a small part $A(z)$ and a big part $B(z)$; $Q = B/(1 - A)$, $b = B(1)$, and a cluster $Y = X + R$ is an independent big jump X (the manuscripts’ J) plus a geometric run R of small jumps. $A(z)$, $B(z)$ carry no index: manuscript 30’s $A_m(z)$, $B_m(z)$, $Q_m(z)$, b_m are printed without the m , because $A_m(x)$ is Part I’s generating function of $a_n^{(m)}$. The big jump is printed X because J is the first deficiency of Parts III, IV, VII and VIII; the run R is not the operation R_β .
- *The cutoff.* The cutoff differs between the sources and changes A, B, Q, b, Y , its mean μ , variance σ^2 , and hence the phase $L_m(v) = v/\mu$ and the width $H_m(v) = (\sigma/\mu)\sqrt{L_m(v)}$: manuscript 36 splits Part I’s block weights at $M/2$, with $M = m - d$; manuscripts 33 and 30 split at $m/2$; manuscripts 27, 24 and 22 at a small fixed fraction εm (manuscript 22 for the critical weights b_k). Each section states its cutoff. The mean $v_m = \sum_k k p_k = b\mu$ does not depend on it.
- *Scales.* $w = 1/\log(4r_m)$ (manuscript 24’s $w_m = m/t_m$, the same number); in sections 103.2 and 105 manuscript 36’s $w = 1/\log(4s_{m,d})$ instead. $\Psi_m(h)$ is a scaled logarithmic moment generating function, not Part IV’s profile $\Psi(u)$.
- *Renamed from the sources.* Manuscript 36’s lower kernel $L_m(x)$ and root s_m are Part I’s $K_{m,d}(x)$ and $s_{m,d}$ with $d = \lceil \log_2 m \rceil$, and are printed so; its interval length L is N_I . Manuscript 33’s integer cutoffs D, L (and the D of manuscripts 30, 27, 22) are manuscript 24’s $D_{\text{app}}, L_{\text{app}}$, with $\varepsilon_{D_{\text{app}}} = (D_{\text{app}} + 2)2^{-D_{\text{app}}}$. The tilt intervals $\mathcal{H}$ of manuscripts 36, 27 and 24 are printed $\mathcal{T}$; the letter H is reserved for the width. The Gamma terms $F_{m,v}(\ell)$ of manuscript 27 and $G_{m,v}(j)$ of manuscript 22, the same function, are printed $g_{m,v}(\ell)$; manuscript 27’s rate $J_c(s)$ is $r_c(s)$. The coefficients $C_\ell(v)$, $V_m(v)$, $T_m(v)$ and $\mathcal{S}_m(v)$ of manuscripts 27, 24, 33 and 30 are printed $\mathfrak{C}_\ell(v)$, $\mathfrak{V}_m(v)$, $\mathfrak{T}_m(v)$, $\mathfrak{S}_m(v)$ (not the Catalan numbers C_ℓ , Part I’s component V_m , Part VI’s $T_{u,v}^{(m)}$ or Part V’s survival function $\mathcal{S}$), and manuscript 24’s neighbour set $\mathcal{J}(L)$ is $\mathfrak{J}(L)$. Residual masses and runs $D(h)$, W_h are printed $\mathfrak{D}(h)$, $\mathfrak{W}_h$ (not Part III’s law D or Part IV’s word W). Manuscripts 27 and 22 write β_k for $C_{k-1}4^{-k}$; it is printed b_k , as in Part II, because β_m is Part II’s r_m^{-1} ; their critical split A, B, Q (manuscript 22) or A_0, B_0 (manuscript 27) is printed A_0, B_0, Q_0 , and manuscript 22’s $V(v)$ and $T(u)$ are $\mathfrak{V}_m^0(v)$ and $\mathfrak{t}(u)$. Manuscript 30’s moment sums A_j, B_j are a_j^s, a_j^b . Characteristic functions are $\phi_h, \phi_{X,h}$, and φ is the standard normal density (manuscript 24 writes them the other way round).

No normalization changed: every renamed symbol denotes exactly its source’s quantity. Probability is written $\mathbb{P}$ and expectation $\mathbb{E}$; the manuscripts’ coefficientwise order $\leq_{\text{coeff}}$ is the report’s $\leq$; their thin-spaced differential is written out. Where the sources cite each other, this Part cites the merged statement; where they cite this report, it cites the label. Equations, tables and figures are numbered consecutively with those of Parts I–VIII.

101.4 Host inputs, and what the sources re-prove

The six manuscripts use the following results of Parts I and II; they are printed there and cited here.

$$\frac{1}{1 - K_{m,d}(x)} \leq A_m(x) \quad (\text{propositions 5.2 and 13.1}), \quad (394)$$

$$(1 - \varepsilon_D)C_{k-1} \leq c_{k,D} \leq C_{k-1} \quad (\text{lemma 13.2}), \quad (395)$$

Table 33: Symbols of Part IX. “Source” gives the manuscripts’ symbol; the last column prints the tempting false reading.

Symbol	Meaning in Part IX	Source	Not to be confused with
$\mathfrak{E}_m(n)$	$[x^n] x/(1 - K_m(x))^2$	$E_m(n)$ (all six)	the centered remainder $E_m \rightarrow 2 \log \pi$ (Parts I–II); Part IV’s event $\mathcal{E}$.
$\vartheta(H, u)$	periodic Gaussian (460)	$\Theta(H, u)$ (30, 24)	order notation $\Theta(\cdot)$; Part V’s ϑ_n .
$A(z), B(z), Q, b$	small and big parts of the tilted law, cluster PGF, big mass	same; A_m, B_m, Q_m, b_m (30)	Part I’s $A_m(x)$; the operation A ; b_k .
$Y = X + R$	cluster: big jump plus small run	$Y = J + R$	the first deficiency J ; the operation R_β .
μ, σ^2, ν_m	cluster mean and variance (cutoff-dependent); $\sum k p_k = b\mu$	same; μ_m, σ_m (30)	—
$L_m(v), H_m(v)$	$v/\mu; (\sigma/\mu)\sqrt{L_m(v)}$	same	the append cutoff L_{app} ; Part I’s L_m .
$D_{\text{app}}, L_{\text{app}}$	good-subclass cutoffs	D, L (33); D (30, 27, 22)	Part III’s D ; the phase L_m .
$K_{m,d}, s_{m,d}$	Part I’s block polynomial and root	L_m, s_m (36)	the deficit s_ℓ .
$\mathcal{T}$	compact tilt interval in $(-1, \infty)$	$\mathcal{H}$ (36, 27, 24)	the width H ; Part V’s $\mathcal{H}(p)$.
$\Psi_m(h)$	$\log \mathbb{E} e^{h(Y-m)/w}$	Ψ_m, Ψ	Part IV’s $\Psi(u)$.
$\mathfrak{g}_{m,v}(\ell); s_\ell$	Gamma term (490); deficit $\ell - v/m$	$F_{m,v}$ (27), $G_{m,v}$ (22)	Part I’s $F_{p,q}; G_q$ (33).
$r_c(s)$	rate $s/2 - c - c \log(s/(2c))$	$J_c(s)$ (27)	the first deficiency J .
$\mathfrak{C}_\ell, \mathfrak{B}_m, \mathfrak{T}_m, \mathfrak{S}_m, \mathfrak{J}$	count- ℓ coefficient, squared coefficient, weighted count sum, saddle sum, neighbour set	$C_\ell, V_m, T_m, S_m, \mathcal{J}$	Catalan C_ℓ ; Part I’s V_m ; Part II’s $T_m (= \Delta_m)$; Part V’s S .
$D(h), W_h$	residual mass and residual runs	$D(h), W_h, W$	Part III’s D ; Part IV’s word W .
$b_k; A_0, B_0, Q_0$	$C_{k-1}4^{-k}$; critical split	$\beta_k; A, B, Q$ (22)	Part II’s root $\beta_m = r_m^{-1}$.
$\phi_h; \varphi$	characteristic function; normal density	24: $\varphi_h; \phi$	—
$\tau, R_{m,\ell}(s)$	transition centre; adjacent Gamma ratio	same	Part VII’s child τ ; the operation R_β .

$$0 \leq s_{m,d} - r_m = O((\log m)^2/m^2), \quad s_{m,d}^{-1} \leq \alpha_m < r_m^{-1} \quad (\text{theorem 15.2, (53)}), \quad (396)$$

$$t_m = \frac{1}{2} \log m + \log \log m + \frac{1}{2} \log \pi + O\left(\frac{\log \log m}{\log m}\right) \quad ((75) \text{ at } J = 0). \quad (397)$$

Here $d = \lceil \log_2 m \rceil$ in (394) and (396) ((63)), and (395) holds uniformly in k and D , with $\varepsilon_D = (D+2)2^{-D} = O(\log m/m)$ at $D = d$; it follows from the Catalan-triangle tail $\sum_{j>D} j2^{-j} = (D+2)2^{-D}$. The exact endpoint recurrence is Part I's proposition 2.1, (8), with the block counts (5), (7): $c_{1,p} = 1$ and $c_{k,p} = \sum_{j=1}^{\min(p,k-1)} \frac{j}{k-1} \binom{2k-j-3}{k-2}$ for $k \geq 2$, which counts $\sigma \in \text{Av}_{k-1}(132)$ with $k - \sigma_1 \leq p$ (so $c_{k,\infty} = C_{k-1}$). The spectral corridor $0 \leq \log(r_m^{-1}/\alpha_m) = O((\log m)^2/m^2)$ is (37). These are the injective block bound (skew concatenations of blocks ending at their maxima, of lengths at most $m - d$ and first deficiency at most d ; every join has jump at most m , and a block ending at its maximum has no internal skew cut, so the construction is injective), the uniform Catalan coefficient bound, the quantitative root corridor and the scalar-root expansion; the manuscripts invoke them as established results, not rederived.

Remark 101.1 (Merge note: credits the sources omit, and one retargeted citation). Four further results of Part II are re-proved in the manuscripts without credit; Part IX prints the re-proofs, with these credits.

- The finite majorant system in the proof of proposition 102.1 (and manuscript 33's grammar of the relaxed class) is Part II's renewal-reset system: lemma 12.1 with $w_k = C_{k-1}x^k$, $a = x$, that is (49) and (50), read coefficientwise; (47) gives $(I - N)H = (1 - K_m)\mathbf{1}$, which is the identity $G_q = xH_q/(1 - K_m)$ used there. New is the row $q = \infty$ and the closed form $x/(1 - K_m)^2$ as a coefficient majorant.
- The lower-half tilted-mass estimates (in the proofs of section 103.2 and lemmas 103.1 and 103.2) and the upper-half endpoint Riemann sums are the proof of Part II's lemma 14.1 (the same bound $O(m^{-1/4}(\log m)^{3/2})$ at zero tilt), and the limit $B(1) \rightarrow 1/2$ is (67).
- Manuscript 33's nearby-root comparison (483) (and manuscript 30's root shift) re-runs Part II's corridor proof: the removed mass is (66), the slope bound (65) of lemma 15.1, and the mean-value step that of theorem 15.2; only the cutoff changes.
- The manuscripts' source ledgers cite the scalar expansion (397) as theorem 10.3, which is the expansion of $m(4 - \alpha_m)$; the statement for t_m with the rate $O(\log \log m / \log m)$ is (75) at $J = 0$ in proposition 16.1, and corollary 14.2 gives only $o(1)$. The citation is retargeted here.

Manuscript 36's remark that Part II excludes the single-pole series as a coefficient majorant is correct (remark 12.4, remark 69.2).

101.5 An earlier announcement of Part I's theorem

Manuscripts 30, 27, 24 and 22 cite a public announcement by A. C. Li [7], of 8 September 2026, of results on strict fixed-cap growth and descent thresholds, and say that its full manuscript could not be retrieved, so that overlap beyond its abstract is not excluded. Its abstract announces that the exponential growth constants are strictly increasing in m , which is Part I's theorem, eleven days before Part I's date. A dated note in Part I's scope box records this (page 2); it is not a retraction: Part I's proof is independent of the announcement, and the report never claimed exhaustive priority. None of the six manuscripts claims priority over the announcement, and none of their results concerns fixed m .

102 The squared-renewal majorant and its exact relaxed class

Throughout Part IX,

$$K_m(x) = \sum_{k=1}^m C_{k-1}x^k, \quad K_m(r_m) = 1, \quad t_m = m \log(4r_m), \quad v_m = r_m K'_m(r_m). \quad (398)$$

The positive root is unique. For $n \geq 1$ put

$$\mathfrak{E}_m(n) = [x^n] \frac{x}{(1 - K_m(x))^2}. \quad (399)$$

(These are Part II's K_m, r_m, t_m ; (398) is the definition shared by all six manuscripts, and manuscript 24 adds $w_m = m/t_m$.)

Proposition 102.1 (Sharp coefficient majorant; manuscript 36). *For every integer $m \geq 1$,*

$$A_m(x) \leq 1 + \frac{x}{(1 - K_m(x))^2}. \quad (400)$$

Equality holds in every degree at most $m + 1$. In particular,

$$a_n^{(m)} r_m^n \leq r_m n \leq n \quad (n \geq 1). \quad (401)$$

Proof. Let $F_{p,q}(x)$, $p, q \in \{0, \dots, m-1, \infty\}$, count the nonempty admissible permutations with first deficiency at most p and last deficiency at most q (negative thresholds give zero; ∞ imposes no restriction), so that Part I's endpoint recurrence (8) reads $F_{p,q} = x + \sum_{k=1}^m c_{k,p} x^k F_{m-k,q-k} + x F_{p-1,m-1}$ and $A_m(x) = 1 + F_{\infty,\infty}$. (It follows by decomposing at the maximum: a nonempty right child leaves a left block of size $k-1$, while an empty right child appends the maximum.) Drop only the first-deficiency restriction. Coefficientwise induction gives the nonnegative majorant system

$$G_q = x + \sum_{k=1}^q C_{k-1} x^k G_{q-k} + x G_{m-1} \quad (0 \leq q < m), \quad G_\infty = x + K_m G_\infty + x G_{m-1}.$$

Indeed every recursive term loses positive degree, so the induction is well founded. With $H_q(x) = \sum_{j=0}^q C_j x^j$, the finite Catalan renewal identity gives $G_q = H_q(x)(x + x G_{m-1})$. Since $x H_{m-1} = K_m$, this implies $G_{m-1} = K_m/(1 - K_m)$ and $G_\infty = x/(1 - K_m)^2$. All permutations of length at most $m+1$ satisfy the cap; also, through those degrees, K_m can be replaced by $xC(x)$ and $1 + x/(1 - xC(x))^2 = C(x)$. This proves the equality statement.

The coefficients of $P_m(z) = K_m(r_m z)$ form a probability law on positive integers. Thus $u_j = [z^j](1 - P_m(z))^{-1}$ is a renewal hitting probability and $0 \leq u_j \leq 1$. Taking the coefficient of degree n in (400) after replacing x by $r_m z$ gives $a_n^{(m)} r_m^n \leq r_m \sum_{j=0}^{n-1} u_j u_{n-1-j} \leq r_m n$. Finally $r_m \leq 1$. $\square$

The squared denominator is essential to this argument; Part II explicitly excludes the single-pole series $(1 - K_m)^{-1}$ as a coefficient majorant (remark 12.4). No claim about the exact denominator or pole order of A_m follows from (400).

[Merge note. The “finite Catalan renewal identity” is Part II's lemma 12.1 with $w_k = C_{k-1} x^k$, $a = x$, in the form (50); the system $(G_q)_{q < m}$ is $x(I - N)^{-1} \mathbf{1}$ for Part II's reset matrix N (remark 101.1). The majorant (400) is new as a coefficient statement and is strictly sharper than Part I's (17), $1 + x/(1 - x - K_m)$, and Part VI's (322): it is exact through degree $m+1$ and grows like r_m^{-n} instead of q_m^{-n} (compare remark 12.3). A ProveIt check made for this write-up confirmed (400) against exact counts for $m \leq 14$, $n \leq 60$, with equality exactly through $n = m+1$, and (401) there.]

102.1 The relaxed class counted by the envelope

Manuscript 33 identifies exactly which objects the envelope counts. Define a class $\mathcal{G}_M$ of nonempty 132-avoiders by their unique decomposition at the maximum. A singleton is allowed. If the right child is nonempty, its left decoration σ has size $k-1$ with $1 \leq k \leq M$, is an arbitrary 132-avoider, and the right child lies in $\mathcal{G}_M$. The jump from the maximum to the first

entry of that child is temporarily unrestricted. If the right child is empty, append the maximum to a nonempty child in $\mathcal{G}_M$ whose last deficiency is at most $M - 1$.

Let G_q count these objects with last deficiency at most q , and use $q = \infty$ for no last restriction. Their exact grammar gives

$$G_q = x + \sum_{k=1}^q C_{k-1} x^k G_{q-k} + x G_{M-1} \quad (0 \leq q < M), \quad G_\infty = x + K_M G_\infty + x G_{M-1}.$$

Writing $H_q = \sum_{j=0}^q C_j x^j$ gives $G_q = H_q(x)(x + x G_{M-1})$, whence

$$G_{M-1} = \frac{K_M}{1 - K_M}, \quad G_\infty = \frac{x}{(1 - K_M)^2}. \quad (402)$$

Thus $|\mathcal{G}_M \cap S_n| = \mathfrak{E}_M(n)$. Actual cap- M avoiders form a subset, so $a_n^{(M)} \leq \mathfrak{E}_M(n)$. This is the squared majorant of [proposition 102.1](#), now viewed as an exact relaxed class; the grammar is the majorant system of its proof, with M for m .

The parser has one recursive spine, with Catalan left decorations attached at its right-child steps. All adjacent jumps already satisfy the cap M except possibly those from a maximum to the first entry of its nonempty right child.

[Merge note. A ProveIt check made for this write-up, coded from the prose above and not from the delivered parser, confirmed $|\mathcal{G}_M \cap S_n| = \mathfrak{E}_M(n)$ for $M \leq 6, n \leq 11$, and that every cap- M avoider lies in $\mathcal{G}_M$ there.]

103 Clusters and a uniform local limit

Cutoff of this section: a fixed fraction εm in lemma 103.1; $m/2$ in lemmas 103.2 and 103.3; $M/2$ on Part I's block weights in section 103.2.

Tilt the Catalan coefficients at r_m :

$$p_k = C_{k-1} r_m^k \quad (1 \leq k \leq m), \quad \lambda = \log(4r_m), \quad w = \lambda^{-1} \sim \frac{2m}{\log m}.$$

The p_k form a probability distribution. For a cutoff fraction $0 < \varepsilon \leq 1/2$, split

$$A(z) = \sum_{k \leq \varepsilon m} p_k z^k, \quad B(z) = \sum_{k > \varepsilon m} p_k z^k, \quad b = B(1) = 1 - A(1), \quad Q(z) = \frac{B(z)}{1 - A(z)}. \quad (403)$$

Let Y have probability generating function Q . It is an independent big jump X plus a geometric run R of small jumps. Put

$$\Psi_m(h) = \log \mathbb{E} e^{h(Y-m)/w}, \quad \mu_h = m + w \Psi'_m(h), \quad \sigma_h^2 = w^2 \Psi''_m(h),$$

so that μ_h and σ_h^2 are the mean and variance of Y under the normalized tilt $e^{h(Y-m)/w}$, and $\mu = \mu_0$, $\sigma = \sigma_0$.

The mean and variance are exactly computable from finite sums (manuscript 30). If $a_j^s = \sum_{k \leq \varepsilon m} k^j p_k$, $a_j^b = \sum_{k > \varepsilon m} k^j p_k$ and $P_j = a_j^s + a_j^b$, then

$$\mu = \frac{P_1}{b}, \quad \sigma^2 = \frac{P_2}{b} + \frac{(a_1^s)^2 - (a_1^b)^2}{b^2}, \quad (404)$$

which follows from $\mathbb{E} X = a_1^b/b$, $\text{Var } X = a_2^b/b - (a_1^b)^2/b^2$, $\mathbb{E} R = a_1^s/b$ and $\text{Var } R = a_2^s/b + (a_1^s)^2/b^2$. Put

$$v_m = \sum_k k p_k = b \mu, \quad L_m(v) = \frac{v}{\mu}, \quad H_m(v) = \frac{\sigma}{\mu} \sqrt{L_m(v)}. \quad (405)$$

This is a scalar renewal cluster, not an asserted statistic of the conditioned permutation.

Lemma 103.1 (Uniform compact tilts; manuscripts 27 and 24). *For any fixed compact interval $\mathcal{T} \subset (-1, \infty)$, choose $0 < \eta < 1 + \inf \mathcal{T}$ and then $\varepsilon > 0$ with $\varepsilon(1 + \sup \mathcal{T} + \eta) < 1$. Uniformly for $h \in \mathcal{T}$, with any fixed number of derivatives,*

$$\Psi_m(h) \rightarrow -\log(1+h), \quad \Psi'_m(h) \rightarrow -\frac{1}{1+h}, \quad \Psi''_m(h) \rightarrow \frac{1}{(1+h)^2}. \quad (406)$$

Under the normalized tilt $e^{h(Y-m)/w}$, the big deficit $(m - X_h)/w$ tends to $\text{Exp}(1+h)$, the small run divided by w tends to zero, there is a uniform exponential moment for $|Y_h - m|/w$, and, for sums $S_\ell = Y_{h,1} + \dots + Y_{h,\ell}$ of independent copies,

$$\sup_{h \in \mathcal{T}, j \in \mathbb{Z}} \left| \sigma_h \sqrt{\ell} \mathbb{P}_h(S_\ell = j) - \varphi\left(\frac{j - \ell \mu_h}{\sigma_h \sqrt{\ell}}\right) \right| \rightarrow 0 \quad (m, \ell \rightarrow \infty), \quad (407)$$

with $\varphi(x) = (2\pi)^{-1/2} e^{-x^2/2}$, and the all- ℓ maximal-atom estimate

$$\sup_{h \in \mathcal{T}, j} \mathbb{P}_h(S_\ell = j) \leq \frac{C}{w\sqrt{\ell}} \quad (\ell \geq 1). \quad (408)$$

Proof. The increase in the lower Catalan sum under the largest required tilt is at most

$$C\lambda e^{(1+\sup \mathcal{T} + \eta)\lambda \varepsilon m} \sqrt{m} = O\left(m^{-1/2 + \varepsilon(1+\sup \mathcal{T} + \eta)/2} (\log m)^{1 + \varepsilon(1+\sup \mathcal{T} + \eta)}\right) = o(1).$$

Thus $A(e^{(h+\eta)/w}) \rightarrow 1/2$ uniformly, and the same holds at h . The tilted generating function factors exactly as

$$\frac{Q(e^{h/w}z)}{Q(e^{h/w})} = \frac{B(e^{h/w}z)}{B(e^{h/w})} \frac{1 - A(e^{h/w})}{1 - A(e^{h/w}z)}. \quad (409)$$

So $Y_h = X_h + R_h$ with independent summands: the big jump is normalized by $B(e^{h/w})$, and the run has success probability $1 - A(e^{h/w})$. The normalized big deficit $(m - X_h)/w$ tends to $\text{Exp}(1+h)$ with uniform exponential moments. Indeed, writing $k = m - u$ and using $C_{k-1}4^{-k} \sim (4\sqrt{\pi})^{-1}k^{-3/2}$, endpoint Riemann sums give

$$\mathbb{P}_h(X_h = m - u) = \frac{1 + h + o(1)}{w} e^{-(1+h)u/w}$$

for bounded u/w , and an exponential upper bound throughout $0 \leq u < (1 - \varepsilon)m$. The small-run exponential moment $[1 - A(e^{h/w})]/[1 - A(e^{(h+\eta)/w})]$ tends to one. These facts prove (406), including derivatives and the moment margin.

Here are the necessary smoothing details. On $[\varepsilon m, m]$, the ideal tilted weights increase for large m , with consecutive ratio $e^{(1+h)/w}(2k-1)/(2k+2) > 1$. Their normalized maximum is $O(1/w)$. Abel summation gives the big-jump characteristic bound $C/[w|\sin(t/2)|]$. Scaled-frequency convergence and this bound give a uniform gap $\rho < 1$ on $\delta/w \leq |t| \leq \pi$. The cluster inherits it, and its maximal atom is $O(1/w)$. Parseval then bounds its squared characteristic integral by C/w . Near zero the uniform moments and positive variance give a Gaussian modulus bound. Fourier inversion proves (408): for $\ell \geq 2$ the outer integral is at most $C\rho^{\ell-2}/w$, and for $\ell = 1$ use the big-jump maximal atom. The central integral has scale $1/(w\sqrt{\ell})$.

The centered characteristic expansion and the uniform third moment give (407) as $m, \ell \rightarrow \infty$. The scaled outer integral is $O(\sqrt{\ell} \rho^{\ell-2})$, with no remaining m factor. This proves the claimed uniform local limit. $\square$

The smaller cutoff permits arbitrarily large *fixed* positive tilts. A cutoff at $m/2$ would not suffice for all compact intervals of $n/(m \log m)$ and all competing phases (manuscript 27); it does suffice for tilts with $\sup \mathcal{T} < 1$, the case of manuscripts 33 and 30 below.

[Merge note. Manuscript 24's lem:11t is this lemma restated with the same hypotheses and the same displayed bound, and its proof is the same; it says so ("established in the logarithmic and integer-count companions; the proof is included in the form needed here"). Manuscript 24 states (406) "with any fixed number of derivatives" and words the high-frequency step as follows: the limiting characteristic function on the scaled frequency axis has a strict modulus gap on every compact annulus away from zero, and with the Abel bound this gives a fixed gap outside a central interval of width $O(w^{-1})$; the geometric-run factor has modulus at most one. Manuscript 30 restates the conclusions for the cutoff $m/2$ and two-sided tilts $|h| \leq h_* < 1$, with a summary of the same mechanism ("these are not inferred from bounded moments alone"), and cites manuscript 33 for the proofs, which follow. The method appears first in manuscript 36 (section 103.2); its proof there is the most detailed.]

103.1 The scales at the half-cap cutoff and a summable deviation bound

Cutoff of this subsection: $\varepsilon = 1/2$ (manuscripts 33 and 30).

Lemma 103.2 (Cluster scales and uniform compact tilts; manuscript 33). *One has $b \rightarrow 1/2$ and*

$$\mu = m - w + o(w), \quad \sigma^2 \sim w^2, \quad v_m = b\mu \sim m/2. \quad (410)$$

Fix $0 < h_ < 1$. Tilt Y by $\exp(h(Y - m)/w)$ and normalize, for $|h| \leq h_*$. Uniformly in that family, the big-jump deficit divided by w tends to $\text{Exp}(1 + h)$, the small run divided by w tends to zero in every fixed moment, and there is a bounded exponential moment on the w scale. The variance divided by w^2 is bounded above and below by positive constants.*

Proof. Stirling's formula gives $C_{k-1}4^{-k} = (4\sqrt{\pi})^{-1}k^{-3/2}(1 + O(k^{-1}))$ and $\sum_{k \geq 1} C_{k-1}4^{-k} = 1/2$. Choose $\eta > 0$ with $h_* + \eta < 1$. Uniformly for $|h| \leq h_*$,

$$A(e^{h/w}) \rightarrow \frac{1}{2}, \quad A(e^{(h+\eta)/w}) \rightarrow \frac{1}{2}.$$

Indeed the increase over the untilted partial Catalan sum is bounded, using $e^u - 1 \leq ue^u$, by

$$C\lambda e^{(1+h_*+\eta)\lambda m/2} \sqrt{m} = O\left(m^{-(1-h_*-\eta)/4} (\log m)^{(3+h_*+\eta)/2}\right) = o(1).$$

The lower estimate follows because $1 + h > 0$. At $h = 0$, normalization then gives $b \rightarrow 1/2$.

The tilted cluster has the exact factorization (409). Consequently $\mathbb{E}_h e^{\eta R_h/w} = [1 - A(e^{h/w})]/[1 - A(e^{(h+\eta)/w})] \rightarrow 1$. This proves the small-run moment assertion.

For $k = m - u > m/2$, the big-jump weights before normalization are

$$\frac{1 + o(1)}{4\sqrt{\pi}} m^{-3/2} e^{(1+h)m/w} e^{-(1+h)u/w} (1 - u/m)^{-3/2},$$

uniformly on that half interval. Endpoint Riemann sums therefore give $\mathbb{P}_h(X_h = m - u) = (1 + h + o(1))w^{-1}e^{-(1+h)u/w}$ for bounded u/w , uniformly in h . Throughout the half interval this probability is bounded by $Cw^{-1}e^{-(1-h_*)u/w}$. These estimates give the uniform exponential deficit limit and moments. Independence yields (410), except for its exact identity, which follows from $\mathbb{E}X = B'(1)/b$ and $\mathbb{E}R = A'(1)/b$. $\square$

The next lemma includes the factor needed to sum probabilities over the number of clusters.

Lemma 103.3 (Local limit and summable deviation bound; manuscript 33). *As both m and ℓ tend to infinity,*

$$\sup_{j \in \mathbb{Z}} \left| \sigma \sqrt{\ell} \mathbb{P}(Y_1 + \cdots + Y_\ell = j) - \varphi\left(\frac{j - \ell\mu}{\sigma \sqrt{\ell}}\right) \right| \rightarrow 0, \quad (411)$$

where the Y_i are independent copies of Y . For all sufficiently large m , all $\ell \geq 1$, and every integer j ,

$$\mathbb{P}(Y_1 + \cdots + Y_\ell = j) \leq \frac{C}{w\sqrt{\ell}} \exp\left\{-c \min\left(\frac{(j - \ell\mu)^2}{\ell w^2}, \frac{|j - \ell\mu|}{w}\right)\right\}. \quad (412)$$

Proof (third route to the local limit; the proof of (412)). Here are the uniform Fourier estimates, including their high-frequency justification. For every tilt in a smaller fixed interval $|h| \leq h_* < 1$, the ideal big weights have consecutive ratio $e^{(1+h)\lambda}(2k-1)/(2k+2) > 1$ on $m/2 < k < m$, for large m . Their normalized maximum is $O(1/w)$. The zero-extended sequence thus has total variation $O(1/w)$; Abel summation gives

$$|\phi_{X,h}(t)| \leq \frac{C}{w|\sin(t/2)|}, \quad 0 < |t| \leq \pi.$$

The deficit limit at $t = u/w$ gives compact convergence of its modulus to $(1 + u^2/(1+h)^2)^{-1/2}$. Combining this on $\delta \leq |u| \leq T$ with the Abel bound beyond T/w proves, for each small fixed $\delta > 0$,

$$\sup_{h, \delta/w \leq |t| \leq \pi} |\phi_h(t)| \leq \rho_\delta < 1. \quad (413)$$

The independent small-run factor has modulus at most one, so the cluster inherits the big-jump gap. Its maximal atom is $O(1/w)$, and Parseval gives $\int_{-\pi}^{\pi} |\phi_h(t)|^2 dt \leq C/w$. Near zero, the uniform moments and variance give $|\phi_h(t)| \leq e^{-cw^2t^2}$ for $|t| \leq \delta/w$. Fourier inversion now yields, uniformly in the tilt,

$$\sup_j \mathbb{P}_h(Y_1 + \cdots + Y_\ell = j) \leq \frac{C}{w\sqrt{\ell}} \quad (\ell \geq 1). \quad (414)$$

For $\ell \geq 2$, the outer integral is at most $C\rho_\delta^{\ell-2}/w$, bounded by $C/(w\sqrt{\ell})$; the central Gaussian integral has the same bound. For $\ell = 1$, use the big-jump maximal atom.

The centered characteristic expansion has a uniform third-moment remainder. At scale $\sigma\sqrt{\ell}$ it converges on compact sets to $e^{-v^2/2}$ and is dominated by a Gaussian. The scaled outer Fourier integral is at most $C\sqrt{\ell} \rho_\delta^{\ell-2} \rightarrow 0$, with no remaining m factor. This proves (411), uniformly in the target phase.

Finally put $Z = (Y - \mu)/w$ and $K(h) = \log \mathbb{E}e^{hZ}$. Lemma 103.2 gives $K(0) = K'(0) = 0$ and $K''(h) \leq C$ on a fixed two-sided tilt interval. Thus $K(h) \leq Ch^2$. Undoing the tilt in (414) bounds the probability by

$$\frac{C}{w\sqrt{\ell}} \exp\{\ell K(h) - h(j - \ell\mu)/w\}.$$

Choose the sign of h to match $j - \ell\mu$ and its magnitude to be the smaller of a fixed positive constant and $|j - \ell\mu|/(2C\ell w)$. This proves (412). $\square$

[Merge note. Lemma 103.2 and the local limit (411) are the cases $\varepsilon = 1/2$, $\mathcal{T} = [-h_*, h_*]$ of lemma 103.1 (admissible because $h_* < 1$), proved by the same method; the exact identity $v_m = b\mu$ and the pointwise deviation bound (412) are new in manuscript 33. Manuscript 30 restates (411) and (412) for its sums S_ℓ (its eq:llt and eq:localtail), together with the scales (410) and $w \sim 2m/\log m$ (its eq:scales), and says “the complete estimates are proved in” manuscript 33; both are replaced by these labels. A ProvelT check made for this write-up found the finite scales converging slowly: at the cutoff $m/2$ and $m = 10^3, \dots, 10^6$, $b = 0.43\text{--}0.49$, $(\mu - m)/w = -0.63$ to -1.06 (limit -1), $\sigma/w = 1.27$ to 1.20 (limit 1) and $v_m/m = 0.39\text{--}0.44$ (limit $1/2$).]

103.2 Negative tilts on Part I's block kernel

Weights and cutoff of this subsection (manuscript 36): Part I's block weights $c_{k,d}s_{m,d}^k$ on $k \leq M$, split at $M/2$, with $w = 1/\log(4s_{m,d})$. Take

$$d = \lceil \log_2 m \rceil, \quad M = m - d, \quad K_{m,d}(x) = \sum_{k=1}^M c_{k,d} x^k, \quad K_{m,d}(s_{m,d}) = 1, \quad (415)$$

Part I's block polynomial (18) and its root (manuscript 36 writes L_m and s_m). All subsequent statements involving these quantities concern sufficiently large m . Set

$$\lambda = \log(4s_{m,d}), \quad w = \lambda^{-1}, \quad p_k = c_{k,d}s_{m,d}^k \quad (1 \leq k \leq M), \quad b_k = C_{k-1}4^{-k}. \quad (416)$$

The p_k sum to one. The inherited bounds (395)–(397) imply

$$M\lambda = t_m + o(1), \quad w \sim \frac{2m}{\log m}, \quad M \sim m, \quad p_k = (1 + O(\varepsilon_d))b_k e^{\lambda k} \quad (417)$$

uniformly in k , where $b_k = (4\sqrt{\pi})^{-1}k^{-3/2}(1 + O(k^{-1}))$ and $\sum_{k \geq 1} b_k = 1/2$. Define $A(z) = \sum_{k \leq M/2} p_k z^k$, $B(z) = \sum_{M/2 < k \leq M} p_k z^k$ and $Q = B/(1 - A)$. Here and below the integer endpoints mean the corresponding floor or ceiling. Since $A(1) + B(1) = 1$, Q is a probability generating function, of $Y = X + R$ as before. In particular,

$$\frac{1}{1 - A - B} = \frac{1}{1 - A} \frac{1}{1 - Q}. \quad (418)$$

Every coefficient is nonnegative, and the first factor has constant term one.

We establish all estimates simultaneously for a compact family of negative tilts. Fix $\mathcal{T} = [h_0, 0]$, $-1 < h_0 < 0$. Under $\mathbb{P}_h$, tilt Y by $\exp(h(Y - M)/w)$ and normalize. Its generating function factors exactly as

$$Q_h(z) = \frac{Q(e^{h/w}z)}{Q(e^{h/w})} = \frac{B(e^{h/w}z)}{B(e^{h/w})} \frac{1 - A(e^{h/w})}{1 - A(e^{h/w}z)}. \quad (419)$$

Thus $Y_h = X_h + R_h$ with independent summands. The big jump is normalized by $B(e^{h/w})$, whereas the geometric run has success probability $1 - A(e^{h/w})$. These quantities need not be equal after tilting.

Lemma 103.4 (Uniform tilted cluster scales; manuscript 36). *Uniformly for $h \in \mathcal{T}$,*

$$\frac{M - X_h}{w} \implies \text{Exp}(1 + h), \quad \frac{R_h}{w} \longrightarrow 0 \quad \text{in every fixed moment}, \quad (420)$$

$$\mu_h := \mathbb{E}_h Y = M - \frac{w}{1 + h} + o(w), \quad \frac{\sigma_h^2}{w^2} := \frac{\text{Var}_h Y}{w^2} \longrightarrow \frac{1}{(1 + h)^2}. \quad (421)$$

For some $\theta_0 > 0$ depending only on $\mathcal{T}$,

$$\sup_{m \text{ large}} \sup_{h \in \mathcal{T}} \mathbb{E}_h e^{\theta_0 |Y - M|/w} < \infty, \quad \sup_j \mathbb{P}_h(Y = j) \leq \frac{C_{\mathcal{T}}}{w}. \quad (422)$$

In particular $A(1) \rightarrow 1/2$ and $B(1) \rightarrow 1/2$.

Proof. For any fixed $0 < \eta < 1$, the lower-half sum satisfies

$$A(e^{h/w}) \rightarrow \frac{1}{2}, \quad A(e^{(h+\eta)/w}) \rightarrow \frac{1}{2} \quad (423)$$

uniformly in $h \in \mathcal{T}$. Indeed the untilted partial Catalan sum tends to $1/2$. Its increase under any of these tilts is bounded by

$$C\lambda e^{(1+\eta)\lambda M/2} \sum_{k \leq M/2} k^{-1/2} = O\left(M^{-(1-\eta)/4} (\log M)^{(3+\eta)/2}\right) = o(1).$$

We used $e^u - 1 \leq ue^u$, (417), and the uniform relative perturbation (395). The lower bound follows from $1 + h \geq 1 + h_0 > 0$. Equation (419) now gives

$$\mathbb{E}_h e^{\eta R_h/w} = \frac{1 - A(e^{h/w})}{1 - A(e^{(h+\eta)/w})} \rightarrow 1.$$

Since $R_h \geq 0$, this proves its asserted moment convergence and an exponential bound.

For the big jump, write $k = M - u$, $0 \leq u < M/2$. Uniformly on that interval,

$$p_{M-u} e^{h(M-u)/w} = \frac{1 + o(1)}{4\sqrt{\pi}} M^{-3/2} e^{(1+h)M/w} e^{-(1+h)u/w} (1 - u/M)^{-3/2}.$$

The factor $(1 - u/M)^{-3/2}$ lies between 1 and $2^{3/2}$. Endpoint Riemann sums and $w/M \rightarrow 0$ therefore give, uniformly for bounded u/w and for $h \in \mathcal{T}$,

$$\mathbb{P}_h(X_h = M - u) = \frac{1 + h + o(1)}{w} e^{-(1+h)u/w}.$$

Throughout the entire half interval it is bounded by $Cw^{-1}e^{-(1+h_0)u/w}$. This proves the exponential deficit limit with uniform exponential moments at every rate below $1 + h_0$. It also proves that the maximal big-jump atom is $O(1/w)$. Convolution with the probability law of R_h preserves that bound. Independence and the two moment conclusions give (421)–(422). Finally (423) at $h = 0$ and normalization give the last statement. $\square$

The changing lattice law has width of order w , which diverges. A moment bound alone would not control its high frequencies. The next proof supplies that control explicitly.

Lemma 103.5 (Cluster local limit, uniformly under compact tilts; manuscript 36). *Let $Y_{h,1}, \dots, Y_{h,\ell}$ be independent copies of Y_h . As both $m \rightarrow \infty$ and $\ell \rightarrow \infty$, with no further relation required,*

$$\sup_{\substack{h \in \mathcal{T} \\ j \in \mathbb{Z}}} \left| \sigma_h \sqrt{\ell} \mathbb{P}_h \left(\sum_{i=1}^{\ell} Y_{h,i} = j \right) - \varphi \left(\frac{j - \ell \mu_h}{\sigma_h \sqrt{\ell}} \right) \right| \rightarrow 0. \quad (424)$$

Proof (second route to the local limit, in full detail). Let $\phi_h(t) = \mathbb{E}_h e^{itY}$ and $\phi_{X,h}(t) = \mathbb{E}_h e^{itX_h}$. The ideal big-jump weights are $b_k e^{(1+h)\lambda k}$. Their exact consecutive ratio is

$$e^{(1+h)\lambda} \frac{2k-1}{2k+2} > 1 \quad (M/2 < k < M)$$

for sufficiently large m , uniformly in $h \in \mathcal{T}$. After normalization their maximum is $O(1/w)$, by the same endpoint sum as in lemma 103.4. Their zero-extended sequence consequently has total variation at most twice its maximum. Abel summation gives

$$|\phi_{X,h}(t)| \leq \frac{C_{\mathcal{T}}}{w|\sin(t/2)|} + C_{\mathcal{T}} \varepsilon_d \quad (0 < |t| \leq \pi). \quad (425)$$

The last term follows because a uniform relative weight error $O(\varepsilon_d)$ changes the normalized probability law by $O(\varepsilon_d)$ in total variation. This remains true even if its unnormalized total mass tends to zero.

At $t = u/w$, the deficit limit gives uniform compact convergence of the modulus to $(1 + u^2/(1 + h)^2)^{-1/2}$. Fix a sufficiently small $\delta > 0$. On $\delta \leq |u| \leq T$ this yields a bound strictly below one, uniformly in h . For $|t| \geq T/w$, choose T large and use (425), with $|\sin(t/2)| \geq |t|/\pi$ on $[-\pi, \pi]$. Thus

$$\sup_{\substack{h \in \mathcal{T} \\ \delta/w \leq |t| \leq \pi}} |\phi_{X,h}(t)| \leq \rho_\delta < 1 \quad (426)$$

for all sufficiently large m . The independent factorization (419) gives $|\phi_h(t)| \leq |\phi_{X,h}(t)|$, so the cluster inherits this gap.

The maximal-atom bound in (422) and Parseval imply

$$\int_{-\pi}^{\pi} |\phi_h(t)|^2 dt = 2\pi \sum_j \mathbb{P}_h(Y = j)^2 \leq \frac{C_{\mathcal{T}}}{w}. \quad (427)$$

Consequently the outer part of Fourier inversion for the ℓ -fold sum is bounded, after multiplication by $\sigma_h \sqrt{\ell}$, by

$$\sigma_h \sqrt{\ell} \int_{\delta/w \leq |t| \leq \pi} |\phi_h(t)|^\ell dt \leq C_{\mathcal{T}} \sqrt{\ell} \rho_\delta^{\ell-2} \longrightarrow 0. \quad (428)$$

There is no remaining m -dependent prefactor.

For $|t| \leq \delta/w$, the uniform third moment and the variance bounds give $|\phi_h(t)| \leq e^{-cw^2 t^2}$ after reducing δ if necessary. The centered expansion, with $v = \sigma_h \sqrt{\ell} t$, gives uniformly for bounded v and all h ,

$$(e^{-it\mu_h} \phi_h(t))^\ell = \exp\left(-\frac{v^2}{2} + O_{\mathcal{T}}(|v|^3/\sqrt{\ell})\right).$$

The preceding Gaussian modulus bound supplies an integrable dominating function. Hence the central Fourier integrand converges in L^1 to $e^{-v^2/2}$. Together with (428) and the Gaussian tail this proves (424); the error is independent of the target Fourier phase and therefore uniform in j . $\square$

[Merge note. Lemmas 103.4 and 103.5 are the variant of lemma 103.1 that the logarithmic bridge needs: Part I's block weights $c_{k,d} s_{m,d}^k$ instead of $C_{k-1} r_m^k$ (a uniform relative perturbation $O(\varepsilon_d)$, which is the origin of the extra term in (425)), cut at $M/2$, and negative tilts only. The lower-half estimate and the upper-half Riemann sum in the proof of lemma 103.4 are Part II's proof of lemma 14.1 at a tilt, and $B(1) \rightarrow 1/2$ is (67) (remark 101.1). Manuscript 33's proofs (lemmas 103.2 and 103.3) repeat several sentences of these nearly verbatim.]

104 The good-subclass inequality and the transfer to actual avoiders

Recall from section 102.1 that the actual class lies inside the relaxed class, so $a_n^{(m)} \leq \mathfrak{E}_m(n)$ for every n . The converse direction is manuscript 33's.

Lemma 104.1 (Good objects and two exact loss bounds; manuscript 33). *Let $D_{\text{app}}, L_{\text{app}}$ be positive integers, set $d = D_{\text{app}} + L_{\text{app}}$ and $M = m - d \geq 1$, and suppose $n > L_{\text{app}}$. Then*

$$(1 - n\varepsilon_{D_{\text{app}}})\mathfrak{E}_M(n) - n\mathfrak{E}_M(n - L_{\text{app}}) \leq a_n^{(m)} \leq \mathfrak{E}_m(n), \quad \varepsilon_{D_{\text{app}}} = (D_{\text{app}} + 2)2^{-D_{\text{app}}}. \quad (429)$$

Proof. The upper bound was shown above. Call an envelope object of $\mathcal{G}_M$ good if every left decoration has first deficiency $k - \sigma_1 \leq D_{\text{app}}$ (zero for the empty decoration), and every consecutive run of append steps on the spine has length at most L_{app} . Each spine subtree of a good object has first deficiency at most $D_{\text{app}} + L_{\text{app}}$: after its initial append run, a right-child step has first deficiency at most D_{app} , or a singleton has deficiency zero; each append raises it by one.

Every formerly unrestricted jump is therefore at most $k + (D_{\text{app}} + L_{\text{app}}) \leq M + d = m$. Append jumps are already at most M , and internal decoration jumps are smaller than M . The underlying permutation is unchanged, so good objects inject into actual cap- m avoiders.

Fix the spine shape and its step costs. Its last-deficiency conditions depend only on those costs and the positions of append steps, not on any decoration's internal shape. Each decoration independently retains at least the fraction $1 - \varepsilon_{D_{\text{app}}}$ by (395). There are at most n decorations; the union bound shows that at most $n\varepsilon_{D_{\text{app}}} \mathfrak{E}_M(n)$ objects fail this restriction.

For an append run longer than L_{app} , choose the first such maximal run in spine order. Delete its outermost L_{app} append nodes and mark the first surviving append node. At least one append remains. Both the old and new subtrees have last deficiency zero. Every ancestor's last-deficiency condition is unchanged: a right-child step adds its fixed cost, and an append resets the deficiency to zero. The reduced object therefore lies in $\mathcal{G}_M$ and has size $n - L_{\text{app}}$. Reinserting L_{app} append nodes immediately above the mark recovers the original uniquely. There are at most n possible marked spine nodes, giving at most $n\mathfrak{E}_M(n - L_{\text{app}})$ bad objects. This includes terminal runs, since at least one append remains above the terminal singleton. Subtracting both excluded families proves (429). $\square$

The two losses have different origins: restricting the first deficiency of every Catalan decoration loses at most $n\varepsilon_{D_{\text{app}}}$ by the endpoint law, and the long append runs are removed by a marked injection (manuscript 24's summary). The bound needs no uniform fixed-cap Perron assumption (manuscript 27).

[Merge note. Manuscripts 30, 27, 24 and 22 restate (429) (each as its eq:good) from manuscript 33, with a one-paragraph proof sketch of the same two losses; manuscript 24 keeps $D_{\text{app}}, L_{\text{app}}$ general, the others fix them as in (430). Those restatements are replaced by this label. A ProveIt check made for this write-up tested (429) against exact counts for $3 \leq m \leq 16$, all $D_{\text{app}}, L_{\text{app}} \geq 1$ with $M \geq 1$, and $L_{\text{app}} < n \leq 45$ (22,820 cases, no violation); at such sizes the lower bound is weak (positive in only 296 of them).]

Choose

$$D_{\text{app}} = L_{\text{app}} = \lceil 3 \log_2 m \rceil, \quad d = D_{\text{app}} + L_{\text{app}}, \quad M = m - d. \quad (430)$$

Then $\varepsilon_{D_{\text{app}}} = O(\log m / m^3)$, and $r_m^{L_{\text{app}}} = O(m^{-6})$. So the squeeze (429) gives $a_n^{(m)} \sim \mathfrak{E}_m(n)$ as soon as three things are known in the range under study:

$$n\varepsilon_{D_{\text{app}}} \rightarrow 0, \quad \frac{\mathfrak{E}_{m-d}(n)}{\mathfrak{E}_m(n)} \rightarrow 1, \quad \frac{n\mathfrak{E}_M(n - L_{\text{app}})}{\mathfrak{E}_M(n)} \rightarrow 0. \quad (431)$$

The first holds whenever $n(\log m)^2 / m^2 \rightarrow 0$. The third follows from the second at both lengths and the shift estimate $\mathfrak{E}_m(n - j) / \mathfrak{E}_m(n) \sim r_m^j$ for $j = O(\log m)$, since $nr_m^{L_{\text{app}}} \rightarrow 0$; manuscript 22 writes the shift as 4^{-j} , which is equivalent because $jt_m / m \rightarrow 0$. The second is cap shortening.

Proposition 104.2 (Cap shortening; manuscripts 33, 30, 27, 24 and 22). *Let $d = O(\log m)$. In each of the ranges of theorems 106.1, 108.2, 109.1, 109.3 and 109.4, under the upper condition $n(\log m)^2 / m^2 \rightarrow 0$ where it is stated,*

$$\frac{\mathfrak{E}_{m-d}(n)}{\mathfrak{E}_m(n)} \rightarrow 1, \quad \frac{\mathfrak{E}_m(n - j)}{\mathfrak{E}_m(n)} \sim r_m^j \quad (j = O(\log m)).$$

The statement is proved three times, by three different routes, each printed where its regime is treated:

1. *Root shift and phase stability* (manuscripts 33 and 30): the roots r_{m-d} and r_m differ by $O(d \log m / m^2)$, and the explicit equivalent of the envelope, including its theta phase, is stable under that shift (section 108.4).

2. *A conditional endpoint layer* (manuscripts 27 and 24): at the exact centering tilt, the probability that one of the ℓ big jumps lies in the forbidden layer above $m - d$ is $O(\ell d/w)$ (section 106.2).
3. *Gamma terms* (manuscript 22): the explicit Gamma terms change by a relative $o(1)$ when the deficit is shifted exactly (section 109.6).

With proposition 104.2 and (430), (431) holds in each of those ranges, and $a_n^{(m)} \sim \mathfrak{E}_m(n)$ there; the marked endpoint loss is $O(n(\log m)^2/m^2)$ (manuscript 24). Catalan Stirling asymptotics $C_n \sim 4^n/(\sqrt{\pi} n^{3/2})$ then turn each envelope equivalent into an equivalent for $p_n(m)$.

105 The uniform logarithmic bridge

Source: manuscript 36. Its cluster is that of section 103.2. Part II, in its research direction “Joint growth of length and adjacency bound” (section 19.5), asks for a bridge between fixed- m spectral growth and growing-cap regimes, and warns that its fixed- m multiplicative constants are not controlled uniformly (remark 11.2). Manuscript 36 answers that question on the following range.

Theorem 105.1 (Uniform logarithmic bridge; manuscript 36). *For each fixed $c_0 > 0$, uniformly for sufficiently large integers m and all integers $n \geq c_0 m \log m$,*

$$\log p_n(m) = -\frac{n}{m} t_m + O_{c_0} \left(\frac{n(\log m)^2}{m^2} + \log n \right) \quad (432)$$

$$= n \log(\alpha_m/4) + O_{c_0} \left(\frac{n(\log m)^2}{m^2} + \log n \right). \quad (433)$$

Consequently, on the same range,

$$-\frac{m}{n} \log p_n(m) = \frac{1}{2} \log m + \log \log m + \frac{1}{2} \log \pi + O_{c_0} \left(\frac{\log \log m}{\log m} + \frac{m \log n}{n} \right). \quad (434)$$

Corollary 105.2 (The sharp constant; manuscript 36). *Along every sequence $m \rightarrow \infty$ with $n/(m \log m) \rightarrow \infty$,*

$$-\frac{m}{n} \log p_n(m) - \frac{1}{2} \log m - \log \log m \rightarrow \frac{1}{2} \log \pi.$$

On the uniform range $n \geq c_0 m(\log m)^2$, the error in (434) is $O_{c_0}(\log \log m / \log m)$.

The sharp constant is not asserted uniformly at the lower edge $n \asymp m \log m$: there $(m/n) \log n$ can remain of order one. A polynomial prefactor could contribute at that order. These are logarithmic estimates, not a probability equivalent or a uniform leading prefactor. Thus no substitution into a nonuniform fixed-cap Perron equivalent is needed. (On $n(\log m)^2/m^2 \rightarrow 0$ the relative equivalents of sections 106 and 108 sharpen (432); above that band, for $n \gtrsim m^2/(\log m)^2$, theorem 105.1 is the only statement of Part IX about the actual class.)

105.1 Exact centering and the proof of the bridge

Fix $c_0 > 0$ and let $n \geq c_0 m \log m$. Write $\mu_0 = \mathbb{E}Y$ and choose $\ell = \lceil n/\mu_0 \rceil$. Then $\ell \asymp n/m$, in particular $\ell \geq c \log m$, and

$$0 \leq \frac{\mu_0 - n/\ell}{w} < \frac{\mu_0}{w\ell} \leq C \frac{\log m}{\ell} \leq C_{c_0}. \quad (435)$$

Only a negative tilt is needed. Define the scaled log moment generating function $\Psi_m(h) = \log \mathbb{E} \exp(h(Y - M)/w)$ and $x = (n/\ell - M)/w$. Then $\Psi'_m(h) = (\mu_h - M)/w$ and $\Psi''_m(h) = \sigma_h^2/w^2$. At $h = 0$ the derivative tends to -1 . Choose $h_0 = h_0(c_0) > -1$ sufficiently close to -1 . By (421), its

limiting derivative $-1/(1 + h_0)$ lies below all targets x allowed by (435). The variance is strictly positive, so for every sufficiently large m and all allowed n there is a unique $h \in [h_0, 0]$ such that

$$\Psi'_m(h) = x, \quad \ell \mu_h = n. \quad (436)$$

The mean n/ℓ need not be integral; continuity of the tilt mean permits exact centering of the sum at the integer n .

Lemma 103.5 at its center now yields

$$\mathbb{P}_h \left(\sum_{i=1}^{\ell} Y_{h,i} = n \right) \geq \frac{c_{c_0}}{w\sqrt{\ell}}. \quad (437)$$

To undo the tilt, let $a = \Psi'_m(0)$. On the fixed interval $[h_0, 0]$, lemma 103.4 gives $\Psi''_m \geq v_0 > 0$ uniformly for large m . Since $\Psi_m(0) = 0$,

$$0 \leq hx - \Psi_m(h) \leq h(x - a) - \frac{v_0 h^2}{2} \leq \frac{(x - a)^2}{2v_0} \leq C_{c_0} \left(\frac{\log m}{\ell} \right)^2.$$

The first inequality is convex duality at $\Psi'_m(h) = x$. Multiplying (437) by the inverse tilting factor gives

$$\mathbb{P} \left(\sum_{i=1}^{\ell} Y_i = n \right) \geq \frac{c_{c_0}}{w\sqrt{\ell}} \exp \left(-C_{c_0} \frac{(\log m)^2}{\ell} \right). \quad (438)$$

The coefficient of degree n in $(1 - K_{m,d}(s_{m,d}z))^{-1} = (1 - A(z) - B(z))^{-1}$ dominates the probability in (438), by (418). The injective block bound (394) and proposition 102.1 therefore prove the explicit two-sided estimate

$$\frac{c_{c_0} s_{m,d}^{-n}}{w\sqrt{n/m}} \exp \left(-C_{c_0} \frac{m(\log m)^2}{n} \right) \leq a_n^{(m)} \leq n r_m^{-n}. \quad (439)$$

Here constants absorb the bounded comparison of ℓ and n/m . On the stated range the extra negative logarithm is $O_{c_0}(\log m)$. The denominator contributes $O(\log n)$, and (396) implies $\log(s_{m,d}/r_m) = O((\log m)^2/m^2)$. Taking logarithms in (439), and using $\log C_n = n \log 4 - \frac{3}{2} \log n + O(1)$, proves (432). The rate sandwich in (396) proves (433).

Multiplication by m/n and the scalar expansion (397) prove (434); the term $(\log m)^2/m$ is absorbed in its first error. If $N = n/m$ and $N/\log m \rightarrow \infty$, then

$$\frac{m \log n}{n} = \frac{\log m + \log N}{N} \rightarrow 0,$$

proving corollary 105.2. On $n \geq c_0 m (\log m)^2$, the decreasing function $\log n/n$ has its maximum at the lower endpoint for large m , and the same error is $O_{c_0}(1/\log m)$.

[Merge note. Two details checked for this write-up. The lower bound $s_{m,d}^{-1} \leq \alpha_m < r_m^{-1}$ of (396) is Part II's (53) with $\beta_m = r_m^{-1}$ ((36)); and the remainder $t_m - (\frac{1}{2} \log m + \log \log m + \frac{1}{2} \log \pi)$ is 0.207, 0.196, 0.204, 0.213 at $m = 64, 256, 1024, 4096$, the size of Part II's first correction $\mathcal{P}_1(z)/\Lambda$ ((41)), so the convergence in corollary 105.2 is very slow. Manuscript 36 writes L for n/m here; it is printed N .]

105.2 An elementary two-term alternative

There is also a coefficient-composition proof on the same n range which avoids Fourier analysis, at the cost of losing the constant term.

Proposition 105.3 (manuscript 36). *For every fixed $c_0 > 0$, uniformly on $n \geq c_0 m \log m$ as $m \rightarrow \infty$,*

$$\log p_n(m) = -\frac{n}{2m} \log m - \frac{n}{m} \log \log m + O_{c_0} \left(\frac{n}{m} + \log n \right).$$

Proof. The endpoint calculation in lemma 103.4 at $h = 0$ shows that every integer $k \in [M - Tw, M]$ has $p_k \geq c_T/w$, for each fixed $T > 0$ and all sufficiently large m . Choose a large $T = T(c_0)$ and an integer interval I of length $N_I \asymp Tw$ ending at M . If c_I is its midpoint, choose j nearest to n/c_I . Then $j \asymp n/m$ and

$$|n/j - c_I| \leq Cm^2/n \leq C_{c_0}w.$$

Taking T sufficiently large ensures that $c = \lfloor n/j \rfloor$ is in the middle half of I . Let $R = n - jc$, so $0 \leq R < j$. For a sufficiently small fixed fraction K of N_I , choose integer offsets $z_i \in [-K, K]$ with $\sum_{i=1}^j z_i = 0$. Then

$$k_i = c + z_i + \mathbf{1}_{\{i \leq R\}} \quad (1 \leq i \leq j)$$

are valid jumps in I , sum exactly to n , and determine the offsets uniquely.

The convolution of symmetric unimodal integer sequences remains symmetric unimodal. This follows by expressing each sequence as a nonnegative combination of centered interval indicators and convolving; the convolution of two such indicators is symmetric unimodal. Hence the central coefficient of the j -fold uniform sum on $[-K, K]$ is maximal and at least $(2K + 1)^j / (2jK + 1)$. The tilted renewal coefficient is consequently at least

$$\frac{(2K + 1)^j (c_T/w)^j}{2jK + 1} \geq \frac{(c'_T)^j}{2jK + 1}.$$

Its logarithm is bounded below by $-C_{c_0}n/m - C \log n$. Combine the lower renewal bound with (401), (396), and $t_m = \frac{1}{2} \log m + \log \log m + O(1)$. $\square$

106 One saddle sum for the whole growing range

Source: manuscript 24; the compact- $n/(m \log m)$ case is manuscript 27's (section 109.1). Cutoff: a small fixed εm , as in lemma 103.1. Part II asks for joint growth of m and n and warns that fixed-cap spectral asymptotics cannot simply be substituted along a moving cap. The results of manuscripts 33, 30 and 27 cover separate scales; the following formula includes these scales and the band between $m \log m$ and $m(\log m)^2$.

Fix $c_0 > 0$. A fixed compact interval $\mathcal{T} = [h_-, h_+] \Subset (-1, \infty)$, with $h_- < 0 < h_+$, and a sufficiently small fixed $\varepsilon > 0$, both depending only on c_0 , are chosen below. With the split (403) at εm , put

$$\Psi_m(h) = \log Q(e^{h/w}) - \frac{hm}{w}, \quad \mu_h = m + w\Psi'_m(h), \quad \sigma_h^2 = w^2\Psi''_m(h), \quad D(h) = (1 - A(e^{h/w}))^{-2}. \quad (440)$$

Here Q is a probability generating function; its h/w exponential tilt has mean μ_h and variance σ_h^2 . For $v = n - 1$, define

$$\begin{aligned} \mathcal{I}_m(v) &= \{\ell \in \mathbb{Z}_{\geq 1} : \mu_{h_-} \leq v/\ell \leq \mu_{h_+}\}, \\ x_\ell &= (v/\ell - m)/w, \quad \Psi'_m(h_\ell) = x_\ell, \\ \Phi_v(\ell) &= \ell\{h_\ell x_\ell - \Psi_m(h_\ell)\}. \end{aligned} \quad (441)$$

Every retained saddle h_ℓ exists uniquely. All parameters in these definitions are exact finite- m quantities. (Φ_v is an action, not Part I's shift Φ .)

Theorem 106.1 (Uniform positive saddle sum; manuscript 24). *The constants $\mathcal{T}, \varepsilon$ can be chosen so that, uniformly for all integers $n \geq c_0 m \log m$ as $m \rightarrow \infty$,*

$$\mathfrak{E}_m(n) \sim r_m^{1-n} \mathfrak{S}_m(n-1), \quad \mathfrak{S}_m(v) = \sum_{\ell \in I_m(v)} \frac{(\ell+1)D(h_\ell)e^{-\Phi_v(\ell)}}{\sigma_{h_\ell} \sqrt{2\pi\ell}}. \quad (442)$$

There is no upper restriction on n in this scalar statement. Along every sequence satisfying

$$n \geq c_0 m \log m, \quad \frac{n(\log m)^2}{m^2} \longrightarrow 0, \quad (443)$$

one further has

$$a_n^{(m)} \sim \mathfrak{E}_m(n), \quad p_n(m) \sim \sqrt{\pi} n^{3/2} r_m e^{-nt_m/m} \mathfrak{S}_m(n-1). \quad (444)$$

The number of summands in (442) may grow. Uniform relative error for each positive summand permits this growth. Different sufficiently generous fixed choices of $\mathcal{T}$ and sufficiently small ε give equivalent expressions, since they approximate the same exact coefficient. No truncated root expansion is substituted into the exponential. The scalar inputs are $t_m = \frac{1}{2} \log m + \log \log m + \frac{1}{2} \log \pi + o(1)$ (Part II's (62)) and $w \sim 2m/\log m$, and the analytic input is lemma 103.1.

106.1 Relative control of every omitted count

Write $\mu = \mu_0$ and $L = v/\mu$. Under the lower bound on n , one has $L \geq c_* \log m$ and $t_m/L \leq C(c_0)$.

A competitor chosen before the tail interval. Let $\ell_0 = \lceil L \rceil$. Its centered mean displacement satisfies

$$0 \leq \Psi'_m(0) - \frac{v/\ell_0 - m}{w} \leq \frac{\mu}{w\ell_0} \leq C(c_0).$$

First choose a negative interval $\mathcal{T}_c = [h_c, 0] \Subset (-1, \infty)$ from the limiting function $-\log(1+h)$, with strict margins sufficient to contain these displacements. Choose all constants in the following quadratic bound from this limiting interval, also with strict margins. They then hold for every sufficiently small final cutoff by lemma 103.1; the choice is independent of the larger interval used for the tails. The Legendre transform has second derivative $1/\Psi''_m$, bounded on this interval. Thus the exact centered saddle of ℓ_0 satisfies

$$\Phi_v(\ell_0) \leq C_c \frac{t_m^2}{L}. \quad (445)$$

The centered local limit, including the residual convolution below, provides a positive coefficient contribution at least

$$c \frac{\sqrt{L}}{w} \exp\left(-C_c \frac{t_m^2}{L}\right). \quad (446)$$

This is a local calculation for one count, independent of any tail conclusion.

Since $t_m^2/L \leq C_* L$ uniformly in the theorem's range, choose a large fixed $D_* > 4C_c C_* + 20$. The limiting cost $i(h) = \log(1+h) - h/(1+h)$ tends to infinity as either $h \downarrow -1$ or $h \rightarrow \infty$. Choose $h_- < h_c$ and $h_+ > 0$ with $i(h_\pm) > 4D_*$, and finally choose ε sufficiently small for this $\mathcal{T}$, with a slight surrounding moment margin. For large m ,

$$h_\pm \Psi'_m(h_\pm) - \Psi_m(h_\pm) > 3D_*. \quad (447)$$

The exact decomposition and its tails. The squared-renewal identity is

$$\begin{aligned} \mathfrak{B}_m(v) &= [z^v](1 - K_m(r_m z))^{-2} = \sum_{\ell \geq 0} (\ell+1) \mathfrak{E}_\ell(v), \\ \mathfrak{E}_\ell(v) &= [z^v](1 - A(z))^{-2} Q(z)^\ell. \end{aligned} \quad (448)$$

For $h \in \mathcal{T}$, let W_h have generating function $((1 - A(e^{h/w}))(1 - A(e^{h/w}z)))^2$. It represents the two residual small runs and has a uniform exponential moment on scale w . Exact tilting gives

$$\mathfrak{E}_\ell(v) = D(h) \exp\{\ell \log Q(e^{h/w}) - hv/w\} \mathbb{P}_h(S_\ell + W_h = v). \quad (449)$$

Both $D(h)$ and its reciprocal are uniformly bounded.

The retained real interval is $[\ell_+, \ell_-]$, where $\ell_\pm = v/\mu_{h_\pm}$. Both endpoints are $(1+o(1))L$, uniformly for arbitrarily large v . At either endpoint the exponential factor in (449) is at most e^{-2D_*L} by (447). For $\ell < \ell_+$, use h_+ . The factor increases geometrically with ℓ , because $\log Q(e^{h_+/w}) = h_+ t_m + \Psi_m(h_+) \rightarrow +\infty$. For $\ell > \ell_-$, use h_- and obtain a decreasing geometric factor, whose logarithmic ratio tends to $-\infty$. The maximal-atom bound, after convolution with W_h , consequently gives

$$\sum_{\substack{\ell \geq 1 \\ \ell \notin \mathcal{I}_m(v)}} (\ell + 1) \mathfrak{E}_\ell(v) \leq C \frac{\sqrt{L}}{w} e^{-2D_*L}. \quad (450)$$

The geometric sums introduce no growing count factor. In the lower tail $\sqrt{\ell} \leq C\sqrt{L}$, and in the upper tail $\sum_{j \geq 0} \sqrt{\ell_- + j} e^{-ct_m j} \leq C\sqrt{L}$. The $\ell = 0$ term is at most $D(h_+)e^{-h_+v/w}$. Relative to (446), this also tends to zero uniformly, since v/w has order Lt_m and $\log w = O(L)$. Thus all omitted counts are relatively negligible. This includes $\ell m \leq v$, although individual clusters and residual runs can exceed m ; their overflow is controlled on the coefficient scale.

Summing the retained local limits. Every retained ℓ is asymptotic to L uniformly, and hence tends to infinity. At its exact centering tilt, $\ell \mu_{h_\ell} = v$. In (449), the shift $W_{h_\ell}/(\sigma_{h_\ell} \sqrt{\ell})$ tends to zero in probability uniformly. Averaging (407) therefore proves

$$\mathfrak{E}_\ell(v) = \frac{D(h_\ell) e^{-\Phi_v(\ell)}}{\sigma_{h_\ell} \sqrt{2\pi\ell}} (1 + o(1)) \quad (451)$$

with a uniformly relative error. Indeed the expected Gaussian factor tends to $\varphi(0) > 0$, and the local-limit error is absolute on that same scale. All terms are positive, so summation preserves the relative error regardless of the size of $\mathcal{I}_m(v)$. Together with (450) and $\mathfrak{E}_m(n) = r_m^{1-n} \mathfrak{B}_m(n-1)$, this proves the scalar part of theorem 106.1.

106.2 Transfer to the avoidance class: the endpoint-layer route

The comparison of the coefficients is direct, without a nearby-root or phase estimate. For $d = O(\log m)$, forbidding all scalar letters greater than $m - d$ changes $\mathfrak{E}_m(n)$ exactly to $\mathfrak{E}_{m-d}(n)$ (use the positive two-renewal-word interpretation of $(1 - K_m)^{-2}$; in the cluster decomposition only the ℓ big jumps can violate the restriction for large m). At any retained centered tilt, the endpoint atom bound gives $\mathbb{P}_h(X > m - d) \leq Cd/w$. After marking one such jump, the remaining $\ell - 1$ full clusters and all small runs have maximal atom $C/(w\sqrt{\ell})$. The unrestricted centered probability in (449) is at least $c/(w\sqrt{\ell})$ by (451). Hence

$$\mathbb{P}(\text{a forbidden jump} \mid \text{total } v, \text{ count } \ell) \leq C\ell d/w \leq \frac{Cnd}{mw}. \quad (452)$$

Tilting does not alter this conditional law, because the total size is fixed. The same bound survives the positive count sum; the already omitted tails are negligible. Under (443), it follows that

$$\mathfrak{E}_{m-d}(n)/\mathfrak{E}_m(n) \longrightarrow 1. \quad (453)$$

For an integer target shift $j = O(\log m)$, keep the original centering tilts in (449). The target moves by $o(w)$ and the untilting ratio is $e^{h_\ell j/w} = 1 + o(1)$. The local probabilities have ratio

$1 + o(1)$, uniformly. The omitted-tail estimates at $v - j$ use the same old cutoffs; their factors change only by $e^{h_{\pm}j/w} = 1 + o(1)$. Thus

$$\mathfrak{E}_m(n - j)/\mathfrak{E}_m(n) \sim r_m^j. \quad (454)$$

Apply (453) at both lengths when needed. Finally take $D_{\text{app}} = L_{\text{app}} = \lceil 3 \log_2 m \rceil$ in (429). The upper condition in (443) makes $n \varepsilon_{D_{\text{app}}} \rightarrow 0$ and $nr_m^{L_{\text{app}}} \rightarrow 0$. Equations (453)–(454) squeeze $a_n^{(m)}/\mathfrak{E}_m(n)$ to one. Catalan Stirling asymptotics give (444). The marked endpoint loss is $O(n(\log m)^2/m^2)$.

[Merge note. This is route 2 of proposition 104.2. Manuscript 27 proves the same two statements (its eq: shortcap, eq: layer, eq: shortshift) by the same argument at $n \asymp m \log m$, where its bound (452) reads $C \ell d/w = O((\log m)^3/m)$; it adds that a slight enlargement of the retained window handles the endpoints of the window, whose terms already have the strict tail margins. Those labels are replaced by these.]

107 Two exact actions in the intermediate band

Source: manuscript 24. Cutoff as in section 106. Set

$$b = B(1), \quad \mu = \mu_0, \quad \sigma = \sigma_0, \quad v = b\mu = \sum_{k=1}^m kp_k, \quad L = \frac{v}{\mu}, \quad H = \frac{\sigma}{\mu} \sqrt{L}. \quad (455)$$

The width H is distinct from the tilt interval $\mathcal{T}$. The basic scales are $b \rightarrow 1/2$, $\mu = m - w + o(w)$, and $\sigma \sim w$.

Theorem 107.1 (The intermediate two-action formula; manuscript 24). *If*

$$m \log m \ll n \ll m(\log m)^2, \quad (456)$$

let $\mathfrak{J}(L)$ be the set of distinct integers $\{\lfloor L \rfloor, \lceil L \rceil\}$. Then, uniformly along such sequences,

$$\mathfrak{E}_m(n) \sim a_n^{(m)} \sim r_m^{1-n} \frac{\sqrt{L}}{b^2 \sigma \sqrt{2\pi}} \sum_{j \in \mathfrak{J}(L)} e^{-\Phi_v(j)}. \quad (457)$$

Equivalently,

$$\frac{a_n^{(m)}}{(n/m^2)r_m^{-n}} \sim \frac{1}{H\sqrt{2\pi}} \sum_{j \in \mathfrak{J}(L)} e^{-\Phi_v(j)}. \quad (458)$$

At integral L the shared neighbor is counted once.

Proof. Here $L/t_m \rightarrow \infty$ and $H \rightarrow 0$. The centering tilts of the one or two neighbors tend uniformly to zero, since their scaled mean displacements are $O(t_m/L) = o(1)$. Extend Φ_v to the retained real interval. Differentiation gives

$$\Phi'_v(\ell) = -\log Q(e^{h_{\ell}/w}), \quad \Phi''_v(\ell) = \frac{v^2}{\ell^3 \sigma_{h_{\ell}}^2} > 0, \quad \Phi_v(L) = \Phi'_v(L) = 0. \quad (459)$$

Throughout that interval, Φ''_v is between positive constant multiples of H^{-2} .

Put $j_- = \lfloor L \rfloor$ and $j_+ = \lceil L \rceil$. For $\ell \leq j_- - 1$, integrating the curvature bound gives

$$\Phi_v(\ell) - \Phi_v(j_-) \geq \frac{c}{H^2} ((L - \ell)^2 - (L - j_-)^2) \geq \frac{c(j_- - \ell)^2}{H^2}.$$

The corresponding bound on $\ell \geq j_+ + 1$ is relative to j_+ . All retained prefactors are uniformly comparable. The omitted Gaussian lattice tails are therefore bounded by Ce^{-c/H^2} times the sum of the neighboring terms, and hence by $2Ce^{-c/H^2}$ times the larger neighboring term. This comparison is uniform in the fractional phase, including integral L .

The one or two neighboring prefactors in (442) are each asymptotic to $b^{-2}\sqrt{L}/(\sigma\sqrt{2\pi})$; other fixed-tilt prefactors are only uniformly comparable. The scalar tails were already excluded, proving (457). Condition (456) implies the actual-class upper bound. Dividing by $r_m^{1-n}v/v^2$ gives exactly the factor $1/(H\sqrt{2\pi})$; also $r_m m^2/v^2 \rightarrow 1$ and $v/n \rightarrow 1$. This proves (458). $\square$

The exact actions determine dominance and transitions. Their higher Taylor terms can remain relevant even though the centering tilts tend to zero. The next section identifies when the quadratic approximation is relatively valid.

108 Gaussian periodization, the theta boundary and the cubic boundary

Sources: manuscript 24 (the base statement), manuscript 30 (the theta-modulated equivalent above $m(\log m)^2$, its lattice-sum proof and its boundary limits) and manuscript 33 (the unmodulated equivalent). Cutoff: the cluster parameters μ, σ and hence L_m, H_m of (405) are computed with one fixed cutoff, the small εm of section 106 in manuscript 24's statements and $m/2$ in manuscripts 30 and 33's; each statement holds with its source's cutoff (remark 108.4).

For $H > 0$, let

$$\vartheta(H, u) = \frac{1}{H} \sum_{j \in \mathbb{Z}} \varphi\left(\frac{j - u}{H}\right) = \sum_{j \in \mathbb{Z}} e^{-2\pi^2 H^2 j^2} e^{2\pi i j u}, \quad \varphi(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}. \quad (460)$$

This is the classical periodic Gaussian, equivalently a Jacobi theta function [9]. It is positive and has period one in u .

Lemma 108.1 (Bounds on the periodic Gaussian; manuscript 30). *For every $h_0 > 0$, $\vartheta(H, u)$ is bounded above and bounded away from zero uniformly for $H \geq h_0$ and real u . Moreover*

$$\sup_{H \geq h_0, u \in \mathbb{R}} \{|\partial_u \vartheta(H, u)| + |H \partial_H \vartheta(H, u)|\} < \infty, \quad (461)$$

and $\vartheta(H, u) \rightarrow 1$ uniformly in u as $H \rightarrow \infty$.

Proof. Computing the Fourier coefficients of the first series in (460) gives $e^{-2\pi^2 H^2 j^2}$, proving the second representation. Absolute Gaussian convergence justifies every interchange and differentiation. That representation gives the upper and derivative bounds and the uniform limit. For a lower bound use positivity and compactness on $[h_0, H_0] \times (\mathbb{R}/\mathbb{Z})$, then the uniform limit on $H \geq H_0$. $\square$

Theorem 108.2 (Gaussian periodization, its range, and the cubic boundary; manuscripts 24, 30 and 33).

(i) (manuscript 24) *If $n/[m(\log m)^{3/2}] \rightarrow \infty$, then*

$$\mathfrak{E}_m(n) \sim \frac{r_m n}{v_m^2} r_m^{-n} \vartheta(H_m(n), L_m(n)). \quad (462)$$

(manuscript 30, cutoff $m/2$) *In particular, for every fixed $c_0 > 0$, (462) holds uniformly over integers $n \geq c_0 m(\log m)^2$ as $m \rightarrow \infty$; there is no upper restriction on n in this statement.*

(ii) (manuscripts 24 and 30) Along every integer sequence $m \rightarrow \infty$ satisfying, for a fixed $c_0 > 0$,

$$n \geq c_0 m (\log m)^2, \quad \frac{n (\log m)^2}{m^2} \rightarrow 0, \quad (463)$$

and more generally whenever $n/[m(\log m)^{3/2}] \rightarrow \infty$ and $n(\log m)^2/m^2 \rightarrow 0$, one has

$$a_n^{(m)} \sim \mathfrak{E}_m(n) \sim \frac{n}{m^2} r_m^{-n} \mathfrak{D}(H_m(n), L_m(n)), \quad (464)$$

$$p_n(m) \sim \frac{\sqrt{\pi} n^{5/2}}{m^2} e^{-nt_m/m} \mathfrak{D}(H_m(n), L_m(n)). \quad (465)$$

(iii) (manuscript 33, cutoff $m/2$) Along every integer sequence $m \rightarrow \infty$ satisfying

$$\frac{n}{m(\log m)^2} \rightarrow \infty, \quad \frac{n(\log m)^2}{m^2} \rightarrow 0, \quad (466)$$

one has

$$a_n^{(m)} \sim \mathfrak{E}_m(n) \sim \frac{r_m n}{v_m^2} r_m^{-n} \sim \frac{n}{m^2} r_m^{-n}, \quad (467)$$

$$p_n(m) \sim \frac{\sqrt{\pi} n^{5/2}}{m^2} \exp(-nt_m/m) \sim \frac{\sqrt{\pi} n^{5/2}}{m^2} \left(\frac{\alpha_m}{4}\right)^n. \quad (468)$$

The scalar envelope equivalent $\mathfrak{E}_m(n) \sim r_m n v_m^{-2} r_m^{-n}$ requires only the first condition in (466).

(iv) (manuscript 24) At $n/[m(\log m)^{3/2}] \rightarrow c \in (0, \infty)$, let $L = L_m(n-1)$, $H = H_m(n-1)$ and $\delta_j = j - L$. In (457), the neighboring sum may instead be written

$$\sum_{j \in \mathfrak{S}(L)} \exp\left(-\frac{\delta_j^2}{2H^2} + \frac{\delta_j^3}{24c^2}\right) \quad (469)$$

with relative error $o(1)$. There are integer sequences at this scale for which the ratio of the exact coefficient to the Gaussian expression (462) tends to

$$\cosh\left(\frac{1}{192c^2}\right) > 1. \quad (470)$$

Thus the lower scale $m(\log m)^{3/2}$ is a genuine obstruction to a uniform unmodified Gaussian equivalent.

Both theta arguments use the exact parameters (404). An asymptotic replacement for μ can lose the phase modulo one. Likewise the complete scalar t_m is retained in the exponential: multiplying the displayed error of (397) by n/m would not yield an $o(1)$ exponent throughout the band (466), so the expansion is used only to control scales (manuscript 33). The factor $n^{5/2}/m^2$ has been obtained from uniform coefficient analysis and a retained class, not from a fixed- m Perron approximation. In (iii), the second form follows from the spectral corridor $0 \leq \log(r_m^{-1}/\alpha_m) = O((\log m)^2/m^2)$ ((37)), whose product with n tends to zero under (466). Manuscript 33 does not assert a fixed- m prefactor or a relative equivalent throughout $m = o(n)$. Part (iii) is the case $H \rightarrow \infty$ of (ii), because $\mathfrak{D} \rightarrow 1$ uniformly (lemma 108.1); and the stricter lower condition of (iii) has mathematical content, by corollary 108.3 below (manuscript 30).

Corollary 108.3 (Every phase occurs; manuscript 30). Suppose $n/[m(\log m)^2] \rightarrow c \in (0, \infty)$ and $L_m(n) \rightarrow u$ modulo one, with the cutoff $m/2$. Then

$$\frac{a_n^{(m)}}{(n/m^2)r_m^{-n}} \rightarrow \mathfrak{D}(2\sqrt{c}, u) = \sum_{j \in \mathbb{Z}} e^{-8\pi^2 c j^2} e^{2\pi i j u}. \quad (471)$$

The same limiting factor is obtained by normalizing $p_n(m)$ by $\sqrt{\pi} n^{5/2} m^{-2} e^{-nt_m/m}$. Every $u \in \mathbb{R}/\mathbb{Z}$ is attained by an integer sequence.

Proof. The scales (410) give $H_m(n) \rightarrow 2\sqrt{c}$, while the upper condition in (463) holds automatically. Apply part (ii) of theorem 108.2 and continuity of ϑ . For a prescribed phase, take n_m to be a nearest integer to

$$\mu \left\{ \left\lfloor \frac{cm(\log m)^2}{\mu} \right\rfloor + u \right\}, \quad 0 \leq u < 1.$$

Its change from $cm(\log m)^2$ is $O(m)$, so the normalized length tends to c . Also n_m/μ has the required phase with error $O(1/\mu)$. $\square$

For any fixed $c > 0$, the limiting amplitudes at phases 0 and $1/2$ differ by

$$4 \sum_{\substack{j \geq 1 \\ j \text{ odd}}} e^{-8\pi^2 c j^2} > 0. \quad (472)$$

In particular $\vartheta(2\sqrt{c}, 0) > 1$. Thus the unmodulated equivalent fails at the order $m(\log m)^2$: its normalized ratio has distinct subsequential limits. If instead $n/[m(\log m)^2] \rightarrow \infty$, then $H_m(n) \rightarrow \infty$ and $\vartheta \rightarrow 1$ uniformly in phase, recovering the relative equivalent (iii). The same boundary limit follows from (i)–(ii) with manuscript 24's cutoff, up to the phase convention of remark 108.4; manuscript 24 does not state it.

Remark 108.4 (Merge note: the cutoff fixes the phase, and the effects are small). Neither manuscript says so, but the phase $L_m(n) = n/\mu$ depends on the cluster cutoff: $v_m = b\mu$ does not, but μ and σ do. Manuscript 24 computes them at a small εm , manuscript 30 at $m/2$. Both theta formulas are equivalents of the same $\mathfrak{E}_m(n)$, so they agree asymptotically (the phase difference is $O(cm^{-1/4}(\log m)^{3/2})$ at $n = cm(\log m)^2$), and each theorem is true with its own cutoff; but at feasible sizes the phase is not canonical. A ProveIt check made for this write-up found $L_{m/2}(n) - L_{m/16}(n) = -11.4c, -9.2c, -6.7c, -4.7c$ at $n = cm(\log m)^2$ and $m = 10^3, 10^4, 10^5, 10^6$. Manuscript 24's own remark, “each formula evaluates the same scalar coefficient”, is printed in section 111. The effects themselves are small: $\vartheta(2\sqrt{c}, 0) - 1 \approx 2e^{-8\pi^2 c}$ is $0.995, 0.039, 7.4 \cdot 10^{-4}, 5.4 \cdot 10^{-9}$ and 10^{-34} at $c = 0.01, 0.05, 0.1, 0.25$ and 1 , so the modulation is numerically visible only for $c \lesssim 0.1$; and $\cosh(1/(192c^2)) - 1 \approx 1.4 \cdot 10^{-5}c^{-4}$. The convergence of the constants is slow: at the cutoff $m/16$ the ratio κ_3/w^3 of (477) is $-4.2, -5.8, -5.8, -5.2, -4.5$ at $m = 10^3, \dots, 10^7$, and at the cutoff $m/2$ it is $+2.6, +1.6, +0.7, -0.1, -0.8$, against the limit -2 ; $\Phi_v'''(L)$ is 68–88% of its limit $-1/(4c^2)$ for $m \leq 16384$. At finite m the exact action (457) should be used rather than the limiting cubic weight. None of this contradicts the theorem, whose statements are limits.

108.1 Width bounded below: retaining the discrete cluster-count sum

Source: manuscript 30 (its proof of part (i) above $m(\log m)^2$); cutoff $m/2$, with lemmas 103.2 and 103.3 as inputs. Manuscript 24 cites this proof for H bounded below and says it applies to its fixed cutoff as well. Define

$$\mathfrak{T}_m(v) = \sum_{\ell \geq 0} (\ell + 1) \mathbb{P}(S_\ell = v),$$

with $S_0 = 0$ and $S_\ell = Y_1 + \dots + Y_\ell$. Fix $c_0 > 0$ and suppose $v \geq c_0 m(\log m)^2$. Write $L = L_m(v)$ and $H = H_m(v)$. By (410), $L \geq c(\log m)^2$ and $H \geq h_0(c_0) > 0$ for large m . For fixed $A > 0$ and $|\ell - L| \leq AH$, the local limit gives

$$\mathbb{P}(S_\ell = v) = \frac{\varphi((\ell - L)/H) + o(1)}{\sigma\sqrt{L}}$$

uniformly. Here $\ell/L \rightarrow 1$ because $H/L = O(1/(\sqrt{L} \log m))$. After multiplication by $\ell + 1 \sim L$ and division by $v/\mu^2 = L/\mu$, the window sum is

$$\frac{1}{H} \sum_{|\ell - L| \leq AH} \varphi((\ell - L)/H) + o(1). \quad (473)$$

There are $O(AH + 1)$ summands, so the accumulated local-limit error is at most $o(1)(A + 1/h_0)$. No assumption that H diverges is required.

We give the tail details to justify using the whole Gaussian sum. Set $\gamma = \mu/w \sim (\log m)/2$. On $L/2 \leq \ell \leq 2L$, the exponent in (412) is at most $-c \min\{\gamma^2(\ell - L)^2/L, \gamma|\ell - L|\}$. Beyond AH and up to L/γ , its normalized sum has a Gaussian tail that tends to zero as $A \rightarrow \infty$, uniformly for $H \geq h_0$. Beyond L/γ , the geometric tail starts at e^{-cL} and is negligible. For $\ell < L/2$, the normalized sum is at most $C\gamma\sqrt{L}e^{-c\gamma L}$. For $\ell > 2L$, the bound is a weighted geometric sum with exponent at most $-c\gamma\ell$ and is negligible too. The tails of the Gaussian sum in (473) also vanish uniformly for $H \geq h_0$: one may compare its terms beyond AH with the Gaussian integral, allowing one extra lattice interval of length at most $1/h_0$ in the rescaled variable. Taking the joint limit with A fixed and then $A \rightarrow \infty$ proves

$$\mathfrak{T}_m(v) = \frac{v}{\mu^2} \{\vartheta(H_m(v), L_m(v)) + o(1)\}, \quad (474)$$

uniformly for $v \geq c_0 m(\log m)^2$.

The two residual small runs do not change the limiting phase. Let W have generating function $[b/(1 - A(z))]^2$. Then $\mathbb{E}W = O(w)$, $\mathbb{E}e^{\theta W/w} \leq C$ for some fixed $\theta > 0$, and

$$[z^v](1 - K_m(r_m z))^{-2} = b^{-2} \mathbb{E} \mathfrak{T}_m(v - W). \quad (475)$$

On $W \leq v/2$, use (474) with lower constant $c_0/2$. The phase and logarithmic-width changes satisfy

$$|L_m(v - W) - L_m(v)| = \frac{W}{\mu}, \quad \left| \log \frac{H_m(v - W)}{H_m(v)} \right| \leq \frac{W}{v}.$$

Lemma 108.1 bounds their expected effect by $O(w/\mu + w/v) = o(1)$. Also $\mathbb{E}[(v - W)\mathbf{1}_{\{W \leq v/2\}}] \sim v$. For $W > v/2$, use its exponential tail and $\mathfrak{T}_m(u) \leq C(1 + (u/m)^2)$ for $0 \leq u \leq v$, which follows from $Y > m/2$. After division by v/m^2 , this contribution still tends uniformly to zero because $v/m \geq c_0(\log m)^2$ and the exponential is $e^{-c(v/m)\log m}$. Since $v_m = b\mu$, equation (475) becomes

$$[z^v](1 - K_m(r_m z))^{-2} = \frac{v}{v_m^2} \{\vartheta(H_m(v), L_m(v)) + o(1)\}. \quad (476)$$

Now use $v = n - 1$ in $\mathfrak{E}_m(n) = r_m^{1-n}[z^{n-1}](1 - K_m(r_m z))^{-2}$. Replacing $n - 1$ by n changes the phase by $1/\mu$ and the log width by $O(1/n)$, hence changes ϑ by $o(1)$. Its positive lower bound turns this additive expansion into the relative equivalent (462) for $n \geq c_0 m(\log m)^2$.

[Merge note. Manuscript 24's account of this case, for its fixed cutoff, is a summary of the same steps: central counts $|\ell - L| \leq AH$ give a normalized Gaussian sum with error $o(1)(A + H^{-1})$; compact two-sided tilts and (408) give the summable pointwise bound (412), which controls all tails as $A \rightarrow \infty$; the two residual runs have an exponential moment on scale w and an expected phase shift $O(w/\mu) = o(1)$; and the Fourier series in (460) gives bounded derivatives and a positive lower bound for ϑ when H is bounded below, so the additive approximation is relative. Manuscript 24 is therefore not self-contained for H bounded below; this subsection is the full proof of that regime.]

108.2 Vanishing width: Taylor expansion of the action

Source: manuscript 24. Cutoff εm . Let $\kappa_{3,h}$ and $\kappa_{4,h}$ be the third and fourth cumulants under tilt h/w , and write $\kappa_3 = \kappa_{3,0}$. The limiting centered fluctuation is the negative of a centered exponential, so lemma 103.1 gives

$$\frac{\kappa_3}{w^3} \rightarrow -2, \quad \frac{\sigma}{w} \rightarrow 1, \quad \frac{\mu}{w} \sim \frac{1}{2} \log m. \quad (477)$$

In particular $\Psi'''_m = \kappa_{3,h}/w^3$. Since $h'_\ell = -vw/(\ell^2\sigma_h^2)$ and $d\sigma_h^2/dh = \kappa_{3,h}/w$, differentiating (459) gives exactly

$$\Phi'''_v(L) = -\frac{3\mu^2}{L^2\sigma^2} + \frac{\mu^3\kappa_3}{L^2\sigma^6}. \quad (478)$$

One more differentiation gives, with $h = h_\ell$,

$$\Phi''''_v(\ell) = \frac{12v^2}{\ell^5\sigma_h^2} - \frac{8v^3\kappa_{3,h}}{\ell^6\sigma_h^6} - \frac{v^4\kappa_{4,h}}{\ell^7\sigma_h^8} + \frac{3v^4\kappa_{3,h}^2}{\ell^7\sigma_h^{10}}. \quad (479)$$

For $|\ell - L| \leq C$ and $L/\log m \rightarrow \infty$, these tilts tend uniformly to zero. Their cumulants obey $\kappa_{j,h} = O(w^j)$ and $\sigma_h \asymp w$, so

$$\sup_{|\ell-L| \leq C} |\Phi''''_v(\ell)| = O\left(\frac{(\log m)^4}{L^3}\right). \quad (480)$$

Suppose $H \rightarrow 0$ and $L/(\log m)^{3/2} \rightarrow \infty$. For each relevant neighbor, the cubic Taylor term is $O((\log m)^3/L^2) = o(1)$, and (480) makes the fourth-order remainder $o(1)$. Thus the action can be replaced relatively by $\delta_j^2/(2H^2)$. All other Gaussian lattice terms are relatively negligible by the same quadratic curvature comparison used in the proof of theorem 107.1. Every sequence has subsequences with $H \rightarrow 0$ or H bounded below, which with section 108.1 proves the full range of part (i). Replacing $v = n - 1$ by n in the two Gaussian neighbors changes their logarithms by $O((\log m)^2/(mL)) = o(1)$ when $H \rightarrow 0$. This can also be seen from the entire Gaussian lattice sum, so a change in the identity of a nearest neighbor causes no problem. For H bounded below, use the bounded theta derivatives. This proves (462); theorem 106.1 gives the actual-class transfer of part (ii) by the endpoint-layer route (section 106.2), and section 108.4 gives it by the root-shift route.

108.3 Diverging width: the window integral

Source: manuscript 33, a second route to part (iii); cutoff $m/2$.

Proposition 108.5 (The squared-renewal equivalent; manuscript 33). *Uniformly along integer sequences with $m \rightarrow \infty$ and $v/[m(\log m)^2] \rightarrow \infty$,*

$$\mathfrak{I}_m(v) \sim v/\mu^2, \quad [z^v](1 - K_m(r_m z))^{-2} \sim v/v_m^2. \quad (481)$$

Consequently the scalar envelope equivalent of part (iii) of theorem 108.2 holds under its stated one-sided condition.

Proof. Put $L = v/\mu$ and $H = (\sigma/\mu)\sqrt{L}$. Then H is comparable to $\sqrt{L}/\log m$ and tends to infinity. For fixed $A > 0$ and $|\ell - L| \leq AH$, the local limit (411) gives uniformly

$$\mathbb{P}(Y_1 + \cdots + Y_\ell = v) = \frac{\varphi(-(\ell - L)/H) + o(1)}{\sigma\sqrt{L}}.$$

Since $\ell + 1 \sim L$, the sum over that window, divided by v/μ^2 , tends to $\int_{-A}^A \varphi(u) du$. The mesh is $1/H$, and the normalization is $LH/(\sigma\sqrt{L}) = L/\mu = v/\mu^2$.

For completeness, the remaining cluster counts are uniformly summable. Set $\gamma = \mu/w \sim (\log m)/2$. On $L/2 \leq \ell \leq 2L$, the exponent in (412) is at most $-c \min\{\gamma^2(\ell - L)^2/L, \gamma|\ell - L|\}$. For $AH < |\ell - L| \leq L/\gamma$, this gives a Gaussian tail after summation, with normalized bound $C \int_{cA}^\infty e^{-cu^2} du + o(1)$. For $|\ell - L| > L/\gamma$, the geometric tail starts at e^{-cL} and is negligible. When $\ell < L/2$, the entire contribution is at most $CL^{3/2}w^{-1}e^{-c\gamma L}$; after division by L/μ this is at most

$C\gamma\sqrt{L}e^{-c\gamma L} \rightarrow 0$. For $\ell > 2L$, the exponent is at most $-c\gamma\ell$, and the weighted geometric sum is negligible too. First let m, v tend to infinity with A fixed, and then let A increase. This proves the first equivalent in (481).

The exact factorization

$$(1 - A - B)^{-2} = (1 - A)^{-2}(1 - Q)^{-2} \quad (482)$$

handles the residual small runs. Let W have probability generating function $[b/(1 - A(z))]^2$, the sum of two independent small runs. Lemma 103.2 gives $\mathbb{E}W = O(w)$ and $\mathbb{E}e^{\theta W/w} \leq C$ for some $\theta > 0$. Thus $[z^v](1 - A - B)^{-2} = b^{-2}\mathbb{E}\mathfrak{T}_m(v - W)$, where $\mathfrak{T}_m$ is zero at negative arguments. On $W \leq v/2$, the first equivalent is uniform at every argument $v - W$, and $\mathbb{E}[(v - W)\mathbf{1}_{\{W \leq v/2\}}] \sim v$. For the complement, note $Y > m/2$, so the elementary bound $\mathfrak{T}_m(u) \leq C(1 + (u/m)^2)$ holds for $0 \leq u \leq v$. Its contribution is at most $C(1 + (v/m)^2)e^{-cv/w}$. Dividing by v/m^2 still gives a quantity tending to zero: with $L \asymp v/m$ the exponential is $e^{-cL \log m}$, which dominates the displayed polynomial factor when $L/(\log m)^2 \rightarrow \infty$. Finally $v_m = b\mu$, proving the second equivalent.

The identity $\mathfrak{G}_m(n) = r_m^{1-n}[z^{n-1}](1 - K_m(r_m z))^{-2}$ now gives $\mathfrak{G}_m(n) \sim r_m n v_m^{-2} r_m^{-n}$. $\square$

[Merge note. Section 108.1 is this argument with the discrete Gaussian sum retained instead of summed to one; manuscript 30 says its result recovers manuscript 33's when $H \rightarrow \infty$. A ProvelIt check made for this write-up confirmed the complement estimate: $C(m^2/v + v)e^{-cv/w} \rightarrow 0$ already when $v/m \rightarrow \infty$, so manuscript 33's stated condition is sufficient.]

108.4 Cap shortening by root shift and phase stability

Sources: manuscripts 33 and 30; route 1 of proposition 104.2. Cutoff $m/2$. We need a quantitative comparison of the scalar roots at $M = m - d$ and m , $d = O(\log m)$. The endpoint calculation in lemma 103.2 gives $p_k \leq C \log m/m$ for $m - d < k \leq m$. Consequently $1 - K_M(r_m) = \sum_{k=M+1}^m p_k = O(d \log m/m)$. Since $d/w \rightarrow 0$, the retained layer $[m - w, m - d]$ has probability mass bounded below by a positive constant. Hence $K'_M(r_m) \geq cm$. The derivative is increasing on positive arguments, so the mean value theorem gives

$$0 \leq r_M - r_m \leq C \frac{d \log m}{m^2} = O((\log m)^2/m^2). \quad (483)$$

Moreover $v_M/v_m \rightarrow 1$, by $v_j \sim j/2$ and $M/m \rightarrow 1$. (Manuscript 30 states the same bound as $0 \leq r_M - r_m = O(d \log m/m^2)$, with the removed endpoint mass $O(d \log m/m)$ and a retained layer of width comparable to w giving $K'_M(r_m) \geq cm$; that display, its eq:root shift, is replaced by (483). Both re-run Part II's corridor proof, remark 101.1.)

Completion of part (iii) (manuscript 33). With (430), apply proposition 108.5 to (m, n) , (M, n) , and $(M, n - L_{\text{app}})$; all satisfy its lower growth condition. Equation (483) and the upper condition in (466) give

$$\frac{\mathfrak{G}_M(n)}{\mathfrak{G}_m(n)} \rightarrow 1, \quad \frac{\mathfrak{G}_M(n - L_{\text{app}})}{\mathfrak{G}_M(n)} \sim r_M^{L_{\text{app}}} \frac{n - L_{\text{app}}}{n}. \quad (484)$$

Here $r_M^{L_{\text{app}}} = 4^{-L_{\text{app}}} \exp(O(L_{\text{app}} \log m/m)) = O(m^{-6})$. Thus $n\mathfrak{G}_M(n - L_{\text{app}})/\mathfrak{G}_M(n) \rightarrow 0$, and $n\varepsilon_{D_{\text{app}}} \rightarrow 0$ under (466). Lemma 104.1 and the exact upper bound $a_n^{(m)} \leq \mathfrak{G}_m(n)$ prove relative sharpness. The scalar equivalent and $v_m \sim m/2$, $r_m \rightarrow 1/4$ prove (467). Stirling's formula $C_n \sim 4^n/(\sqrt{\pi} n^{3/2})$ then proves (468). In the band, the lower condition makes the cluster-count lattice window diverge, while the upper condition makes the nearby-root replacement harmless.

Phase stability (manuscript 30). The extra issue in transferring the theta-modulated envelope to actual avoiders is stability of its phase.

Lemma 108.6 (Exact-law and moment stability; manuscript 30). For $M = m - d$ as above, with $\mu_M, \sigma_M, L_M, H_M$ the parameters of the cap M (cutoff $M/2$),

$$|\mu_M - \mu_m| = O(d \log m), \quad (485)$$

$$|\sigma_M^2 - \sigma_m^2| = O(md \log m), \quad (486)$$

$$|L_M(n) - L_m(n)| = O(nd \log m / m^2), \quad (487)$$

$$\left| \log \frac{H_M(n)}{H_m(n)} \right| = O(d(\log m)^3 / m). \quad (488)$$

The last two estimates hold at every common positive argument n .

Proof. Write $p_{m,k} = C_{k-1} r_m^k$, and let $T = \sum_{k=M+1}^m p_{m,k}$ be the removed endpoint mass, $O(d \log m / m)$. Extend the cap- M tilted law by zero above M . On the common support, $p_{M,k} = p_{m,k} (r_M / r_m)^k \geq p_{m,k}$. Both laws sum to one, so their ℓ^1 distance is exactly $2T$. Write $\epsilon = d \log m / m$. Their raw moments of degree $j = 1, 2$ therefore differ by $O(m^j \epsilon)$.

The splitting thresholds $M/2$ and $m/2$ differ by $O(d)$ lattice points. Each probability on this middle layer is $O(\log m / m)$: the ideal weights are eventually increasing for $k \geq m/3$ and their endpoint maximum has that bound. The same holds for the cap- M law, or follows from its pointwise root reweighting. Thus the small and big masses differ by $O(\epsilon)$, and their degree- j moment sums by $O(m^j \epsilon)$. The masses b_m, b_M stay bounded away from zero. The finite formulas (404) now imply (485)–(486).

Since $\sigma_m^2 \sim 4m^2 / (\log m)^2$ and $\mu_m \sim m$,

$$\left| \log \frac{\sigma_M}{\sigma_m} \right| = O(d(\log m)^3 / m), \quad \left| \log \frac{\mu_M}{\mu_m} \right| = O(d \log m / m).$$

Using $L_m(n) = n / \mu_m$ gives (487); using $H_m(n) = \sigma_m \sqrt{n} / \mu_m^{3/2}$ gives (488). $\square$

Under (463), with $d = O(\log m)$, both phase changes in lemma 108.6 tend to zero. The derivative bound in the logarithmic width variable is uniform even as H tends to infinity. Thus

$$\frac{\vartheta(H_M(n), L_M(n))}{\vartheta(H_m(n), L_m(n))} \longrightarrow 1. \quad (489)$$

Part (i), (483) and (489) imply $\mathfrak{E}_M(n) / \mathfrak{E}_m(n) \rightarrow 1$. At a fixed cap M , replacing n by $n - L_{\text{app}}$ changes the phase by $O(\log m / m)$ and log width by $O(\log m / n)$, so its theta ratio also tends to one. Therefore

$$\frac{\mathfrak{E}_M(n - L_{\text{app}})}{\mathfrak{E}_M(n)} \sim r_M^{L_{\text{app}}} \frac{n - L_{\text{app}}}{n}.$$

Now $r_M^{L_{\text{app}}} = O(m^{-6})$ and $n \varepsilon_{D_{\text{app}}} \rightarrow 0$ under (463). The squeeze (429) proves $a_n^{(m)} \sim \mathfrak{E}_m(n)$. Combining this with $v_m \sim m/2$, $r_m \rightarrow 1/4$, and Catalan Stirling asymptotics proves part (ii) on (463).

[Merge note. A ProveIt check made for this write-up found the bound (485) generous (the delivered stability rows show a mean shift of about d , e.g. -58.4 for $d = 66$ at $m = 2048$), and the root gap (483) bounded after scaling: $(r_M - r_m)m^2 / (\log m)^2 = 2.72, 1.97, 1.76$ at $m = 128, 512, 2048$.]

108.5 The cubic boundary

Source: manuscript 24; proof of part (iv). At the boundary $L / (\log m)^{3/2} \rightarrow c$, equation (478) tends to $-1/(4c^2)$. Its first term tends to zero and its second has this limit by (477). The fourth-order

remainder is $O((\log m)^{-1/2})$. Consequently, uniformly for bounded δ ,

$$\Phi_v(L + \delta) = \frac{\delta^2}{2H^2} - \frac{\delta^3}{24c^2} + o(1).$$

This proves (469).

Finally choose an integer k_m nearest $c(\log m)^{3/2}$, and choose $v = n - 1$ nearest to $\mu(k_m + 1/2)$. Then $L/(\log m)^{3/2} \rightarrow c$ and $\text{dist}(L, \mathbb{Z} + 1/2) \leq 1/(2\mu) = o(H^2)$. The two Gaussian weights are asymptotically equal, while their cubic multipliers tend to $e^{\pm 1/(192c^2)}$. Their mean is (470). The actual-class upper condition holds automatically at this scale.

The full two-weight expression is needed when phases approach a half-integer at an unspecified rate. For a fixed limiting fractional phase $0 < u < 1/2$, the Gaussian-dominant neighbor instead has multiplier $e^{-u^3/(24c^2)}$; for $1/2 < u < 1$ its multiplier is $e^{(1-u)^3/(24c^2)}$. At integral limiting phase the multiplier is one.

[Merge note. The derivative identities (478) and (479), the limit $\Phi_v'''(L) \rightarrow -1/(4c^2)$, the cubic coefficient, the multipliers $e^{\pm 1/(192c^2)}$ and their mean, and the fixed-phase multipliers were rechecked symbolically for this write-up, and (478) with the exact κ_3 was confirmed against finite differences of the exact action at $m = 1024, 4096, 16384$; see remark 108.4 for the slow approach of the constants.]

109 The Gamma law for few macroscopic clusters

Sources: manuscripts 27 ($n \asymp m \log m$) and 22 ($1 \ll n/m \ll \log m$). Cutoff: a small fixed εm ; manuscript 22 works with the critical weights b_k . Part VIII keeps the number of macroscopic pieces fixed, while theorem 106.1 starts at a fixed positive multiple of $m \log m$. Here the macroscopic count is $\approx (n - 1)/m$, at most of order $\log m$.

Put $v = n - 1$. For an integer ℓ with $s_\ell = \ell - v/m > 0$, define the Gamma term

$$g_{m,v}(\ell) = \frac{(\ell + 1)e^{3s_\ell/2}s_\ell^{\ell-1}}{(2\sqrt{\pi m})^\ell \Gamma(\ell)}, \quad \Gamma(\ell) = (\ell - 1)!. \quad (490)$$

(Manuscript 27's $F_{m,v}(\ell)$ and manuscript 22's $G_{m,v}(j)$ are this function, character for character.) Fix $0 < c_0 < c_1 < \infty$, and define

$$r_c(s) = \frac{s}{2} - c - c \log \frac{s}{2c} \quad (c, s > 0), \quad B_* = 10 + \max_{\substack{c_0 \leq c \leq c_1 \\ 1 \leq s \leq 2}} r_c(s).$$

Choose constants a, b satisfying

$$\begin{aligned} 0 < a < \min(1, c_0), \quad b > \max(2, 4c_1), \\ \inf_{c_0 \leq c \leq c_1} r_c(a) > B_* + 10, \quad \inf_{c_0 \leq c \leq c_1} r_c(b) > B_* + 10. \end{aligned} \quad (491)$$

Such choices exist because $r_c(s) \rightarrow \infty$ uniformly on this compact c interval as $s \downarrow 0$ or $s \rightarrow \infty$. No optimization of a, b is intended. Set $\mathcal{I}_m(v) = \{\ell \geq 1 : \ell \in \mathbb{Z}, a \leq s_\ell \leq b\}$. (This index set is manuscript 27's; manuscript 24's $\mathcal{I}_m(v)$ of (441) is defined by tilts instead.)

Theorem 109.1 (Uniform integer-count equivalent; manuscript 27). *Uniformly over integers m, n with $m \rightarrow \infty$ and*

$$c_m = \frac{n - 1}{m \log m} \in [c_0, c_1], \quad (492)$$

one has

$$a_n^{(m)} \sim \mathfrak{E}_m(n) \sim \frac{4^n}{m} \sum_{\ell \in \mathcal{I}_m(n-1)} g_{m,n-1}(\ell), \quad (493)$$

$$p_n(m) \sim \frac{\sqrt{\pi} n^{3/2}}{m} \sum_{\ell \in \mathcal{I}_m(n-1)} g_{m,n-1}(\ell). \quad (494)$$

The number of positive summands is bounded independently of m, n .

The full finite sum is important near a count transition. Its coefficients are elementary apart from the factorial notation, and no Gaussian theta approximation is extrapolated to this scale.

Corollary 109.2 (Phase-dependent rate; manuscript 27). *If $c_m \rightarrow c > 0$ and $v/m \rightarrow f$ modulo one, then*

$$\frac{1}{\log m} \log \frac{a_n^{(m)}}{(n/m^2)r_m^{-n}} \rightarrow - \min_{\substack{j \in \mathbb{Z} \\ j-f > 0}} r_c(j-f). \quad (495)$$

The same rate is obtained on normalizing $p_n(m)$ by $\sqrt{\pi} n^{5/2} m^{-2} e^{-nt_m/m}$. Adjacent deficits $s, s+1$ tie at leading order exactly when

$$s = s_*(c) := \frac{1}{e^{1/(2c)} - 1}. \quad (496)$$

Every fixed phase can be realized by integer lengths, for example by taking v_m to be a nearest integer to $m(\lfloor c \log m \rfloor + f)$. The minimum in (495) is zero precisely when $2c + f$ is integral. At other phases it is positive, so the normalized count has an additional polynomial decay in m .

For the range below $m \log m$ put

$$\rho = \frac{v}{m}, \quad \ell = \lceil \rho \rceil, \quad s = \ell - \rho = s_\ell. \quad (497)$$

The statements below exclude integral ρ and thus have $0 < s < 1$.

Theorem 109.3 (One minimal macro count; manuscript 22). *Fix $0 < \delta < 1/2$. Uniformly along integer sequences with*

$$m \rightarrow \infty, \quad \ell \rightarrow \infty, \quad \ell = o(\log m), \quad \delta \leq s \leq 1 - \delta,$$

one has

$$a_n^{(m)} \sim \mathfrak{E}_m(n) \sim \frac{4^n}{m} g_{m,v}(\ell), \quad p_n(m) \sim \frac{\sqrt{\pi} n^{3/2}}{m} g_{m,v}(\ell). \quad (498)$$

The first count ℓ is the smallest number of letters of size at most m that can reach v , apart from the small-run contribution. The proof quantitatively excludes the small-run overflows that might replace one of these letters. The exact ratio of two adjacent Gamma terms is

$$R_{m,\ell}(s) = \frac{g_{m,v}(\ell+1)}{g_{m,v}(\ell)} = \frac{\ell+2}{\ell+1} e^{3/2} \frac{s}{2\ell\sqrt{\pi m}} \left(1 + \frac{1}{s}\right)^\ell, \quad (499)$$

with $v = m(\ell - s)$ in the ratio and $s = s_\ell$. (This is manuscript 27's eq:ratio and manuscript 22's eq:R; both delivered suites check it in exact rationals.)

Theorem 109.4 (The shrinking transition; manuscript 22). *Assume $m \rightarrow \infty, \ell \rightarrow \infty, \ell = o(\log m)$ and $0 < s < 1$. Uniformly whenever $R_{m,\ell}(s)$ stays in a fixed compact subset of $(0, \infty)$,*

$$\begin{aligned} a_n^{(m)} &\sim \mathfrak{E}_m(n) \sim \frac{4^n}{m} (g_{m,v}(\ell) + g_{m,v}(\ell+1)), \\ p_n(m) &\sim \frac{\sqrt{\pi} n^{3/2}}{m} (g_{m,v}(\ell) + g_{m,v}(\ell+1)). \end{aligned} \quad (500)$$

For all sufficiently large m along this count regime, there is a unique $\tau = \tau_{m,\ell} \in (0, 1)$ with $R_{m,\ell}(\tau) = 1$. It satisfies

$$\begin{aligned} \tau^{\ell-1} &= \frac{e^{3/2}(\ell+2)}{2\ell(\ell+1)\sqrt{\pi m}}(1+\tau)^\ell, \\ \tau &\sim \left(\frac{e^{3/2}(\ell+2)}{2\ell(\ell+1)\sqrt{\pi m}} \right)^{1/(\ell-1)}, \quad \log \frac{1}{\tau} \sim \frac{\log m}{2(\ell-1)}, \quad m\tau = m^{1-o(1)}. \end{aligned} \quad (501)$$

For $s = \tau e^{x/(\ell-1)}$, uniformly for x in a fixed compact real interval,

$$R_{m,\ell}(s) \rightarrow e^{-x}. \quad (502)$$

Thus the two positive terms in (500) have limiting shares $1/(1+e^{-x})$ and $1/(1+e^x)$.

The transition width is of order τ/ℓ in s , or $m\tau/\ell$ in v . The explicit equivalent for τ alone need not resolve that smaller width, so the exact root is retained. The shares above describe contributions to the equivalent, not an unproved statistic of a conditioned permutation. Theorem 109.4 is a specified below-integer window; it makes no uniform assertion over every phase tending to an integer.

Remark 109.5 (Merge note: one Gamma law on adjacent ranges). Theorems 109.1, 109.3 and 109.4 are one formula, the Gamma terms (490), on disjoint adjacent ranges: counts $\ell \asymp \log m$ (a bounded number of terms, selected by the rate r_c) and counts $\ell \rightarrow \infty$, $\ell = o(\log m)$ (one or two terms, with phase restrictions). They are consistent at the seam: with $\ell = c \log m$, manuscript 22's transition centre τ and manuscript 27's leading-order tie $s_*(c)$ agree to leading exponential order ($\log \tau \sim -\log m/(2(\ell-1))$, $\log s_* \sim -1/(2c)$); a ProveIt check made for this write-up found $\tau/s_*(\ell/\log m) = 0.24, 0.34, 0.44, 0.43, 0.60$ at $m = 10^{30}, \dots, 10^{106}$, a ratio that tends to one only like $\exp(O(\log \ell/\ell + 1/(c\ell)))$. So the transition of theorem 109.4 is the $\ell \rightarrow \infty$, $c \rightarrow 0$ shadow of the tie in corollary 109.2. At $n \asymp m \log m$ the Gamma sum (493) and manuscript 24's saddle sum (442) are two expressions of one coefficient: both are equivalents of $\mathfrak{G}_m(n)$, and manuscript 27's (503) and section 109.2 evaluate the same count coefficients $\mathfrak{G}_\ell(v)$; no term-by-term identity of the two sums is proved (manuscript 24's sentence "a common expression for the earlier finite Gamma sum", printed in section 111, is to be read so).

Convergence in theorem 109.1 is slow, and manuscript 27 does not quantify it. The first equivalence of the composition estimate (509) carries a relative error $\exp(O(s_\ell^2/\ell))$, which is $1 + O(1/\log m)$ at this scale, and the small runs add a correction of order $\ell^{3/2}(ms_\ell)^{-1/2}$. A ProveIt check made for this write-up computed exact envelopes against the Gamma sum at $n = cm \log m$: at $c = 1$ the ratio is 1.32, 1.87, 2.24, 2.39, 2.40, 2.35, 2.06, 1.93 for $m = 64, 128, 256, 512, 1024, 4096, 16384, 65536$, and at $c = 1/2$ between 1.16 and 1.66; a Monte Carlo evaluation of the composition factor reproduces the $1 + O(1/\log m)$ correction. The theorem stays asymptotic and is not contradicted.

109.1 A finite saddle sum with controlled overflow

Source: manuscript 27; the compact- c_m case of theorem 106.1, proved with its own tail argument by the rate r_c , and the first step of the proof of theorem 109.1. Choose a compact interval $\mathcal{T} \subset (-1, \infty)$ containing all $2c/s - 1$ for $c \in [c_0, c_1]$, $s \in [a, b]$, in its interior. Choose ε for lemma 103.1; it may be reduced further below. Because (491) is strict, these choices and margins persist on a fixed small neighborhood of the c interval.

With the squared-renewal decomposition (448) and the exact tilting identity (449) (the two residual small runs are included in this identity; their total mass $D(h)$ is uniformly bounded, and W_h has a uniform exponential moment on scale w), put, for each retained $\ell \in \mathcal{I}_m(v)$, $x_\ell = (v/\ell - m)/w$. Uniformly in this set, $x_\ell \rightarrow -s_\ell/(2c_m)$. The strict convexity and limits in (406)

give a unique exact $h_\ell \in \mathcal{T}$ with $\Psi'_m(h_\ell) = x_\ell$, $\ell\mu_{h_\ell} = v$. With $I_\ell = h_\ell x_\ell - \Psi_m(h_\ell)$, the centered local limit in (449) yields

$$\mathfrak{E}_\ell(v) \sim \frac{D(h_\ell)e^{-\ell I_\ell}}{\sigma_{h_\ell}\sqrt{2\pi\ell}}, \quad \frac{\ell I_\ell}{\log m} - r_{c_m}(s_\ell) \longrightarrow 0 \quad (503)$$

uniformly. To justify the residual convolution relatively, the error in (407) is uniform in its target. Its Gaussian factor at $-W_h/(\sigma_h\sqrt{\ell})$ tends in expectation to $\varphi(0)$, by the moment bound and $\ell \rightarrow \infty$. This calculation also supplies the positive competitor $\ell = \lfloor v/m \rfloor + 2$, whose deficit lies in $(1, 2]$ and whose rate is at most $B_* - 10 + o(1)$. (ℓI_ℓ is manuscript 24's action $\Phi_v(\ell)$.)

We now exclude *all* other counts on the coefficient scale. For $s_\ell < a$ use $h_+(c_m) = 2c_m/a - 1 > 0$. For $\ell \geq 1$, (408) bounds the probability in (449) by $C/(w\sqrt{\ell})$. The factor $Q(e^{h_+/w})^\ell e^{-h_+v/w}$ grows geometrically in ℓ , since $Q(e^{h_+/w}) > 1$ for large m . At its largest tail integer, $s_\ell \in [a - 1, a)$ and

$$\frac{1}{\log m} \log\{Q(e^{h_+/w})^\ell e^{-h_+v/w}\} \leq -r_{c_m}(a) + o(1).$$

Indeed its exponent is $\ell\Psi_m(h_+) + h_+s_\ell m/w$, and its limiting cost is $c_m \log(1 + h_+) - h_+s_\ell/2 \geq r_{c_m}(a)$. The weighted geometric sum preserves this bound. It includes every $s_\ell \leq 0$, even though Y can exceed m . The residual-only $\ell = 0$ term is at most $D(h_+)e^{-h_+v/w} = O(e^{-c(\log m)^2})$.

For $s_\ell > b$, use $h_-(c_m) = 2c_m/b - 1 < 0$. Now $Q(e^{h_-/w}) < 1$; the first tail term has cost at least $r_{c_m}(b) \log m + o(\log m)$, and the rest is a weighted geometric tail. The margins in (491), compared with the positive competitor, make both omitted tails relatively negligible. This quantitatively rules out large small-run overflows. Equations (448), (449) and (503) therefore already give a uniform positive finite exact-MGF saddle sum.

[Merge note. Manuscript 27's eq: tm (the scalar expansion), its eq: decomp and its eq: exact tilt are (397), (448) and (449), and are replaced by them. Manuscript 24's section 106.1 re-runs this section with a growing count range and a centered competitor in place of the rate margins.]

109.2 Small runs and compositions

Source: manuscript 22, with manuscript 27's reduction as its compact-deficit case. Critical weights $b_k = C_{k-1}4^{-k}$, cutoff εm . Choose a sufficiently small fixed $\varepsilon > 0$ later, and write

$$A_0(z) = \sum_{k \leq \varepsilon m} b_k z^k, \quad B_0(z) = \sum_{\varepsilon m < k \leq m} b_k z^k, \quad Q_0(z) = \frac{B_0(z)}{1 - A_0(z)}. \quad (504)$$

These are critical Catalan weights, and $Q_0(1)$ is not one. The exact positive decomposition is

$$\begin{aligned} \mathfrak{B}_m^0(v) &= [z^v](1 - A_0(z) - B_0(z))^{-2} \\ &= \sum_{j \geq 0} (j+1)[z^v](1 - A_0(z))^{-j-2} B_0(z)^j, \quad \mathfrak{E}_m(n) = 4^{n-1} \mathfrak{B}_m^0(n-1). \end{aligned} \quad (505)$$

There are j big letters and $j+2$ small runs in the j th term. (Every count's coefficient here is $(4r_m)^{-v}$ times its tilted coefficient $\mathfrak{E}_j(v)$, so the relative tail estimates of section 109.1 persist.)

Lemma 109.6 (Small-run domination; manuscript 22). *Let U be the total of N independent normalized small runs, with generating function $[(1 - A_0(1))/(1 - A_0(z))]^N$. Uniformly for $N = O(\ell)$, $\ell = o(\log m)$, one has*

$$\mathbb{P}(U = u) \leq \frac{CN}{(u+1)^{3/2}} \quad (u \geq 0). \quad (506)$$

Moreover $\mathbb{E}e^{\gamma U} \rightarrow 1$ in either of the following cases:

$$\gamma m = O(\ell), \quad \gamma \geq 0; \quad \text{or} \quad \gamma = T \frac{\log m}{m}, \quad T > 0 \text{ fixed}, \quad T\varepsilon < 1/4.$$

Proof. The full critical Catalan series is

$$A_\infty(z) = \frac{1 - \sqrt{1 - z}}{2}, \quad \left(\frac{1/2}{1 - A_\infty(z)} \right)^N = \left(\frac{C(z/4)}{2} \right)^N.$$

The classical Catalan-power identity, obtained by Lagrange inversion, gives the exact probability

$$\mathbb{P}(U_\infty = u) = 2^{-N} 4^{-u} \frac{N}{2u + N} \binom{2u + N}{u}. \quad (507)$$

The maximal binomial coefficient in row $2u + N$ is at most $C2^{2u+N}/\sqrt{2u + N}$. Thus (507) is bounded by $CN/(u + 1)^{3/2}$. Also $A_0(1) = 1/2 + O(m^{-1/2})$, and coefficientwise domination gives $\mathbb{P}(U = u) \leq [2(1 - A_0(1))]^N \mathbb{P}(U_\infty = u)$. The prefactor is $1 + o(1)$ for the stated range of N , proving (506).

For every $\gamma \geq 0$,

$$A_0(e^\gamma) - A_0(1) \leq \gamma e^{\gamma \varepsilon m} \sum_{k \leq \varepsilon m} kb_k \leq C\gamma e^{\gamma \varepsilon m} \sqrt{m}. \quad (508)$$

In the first case, multiplying this bound by $N = O(\ell)$ gives $m^{-1/2+o(1)} \rightarrow 0$. In the second case it gives $O(\ell \log m m^{-1/2+T\varepsilon}) \rightarrow 0$. In both cases $A_0(e^\gamma) < 1$ eventually and

$$\log \mathbb{E} e^{\gamma U} = N \log \frac{1 - A_0(1)}{1 - A_0(e^\gamma)} = O(N[A_0(e^\gamma) - A_0(1)]) \rightarrow 0.$$

This controls all the runs together, including rare large totals. $\square$

Let $j \rightarrow \infty$, $j = O(\ell)$, and put $\Delta = jm - v = ms_j > 0$. Write a big letter as $m - d_i$, where the nonnegative deficits sum to Δ .

Lemma 109.7 (Composition estimate; manuscript 22). *Suppose s_j is bounded above, $j^2/\Delta \rightarrow 0$, and either s_j is bounded away from zero or $s_j \rightarrow 0$. Then*

$$\begin{aligned} [z^v]B_0(z)^j &\sim b_m^j e^{3s_j/2} \binom{\Delta + j - 1}{j - 1} \\ &\sim (4\sqrt{\pi})^{-j} m^{-3j/2} e^{3s_j/2} \frac{(ms_j)^{j-1}}{\Gamma(j)}. \end{aligned} \quad (509)$$

For every $0 \leq u \leq v$, a global ratio bound is

$$\frac{[z^{v-u}]B_0(z)^j}{[z^v]B_0(z)^j} \leq C e^{C_\varepsilon u/m} \left(1 + \frac{u}{\Delta}\right)^{j-1}. \quad (510)$$

Proof. On legal compositions $d_i < (1 - \varepsilon)m$, Catalan Stirling estimates give

$$\prod_i \frac{b_{m-d_i}}{b_m} = \prod_i (1 - d_i/m)^{-3/2} (1 + O(j/(\varepsilon m))).$$

The product is bounded by $e^{C_\varepsilon s_j}$. Under the uniform weak compositions of Δ , for fixed $\eta > 0$,

$$\mathbb{P}(\max_i d_i \geq \eta m) \leq j \left(1 - \frac{\lceil \eta m \rceil}{\Delta + j - 1}\right)^{j-1}$$

when the event is possible. For s_j in a compact positive interval this tends to zero, and illegal compositions have probability $o(1)$. The logarithm of the product therefore tends to $3s_j/2$. For

$s_j \rightarrow 0$, every composition is legal eventually and $\max_i d_i/m \leq s_j \rightarrow 0$, which gives the same conclusion directly. Bounded domination proves the first equivalent. The last approximation has error $O(j^2/\Delta)$; the Catalan normalization contributes $O(j/m)$.

For the global ratio, bound the numerator by $b_m^j e^{C_\varepsilon(\Delta+u)/m} \binom{\Delta+u+j-1}{j-1}$. The denominator in (509) is bounded below by a fixed positive multiple of $b_m^j \binom{\Delta+j-1}{j-1}$. The exact binomial ratio is at most $(1 + u/\Delta)^{j-1}$. The bounded value of s_j absorbs the remaining constant. Impossible coefficients are zero. $\square$

If s_j stays in a fixed compact positive interval, apply lemma 109.6 with $\gamma = Cj/m$ sufficiently large to dominate the exponent in (510). For U equal to the total of $j + 2$ normalized small runs, $\mathbb{E}e^{\gamma U} \rightarrow 1$. When $\gamma u \rightarrow 0$, the composition formula at $\Delta + u$ and Δ makes the ratio tend to one. The exponential bound provides uniform integrability. Consequently

$$[z^v](1 - A_0)^{-j-2} B_0^j \sim (1 - A_0(1))^{-j-2} [z^v] B_0^j \sim 2^{j+2} [z^v] B_0^j. \quad (511)$$

Using (505), the contribution of this count to $\mathfrak{E}_m(n)$ is exactly the equivalent $(4^n/m) \mathfrak{g}_{m,v}(j)$. For theorem 109.3, take $j = \ell$; its compositions even satisfy $m - d_i \geq \delta m$ automatically.

The compact-deficit case (manuscript 27): reducing the retained terms of section 109.1 to Gamma factors. For a retained ℓ of theorem 109.1, $\ell = O(\log m)$, $\Delta = \ell m - v = m s_\ell$ and $m - d_i > \varepsilon m$, so (509) holds: under uniformly chosen compositions the maximum d_i/m tends to zero in probability, uniformly for $s_\ell \in [a, b]$; on legal compositions $\prod_i (1 - d_i/m)^{-3/2} \leq e^{C_\varepsilon \Delta/m} \leq e^{C_\varepsilon b}$, and its logarithm tends to $3s_\ell/2$ on the small-maximum event; the last errors are $O(\ell/m)$ and $O(\ell^2/\Delta)$, both $o(1)$. All $\ell + 2$ small runs must still be controlled relatively. Since $A_0(1) = 1/2 + O(m^{-1/2})$, their total mass is $(1 - A_0(1))^{-\ell-2} \sim 2^{\ell+2}$. Let U be the sum of the corresponding normalized runs. Choose $G > 4(c_1 + 1)/a$, put $\gamma_m = G \log m/m$, and reduce ε so $\varepsilon G < 1/4$. Then $A_0(e^{\gamma_m}) - A_0(1) \leq C \gamma_m e^{\gamma_m \varepsilon m} \sqrt{m} = O(m^{-1/2+\varepsilon G} \log m)$. Because $\ell = O(\log m)$, it follows that

$$\mathbb{E}e^{\gamma_m U} \rightarrow 1 \quad (512)$$

uniformly (the second case of lemma 109.6). For every $0 \leq u \leq v$, there is a global bound

$$\frac{[z^{v-u}] B_0(z)^\ell}{[z^v] B_0(z)^\ell} \leq C e^{\gamma_m u}. \quad (513)$$

The numerator is at most $b_m^\ell e^{C_\varepsilon(\Delta+u)/m} \binom{\Delta+u+\ell-1}{\ell-1}$. The denominator is at least a fixed positive fraction of $b_m^\ell \binom{\Delta+\ell-1}{\ell-1}$, using the composition estimate and $b_{m-d} \geq b_m$. The binomial ratio is at most $e^{\ell u/\Delta}$. For large m , $\ell/\Delta \leq 2c_1 \log m/(am)$, so G absorbs this exponent and $C_\varepsilon u/m$. When $\gamma_m u \rightarrow 0$, the uniform formula (509) at the slightly shifted deficit makes the ratio in (513) tend to one. Equation (512) implies both this convergence in probability at $u = U$ and uniform integrability under the bound (513). Hence (511) holds; this controls the rare large totals as well as the typical small ones. Multiply by $\ell + 1$ and by 4^{n-1} . The constant is

$$4^{n-1} 2^{\ell+2} (4\sqrt{\pi})^{-\ell} m^{-3\ell/2} m^{\ell-1} = \frac{4^n}{m} (2\sqrt{\pi m})^{-\ell}.$$

Together with the finite-window tail bounds of section 109.1, this proves the scalar equivalent in (493); the transfer to $a_n^{(m)}$ is section 104 with route 2 of proposition 104.2 (section 106.2), since $n \varepsilon D_{\text{app}} \rightarrow 0$ and $n r_m^{L_{\text{app}}} \rightarrow 0$ under (492). Catalan Stirling asymptotics then give (494).

[Merge note. Manuscript 27's eq:macro and eq:smallconvolution are (509) and (511), and are replaced by them. Manuscript 22's lemmas generalize manuscript 27's estimates: they allow $s_j \rightarrow 0$ and give the sharper ratio bound $(1 + u/\Delta)^{j-1}$ in place of $e^{\ell u/\Delta}$, and lemma 109.6 adds the exact Catalan-power law.]

109.3 Endpoint-scale atoms and the count tails

Source: manuscript 22. For $j \in \{\ell, \ell + 1\}$ with s_j in a fixed compact positive interval, use the critical tilt $\gamma = j/(ms_j)$. Here $\gamma m \asymp \ell \rightarrow \infty$ and $\gamma m = o(\log m)$. Endpoint summation gives

$$A_0(e^\gamma) = \frac{1}{2} + o(1), \quad B_0(e^\gamma) = \frac{b_m e^{\gamma m}}{\gamma} (1 + O(\ell^{-1})), \quad q_\gamma = Q_0(e^\gamma) = m^{-1/2+o(1)}. \quad (514)$$

Indeed, set $k = m - d$ and sum $e^{-\gamma d} (1 - d/m)^{-3/2}$. The first relative correction is $O((\gamma m)^{-1})$; the lower-cutoff tail is exponentially small in ℓ . The small-series error, even multiplied by ℓ , is $o(1)$ by (508).

Normalize $Q_0(e^\gamma z)/Q_0(e^\gamma)$ into a big jump followed by a geometric small run. The big weights increase on $[\varepsilon m, m]$, have largest atom $O(\gamma)$, and satisfy $\gamma(m - X) \Rightarrow \text{Exp}(1)$ with a uniform exponential-moment margin. Abel summation gives the characteristic bound $C\gamma/|\sin(\theta/2)|$. The central quadratic bound, the strict scaled-frequency modulus gap, and Parseval's $O(\gamma)$ squared-modulus integral therefore imply

$$\sup_u \mathbb{P}_\gamma(S_k = u) \leq \frac{C\gamma}{\sqrt{k}} \quad (k \geq 1). \quad (515)$$

To see the uniformity, the Fourier integral in the central interval is $O(\gamma/\sqrt{k})$. Outside it, a fixed gap $\varrho < 1$ and Parseval bound the integral by $C\gamma\varrho^{k-2}$ for $k \geq 2$; $k = 1$ follows from the largest atom. The small-run characteristic factor has modulus at most one. Thus the bound is proved at the new scale $1/\gamma$, without assuming a previously fixed tilt range remains uniform. (This is the method of lemma 103.1, re-proved at a non-compact scale.)

The explicit composition term and Stirling's formula also give

$$(j+1)[z^v](1 - A_0)^{-j-2} B_0^j \geq c\gamma\sqrt{j} e^{-\gamma v} q_\gamma^j. \quad (516)$$

In fact $q_\gamma^j e^{-\gamma v}$ differs by bounded positive factors from $[(2\sqrt{\pi m})^{-1} s_j/j]^j e^j$; multiplying the Gamma coefficient by $j+1$ supplies the factor of order $\gamma\sqrt{j}$. Only comparability is needed here.

Extra counts. Take $\gamma = \ell/(ms)$ in the bounded-phase case. Exact tilting, (515), and a positive geometric sum give

$$\sum_{j \geq \ell+1} (j+1)[z^v](1 - A_0)^{-j-2} B_0^j \leq C\gamma\sqrt{\ell} e^{-\gamma v} q_\gamma^{\ell+1}.$$

The normalized two residual runs do not increase the maximal atom, and their normalizing factor stays bounded. Compared with (516), the total extra-count contribution is $O(q_\gamma) = m^{-1/2+o(1)}$.

Too few counts. Use a different tilt, $\gamma_T = T \log m/m$, with fixed $T > 4/\delta$, and choose ε small enough that $T\varepsilon < 1/4$. The small denominator is bounded away from zero, while endpoint summation gives $Q_0(e^{\gamma_T}) \asymp m^{T-1/2}/\log m \rightarrow \infty$. (Manuscript 22 writes $\Theta(m^{T-1/2}/\log m)$, order notation.) The positive-coefficient Cauchy bound yields

$$\sum_{0 \leq j \leq \ell-1} (j+1)[z^v](1 - A_0)^{-j-2} B_0^j \leq C\ell e^{-\gamma_T v} Q_0(e^{\gamma_T})^{\ell-1}. \quad (517)$$

This includes the residual-only term $j = 0$. Comparing its logarithm with the main term in (511), using $v/m = \ell - s$, gives

$$\log \frac{\text{bound in (517)}}{\text{main term}} \leq \left(\frac{3}{2} - T(1-s)\right) \log m + \ell \log \frac{\ell}{\log m} + O_\delta(\ell + \log \log m). \quad (518)$$

Since $1 - s \geq \delta$ and $\ell = o(\log m)$, this is bounded above by a negative fixed multiple of $\log m$. It quantitatively excludes every overflow capable of replacing a missing big letter. The scalar assertion of theorem 109.3 follows.

109.4 The transition location

Source: *manuscript 22*. Differentiation of (499) gives

$$\frac{d}{ds} \log R_{m,\ell}(s) = \frac{1+s-\ell}{s(1+s)} < 0 \quad (0 < s < 1, \ell \geq 2).$$

The ratio tends to infinity at zero, while $R_{m,\ell}(1) = m^{-1/2+o(1)} \rightarrow 0$. This proves eventual existence and uniqueness of τ .

When R stays in a fixed compact positive interval, taking logarithms gives

$$\begin{aligned} (\ell - 1) \log(1/s) &= \frac{1}{2} \log m + \log(2\ell\sqrt{\pi}) - \frac{3}{2} \\ &\quad - \log \frac{\ell+2}{\ell+1} - \ell \log(1+s) + O(1). \end{aligned} \quad (519)$$

First this forces $s \rightarrow 0$, and then

$$\log(1/s) \sim \frac{\log m}{2(\ell-1)}, \quad s = m^{-o(1)}, \quad ms = m^{1-o(1)}, \quad \ell^2/(ms) \rightarrow 0.$$

The composition lemma now applies to $j = \ell$ and $j = \ell + 1$, whose deficit is $1 + s$. A further consequence of the upper bound on R is

$$\frac{1}{\sqrt{m} s^{\ell-1}} \leq C\ell. \quad (520)$$

The factor $(1+s)^\ell$ in (499) only improves this estimate.

At $R(\tau) = 1$, its exact equation is the first line of (501). Dividing by the $(\ell-1)$ st root of its constant gives the factor $(1+\tau)^{\ell/(\ell-1)} \rightarrow 1$, proving the stated explicit equivalent. Finally

$$\frac{d \log R}{d \log s} = 1 - \frac{\ell}{1+s}.$$

Integrating over $s = \tau e^{x/(\ell-1)}$ proves (502), uniformly on bounded x intervals. The term $\ell \log(1+\tau)$ can still matter for locating the narrower transition; the exact root is used throughout.

109.5 Small runs at a shrinking deficit

Source: *manuscript 22*. The remaining delicate step is (511) at $j = \ell$ when $s \rightarrow 0$ in theorem 109.4. Write

$$t(u) = \frac{[z^{v-u}]B_0^\ell}{[z^v]B_0^\ell}, \quad u_0 = \frac{ms}{\ell^2},$$

and let U comprise $\ell + 2$ normalized small runs. Here $u_0 = m^{1-o(1)}$. For $0 \leq u \leq u_0$, the composition formula and the binomial ratio give $t(u) \rightarrow 1$ uniformly, because $\ell u/(ms) \leq 1/\ell$. Also $\mathbb{P}(U > u_0) \leq C\ell/\sqrt{u_0} \rightarrow 0$ by (506). We prove that the weighted contribution on this exceptional event tends to zero.

Totals between the typical range and the deficit. For $u_0 < u \leq ms$, equation (510) gives $t(u) \leq C2^\ell$. Thus

$$\mathbb{E}[t(U); u_0 < U \leq ms] \leq \frac{C\ell^2 2^\ell}{\sqrt{ms}} = m^{-1/2+o(1)} \rightarrow 0. \quad (521)$$

This uses both $\ell = o(\log m)$ and $s = m^{-o(1)}$.

Totals up to a small fixed fraction of the cap. Fix $0 < \varepsilon' < 1/4$. On $ms < u \leq \varepsilon'm$, $(1 + u/(ms))^{\ell-1} \leq (2u/(ms))^{\ell-1}$. The other exponential in (510) is bounded by a fixed constant. Using (506), summing $u^{\ell-5/2}$, and then applying (520) gives

$$\mathbb{E}[t(U); ms < U \leq \varepsilon'm] \leq \frac{C(2\varepsilon')^\ell}{\sqrt{m}s^{\ell-1}} \leq C\ell(2\varepsilon')^\ell \rightarrow 0. \quad (522)$$

Constants here may depend on ε' and the final fixed ε .

Larger totals. Choose fixed $T_* > 2/\varepsilon'$. For $u \geq \varepsilon'm$, the function $\log(1 + u/(ms))/u$ is decreasing. Equation (519) consequently implies, for all large m ,

$$C_\varepsilon u/m + (\ell - 1) \log(1 + u/(ms)) \leq T_*(\log m)u/m. \quad (523)$$

Indeed, at $u = \varepsilon'm$ the logarithmic term is at most $(1/2 + o(1)) \log m$: use (519) and $(\ell - 1) \log(s + \varepsilon') \leq 0$ eventually. The right side there is $T_* \varepsilon' \log m$, and the fixed C_ε is absorbed as m grows. Now choose ε sufficiently smaller than ε' that $T_* \varepsilon < 1/4$. Lemma 109.6 gives $\mathbb{E} \exp(T_*(\log m)U/m) \rightarrow 1$. Combining (510) and (523),

$$\mathbb{E}[t(U); U > \varepsilon'm] \leq C \mathbb{E}[e^{T_*(\log m)U/m}; U > \varepsilon'm] \rightarrow 0. \quad (524)$$

For the last implication, on this event the exponential minus one is at least $1 - m^{-T_* \varepsilon'}$ times the exponential itself. The expected exponential minus one tends to zero.

Equations (521)–(524) prove uniform integrability and hence $\mathbb{E} t(U) \rightarrow 1$. This proves (511) at the shrinking deficit, including every residual small run.

109.6 Other counts in the transition, and the Gamma-term route to the actual class

Source: manuscript 22; route 3 of proposition 104.2. For $j = \ell + 1$, the deficit $1 + s$ lies in a fixed compact positive interval. The ordinary composition and small-run argument applies. Use $\gamma = (\ell + 1)/(m(1 + s))$ in the endpoint estimate: all $j \geq \ell + 2$ are smaller by $m^{-1/2+o(1)}$ than this second main term. For $j \leq \ell - 1$, choose a fixed $T > 4$, enlarged if necessary to satisfy the preceding large-total argument, and then make ε smaller. The calculation of (518), now relative to the second Gamma term, gives

$$(2 - T(1 - s)) \log m + (\ell - 1) \log \frac{\ell}{\log m} + O(\ell + \log \log m).$$

Since $s \rightarrow 0$, this is again bounded above by a negative fixed multiple of $\log m$. All counts other than $\ell, \ell + 1$ are negligible relatively, proving the scalar assertion in theorem 109.4.

We finish the actual-class transfer with lemma 104.1. Let $d = O(\log m)$ and $M = m - d$. At the same v , the new deficit is exactly

$$s_M = \ell - v/M = s - \frac{(\ell - s)d}{M}.$$

In theorem 109.3, the count ℓ stays unchanged and the phase remains in a slightly enlarged compact subinterval of $(0, 1)$. The logarithmic change in the Gamma term is $O(\ell^2 d/m) = o(1)$, giving $\mathfrak{E}_M(n)/\mathfrak{E}_m(n) \rightarrow 1$. In theorem 109.4, one has $|s_M - s| = O(\ell d/m) = o(s/\ell)$, since $ms = m^{1-o(1)}$. The count again stays unchanged, both Gamma terms change relatively by $1 + o(1)$, and their ratio stays in a slightly enlarged compact positive interval. Their logarithmic changes are $O(\ell^2 d/(ms)) = o(1)$. Thus $\mathfrak{E}_M(n)/\mathfrak{E}_m(n) \rightarrow 1$ in this window as well. A target shift of $L_{\text{app}} = O(\log m)$ preserves the same conditions and gives $\mathfrak{E}_M(n - L_{\text{app}})/\mathfrak{E}_M(n) \sim 4^{-L_{\text{app}}}$. Take $D_{\text{app}} = L_{\text{app}} = \lceil 3 \log_2 m \rceil$ in (429). Since $n = O(m \log m)$, both loss factors tend to zero. This proves $a_n^{(m)} \sim \mathfrak{E}_m(n)$ and completes both theorems after Catalan Stirling asymptotics.

The transition is realizable with integer lengths. For any integer sequence $\ell \rightarrow \infty$, $\ell = o(\log m)$, choose v nearest $m(\ell - \tau e^{x/(\ell-1)})$. Rounding changes s by $O(m^{-1})$, negligible compared with τ/ℓ because $m\tau = m^{1-o(1)}$. The resulting minimal count is ℓ and its ratio tends to e^{-x} .

109.7 The integer-count transition

Source: manuscript 27; proof of corollary 109.2. The finite positive saddle sum and (503) prove the rate statement (495). Relative to n/m^2 , the remaining prefactors have logarithm $o(\log m)$. Deficits tending to zero cause no exception: their coefficients were uniformly suppressed by the positive-tilt bound.

The function r_c is strictly convex and minimized at $s = 2c$, while

$$r_c(s+1) - r_c(s) = \frac{1}{2} - c \log(1 + 1/s).$$

This difference is negative below $s_*(c)$, zero there, and positive above. Hence the minimizing positive deficit is the first lattice point above $s_*(c)$, except that if $s_*(c)$ itself is on the lattice, it and its successor both minimize. This proves corollary 109.2.

At a fixed positive distance from a limiting leading-rate tie, only the minimizing summand survives relatively, uniformly on compact parameter sets with that separation. At a limiting tie, all other summands are polynomially smaller in m ; the two adjacent terms must be retained with their exact ratio (499). If this ratio tends to $R \in [0, \infty]$, their shares in the equivalent tend to $1/(1+R)$ and $R/(1+R)$, with the natural endpoint conventions. These are shares of the explicit sum, not an unproved identification of a statistic of the conditioned permutation. The prefactor in (499) also explains why a leading-rate tie does not force equal terms. The term $-\log \ell$ in its logarithm can displace the finite- m transition on a finer scale than the leading threshold (496).

[Merge note. The tie criterion was rechecked for this write-up: $r_c(s+1) - r_c(s)$ vanishes exactly at $s_*(c)$, and on a grid of 1,600 pairs (c, f) the minimum in (495) is zero exactly when $2c + f$ is integral, with the minimizer the first lattice point at or above $s_*(c)$. At $(c, s) = (1, 0.7)$ the normalized logarithm approaches $-r_c(s)$ only at the rate $O(\log \log m / \log m)$: -0.406 against -0.392 at $m = 10^{50}$, -0.40013 against -0.39962 at $m = 10^{2000}$.]

110 Checks, data and reproducibility

110.1 What the delivered programs check

Each manuscript ships a standard-library exact checker or an optional floating-point diagnostic, or both. They check finite statements and record finite values; none of them proves an asymptotic statement, and none counts actual avoiders at the large lengths of the theorems.

- *Manuscript 36.* The exact standard-library checker verifies 1,600 upper-envelope coefficients ($1 \leq m \leq 20, 1 \leq n \leq 80$), 1,520 lower-envelope coefficients, 230 equalities with Catalan counts through degree $m+1$, and 32,895 instances of the uniform coefficient bound through $m = 256$. It uses the credited exact endpoint implementation from the report's polynomial-rarity work (Part VI). The recorded integers and hashes make these finite regressions reproducible; they do not replace the proof or certify an asymptotic cutoff. An optional floating-point diagnostic evaluates the scalar roots and clustered means and variances (table 34). The limiting values are $1/2, 1, 1$. These diagnostics illustrate slow convergence and are not proof evidence for a uniform error rate.
- *Manuscript 33.* The exact standard-library checker reconstructs all 132-avoiders through size nine and their relaxed parsers. It verifies 6,917 parser round trips, 5,766 exact spine-shape factorizations, 17,298 restricted-decoration factorizations, 53,563 retained cap inequalities, and 16,932 marked append deletions. The 405 parameter rows and their hash are retained in the package. The inherited Catalan generator and endpoint model are credited separately. A second, separately implemented permutation-level replay through size ten, importing nothing from the first checker or the model, checks 50 envelope equalities, 450 bad-family parameter cases and 126,565 retained cap inequalities. These tests check the combinatorial normalization and injections; no finite computation is used to infer an asymptotic statement.

- *Manuscript 30.* The optional numerical checker compares the Gaussian and Fourier forms of ϑ at twenty parameter pairs, checks the exact total-variation identity for six neighboring-cap pairs, and records thirty-six finite squared-envelope values computed by the scalar recurrence. It uses high-precision and ordinary floating-point arithmetic and is explicitly diagnostic. No numerical error enclosure or asymptotic cutoff is claimed; the theorem follows from the written uniform estimates and exact inclusions.
- *Manuscript 27.* The accompanying checker verifies two hundred instances of the adjacent-ratio identity (499) by exact rational arithmetic after removing the common exponential and square-root factors. It also records thirty-six scalar-envelope values and their finite Gamma-sum comparisons. The latter use floating-point arithmetic and can converge slowly; no finite approximation bound or effective asymptotic threshold is asserted. They are diagnostics, not evidence replacing the uniform proof.
- *Manuscript 24.* The packaged standard-library checker differentiates the exact second derivative twice, verifies (478)–(479), checks the constants $1/24$ and $1/192$, and performs 54 rational checks of the normalized prefactor. A separate optional NumPy/SciPy diagnostic compares the positive saddle sum with finite scalar coefficients obtained from the renewal recurrence (twenty rows, $m = 128, \dots, 1024$, tilt interval $[-0.8, 1]$, cutoff $\lfloor m/64 \rfloor$; saddle-to-scalar ratios between about 0.915 and 1.103). Those numerical rows are not a proof, do not count actual avoiders at large lengths, and assert no finite error bound.
- *Manuscript 22.* The package contains a standard-library checker for 610 Catalan-power and probability-normalization identities and 40 exact adjacent-Gamma ratios. All checks remain active under optimized Python. A separate optional NumPy/SciPy script compares the Gamma expressions with finite scalar coefficients computed from a positive renewal recurrence, for $m = 128, 512, 2048, 4096$ at ordinary and near-transition phases (twenty rows). These diagnostics neither count actual avoiders at those lengths nor provide a finite error bound; the samples are visibly pre-asymptotic, their single-Gamma-to-exact ratios lying between about 0.076 and 0.519 and their two-Gamma ratios between 0.358 and 0.656.

m	$A(1)$	$(M - \mu_0)/w$	σ_0/w
256	0.596383	0.404554	1.300960
1024	0.570264	0.632334	1.268606
4096	0.549363	0.790077	1.249886

Table 34: Manuscript 36’s cluster diagnostic, in the setting of section 103.2 (block weights, cutoff $M/2$, $h = 0$); the limits are $1/2, 1, 1$ (lemma 103.4). Rounded floating-point values from the shipped diagnostic record, reproduced to all printed digits for this write-up.

[Merge note on the checks. (i) Several diagnostic families lie outside the theorems’ ranges: manuscript 30’s thirty-six envelope rows use a fixed mean count $L \leq 16.5$, so $n/[m(\log m)^2]$ is only about 0.02–0.28 and decreases with m (their ratios to the theta formula, 1.04–1.28, drift upwards); manuscript 27’s six phase-zero rows with $v = m$ have $c_m = 1/\log m \rightarrow 0$ and lowest count $\ell = 1$ at zero deficit, and their ratios (2.30 to 22.05 for $m = 64, \dots, 2048$) reflect the fixed-count regime outside theorem 109.1; and manuscript 22’s row labelled `transition_x2` at $m = 128$ has count 3 and deficit 0.18, because the target $\tau e^{2/(\ell-1)}$ exceeds one there, so it is not a transition sample and the checks README’s “their counts are only four or five” is false for that row. (ii) Some exact checks are algebraic identities by construction: manuscript 24’s 54 prefactor normalizations and manuscript 27’s 200 ratios compare two rearrangements of one rational expression (manuscript 24’s derivative identities are the substantive check); manuscript 22’s 610 rows check each Catalan-power coefficient and its probability scaling, not that the probabilities sum to one. (iii) The checks were rerun on copies for the placement: every suite passed, and the regenerated outputs equal the delivered ones apart from line endings, recorded library versions

and floating-point noise. (iv) A ProveIt check made for this write-up compared exact envelopes with manuscript 30's theta formula at $n = cm(\log m)^2$ for m up to 65536: the ratios decrease monotonically towards one (1.07 at $c = 0.1$, 1.03 at $c = 0.25$, 1.007 at $c = 1$, all at $m = 65536$).]

110.2 Rebuilding and rerunning

Part IX's files carry the prefixes 10-logarithmic-bridge-, 11-relative-asymptotic-, 12-theta-boundary-, 13-integer-count-, 14-saddle-sum- and 15-growing-macro-counts-:

code/10-...-verify_envelopes.py, probe_cluster.py	ms. 36: exact envelopes; cluster
diagnostics	
code/11-...-verify_good_subclass.py,	
independent_replay.py	ms. 33: relaxed class, good
subclass; replay	
code/12-...-theta_diagnostics.py	ms. 30: theta and envelope
diagnostics	
code/13-...-verify_summand_ratio.py,	
gamma_sum_diagnostics.py	ms. 27: ratio identity; Gamma-sum
diagnostics	
code/14-...-verify_saddle_algebra.py,	
saddle_diagnostics.py	ms. 24: action derivatives; saddle
diagnostics	
code/15-...-verify_exact_identities.py,	
scalar_diagnostics.py	ms. 22: Catalan powers, ratios;
diagnostics	
code/1N-...-build.sh	each manuscript's build of its own
.tex	
data/1N-...-*.json, data/11-...-*.txt	recorded runs and delivery
validation records	
1N-...-SOURCES.md, 1N-...-CHECKS.md	source ledgers; the delivered
checks READMEs	

The programs are shipped byte-identical to the delivery and still use the delivered names: each writes its record beside itself under its delivered name; the checkers of manuscripts 36 and 33 import `model`, which in this directory would load Part I's `different_code/model.py`, and manuscript 36's also checks the SHA-256 of a `model.py` beside itself; each `build.sh` builds the unshipped manuscript. The report's README gives a recipe that restores the delivered layout in a scratch copy, with Part VI's `code/07-polynomial-rarity-model.py` as `model.py`. The exact checkers need only Python's standard library; the diagnostics need NumPy and SciPy, and manuscript 30's also mpmath.

111 Attribution, scope and non-claims

The six manuscripts state their attributions and limits as follows; every sentence is kept, and the bracketed pointers are the merge's.

Manuscript 36. The fixed-cap enumeration setting is studied by Nadler [1] and Mayama-Akita [2]. The endpoint recurrence, injective block kernel, coefficient defect, root corridor, and inverse-logarithmic scalar expansion used here belong to this report (section 101.4). Its contribution is the squared-renewal majorant and the uniform coefficient analysis giving the joint-growth logarithmic bridge. The Fourier and exponential-tilting methods are classical tools; the needed triangular-array estimates are proved rather than assumed from a fixed-law theorem. The note answers a stated repository question and makes no worldwide priority claim. No effective threshold for m , relative probability asymptotic, leading polynomial prefactor, or uniform extension of fixed-index crossover formulas is established. In particular, the squared-renewal envelope has not been proved relatively sharp [by manuscript 36; manuscript 33 proves

it on the band (466), and section 104 wherever $n(\log m)^2/m^2 \rightarrow 0$]. The proof is ordinary mathematics, not formally verified or externally refereed.

Manuscript 33. This report supplies the endpoint recurrence, uniform Catalan-triangle tail, scalar-root expansion, and spectral corridor. Manuscript 36 supplies the squared majorant and the clustered Fourier method at logarithmic precision. The new steps are the weighted cluster-count summation giving the squared-renewal equivalent and the exact good-subclass losses making that envelope relatively sharp. The underlying enumeration context is given by Nadler [1] and Mayama–Akita [2]. All analytic estimates required for the triangular-array equivalent are included. No effective numerical threshold, error rate for the relative equivalent, full fixed-cap prefactor, or extension outside (466) is asserted. In particular, the lower condition makes the cluster-count lattice window diverge, while the upper condition makes the nearby-root replacement harmless. The proof is ordinary mathematics, not formally verified or externally refereed, and no worldwide priority claim is made. [The band (466) is purely asymptotic: $m(\log m)^2 < m^2/(\log m)^2$ only when $m > (\log m)^4$, which fails for $5 \leq m \leq 5503$; the ratio of its ends is 1.39 at $m = 10^4$ and 27 at $m = 10^6$, so no computation can exercise it.]

Manuscript 30. The Gaussian periodization and Jacobi identity are classical [9]. The root and endpoint inputs are credited to this report; the uniform local limit, pointwise bound, and good-subclass inequality were established by manuscript 33, with their required forms and mechanisms stated. The contribution of this continuation is the retained lattice modulation, its stability under the cap shift, and the phase-dependent boundary limits. A related announcement by Li [7] concerns strict growth and descent thresholds. Its full manuscript could not be retrieved during this check, so overlap beyond its public abstract has not been excluded. The priority claim is source-relative, not exhaustive. The result is uniform for each fixed positive lower constant c_0 . It makes no uniform assertion as c_0 decreases to zero, no claim in $n = o(m(\log m)^2)$ [manuscript 24 covers $m(\log m)^{3/2} \ll n$], and no extension of the upper band condition. It is ordinary mathematical research, not formally verified or externally refereed.

Manuscript 27. This report supplies the endpoint recurrence, uniform first-deficiency tail, and scalar-root estimates. Manuscripts 33 and 30 supply the exact good-subclass framework and the clustered Fourier method [which first appears in manuscript 36, uncredited by manuscript 27]. The new contributions are the compact-tilt finite-count selection at the lower scale, the relatively valid Gamma reduction including all small runs, the direct conditional endpoint-layer transfer, and the integer transition law. A related public announcement by Li [7] was inspected, but its full manuscript could not be retrieved; overlap beyond its abstract has not been excluded. The priority claim is source-relative. All constants may depend on the fixed compact c interval. No uniform claim is made as $c_0 \downarrow 0$ or $c_1 \rightarrow \infty$, and no exact finite coefficient formula is asserted. The Gaussian theta expression is not used outside its proved range. The proof is ordinary mathematics, not formally verified or externally refereed.

Manuscript 24. The exact sum (442) is a common expression for the earlier finite Gamma sum at $n \asymp m \log m$, the present two-action regime, and Gaussian periodization above its now sharpened threshold [as equivalents of one coefficient; no term-by-term identity with the Gamma sum is proved, remark 109.5]. It does not extrapolate the elementary Gamma formula to an unproved growing-count range. The cutoff used here can differ from the half-cap cutoff in the theta report; each formula evaluates the same scalar coefficient [remark 108.4]. Every Gaussian phase stated here uses its specified exact cluster law. The analytical argument uses uniform Fourier smoothing, relative tail estimates and the exact good-subclass inequality [and, for H bounded below, manuscript 30's lattice-sum proof, section 108.1]. The saddle-point and Gaussian-periodization methods are classical [8, 9]; the claims concern their uniform implementation and combinatorial transfer for this particular problem. The related public announcement [7] states fixed-cap growth and descent-threshold results; its full linked manuscript was not retrieved, so overlap beyond that abstract is not excluded. These are source-relative research claims. The proofs are not formally verified or externally refereed. No uniformity as $c_0 \downarrow 0$ is asserted.

Manuscript 22. The main new estimate is the relative control of small runs at a shrinking deficit. It uses the exact Catalan-power law and the full three-range argument, rather than moment bounds alone. The Fourier maximal-atom bound is reproved at the changing endpoint scale. These facts distinguish the lower range from the compact- $n/(m \log m)$ theorem, and no compact-tilt theorem at $n \asymp m \log m$ is extrapolated to it. Lagrange inversion and coefficient saddle estimates are classical tools [8]; no novelty is claimed for those general methods. The conclusions are relative to this report and the preceding companions. The related announcement [7] states fixed-cap growth and descent-threshold results, but its full linked manuscript was not retrieved; overlap beyond its abstract is not excluded. The arguments are not formally verified or externally refereed. Uniformity as the phase margin δ tends to zero is not asserted in theorem 109.3, and theorem 109.4 retains its explicit compact-ratio restriction.

112 Where the merge had to choose

- *One Part, not six.* The six main theorems nest or continue one another (manuscript 33's theorem is the $H \rightarrow \infty$ case of manuscript 30's, which is a sub-range of manuscript 24's Gaussian theorem; manuscripts 27 and 22 prove one Gamma formula on adjacent ranges), and the cluster local limit, the squared majorant, the good-subclass inequality and the cap-shortening estimate are each proved in two to five of them. Six Parts would have restated one another; one Part prints each statement once. The base of the Gaussian statements is manuscript 24 (weakest hypotheses), of the Gamma law manuscripts 27 and 22 jointly. Part IX follows Part VIII and precedes Part I's appendices; no number of Parts I–VIII changed.
- *Order.* Shared tools first (sections 102 to 104), then the theorems from coarse to fine precision and from large to small n/m (sections 105 to 109), then the checks, the sources' own scope statements, this list and the questions. The manuscripts' title pages and abstracts are replaced by the opening note of Part IX, section 101.1 and a paragraph of the report's abstract; every claim of each abstract is stated in a theorem or remark.
- *Theorems printed once.* Theorem 108.2 combines manuscript 24's thm:cubic, manuscript 30's thm:envelope and thm:main and manuscript 33's thm:main, each part naming its source and keeping its displays and labels; the three replaced theorem labels are jg:tb:thm:envelope, jg:tb:thm:main, jg:ra:thm:main (not defined). Manuscript 27's intermediate saddle sum is the compact case of theorem 106.1 and is kept with its own tail argument (section 109.1).
- *Proofs printed once, with marked second routes.* The cluster local limit: the general lemma of manuscripts 27 and 24 (lemma 103.1) with its proof; manuscript 36's detailed proof for its variant (lemma 103.5, "second route") and manuscript 33's proof with the pointwise bound (lemma 103.3, "third route"); manuscript 30's summary and manuscript 24's identical proof are replaced by merge notes. The good-subclass inequality: manuscript 33's proof; the four restatements with sketches are references. The squared majorant: manuscript 36's proof, with manuscript 33's grammar as its exact-class reading. Cap shortening: one statement (proposition 104.2) and three routes, each printed in full where its regime is treated. The Gaussian theorem: three regimes of H , three proofs (sections 108.1 to 108.3). The composition estimate: manuscript 22's lemma, with manuscript 27's reduction as its compact-deficit case.
- *Credits added.* Remark 101.1: Part II's lemma 12.1, lemma 14.1 and lemma 15.1, re-proved without credit, and the retargeted citation (75) for the rate in t_m .
- *Additions of the merge.* The convention and range tables, the remarks 101.1, 108.4 and 109.5, the merge notes, the statement (431) and proposition 104.2 collecting what the sources prove five times, and the numerical findings of ProveIt checks made for this write-up (none shipped). No step of a manuscript was changed; manuscript 22's $\Theta(m^{T-1/2}/\log m)$ is printed with $\asymp$ to keep Θ free, which is the same statement.
- *Notation.* Every rename is listed in section 101.3 and table 33; no normalization changed. The cutoff of each statement is the source's and is stated.
- *Citations.* The manuscripts cite this report as an external item ("combined article at commit

7421a4ca60fd”, “Parts I–II”); every such citation is printed as a reference by label. Their citations of one another (by title, or by package name) are printed as references to the merged statements. Their Nadler and Mayama–Akita entries are the report’s; added to the bibliography are [7], [8] and [9].

- *Labels.* The six manuscripts carry 192 labels, all bare generic names (thm:main four times, eq:11t five times, ...; manuscript 36’s prop:upper and eq:endpoint would even collide with Part I’s). Those that name printed material are kept with the prefix jg: and a sub-prefix naming the manuscript. Labels of duplicates printed once, and of displays that restate this report, are replaced by the printed label; the report’s README lists them.
- *Research questions.* None of the manuscripts has a section of questions; their non-claims and the gaps between their ranges are turned into section 113.
- *Stale sentences elsewhere.* The abstract gained a paragraph; dated pointers to Part IX were added where Parts II, III, IV, VI and VIII ask for growing caps or the bridge to the fixed- m regime, and to Part I’s section B; Part II’s research direction 19.5 received the label jg:sec:crem-joint; Part I’s scope box received the dated note on Li’s announcement. Their original wording is kept.

113 Further research questions

The questions below are the merge’s, drawn from the six manuscripts’ non-claims and from the gaps between their ranges.

113.1 Bounded counts with a growing cap

Find the asymptotics when $\lceil (n-1)/m \rceil$ stays bounded while $m \rightarrow \infty$ and m/n does not converge, between Part VIII’s fixed- θ theorems (theorems 91.1 and 91.2) and theorem 109.3, which needs $\ell \rightarrow \infty$.

113.2 Phases near integers

Theorem 109.3 keeps the phase s in $[\delta, 1-\delta]$ and theorem 109.4 keeps the adjacent ratio in a compact subset of $(0, \infty)$. Phases $s = \tau e^{x/(\ell-1)}$ with $|x| \rightarrow \infty$ but $s \rightarrow 0$, and phases just above an integer, are covered by neither.

113.3 Uniformity in the fixed constants

Every theorem of Part IX is uniform only for fixed c_0 (and c_1, δ). Make the dependence explicit as $c_0 \downarrow 0$, $c_1 \rightarrow \infty$ or $\delta \rightarrow 0$, and join the Gamma law of section 109 continuously to the saddle sum of section 106.

113.4 The hand-off to the fixed-cap Perron regime

Every relative statement about $a_n^{(m)}$ needs $n(\log m)^2/m^2 \rightarrow 0$; for larger n only the logarithmic bridge holds. Find a relative equivalent, with its prefactor, when $n(\log m)^2/m^2 \not\rightarrow 0$, and decide whether the squared envelope stays relatively sharp there. This is the part of Part II’s direction 19.5, Part III’s section 32.10 and Part VI’s section 74.6 that Part IX leaves open; it meets Part II’s section 19 questions on poles and a uniform spectral gap.

113.5 Error rates and effective thresholds

None of the six manuscripts gives an error rate for a relative equivalent or an effective threshold in m . The numerical findings recorded above (the $1 + O(1/\log m)$ correction in the Gamma law, the slow cumulant limits at the cubic boundary, the cutoff dependence of the theta phase) indicate where rates are needed.

113.6 The conditioned permutation

The shares in theorem 109.4 and corollary 109.2 and the cluster counts of section 108 are contributions to an equivalent, not statistics of a conditioned permutation. Identify the number and sizes of the macroscopic pieces of a uniform avoider conditioned on $M_n \leq m$ in these regimes, as Part VIII does for fixed θ (corollary 98.1).

A Exact small-bound algebra

For $m = 3$, the certified generating function is $A_3(x) = P_3(x)/Q_3(x)$, where

$$\begin{aligned} P_3(x) &= 1 - x - x^2 + x^3 + 4x^4 - 4x^6 - 3x^7 - 3x^8 \\ &\quad - 5x^9 - 3x^{10} + 2x^{11} + 2x^{12}, \\ Q_3(x) &= (x - 1)(x + 1)(x^5 - x^4 - x^3 + x^2 - 2x + 1) \\ &\quad \cdot (x^6 + x^5 + x^4 + 2x^3 + x^2 - 1). \end{aligned}$$

For $m = 4$, the denominator factors as

$$\begin{aligned} Q_4(x) &= (x - 1) D_U(x) D_V(x), \\ D_U(x) &= x^{10} + x^8 + x^7 - 6x^5 - x^4 - x^3 - x^2 - x + 1, \\ D_V(x) &= x^{12} - x^{11} - x^{10} - x^9 + 5x^8 + 4x^7 + 4x^6 \\ &\quad + 4x^5 + 5x^4 + 3x^3 + x^2 - 1. \end{aligned}$$

Both displayed full denominators have constant term one. The numerator of A_4 , listed from constant term upwards, has coefficient vector

$$(1, -1, -1, -1, 2, 9, 5, -3, -14, -10, -13, -23, \\ -31, -11, 21, 32, 21, 1, -3, -5, -3, -3).$$

It has degree 21 and Q_4 has degree 23. Thus the usual coefficient recurrence from $Q_4 A_4 = P_4$ is valid for all $n \geq 23$ when written using 23 preceding terms. The complete ascending coefficient arrays, full state certificates, and exact checks are in the JSON files.

B Reproduction and certificate formats

The archive is arranged as follows:

article.tex, article.pdf	Main editable source and report
README.md, requirements.txt	Reproduction instructions
code/model.py	Exact endpoint model and combinatorics
code/compute.py	Numerical proposals and exact symbolic solver
code/verify.py	Standard-library independent exact verifier
code/plot_results.py	Recreates the two separate figures
data/sequences.csv	$n = 0..200$, $m = 1..12$
data/growth_constants.csv	Components for $m = 2..20$
data/perron_certificates.json	Rational bounds and positive integer vectors
data/symbolic_certificate_m*.json	Polynomial state certificates, $m = 1..4$
data/generating_functions.json	Reduced rational functions, $m = 1..4$
data/large_m_scalar_bounds.csv	Scalar evaluations through $m = 1000000$

[Added 28 September 2026, batch 38: Part II's code, data and figures carry the prefix 02-centered-remainder- and are listed in section 20.2. Added 29 September 2026, batch 43: Part III's carry the prefix 03-largest-jump- and are listed in section 31.4. Added 29 September 2026, batch 52: Part IV's carry the prefixes 04-macroscopic-jump- and 05-macroscopic-deficits- and are listed in section 46.8. Added 30 September 2026, batch 64: Part V's carry the prefix 06-critical-moments- and are listed in section 61.4. Added 30 September 2026, batch 68: Part VI's carry the prefix 07-polynomial-rarity- and are listed in section 73.4. Added 1 October 2026, batch 71: Part VII's carry the prefix 08-half-size-crossover- and are listed in section 88.2. Added 1 October 2026, batch 72: Part VIII's carry the prefix 09-general-crossovers- and are listed in section 99.2. Added 1 October 2026, batch 72: Part IX's carry the prefixes 10-logarithmic-bridge- to 15-growing-macro-counts- and are listed in section 110.2.]

The primary reproducibility command needs no third-party Python packages:

```
python code/verify.py
```

To regenerate numerical and symbolic proposals, install the packages in `requirements.txt` and run

```
python code/compute.py --max-m 20 --symbolic-max 4 --large-scalars
python code/verify.py
python code/plot_results.py
```

The generator deliberately raises an exception if it cannot convert a numerical proposal to a valid exact certificate. A floating-point result alone is never accepted as a proof of a spectral inequality.

To build the report from the supplied figures:

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The third pass settles page references and the table of contents if needed. The TeX source uses standard packages from a current full TeX Live installation; no font files are distributed.

The exact recurrence uses $(m + 1)^2$ endpoint states and at most $m + 1$ terms per state. Computing all coefficients through length N therefore uses $O(Nm^3)$ integer multiply-add operations, after coefficient precomputation, and $O(Nm^2)$ stored integers in the supplied straightforward implementation. These are arithmetic-operation counts, not bit-complexity bounds. Exact values have growing bit length. A rolling buffer can reduce storage when the complete coefficient table is not needed.

Perron certificates store integers p, D for each radius endpoint p/D , and a positive integer vector in the state order generated by the verifier. Symbolic certificates store all polynomial coefficients in ascending degree. The constant term of each common denominator is checked to be one. The certificate identities prove infinitely many coefficients because the formal system has a unique solution; the finite coefficient checks are additional consistency tests.

C Source status and proof audit

The literature check was performed on September 19, 2026. The inspected primary items are the arXiv v1 versions listed below, including Nadler’s Conjecture 2 and Mayama–Akita’s Theorem 4.14 and Problem P3. The latter paper’s displayed numerical table and statement that its component comparison was unproved were also inspected in the PDF. Targeted searches used the paper titles and identifiers together with strict monotonicity, growth constants, and asymptotic refinements. No later solution was located in that search. This establishes the specific source-based status used here; it is not an assertion that every possible publication has been searched.

The logical dependencies of the new claims are particularly short. The strict-growth theorem uses the exact endpoint recurrence, the two irreducible components, their root characterization, the proper-principal Perron lemma, and the explicit shift of thresholds. This report includes proofs of those prerequisites. The asymptotic theorem instead uses the maximum-decomposition majorant, the injective skew-block construction, the first-entry Catalan formula, and the truncated-tail lemma. In particular, no numerical fit, unproved component-ordering pattern, or formal passage from finite checks to all m is used.

The work has not been formalized in a proof assistant. The integer and polynomial certificates are executable verification of the stated finite algebraic claims; they should not be described as a formal verification of the analytic limiting argument or of the all- m theorem.

References

- [1] N. Nadler, *On 132-Avoiding Permutations with an Adjacency Constraint*, arXiv:2604.22135v1, submitted April 24, 2026. <https://arxiv.org/abs/2604.22135>. Conjecture 2 is in Section 5.2.
- [2] T. Mayama and D. Akita, *Finite-state enumeration of adjacency-constrained 132-avoiding permutations*, arXiv:2605.23519v1, submitted May 22, 2026. <https://arxiv.org/abs/2605.23519>. Relevant locations: Theorem 2.4, Proposition 2.7, Theorems 4.12 and 4.14, and Problem P3 in Section 5.
- [3] S. Janson, *Simply generated trees, conditioned Galton–Watson trees, random allocations and condensation*, arXiv:1112.0510, 2011. <https://arxiv.org/abs/1112.0510>. Cited in Part III as background only.
- [4] S. Janson, *Patterns in random permutations avoiding the pattern 132*, arXiv:1401.5679, 2014. <https://arxiv.org/abs/1401.5679>. Cited in Part III as background only.
- [5] P. Flajolet and A. Odlyzko, *Singularity analysis of generating functions*, SIAM Journal on Discrete Mathematics **3**(2) (1990), 216–240. DOI: 10.1137/0403019. Manuscript 02 records that it did not inspect the PDF of this paper (03-largest-jump-SOURCES.md).
- [6] C. Kerriou and P. Mörters, *The fewest-big-jumps principle and an application to random graphs*, Bernoulli **31**(3) (2025), 2525–2543. DOI: 10.3150/24-BEJ1816. Preprint: <https://arxiv.org/abs/2206.14627>. Cited in Part VI as related context only; no theorem of it is used.
- [7] A. C. Li (Alex Chengyu Li), *Strict Growth and Descent Thresholds for Bounded-Gap 132-Avoiders*, working-paper announcement, 8 September 2026. DOI: 10.5281/zenodo.22648521; <https://crabresearch.com/research/bounded-gap-growth?lang=en>. Cited by manuscripts 30, 27, 24 and 22 of batch 72 (Part IX), whose public abstract they inspected on 1 October 2026; the full manuscript was not inspected, by them or for this report.
- [8] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009, Chapter VIII. <https://algo.inria.fr/flajolet/Publications/AnaCombi/anacombi.html>. Cited in Part IX for classical methods only.
- [9] NIST Digital Library of Mathematical Functions, Chapter 20, §20.2 (theta Fourier series) and §20.7 (identities). <https://dlmf.nist.gov/20.2>; <https://dlmf.nist.gov/20.7>. Cited in Part IX for the classical periodic Gaussian only.